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United States Patent	12417462
Kind Code	B2
Date of Patent	September 16, 2025
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System and method for web-based payments

Abstract

A computer implemented method for enabling a payment transaction in a computer system executed at least partially by a computer of a user connected via a communication network to a remote computer comprising a database of payment-related data, said method comprising the steps of: a) accessing the payment-related data of a user from within the database, wherein a web browser extension on the user's computer is invoked by the user clicking on a web browser extension graphical user interface (GUI) widget displayed on the user's computer, wherein secure network access is established to access the user's payment-related data from within the database; b) presenting the user's payment-related data via a pop-up window displayed on the user's computer, without requiring application programming interface integration with a website or a web application of a payee, and without requiring a form re-direct away from the website or the web application of the payee; c) enabling the payment transaction, wherein the web browser extension invokes an auto-complete function to automatically populate fields of a checkout form presented from the website or the web application of the payee with the user's payment-related data.

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Family ID:	1000008818385
Appl. No.:	16/375326
Filed:	April 04, 2019

Prior Publication Data

Document Identifier	Publication Date
US 20220383325 A1	Dec. 01, 2022

Related U.S. Application Data

continuation parent-doc US 15368813 20161205 ABANDONED child-doc US 16375326

Publication Classification

Int. Cl.: G06Q20/40 (20120101)

U.S. Cl.:

CPC G06Q20/4018 (20130101); G06Q20/4012 (20130101); G06Q20/4016 (20130101);

Field of Classification Search

CPC: G06Q (20/4018); G06Q (20/4012); G06Q (20/4016); G06Q (20/20); G06Q (20/204); G07F (7/0873); G07F (7/0886); G07F (7/0893)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This utility patent application is a continuation of utility patent application Ser. No. 15/368,813, filed on Dec. 5, 2016, which claims the benefit of provisional patent application Ser. No. 62/270,320, filed on Dec. 21, 2015, and is a continuation-in-part of utility patent application Ser. No. 14/161,638, filed on Jan. 22, 2014, which is a continuation-in-part of utility patent application Ser. No. 13/089,313, filed on Apr. 19, 2011, which claims the benefit of provisional patent application Ser. No. 61/349,158, filed on May 27, 2010, which is a continuation-in-part of utility patent application Ser. No. 12/790,829, filed on May 30, 2010, which claims the benefit of provisional patent application Ser. No. 61/221,514, filed on Jun. 29, 2009, which is a continuation-in-part of utility patent application Ser. No. 11/978,067, filed on Oct. 27, 2007, which claims the benefit of provisional patent application Ser. No. 60/967,278, filed on Sep. 1, 2007, and is a continuation-in-part of utility patent application Ser. No. 11/904,283, filed on Sep. 25, 2007, which claims the benefit of provisional patent application Ser. No. 60/876,417, filed on Dec. 20, 2006, which is a continuation-in-part of utility patent application Ser. No. 11/899,384, filed on Sep. 4, 2007, which claims the benefit of provisional patent application Ser. No. 60/850,374, filed on Oct. 6, 2006, which is a continuation-in-part of utility patent application Ser. No. 11/823,020, filed on Jun. 25, 2007, which claims the benefit of provisional patent application Ser. No. 60/818,953, filed on Jul. 5, 2006, which is a continuation-in-part of utility patent application Ser. No. 11/804,167, filed on May 16, 2007, which claims the benefit of provisional patent application Ser. No. 60/802,265, filed on May 19, 2006, which is a continuation-in-part of utility patent application Ser. No. 11/636,231, filed on Dec. 6, 2006, which claims the benefit of provisional patent application Ser. No. 60/749,901, filed on Dec. 12, 2005, and is a

continuation-in-part of utility patent application Ser. No. 11/523,222, filed on Sep. 18, 2006, which claims the benefit of provisional patent application 60/723,859, filed on Oct. 4, 2005, which is a continuation-in-part of utility patent application Ser. No. 11/517,873, filed on Sep. 8, 2006, which claims the benefit of provisional patent application Ser. No. 60/721,831, filed on Sep. 28, 2005, which is a continuation-in-part of utility patent application Ser. No. 11/502,472, filed on Aug. 9, 2006, which claims the benefit of provisional patent application Ser. No. 60/708,979, filed on Aug. 16, 2005, which is a continuation-in-part of utility patent application Ser. No. 11/394,417, filed on Mar. 31, 2006, which claims the benefit of provisional patent application Ser. No. 60/669,941, filed on Apr. 9, 2005, which is a continuation-in-part of utility patent application Ser. No. 11/305,448, filed on Dec. 15, 2005, which claims the benefit of provisional patent application Ser. No. 60/650,367, filed on Feb. 2, 2005, all of which are incorporated herein by reference.

FIELD OF THE INVENTION

(1) The invention relates generally to computer methods and systems to process electronic financial transactions. More specifically, this invention relates to processing financial transactions via a Rule-Module Nexus, and a User Account Registry.

SUMMARY OF THE INVENTION

(2) The present invention provides an improved system and method for processing of financial transactions using a Rule-Module Nexus, and a User Account Registry.

(3) The invention describes a method for financial transactions, comprising the steps of: a) registering a user, wherein a rule-module is registered to a user within a rule-module nexus, said registered rule-module further comprising a pattern data associated with an execution command; b) data-storing in a nexus access token, wherein a unique user code of the user is stored in a portable nexus access token; c) processing a financial transaction, using a nexus access token and a user interface apparatus located remotely from the rule-module nexus, comprising any one of the following: (i) verifying a user, wherein a user's authority to access the rule-module nexus is verified online by a verification platform using verification data provided via the user interface apparatus, said verification data comprising a bid unique user code provided directly from the nexus access token, and; (ii) accessing financial accounts, wherein upon the verification platform verifying the user is authorized to access the rule-module nexus, the rule-module nexus enables access to a plurality of proprietary financial accounts of the user for completing the financial transaction; whereby, a nexus access token and a rule-module nexus provide an authorized user access to a plurality of proprietary financial accounts for processing a financial transaction.

(4) In an illustrative embodiment of the method of the invention, the unique user code comprises any one of the following: no data which uniquely and directly identifies a specific financial account of the user; no live financial account data; no payment enabling data; a rule-module nexus routing code; an electronic positioning system code, and; a network routing instruction for processing the financial transaction via the rule-module nexus.

(5) In an illustrative embodiment of the method of the invention, the pattern data comprises any one of the following: a registration personal verification code; a registration secondary personal verification code; a registration unique user code, and; a plurality of proprietary financial accounts.

(6) In an illustrative embodiment of the method of the invention, the user interface apparatus comprises any one of the following: being conjoined with a transaction terminal; being conjoined with the nexus access token, and; being fully integrated with a transaction terminal.

(7) In an illustrative embodiment of the method of the invention, completing the financial transaction comprises the rule-module nexus preserving a processing preference of a party registered with the rule-module nexus, said party comprising any one of the following: an account issuer; a user, and; a payee.

(8) In an illustrative embodiment of the method of the invention, selecting the financial account further comprises any one of the following: data-entering by the user via a touch-screen; data-

entering by the user via a key pad; data-entering by the user via an audio receiver.

(9) In an illustrative embodiment of the method of the invention, preserving the processing preference of a party registered with the rule-module nexus, said preference comprising any one of the following: invoking criteria predetermined by the party for declining the financial transaction; invoking criteria predetermined by the party for approving the financial transaction, and; invoking criteria predetermined by the party for settlement of the financial transaction.

(10) In an illustrative embodiment of the method of the invention, criteria predetermined by the party for settlement of the financial transaction comprises any one of the following: invoking a proprietary network; invoking a discount rate; invoking an interchange fee; invoking a settlement protocol; invoking a surcharge; invoking a processing partner; enabling a payee to electronically store and batch the financial transaction with at least one other financial transaction for later settlement, and; invoking a time period for settlement.

(11) In an illustrative embodiment of the method of the invention, processing a financial transaction further comprises any one of the following: precluding a global execution command from requiring all financial transactions of all users to bypass a processing preference of a party; precluding a global execution command from requiring all financial transactions of all users to invoke a specific processing preference of a specific party; precluding a global execution command from requiring all financial transactions of all users to use a specific merchant service, and; precluding a global execution command from requiring all financial transactions of all users to use a specific merchant product.

(12) In an illustrative embodiment of the method of the invention, a code is identified as compromised based on an occurrence comprising any one of the following: unusual usage of the code; loss of the code; inaccessibility of the code due to nexus access token damage; fraudulent duplication of the code; unauthorized access to the code, and; coercion of the user.

(13) In an illustrative embodiment of the method of the invention, the compromised code comprises any one of the following: a unique user code; a personal verification code; a verification approval code; a user account registry code; a payee account registry code; a unique payee code; an account issuer verification code; a payee verification code, and; a user interface apparatus hardware verification code.

(14) In an illustrative embodiment of the method of the invention, resolving the compromised code comprises any one of the following: deactivating the compromised code and activating a replacement code, and; verifying the user further comprising providing dual personal verification codes.

(15) In an illustrative embodiment of the method of the invention, activating the replacement code comprises any one of the following: data-storing a replacement unique user code in the nexus access token of the user to replace a compromised unique user code stored therein, and; data-storing a replacement unique user code in a new nexus access token of the user, the new nexus access token replacing a nexus access token of the user storing a compromised unique user code.

(16) In an illustrative embodiment of the method of the invention, the plurality comprises two or more.

(17) In an illustrative embodiment of the method of the invention, a proprietary financial account comprises any one of the following: a financial account having an account issuer which is distinct from an account issuer of another financial account; a financial account having a unique financial account identifier; a financial account having a code identifying an account issuer, as approved by the International Organization for Standardization [(Bank Identification Code, or BIC Code) or (Bank Identification Number, or BIN)]; a financial account having a code identifying an account issuer, as approved by the Society for Worldwide Interbank Financial Telecommunications (SWIFT Address or SWIFT Code), and; a financial account having a code identifying an account issuer, as approved by the American Banker's Association (ABA Routing Code).

(18) In an illustrative embodiment of the method of the invention, the verification platform

electronically compares the user's bid unique user code with a user's registration unique user code, and makes a matching determination for verifying the user's authority to access the rule-module nexus, said matching determination comprising any one of the following: making a positive matching determination verifying that the user is authorized to access the rule-module nexus, and; failing to make a positive matching determination, wherein making a negative matching determination is automatic, verifying that the user is not authorized to access the rule-module nexus.

(19) In an illustrative embodiment of the method of the invention, upon a positive matching determination, the verification platform issues a verification approval code invoking a rule-module of the user in the rule-module nexus.

(20) In an illustrative embodiment of the method of the invention, the verification approval code invokes a user account registry code identifying a user account registry, said user account registry code comprising any one of the following: no data directly identifying a specific financial account of the user, and; no data identifying a primary financial account of the user for all financial transactions of the user.

(21) In an illustrative embodiment of the method of the invention, the user account registry comprises a plurality of proprietary financial accounts of the user.

(22) In an illustrative embodiment of the method of the invention, the verification approval code comprises any one of the following: a static code identifying the user account registry; a dynamic code comprising a unique code for storage and retrieval of data relating to processing of the financial transaction, and; a static code and a dynamic code.

(23) In an illustrative embodiment of the method of the invention, a rule-module of the user is modified by parties authorized by the rule-module nexus, said parties comprising any one of the following: the user; the rule-module nexus; a payee; an account issuer, and; a third-party with predetermined authorization.

(24) In an illustrative embodiment of the method of the invention, modifying a rule-module further comprises any one of the following: registering, deleting, adding pattern data; registering, deleting, adding execution commands, and; registering, deleting, adding associations between pattern data and execution commands.

(25) In an illustrative embodiment of the method of the invention, the pattern data comprises any one of the following: personal legal name; a private code; a driver's license number; a unique user code; a physical POS unique user code; a virtual POS unique user code; a personal verification code; a secondary personal verification code; an emergency code; a plurality of proprietary financial accounts; demographic information; an email address; social security number; a mother's maiden name; a facial photograph; an Internet browsing pattern; a telephone number; a mailing address; a purchasing pattern; an authorized subordinated user; electronic data usage patterns; employee status; job title; data on user behavior patterns; a credit score; a digital certificate; a network credential; an Internet protocol address; a digital signature; an encryption key; an instant messaging address; personal medical records; an electronic audible signature, and; an electronic visible signature.

(26) In an illustrative embodiment of the method of the invention, the execution command comprises invoking at least one of the following: accessing the rule-module nexus; accessing a user account registry; accessing a financial account; authorizing a subordinated user to access a financial account of the user; presenting a financial account of the user; completing the financial transaction; authorizing settlement of the online transaction; presenting the pattern data; presenting the execution command; presenting the rule-module; notifying an emergency authority upon rule-module nexus receiving an emergency code of the user; accessing a third-party database; accessing an account issuer database, and; accessing a payee database.

(27) In an illustrative embodiment of the method of the invention, invoking the rule-module comprises any one of the following: accessing a plurality of rule-modules in the rule-module nexus;

accessing a plurality of proprietary financial accounts; authorizing a subordinated account user authority; accessing a third-party computer via the rule-module nexus.

(28) In an illustrative embodiment of the method of the invention, the unique user code comprises any one of the following: a dynamic code which changes periodically based on predetermined criteria synchronized with the verification platform, and; a static code which remains constant based on a predetermined code synchronized with the verification platform.

(29) In an illustrative embodiment of the method of the invention, the personal verification code comprises any one of the following: an alpha-numeric sequence selected by the user; an alpha-numeric sequence selected by the rule-module nexus; an alpha-numeric sequence selected by an account issuer; a visible image; an audible sound.

(30) In an illustrative embodiment of the method of the invention, verifying the user further comprises detecting rule-module nexus fraud, wherein criteria predetermined by a fraud prevention platform are invoked for detecting fraud by the user involving the rule-module nexus, said criteria comprising any one of the following: unusual usage of bid verification data; unusual modifying of a rule-module, and; unusual accessing of a financial account.

(31) In an illustrative embodiment of the method of the invention, upon a determination by the rule-module nexus that the user has committed fraud involving the rule-module nexus, data of the user is registered with a fraud prevention platform, said data comprising any one of the following: a pattern data; an execution command, and; a rule-module.

(32) In an illustrative embodiment of the method of the invention, the private code, registered to the user, distinct from a personal verification code and not used in verifying the user, is presented to the user via the rule-module nexus for verifying to the user that the authentic rule-module nexus has been accessed.

(33) In an illustrative embodiment of the method of the invention, the private code is registered to the user in the rule-module nexus by a party, said party comprising any one of the following: the user; the rule-module nexus, and; an account issuer.

(34) In an illustrative embodiment of the method of the invention, during the financial transaction processing, the emergency code, distinct from a personal verification code and not used in verifying the user, is provided by the user to the user interface apparatus for sending an alert via the rule-module nexus of an emergency comprising any one of the following: the bid verification data being compromised; the nexus access token being compromised, and; the user being coerced.

(35) In an illustrative embodiment of the method of the invention, the emergency code comprises any one of the following: an alternate personal verification code; an alpha-numeric code; a visible image, and; an audible signal.

(36) In an illustrative embodiment of the method of the invention, alerting of emergency invokes an execution command via the rule-module nexus, comprising any one of the following: presenting a visible display of predetermined emergency data to the user; presenting an audible signal of predetermined emergency data to the user; alerting an emergency authority, and; identifying a compromised code.

(37) In an illustrative embodiment of the method of the invention, the visible display comprises any one of the following: a false financial account; a false financial data within a financial account, and; confirming an emergency authority has been contacted.

(38) In an illustrative embodiment of the method of the invention, the audible signal comprises any one of the following: a false financial account; a false financial data within a financial account, and; confirming an emergency authority has been contacted.

(39) In an illustrative embodiment of the method of the invention, resolving the compromised code further comprises deactivating the unique user code and activating a secondary personal verification code.

(40) In an illustrative embodiment of the method of the invention, providing dual personal verification codes, further comprises the rule-module nexus enabling the user on a limited basis to

provide a bid secondary personal verification code in replacement of the user's unique user code.
(41) An illustrative embodiment of the method of the invention further comprises any one of the following: a) calling by the user from a predetermined first phone number to a predetermined second phone number; b) data-entering by the user of the personal verification code of the user;
(42) c) invoking by the user of a secondary personal verification code, and; d) notifying the user that the secondary personal verification code has been activated for providing dual personal verification codes when verifying the user.

(43) In an illustrative embodiment of the method of the invention, invoking by the user further comprises any one of the following: activating a registration secondary personal verification code of the user; creating by the user of a secondary personal verification code, and; accepting by the user of an offered secondary personal verification code.

(44) An illustrative embodiment of the method of the invention further comprises any one of the following: a) emailing by the user from a predetermined internet protocol address to a predetermined web site;

(45) b) data-entering by the user of the personal verification code of the user; c) invoking by the user of a secondary personal verification code, and; d) notifying the user that the secondary personal verification code has been activated for providing dual personal verification codes when verifying the user.

(46) In an illustrative embodiment of the method of the invention, invoking by the user further comprises any one of the following: activating a registration secondary personal verification code of the user; creating by the user of a secondary personal verification code, and; accepting by the user of an offered secondary personal verification code.

(47) In an illustrative embodiment of the method of the invention, providing dual personal verification codes further comprises the bid personal verification code and the bid secondary personal verification code, both provided directly by the user to the user interface apparatus, being electronically compared by the verification platform with a registration personal verification code and a registration secondary personal verification code, to make a matching determination for verifying the user's authority to access the rule-module nexus.

(48) In an illustrative embodiment of the method of the invention, the limited basis comprises any one of the following: a predetermined time period; predetermined financial account access when using the secondary personal verification code; predetermined frequency for usage for using the secondary personal verification code, and; predetermined geographic area for using the secondary personal verification code.

(49) In an illustrative embodiment of the method of the invention, the secondary personal verification code comprises any one of the following: an alpha-numeric sequence selected by the user; an alpha-numeric sequence selected by the rule-module nexus; an alpha-numeric sequence selected by an account issuer; a visible image; an audible sound.

(50) In an illustrative embodiment of the method of the invention, processing the financial transaction further comprises any one of the following: a credit transaction; a debit transaction; a scrip transaction; a rewards transaction; an electronic check transaction; a private label transaction; a stored value transaction; an electronic benefits transfer transaction; a brokerage trade transaction; invoking a surcharge to a transaction based on predetermined criteria; a buyer-seller exchange wherein a user's financial account balance is adjusted and a payee's financial account is correspondingly adjusted; an intra-account transfer transaction between financial accounts of the user without a buyer-seller exchange; redeeming a pre-paid ticket transaction for venue admittance without a buyer-seller exchange, and; redeeming a pre-paid membership benefit transaction for venue admittance without a buyer-seller exchange.

(51) In an illustrative embodiment of the method of the invention, the venue comprises any one of the following: a concert hall; a sports stadium; a movie theatre; a live-action theatre; an airplane; a train; a bus; a boat; a dance club; a restaurant; a garage; an office building; a health club; an

apartment building; a medical facility; a toll booth, and; a dining club.

(52) In an illustrative embodiment of the method of the invention, venue admittance comprises displaying a facial photograph of the user, wherein upon the verification platform making a positive matching determination that the user is authorized to access the rule-module nexus, the rule-module nexus transmits the user's registered facial photograph for display to a third-party present during the financial transaction for visually verifying that the user's actual face is sufficiently similar to the user's displayed facial photograph to permit venue admittance.

(53) In an illustrative embodiment of the method of the invention, accessing a financial account comprises any one of the following: accessing a rule-module associated with a financial account; querying data associated with a financial account; retrieving data from a financial account; querying data of a financial account via accessing a third-party computer; accessing a third-party computer to retrieve data from a financial account; presenting a visible electronic image of a financial account; presenting visible electronic text communication of a financial account; presenting a visible J2ME-enabled communication; presenting an audible signal of a financial account; adjusting the balance in a financial account by making a credit to a financial account; transmitting an electronic positioning system communication; transmitting an electronic navigating system communication; transmitting an electronic mapping communication; an electronic guiding system communication; transmitting an electronic text message of a financial account, and; adjusting the balance in a financial account by making a debit from a financial account.

(54) In an illustrative embodiment of the method of the invention, the data-storing in the nexus access token comprises any one of the following: storing no data which uniquely and directly identifies a specific financial account of the user; storing no live financial account data, and; storing no payment-enabling data.

(55) An illustrative embodiment of the method of the invention further comprises verifying the user, further comprising displaying a facial photograph of the user, wherein upon the verification platform making a positive matching determination that the user is authorized to access the rule-module nexus, the rule-module nexus transmits the registered facial photograph of the user for display to a third-party present during the financial transaction, for visually verifying that the user's actual face is sufficiently similar to the user's displayed facial photograph to permit the financial transaction.

(56) An illustrative embodiment of the method of the invention, further comprises data-storing in a user interface apparatus, wherein a hardware verification code, registered with the rule-module nexus and unique to the user interface apparatus, is stored in the user interface apparatus.

(57) In an illustrative embodiment of the method of the invention, a rule-module is registered to a payee, said registered rule-module comprising a pattern data associated with an execution command.

(58) In an illustrative embodiment of the method of the invention, a pattern data comprises any one of the following: the payee's legal name; a user interface apparatus hardware verification code; an employer identification number; financial account access authorization fields; a payee account registry; a unique payee code; a payee verification code; a transaction terminal identification code; a user interface apparatus identification code; an emergency code; a financial account; an email address; a telephone number; a mailing address; authority of at least one employee of the payee; a digital certificate; a network credential; an Internet protocol address; a digital signature; an encryption key; electronic audible signature, and; an electronic visible signature.

(59) In an illustrative embodiment of the method of the invention, the execution command comprises any one of the following: accessing a user's financial account; processing a user's financial transaction; presenting selected data from user's rule-module data, and; alerting an emergency authority.

(60) In an illustrative embodiment of the method of the invention, processing the financial transaction further comprises verifying the payee, wherein the verification platform electronically

compares bid verification data of the payee with registered verification data of a payee, and makes a matching determination for verifying the payee's authority to access the rule-module nexus, said matching determination comprising any one of the following: making a positive matching determination verifying that the payee is authorized to access the rule-module nexus, and; failing to make a positive matching determination, wherein making a negative matching determination is automatic, verifying that the payee is not authorized to access the rule-module nexus.

(61) In an illustrative embodiment of the method of the invention, the bid verification data comprises any registered pattern data of the payee enabling the verification platform to verify that the payee is authorized to access the rule-module nexus.

(62) In an illustrative embodiment of the method of the invention, upon the verification platform making a positive matching determination of the payee's authority to access the rule-module nexus, an execution command of the payee is invoked, comprising any one of the following: authorizing a field for accessing rule-modules in the rule-module nexus; accepting a subordinated payee; authorizing a field for accessing a third-party computer being accessed by the rule-module nexus, and; invoking a processing preference of the payee.

(63) In an illustrative embodiment of the method of the invention, upon a determination by the rule-module nexus that the payee has committed fraud involving the rule-module nexus, data of the payee data is registered with a fraud prevention platform, said data comprising any one of the following: a pattern data; an execution command, and; a rule-module.

(64) In an illustrative embodiment of the method of the invention, a platform is a computer-based set of related data subject to electronic processing with software using predetermined criteria associated with the rule-module nexus, said processing comprising any one of the following: data storage; data queries; data retrieval, and; data modification.

(65) In an illustrative embodiment of the method of the invention, the rule-module nexus is remotely located from the user, the user interface apparatus, and the nexus access token.

(66) In an illustrative embodiment of the method of the invention, processing the financial transaction further comprises accessing the remotely located rule-module nexus via a network comprising any one of the following: a cable network; a wireless network; a land-line phone network; the Internet; an intranet; a local area network, a wide area network; an electronic positioning network; a satellite network, and; an X.25 network.

(67) In an illustrative embodiment of the method of the invention, processing the financial transaction further comprises transmitting transaction data via the rule-module nexus, said transaction data comprising any one of the following: pricing information; a list of goods and services; a verification data of the user; a verification data of a payee; a date or time; a location of the user interface apparatus; a location of the nexus access token; an electronic positioning code; a unique payee code; a hardware verification code of the user interface apparatus; the name of a payee; an invoice number, and; transaction settlement instruction.

(68) In an illustrative embodiment of the method of the invention, the payee and the account issuer are identical.

(69) In an illustrative embodiment of the method of the invention, the predetermined time period for settlement comprises any one of the following: an immediate adjustment of a balance in the user's financial account; a delayed adjustment of a balance in the user's financial account, and; a time interval for repeated adjustment of a balance in the user's financial account.

(70) In an illustrative embodiment of the method of the invention, an account issuer of a financial account comprises any one of the following: an owner, comprising a party that owns the financial account; a servicer, comprising a party that services the financial account on behalf of an account owner; an agent, comprising a party that acts on behalf of another party associated with the financial account, and; a fiduciary, comprising a party with the fiduciary duty to the financial account.

(71) In an illustrative embodiment of the method of the invention, the account issuer comprises any

one of the following: a bank; a merchant; a scrips provider; credit account organization; the rule-module nexus; a government agency; an insurance company; a brokerage firm; a reward incentives provider; a services provider; a product provider, and; an internet payment provider.

(72) In an illustrative embodiment of the method of the invention, the financial account is a computerized set of related financial data having a predetermined legal monetary value for usage comprising any one of the following: purchasing a product; purchasing of a service; exchanging a product; exchanging a service, and; exchanging for other financial data of equivalent monetary value.

(73) In an illustrative embodiment of the method of the invention, the financial account comprises any: checking account; credit account comprising Visa®, MasterCard® and American Express®; reward incentives account; insurance account; brokerage account; savings account; scrip incentives account; pre-paid account; pre-paid ticket; membership benefits account; private label credit account; services barter account; product barter account, and; internet payment account.

(74) In an illustrative embodiment of the method of the invention, an incentives account comprises financial data, comprising any one of the following: gift certificate units; stored-value units; electronic coupon units having a predetermined monetary value; minutes of telephone calling time; miles towards earning a free airplane flight; accruing units of a predetermined monetary value exchangeable for a product, and; accruing units of a predetermined monetary value exchangeable for a service.

(75) In an illustrative embodiment of the method of the invention, accessing financial accounts further comprises presenting a financial account to the user, comprising any one of the following: a visible display of an electronic visible signature; a visible text message, and; an audible signal of an electronic audible signature.

(76) An illustrative embodiment of the method of the invention, further comprises ranking the plurality of proprietary financial accounts within a user account registry, wherein predetermined criteria is used for tagging and ranking the plurality of proprietary financial accounts in a certain order.

(77) In an illustrative embodiment of the method of the invention, the predetermined criteria for the ranking further comprises any one of the following: improving a transaction benefit for an account issuer, and; improving a transaction benefit for the user, and; improving a transaction benefit for a payee.

(78) In an illustrative embodiment of the method of the invention, improving a transaction benefit further comprises any one of the following: increasing efficiency; increasing speed; increasing profit; increasing security; decreasing cost; increasing reward incentives, and; invoking a surcharge predetermined by the user.

(79) In an illustrative embodiment of the method of the invention, the ranking further comprises presenting to the user a display comprising any one of the following: a default financial account; a plurality of financial accounts, with visibly distinct indicators for the respective rankings of the financial accounts, and; a plurality of financial accounts displayed in sequence as a function of their respective rankings.

(80) In an illustrative embodiment of the method of the invention, the electronic visible signature is an electronic visible image comprising any one of the following: streaming media; a video clip; a moving image; a holographic display; a static display; a dynamic display; an alpha-numeric display, and; a textual display.

(81) In an illustrative embodiment of the method of the invention, ranking proprietary financial accounts further comprises presenting: a plurality of financial accounts, with audibly distinct indicators for the respective rankings of the financial accounts.

(82) In an illustrative embodiment of the method of the invention, the electronic audible signature is an audible signal comprising any one of the following: a musical fragment; speech; phonation, and; a song.

- (83) In an illustrative embodiment of the method of the invention, the surcharge comprises any one of the following: an additional financial amount debited from the financial account selected by the user, and credited to a different financial account.
- (84) In an illustrative embodiment of the method of the invention, the surcharge comprises any one of the following: a fixed financial amount, and; a variable financial amount.
- (85) In an illustrative embodiment of the method of the invention, the variable financial amount comprises a formula for calculating the surcharge as a function of a predetermined factor comprising any one of the following: income of the user; a credit score of the user; a financial amount of the transaction; time; a purchasing frequency of the user; a balance in a financial account of the user; an economic indicator, and; a financial objective of the user.
- (86) In an illustrative embodiment of the method of the invention, processing the financial transaction further comprises the rule-module nexus accessing a third-party computer, comprising any one of the following: verifying a user; detecting a rule-module nexus fraud; registering a user fraud; registering an account issuer fraud; registering a payee fraud; alerting of an emergency; resolving a compromised code; accessing a financial account; settlement of the financial transaction, and; completing the financial transaction.
- (87) In an illustrative embodiment of the method of the invention, accessing financial accounts further comprises verifying resources, wherein upon a selection of a financial account by the user, an electronic determination is made if the selected financial account of the user has sufficient resources for settlement of the financial transaction.
- (88) An illustrative embodiment of the method of the invention further comprises settlement of the financial transaction, comprising any one of the following: invoking a debit of financial data within the selected financial account of the user, and; invoking a credit of financial data within the selected financial account of the user.
- (89) In an illustrative embodiment of the method of the invention, the data-storing in the nexus access token further comprises storing the unique user code in the nexus access token in memory comprising any one of the following: read only memory, and; read and write memory.
- (90) In an illustrative embodiment of the method of the invention, processing the financial transaction further comprises any one of the following: the rule-module nexus accessing a third-party computer; the rule-module nexus accessing the user interface apparatus, and; the rule-module nexus accessing a transaction terminal, and; the rule-module nexus accessing the user interface apparatus.
- (91) In an illustrative embodiment of the method of the invention, accessing financial accounts invokes an execution command comprising any one of the following: querying the user's financial account balance; invoking an authorization a field within the rule-module nexus; invoking a user's rewards program; invoking a micro-merchandising advertising and upsell program; invoking an intelligent tracking and extrapolating agent, and; invoking an automated User notification program.
- (92) In an illustrative embodiment of the method of the invention, the automated User notification program invokes an outgoing communication comprising any one of the following: sending a wireless phone text message; mailing a paper notice; sending a fax; a phone number dialing, and; sending a page.
- (93) In an illustrative embodiment of the method of the invention, the transaction terminal comprises any one of the following: a wireless telephone; a wireless pager; a personal computer; a merchant point of sale register; a vending machine; a venue admittance device; a personal digital assistant; an internet kiosk; a land line telephone; a television, and; a digital music player.
- (94) In an illustrative embodiment of the method of the invention, the transaction terminal further comprises any one of the following: a data-entering touch screen; a data-entering key pad; a visible display screen; an audible signal speaker, and; an audio receiver.
- (95) In an illustrative embodiment of the method of the invention, the user interface apparatus comprises any one of the following: a contactless proximity communications interrogator; an

electronic positioning system communications element; an electronic navigating system communications element; an electronic mapping system communications element; an electronic guiding system communications element; a contactless proximity communications data-reader and data-writer; a magnetic stripe reader; a magnetic ink character reader; a bar code reader; a data-entering touch screen; a data-entering key pad; a visible display screen; an audible signal speaker; an audio receiver.

(96) In an illustrative embodiment of the method of the invention, the nexus access token further comprises communicating capability, comprising any one of the following: electronically readable bar-code imprint; electronically readable alpha-numeric code; radio signal; infrared signal; digital signal; cellular signal; visible display; and vibrational signal; acoustic sound signal; microwave signal; satellite signal; X-ray signal; pulsar signal; television signal; electronically readable magnetic stripe, and; photo signal.

(97) In an illustrative embodiment of the method of the invention, the nexus access token is conjoined with a device comprising any one of the following: wireless telephone, key fob; wireless pager; personal digital assistant; wearable ornament; digital media player; refillable container; removeably implantable computer chip; a user interface apparatus, and; door key.

(98) In an illustrative embodiment of the method of the invention, the wearable ornament comprises any of the following: jewelry, and; clothing.

(99) In an illustrative embodiment of the method of the invention, the refillable container comprises any of the following: a beverage container, and; a gasoline container.

(100) In an illustrative embodiment of the method of the invention, processing the financial transaction further comprises the real-time elapsed in which the transaction terminal remains online communicating with the remotely located rule-module nexus as measured from verifying the user to selecting the financial account.

(101) In an illustrative embodiment of the method of the invention, completing the financial transaction comprises any one of the following: declining the financial transaction, and; settlement of the financial transaction.

(102) In an illustrative embodiment of the method of the invention, declining the financial transaction is invoked by a party comprising any one of the following: the user; an account issuer; a payee, and; the rule-module nexus.

(103) In an illustrative embodiment of the method of the invention, declining the financial transaction is invoked based upon predetermined criteria comprising any one of the following: insufficient financial data within the financial account; a credit score of the user; geography; usage frequency; usage recency; demographics of the user; financial amount of the financial transaction; a user fraud; a payee fraud; an account issuer fraud, and; a compromised code.

(104) In an illustrative embodiment of the method of the invention, accessing the financial account further comprises determining resources via an account issuer, comprising any one of the following: a positive determination wherein the selected financial account has sufficient resources, outputting an approval of the financial account for settlement of the financial transaction; a negative determination wherein the selected financial account has insufficient resources, outputting a decline of the financial account for settlement, whereupon at least one other financial account of the user is automatically displayed to the user by the transaction terminal based upon predetermined criteria.

(105) In an illustrative embodiment of the method of the invention, a rule-module comprises any one of the following: pattern data associated with a plurality of execution commands; a plurality of pattern data associated with an execution command, and; a plurality of pattern data associated with a plurality of execution commands.

(106) In an illustrative embodiment of the method of the invention, the rule-module nexus comprises any one of the following: a master rule-module nexus comprising all pattern data, execution commands, and rule-modules registered by users, payees and by account issuers, and; a

subset rule-module nexus comprising a subset of all pattern data, execution commands, and rule-modules registered by users, by payees, and by account issuers.

(107) In an illustrative embodiment of the method of the invention, the subset rule-module nexus comprises a subset of data selected as a function of predetermined criteria, comprising any one of the following: a credit score of the user; geography; usage frequency; usage recency; demographics of the user; financial amount of the financial transaction; a user fraud; a payee fraud; an account issuer fraud, and; a compromised code.

(108) In an illustrative embodiment of the method of the invention, registering a rule-module further comprises checking user re-registration, wherein the registered rule-module of the user is compared against a previously registered rule-module, wherein a match alerts the rule-module nexus that the user is attempting a re-registration.

(109) An illustrative embodiment of the method of the invention, further comprises notifying a user, wherein upon completing the financial transaction, the transaction terminal presents notification to the user of the financial transaction results, comprising any one of the following: notification of a decline of the financial transaction, and; notification of settlement of the financial transaction.

(110) In an illustrative embodiment of the method of the invention, presenting notification comprises any one of the following: a visible display; an audible signal; and a printed receipt.

(111) In an illustrative embodiment of the method of the invention, the emergency authority comprises any one of the following: a government agency, and; a private entity.

(112) In an illustrative embodiment of the method of the invention, registering a rule-module further comprises aggregating financial account maintenance, wherein the rule-module nexus aggregates maintenance services for the plurality of proprietary financial accounts of the user, said maintenance services comprising any one of the following: invoicing the user; collecting invoice payments from the user; reconciling scrip incentive financial data; reconciling reward incentive financial data; being an agent for intelligent data tracking on behalf of the user, and; being an agent for extrapolating data on behalf of the user.

(113) An illustrative embodiment of the method of the invention, further comprises verifying the user interface apparatus, wherein the verification platform electronically compares a bid hardware verification code of the user interface apparatus with a previously registered hardware verification code, to make a matching determination for verifying the authenticity of the user interface apparatus via the rule-module nexus, said matching determination comprising any one of the following: making a positive matching determination verifying to the rule-module nexus that the user interface apparatus is authentic, and; failing to make a positive matching determination, wherein making a negative matching determination is automatic, verifying to the rule-module nexus that the user interface apparatus is not authentic.

(114) The invention also describes a system for financial transactions, comprising any one of the following:

(115) a) a registration platform, configured within a rule-module nexus to comprise registering a rule-module to a user, said rule-module further comprising a pattern data associated with an execution command; b) a nexus access token, configured to comprise: portability, and; storing a unique user code of the user; c) a financial transaction processing platform, comprising any one of the following: (i) a nexus access token; (ii) a user interface apparatus, conjoined with a transaction terminal and located remotely from the rule-module nexus, configured to gather bid verification data of the user, said verification data comprising a bid unique user code provided directly from the nexus access token, and; (iii) a verification platform configured to verify a user online using the bid verification data, wherein a user's authority to access the rule-module nexus is verified, and; (iv) a user account registry platform configured to access financial accounts via the rule-module nexus, wherein upon the verification platform verifying the user is authorized to access the rule-module nexus, the user is provided online access via the rule-module nexus to a plurality of proprietary

financial accounts of the user for completing the financial transaction; whereby, a nexus access token and a rule-module nexus provide an authorized user access to a plurality of proprietary financial accounts for processing a financial transaction.

(116) In an illustrative embodiment of the system of the invention, the unique user code comprises any one of the following: no data which uniquely and directly identifies a specific financial account of the user; no live financial account data; no payment enabling data; a rule-module nexus routing code; an electronic positioning system code, and; a network routing instruction for processing the financial transaction via the rule-module nexus.

(117) In an illustrative embodiment of the system of the invention, the pattern data comprises any one of the following: a registration personal verification code; a registration secondary personal verification code; a registration unique user code, and; a plurality of proprietary financial accounts.

(118) In an illustrative embodiment of the system of the invention, the user interface apparatus comprises any one of the following: being conjoined with a transaction terminal; being conjoined with the nexus access token, and; being fully integrated with a transaction terminal.

(119) In an illustrative embodiment of the system of the invention, the rule-module nexus is configured to preserve a processing preference of a party registered with the rule-module nexus, said party comprising any one of the following: an account issuer; a user, and; a payee.

(120) In an illustrative embodiment of the system of the invention, the transaction terminal is configured for a financial account to be selected by the user, comprising any one of the following: a touch-screen; a key pad; data—an audio receiver.

(121) In an illustrative embodiment of the system of the invention, preserving the processing preference of the party comprises any one of the following: invoking criteria predetermined by the party for declining the financial transaction; invoking criteria predetermined by the party for approving the financial transaction, and; invoking criteria predetermined by the party for settlement of the financial transaction.

(122) In an illustrative embodiment of the system of the invention, criteria predetermined by the party for settlement of the financial transaction comprises any one of the following: invoking a proprietary network; invoking a discount rate; invoking an interchange fee; invoking a settlement protocol; invoking a surcharge; invoking a processing partner; enabling a payee to electronically store and batch the financial transaction with at least one other financial transaction for later settlement, and; invoking a time period for settlement.

(123) In an illustrative embodiment of the system of the invention, the financial transaction processing platform is configured to comprise any one of the following: precluding a global execution command from requiring all financial transactions of all users to bypass a processing preference of a party; precluding a global execution command from requiring all financial transactions of all users to invoke a specific processing preference of a specific party; precluding a global execution command from requiring all financial transactions of all users to use a specific merchant service, and; precluding a global execution command from requiring all financial transactions of all users to use a specific merchant product.

(124) An illustrative embodiment of the system of the invention, further comprises a compromised code identification platform, configured to identify a code as compromised based on an occurrence comprising any one of the following: unusual usage of the code; loss of the code; inaccessibility of the code due to nexus access token damage; fraudulent duplication of the code; unauthorized access to the code, and; coercion of the user.

(125) In an illustrative embodiment of the system of the invention, the compromised code comprises any one of the following: a unique user code; a personal verification code; a verification approval code; a user account registry code; a payee account registry code; a unique payee code; an account issuer verification code; a payee verification code, and; a user interface apparatus hardware verification code.

(126) An illustrative embodiment of the system of the invention, further comprises a compromised

code resolution platform configured to comprise any one of the following: deactivating the compromised code and activating a replacement code, and; verifying the user by providing dual personal verification codes.

(127) In an illustrative embodiment of the system of the invention, activating the replacement code comprises any one of the following: data-storing a replacement unique user code in the nexus access token of the user to replace a compromised unique user code stored therein, and; data-storing a replacement unique user code in a new nexus access token of the user, the new nexus access token replacing a nexus access token of the user storing a compromised unique user code.

(128) In an illustrative embodiment of the system of the invention, the plurality comprises at least two.

(129) In an illustrative embodiment of the system of the invention, a proprietary financial account comprises any one of the following: a financial account having an account issuer which is distinct from an account issuer of another financial account; a financial account having a unique financial account identifier; a financial account having a code identifying an account issuer, as approved by the International Organization for Standardization [(Bank Identification Code, or BIC Code) or (Bank Identification Number, or BIN)]; a financial account having a code identifying an account issuer, as approved by the Society for Worldwide Interbank Financial Telecommunications (SWIFT Address or SWIFT Code), and; a financial account having a code identifying an account issuer, as approved by the American Banker's Association (ABA Routing Code);

(130) In an illustrative embodiment of the system of the invention, the verification platform further comprises being configured to electronically compare the user's bid unique user code with a user's registration unique user code, and make a matching determination for verifying the user's authority to access the rule-module nexus, said matching determination comprising any one of the following: making a positive matching determination verifying that the user is authorized to access the rule-module nexus, and; failing to make a positive matching determination, wherein making a negative matching determination is automatic, verifying that the user is not authorized to access the rule-module nexus.

(131) In an illustrative embodiment of the system of the invention, upon a positive matching determination, the verification platform issues a verification approval code invoking a rule-module of the user in the rule-module nexus.

(132) In an illustrative embodiment of the system of the invention, the verification approval code invokes a user account registry code identifying a user account registry platform, said user account registry code comprising any one of the following: no data directly identifying a specific financial account of the user, and; no data identifying a primary financial account of the user for all financial transactions of the user.

(133) In an illustrative embodiment of the system of the invention, the account registry platform comprises a plurality of proprietary financial accounts of the user.

(134) In an illustrative embodiment of the system of the invention, the verification approval code comprises any one of the following: a static code identifying the user account registry; a dynamic code comprising a unique code for storage and retrieval of data relating to processing of the financial transaction, and; a static code and a dynamic code.

(135) An illustrative embodiment of the system of the invention, further comprises a rule-module modification platform, configured for a rule-module of the user to be modified by parties authorized by the rule-module nexus, said parties comprising any one of the following: the user; the rule-module nexus; a payee; an account issuer; and a third-party with predetermined authorization.

(136) In an illustrative embodiment of the system of the invention, modifying a rule-module further comprises any one of the following: registering, deleting, adding pattern data; registering, deleting, adding execution commands, and; registering, deleting, adding associations between pattern data and execution commands.

(137) In an illustrative embodiment of the system of the invention, the pattern data comprises any

one of the following: personal legal name; a private code; a driver's license number; a unique user code; a physical POS unique user code; a virtual POS unique user code; a personal verification code; a secondary personal verification code; an emergency code; a plurality of proprietary financial accounts; demographic information; an email address; social security number; a mother's maiden name; a facial photograph; an Internet browsing pattern; a telephone number; a mailing address; a purchasing pattern; an authorized subordinated user; electronic data usage patterns; employee status; job title; data on user behavior patterns; a credit score; a digital certificate; a network credential; an Internet protocol address; a digital signature; an encryption key; an instant messaging address; personal medical records; an electronic audible signature, and; an electronic visible signature.

(138) In an illustrative embodiment of the system of the invention, the execution command comprises invoking at least one of the following: accessing the rule-module nexus; accessing a user account registry; accessing a financial account; authorizing a subordinated user to access a financial account of the user; presenting a financial account of the user; completing the financial transaction; authorizing settlement of the online transaction; presenting the pattern data; presenting the execution command; presenting the rule-module; notifying an emergency authority upon rule-module nexus receiving an emergency code of the user; accessing a third-party database; accessing an account issuer database, and; access a payee database.

(139) In an illustrative embodiment of the system of the invention, invoking the rule-module comprises any one of the following: accessing a plurality of rule-modules in the rule-module nexus; accessing a plurality of proprietary financial accounts; authorizing a subordinated account user authority; accessing a third-party computer via the rule-module nexus.

(140) In an illustrative embodiment of the system of the invention, the unique user code comprises any one of the following: a dynamic code which changes periodically based on predetermined criteria synchronized with the verification platform, and; a static code which remains constant based on a predetermined code synchronized with the verification platform.

(141) In an illustrative embodiment of the system of the invention, the personal verification code comprises any one of the following: an alpha-numeric sequence selected by the user; an alpha-numeric sequence selected by the rule-module nexus; an alpha-numeric sequence selected by an account issuer; a visible image; an audible sound.

(142) In an illustrative embodiment of the system of the invention, the rule-module nexus further comprises a fraud prevention platform configured to invoke criteria predetermined to detecting fraud by the user involving the rule-module nexus, said criteria comprising any one of the following: unusual usage of bid verification data; unusual modifying of a rule-module, and; unusual accessing of a financial account.

(143) An illustrative embodiment of the system of the invention, further comprises a user fraud registration platform configured to determine if the user has committed fraud involving the rule-module nexus, wherein data of the user is registered with a fraud prevention platform, said data comprising any one of the following: a pattern data; an execution command, and; a rule-module.

(144) An illustrative embodiment of the system of the invention, further comprises a rule-module nexus verification platform, configured for the private code, registered to the user, distinct from a personal verification code and not used in verifying the user, to be presented to the user via the rule-module nexus for verifying to the user that the authentic rule-module nexus has been accessed.

(145) In an illustrative embodiment of the system of the invention, the private code is registered to the user in the rule-module nexus by a party, said party comprising any one of the following: the user; the rule-module nexus, and; an account issuer.

(146) An illustrative embodiment of the system of the invention, further comprises an emergency alert platform configured to send an alert via rule-module nexus of an emergency wherein, the emergency code, distinct from a personal verification code and not used in verifying the user, is provided by the user to the user interface apparatus, said emergency comprising any one of the

following: the bid verification data being compromised; the nexus access token being compromised, and; the user being coerced.

(147) In an illustrative embodiment of the system of the invention, the emergency code comprises any one of the following: an alternate personal verification code; an alpha-numeric code; a visible image, and; an audible signal.

(148) In an illustrative embodiment of the system of the invention, the emergency alert platform is configured to invoke an execution command via the rule-module nexus, comprising any one of the following: presenting a visible display of predetermined emergency data to the user; presenting an audible signal of predetermined emergency data to the user; alerting an emergency authority, and; identifying a compromised code.

(149) In an illustrative embodiment of the system of the invention, the visible display comprises any one of the following: a false financial account; a false financial data within a financial account, and; confirming an emergency authority has been contacted.

(150) In an illustrative embodiment of the system of the invention, the audible signal comprises any one of the following: a false financial account; a false financial data within a financial account, and; confirming an emergency authority has been contacted.

(151) In an illustrative embodiment of the system of the invention, the compromised code resolution platform is configured to further comprise deactivating the unique user code and activating a secondary personal verification code.

(152) In an illustrative embodiment of the system of the invention, the rule-module nexus is configured to enable the user to provide a bid secondary personal verification code to the verification platform in replacement of the user's unique user code.

(153) An illustrative embodiment of the system of the invention, further comprises any one of the following: a) a calling platform configured for the user to call from a predetermined first phone number to a predetermined second phone number; b) a data-entering platform configured for the user to enter the personal verification code; c) an invoking platform configured for the user to invoke a secondary personal verification code, and; d) a notification platform for notifying the user that the secondary personal verification code has been activated for providing dual personal verification codes when verifying the user.

(154) In an illustrative embodiment of the system of the invention, the invoking platform is configured to comprise any one of the following: activating a registration secondary personal verification code of the user; creating by the user of a secondary personal verification code, and; accepting by the user of an offered secondary personal verification code.

(155) An illustrative embodiment of the system of the invention, further comprises any one of the following: a) an emailing platform for the user to email from a predetermined internet protocol address to a predetermined web site; b) a data-entering platform configured for the user to enter the personal verification code; c) an invoking platform configured for the user to invoke a secondary personal verification code, and; d) a notification platform for notifying the user that the secondary personal verification code has been activated for providing dual personal verification codes when verifying the user.

(156) In an illustrative embodiment of the system of the invention, the invoking platform is configured to comprise any one of the following: activating a registration secondary personal verification code of the user; creating by the user of a secondary personal verification code, and; accepting by the user of an offered secondary personal verification code

(157) In an illustrative embodiment of the system of the invention, the verification platform is configured to further comprise the bid personal verification code and the bid secondary personal verification code, both provided directly by the user to the user interface apparatus, being electronically compared with a registration personal verification code and a registration secondary personal verification code, to make a matching determination for verifying the user's authority to access the rule-module nexus.

(158) In an illustrative embodiment of the system of the invention, the limited basis comprises any one of the following: a predetermined time period; predetermined financial account access when using the secondary personal verification code; predetermined frequency for usage for using the secondary personal verification code, and; predetermined geographic area for using the secondary personal verification code.

(159) In an illustrative embodiment of the system of the invention, the secondary personal verification code comprises any one of the following: an alpha-numeric sequence selected by the user; an alpha-numeric sequence selected by the rule-module nexus; an alpha-numeric sequence selected by an account issuer; a visible image; an audible sound.

(160) In an illustrative embodiment of the system of the invention, the financial transaction platform is configured to comprise any one of the following: a credit transaction; a debit transaction; a scrip transaction; a rewards transaction; an electronic check transaction; a private label transaction; a stored value transaction; an electronic benefits transfer transaction; a brokerage trade transaction; invoking a surcharge to a transaction based on predetermined criteria; a buyer-seller exchange wherein a user's financial account balance is adjusted and a payee's financial account is correspondingly adjusted; an intra-account transfer transaction between financial accounts of the user without a buyer-seller exchange; redeeming a pre-paid ticket transaction for venue admittance without a buyer-seller exchange, and; redeeming a pre-paid membership benefit transaction for venue admittance without a buyer-seller exchange.

(161) In an illustrative embodiment of the system of the invention, the venue comprises any one of the following: a concert hall; a sports stadium; a movie theatre; a live-action theatre; an airplane; a train; a bus; a boat; a dance club; a restaurant; a garage; an office building; a health club; an apartment building; a medical facility; a toll booth, and; a dining club.

(162) In an illustrative embodiment of the system of the invention, venue admittance comprises displaying a facial photograph of the user, wherein upon the verification platform making a positive matching determination that the user is authorized to access the rule-module nexus, the rule-module nexus transmits the user's registered facial photograph for display to a third-party present during the financial transaction for visually verifying that the user's actual face is sufficiently similar to the user's displayed facial photograph to permit venue admittance.

(163) In an illustrative embodiment of the system of the invention, the account registry platform is configured to comprise any one of the following: querying data associated with a financial account; retrieving data from a financial account; querying data of a financial account via accessing a third-party computer; accessing a third-party computer to retrieve data from a financial account; presenting a visible electronic image of a financial account; presenting visible electronic text communication of a financial account; presenting a visible J2ME-enabled communication; presenting an audible electronic signal of a financial account; adjusting the balance in a financial account by making a credit to a financial account; transmitting an electronic positioning system communication; transmitting an electronic navigating system communication; transmitting an electronic mapping communication; an electronic guiding system communication; transmitting an electronic text message of a financial account, and; adjusting the balance in a financial account by making a debit from a financial account.

(164) In an illustrative embodiment of the system of the invention, the nexus access token is configured to further comprise any one of the following: storing no data which uniquely and directly identifies a specific financial account of the user; storing no live financial account data, and; storing no payment-enabling data.

(165) In an illustrative embodiment of the system of the invention, the verification platform is configured to further comprise displaying a facial photograph of the user, wherein upon the verification platform making a positive matching determination that the user is authorized to access the rule-module nexus, the rule-module nexus transmits the registered facial photograph of the user for display to a third-party present during the financial transaction, for visually verifying that the

user's actual face is sufficiently similar to the user's displayed facial photograph to permit the financial transaction.

(166) In an illustrative embodiment of the system of the invention, the user interface apparatus is configured to comprise storing a hardware verification code, registered with the rule-module nexus and unique to the user interface apparatus.

(167) In an illustrative embodiment of the system of the invention, the registration platform is configured to further comprise registering a rule-module to a payee, said registered rule-module comprising a pattern data associated with an execution command.

(168) In an illustrative embodiment of the system of the invention, the pattern data comprises any one of the following: the payee's legal name; a user interface apparatus hardware verification code; an employer identification number; financial account access authorization fields; a payee account registry code; a unique payee code; a payee verification code; a transaction terminal identification code; a user interface apparatus identification code; an emergency code; a financial account; an email address; a telephone number; a mailing address; authority of at least one employee of the payee; a digital certificate; a network credential; an Internet protocol address; a digital signature; an encryption key; electronic audible signature, and; an electronic visible signature.

(169) In an illustrative embodiment of the system of the invention, the execution command comprises any one of the following: accessing a user's financial account; processing a user's financial transaction; presenting selected data from user's rule-module data, and; alerting an emergency authority.

(170) In an illustrative embodiment of the system of the invention, the verification platform is further configured to verify the payee by electronically comparing bid verification data of the payee with registered verification data of a payee, and makes a matching determination for verifying the payee's authority to access the rule-module nexus, said matching determination comprising any one of the following: making a positive matching determination verifying that the payee is authorized to access the rule-module nexus, and; failing to make a positive matching determination, wherein making a negative matching determination is automatic, verifying that the payee is not authorized to access the rule-module nexus.

(171) In an illustrative embodiment of the system of the invention, the bid verification data comprises any registered pattern data of the payee enabling the verification platform to verify that the payee is authorized to access the rule-module nexus.

(172) In an illustrative embodiment of the system of the invention, upon the verification platform making a positive matching determination of the payee's authority to access the rule-module nexus, an execution command of the payee is invoked, comprising any one of the following: authorizing a field for accessing rule-modules in the rule-module nexus; accepting a subordinated payee; authorizing a field for accessing a third-party computer being accessed by the rule-module nexus, and; invoking a processing preference of the payee.

(173) An illustrative embodiment of the system of the invention, further comprises a fraud prevention platform configured to register data of the payee upon a determination by the rule-module nexus that the payee has committed fraud involving the rule-module nexus, said data comprising any one of the following: a pattern data; an execution command, and; a rule-module.

(174) In an illustrative embodiment of the system of the invention, a platform is a computer-based set of related data subject to electronic processing with software using predetermined criteria associated with the rule-module nexus, said processing comprising any one of the following: data storage; data queries; data retrieval, and; data modification.

(175) In an illustrative embodiment of the system of the invention, the rule-module nexus is remotely located from the user, the user interface apparatus, and the nexus access token.

(176) In an illustrative embodiment of the system of the invention, the financial transaction processing platform is configured to further comprise accessing the remotely located rule-module nexus via a network, comprising any one of the following: a cable network; a wireless network; a

land-line phone network; the Internet; an intranet; a local area network, a wide area network; an electronic positioning network; a satellite network, and; an X.25 network.

(177) In an illustrative embodiment of the system of the invention, the financial transaction processing platform further comprises transmitting transaction data via the rule-module nexus, said transaction data comprising any one of the following: pricing information; a list of goods and services; a verification data of the user; a verification data of a payee; a date or time; a location of the user interface apparatus; a location of the nexus access token; an electronic positioning code; a unique payee code; a hardware verification code of the user interface apparatus; the name of a payee; an invoice number, and; transaction settlement instruction.

(178) In an illustrative embodiment of the system of the invention, the payee and the account issuer are identical.

(179) In an illustrative embodiment of the system of the invention, the predetermined time period for settlement comprises any one of the following: an immediate adjustment of a balance in the user's financial account; a delayed adjustment of a balance in the user's financial account, and; a time interval for repeated adjustment of a balance in the user's financial account.

(180) In an illustrative embodiment of the system of the invention, the account issuer of a financial account comprises any one of the following: an owner, comprising a party that owns a financial account; a servicer, comprising a party that services a financial account on behalf of an account owner; an agent, comprising a party that acts on behalf of another party associated with a financial account, and; a fiduciary, comprising a party with a fiduciary duty to the financial account.

(181) In an illustrative embodiment of the system of the invention, the account issuer comprises any one of the following: a bank; a merchant; a scrips provider; credit account organization; the rule-module nexus; a government agency; an insurance company; a brokerage firm; a reward incentives provider; a services provider; a product provider, and; an internet payment provider.

(182) In an illustrative embodiment of the system of the invention, the financial account is a computerized set of related financial data having a predetermined legal monetary value for usage comprising any one of the following: purchasing a product; purchasing of a service; exchanging a product; exchanging a service, and; exchanging for other financial data of equivalent monetary value.

(183) In an illustrative embodiment of the system of the invention, the financial account comprises any: checking account; credit account comprising Visa®, MasterCard® and American Express®; reward incentives account; insurance account; brokerage account; savings account; scrip incentives account; pre-paid account; pre-paid ticket; membership benefits account; private label credit account; services barter account; product barter account, and; internet payment account.

(184) In an illustrative embodiment of the system of the invention, an incentives account comprises financial data, comprising any one of the following: gift certificate units; stored-value units; electronic coupon units having a predetermined monetary value; minutes of telephone calling time; miles towards earning a free airplane flight; accruing units of a predetermined monetary value exchangeable for a product, and; accruing units of a predetermined monetary value exchangeable for a service.

(185) An illustrative embodiment of the system of the invention is configured to further comprise presenting a financial account to the user, comprising any one of the following: a visible display of an electronic visible signature; a visible text message, and; an audible signal of an electronic audible signature.

(186) In an illustrative embodiment of the system of the invention, the user account registry platform is configured to further comprise ranking the plurality of proprietary financial accounts, wherein predetermined criteria is used for tagging and ranking the plurality of proprietary financial accounts in a certain order.

(187) In an illustrative embodiment of the system of the invention, the predetermined criteria for the ranking further comprises any one of the following: improving a transaction benefit for an

account issuer; improving a transaction benefit for the user, and; improving a transaction benefit for a payee.

(188) In an illustrative embodiment of the system of the invention, improving a transaction benefit further comprises any one of the following: increasing efficiency; increasing speed; increasing profit; increasing security; decreasing cost; increasing reward incentives, and; invoking a surcharge predetermined by the user.

(189) In an illustrative embodiment of the system of the invention, the ranking further comprises presenting to the user a display comprising any one of the following: a default financial account; a plurality of financial accounts, with visibly distinct indicators for the respective rankings of the financial accounts, and; a plurality of financial accounts displayed in sequence as a function of their respective rankings.

(190) In an illustrative embodiment of the system of the invention, the electronic visible signature is an electronic visible image comprising any one of the following: streaming media; a video clip; a moving image; a holographic display; a static display; a dynamic display; an alpha-numeric display, and; a textual display.

(191) In an illustrative embodiment of the system of the invention, ranking proprietary financial accounts further comprises presenting: a plurality of financial accounts, with audibly distinct indicators for the respective rankings of the financial accounts.

(192) In an illustrative embodiment of the system of the invention, the electronic audible signature is an audible signal comprising any one of the following: a musical fragment; speech; phonation, and; a song.

(193) In an illustrative embodiment of the system of the invention, the surcharge comprises any one of the following: an additional financial amount debited from the financial account selected by the user, and credited to a different financial account.

(194) In an illustrative embodiment of the system of the invention, the surcharge comprises any one of the following: a fixed financial amount, and; a variable financial amount.

(195) In an illustrative embodiment of the system of the invention, the variable financial amount comprises a formula for calculating the surcharge as a function of a predetermined factor comprising any one of the following: income of the user; a credit score of the user; a financial amount of the transaction; time; a purchasing frequency of the user; a balance in a financial account of the user; an economic indicator, and; a financial objective of the user.

(196) In an illustrative embodiment of the system of the invention, the financial transaction processing platform is configured to further comprise the rule-module nexus accessing a third-party computer, comprising any one of the following: verifying a user; detecting a rule-module nexus fraud; registering a user fraud; registering a payee fraud; registering an account issuer fraud; alerting of an emergency; resolving a compromised code; accessing a financial account; settlement of the financial transaction, and; completing the financial transaction.

(197) In an illustrative embodiment of the system of the invention, the user account registry platform is configured to further comprise verifying resources, wherein upon a selection of a financial account by the user, an electronic determination is made if the selected financial account of the user has sufficient resources for settlement of the financial transaction.

(198) In an illustrative embodiment of the system of the invention, the financial transaction processing platform is configured to further comprise settlement of the financial transaction, comprising any one of the following: invoking a debit of financial data within the selected financial account of the user, and; invoking a credit of financial data within the selected financial account of the user.

(199) In an illustrative embodiment of the system of the invention, the nexus access token is configured to further comprise any one of the following: read only memory, and; read and write memory.

(200) In an illustrative embodiment of the system of the invention, the financial transaction

processing platform is configured to further comprise any one of the following: the rule-module nexus accessing a third-party computer; the rule-module nexus accessing the transaction terminal, and; the rule-module nexus accessing the user interface apparatus.

(201) In an illustrative embodiment of the system of the invention, the user account registry platform is configured to further comprise accessing financial accounts, comprising any one of the following: querying the user's financial account balance; invoking an authorization a field within the rule-module nexus; invoking a user's rewards program; invoking a micro-merchandising advertising and upsell program; invoking an intelligent tracking and extrapolating agent, and; invoking an automated User notification program.

(202) In an illustrative embodiment of the system of the invention, the automated User notification program invokes an outgoing communication comprising any one of the following: sending a wireless phone text message; mailing a paper notice; sending a fax; a phone number dialing, and; sending a page.

(203) In an illustrative embodiment of the system of the invention, the transaction terminal comprises any one of the following: a wireless telephone; a wireless pager; a personal computer; a merchant point of sale register; a vending machine; a venue admittance device; a personal digital assistant; an internet kiosk; a land line telephone; a television, and; a digital music player.

(204) In an illustrative embodiment of the system of the invention, the transaction terminal further comprises any one of the following: a data-entering touch screen; a data-entering key pad; a visible display screen; an audible signal speaker, and; an audio receiver.

(205) In an illustrative embodiment of the system of the invention, the user interface apparatus comprises any one of the following: a contactless proximity communications interrogator; an electronic positioning system communications element; an electronic navigating system communications element; an electronic mapping system communications element; an electronic guiding system communications element; a contactless proximity communications data-reader and data-writer; a magnetic stripe reader; a magnetic ink character reader; a bar code reader; a data-entering touch screen; a data-entering key pad; a visible display screen; an audible signal speaker; an audio receiver.

(206) In an illustrative embodiment of the system of the invention, the nexus access token further comprises data communications, comprising any one of the following: electronically readable bar-code imprint; electronically readable alpha-numeric code; radio signal; infrared signal; digital signal; cellular signal; visible display; and vibrational signal; acoustic sound signal; microwave signal; satellite signal; X-ray signal; pulsar signal; television signal; electronically readable magnetic stripe, and; photo signal.

(207) In an illustrative embodiment of the system of the invention, the nexus access token is conjoined with a device comprising any one of the following: wireless telephone, key fob; wireless pager; personal digital assistant; wearable ornament; digital media player; refillable container; removeably implantable computer chip; a user interface apparatus, and; door key.

(208) In an illustrative embodiment of the system of the invention, the wearable ornament comprises any of the following: jewelry, and; clothing.

(209) In an illustrative embodiment of the system of the invention, the refillable container comprises any of the following: a beverage container, and; a gasoline container.

(210) In an illustrative embodiment of the system of the invention, the financial transaction processing platform is configured to further comprise measuring the real-time elapsed in which the transaction terminal remains online communicating with the remotely located rule-module nexus from verifying the user to selecting the financial account.

(211) In an illustrative embodiment of the system of the invention, completing the financial transaction comprises any one of the following: declining the financial transaction, and; settlement of the financial transaction.

(212) In an illustrative embodiment of the system of the invention, declining the financial

transaction is invoked by a party comprising any one of the following: the user; an account issuer; a payee, and; the rule-module nexus.

(213) In an illustrative embodiment of the system of the invention, declining the financial transaction is based upon predetermined criteria comprising any one of the following: insufficient financial data within the financial account; a credit score of the user; geography; usage frequency; usage recency; demographics of the user; financial amount of the financial transaction; a user fraud; an account issuer fraud; a payee fraud, and; a compromised code.

(214) An illustrative embodiment of the system of the invention further comprises a resource determination platform configured to determining resources of a financial account via an account issuer, comprising any one of the following: a positive determination wherein the selected financial account has sufficient resources, outputting an approval of the financial account for settlement of the financial transaction; a negative determination wherein the selected financial account has insufficient resources, outputting a decline of the financial account for settlement, whereupon at least one other financial account of the user is automatically displayed to the user by the transaction terminal based upon predetermined criteria.

(215) In an illustrative embodiment of the system of the invention, a rule-module comprises any one of the following: pattern data associated with a plurality of execution commands; a plurality of pattern data associated with an execution command, and; a plurality of pattern data associated with a plurality of execution commands.

(216) In an illustrative embodiment of the system of the invention, the rule-module nexus is configured to comprise any one of the following: a master rule-module nexus comprising all pattern data, execution commands, and rule-modules registered by users, by payees, and by account issuers, and; a subset rule-module nexus comprising a subset of all pattern data, execution commands, and rule-modules registered by users, by payees, and by account issuers.

(217) In an illustrative embodiment of the system of the invention, the subset rule-module nexus is configured to comprise a subset of data selected as a function of predetermined criteria, comprising any one of the following: a credit score of the user; geography; usage frequency; usage recency; demographics of the user; financial amount of the financial transaction; a user fraud; a payee fraud; an account issuer fraud, and; a compromised code.

(218) In an illustrative embodiment of the system of the invention, registering a rule-module further comprises checking user re-registration, wherein the registered rule-module of the user is compared against a previously registered rule-module, wherein a match alerts the rule-module nexus that the user is attempting a re-registration.

(219) An illustrative embodiment of the system of the invention, further comprises a notification platform configured to present notification of the financial transaction results to the user upon completing the financial transaction, comprising any one of the following: notification via the transaction terminal of a decline of the financial transaction, and; notification via the transaction terminal of settlement of the one-line financial transaction.

(220) In an illustrative embodiment of the system of the invention, presenting notification comprises any one of the following: a visible display; an audible signal; and a printed receipt.

(221) In an illustrative embodiment of the system of the invention, the emergency authority comprises any one of the following: a government agency, and; a private entity.

(222) In an illustrative embodiment of the system of the invention, the rule-module nexus further comprises a financial account aggregating maintenance module configured to aggregate maintenance services for the plurality of proprietary financial accounts of the user, said maintenance services comprising any one of the following: invoicing the user; collecting invoice payments from the user; reconciling scrip incentive financial data; reconciling reward incentive financial data; being an agent for intelligent data tracking on behalf of the user, and; being an agent for extrapolating data on behalf of the user.

(223) In an illustrative embodiment of the system of the invention, the verification platform is

configured to further comprise verifying the user interface apparatus by electronically comparing a bid hardware verification code of the user interface apparatus with a previously registered hardware verification code, to make a matching determination for verifying the authenticity of the user interface apparatus via the rule-module nexus, said matching determination comprising any one of the following: making a positive matching determination verifying to the rule-module nexus that the user interface apparatus is authentic, and; failing to make a positive matching determination, wherein making a negative matching determination is automatic, verifying to the rule-module nexus that the user interface apparatus is not authentic.

(224) In another illustrative embodiment of the method of the invention, wherein during verifying the user, the verification data further comprises a bid personal verification code provided directly by the user, wherein the verification platform electronically compares the user's bid unique user code and the user's bid personal verification code with a user's registration unique user code and a user's registration personal verification code, and makes a matching determination for verifying the user's authority to access the rule-module nexus, said matching determination comprising any one of the following: making a positive matching determination verifying that the user is authorized to access the rule-module nexus, and; failing to make a positive matching determination, wherein making a negative matching determination is automatic, verifying that the user is not authorized to access the rule-module nexus.

(225) In another illustrative embodiment of the system the invention, wherein the verification data further comprises a bid personal verification code provided directly by the user, and wherein the verification platform further comprises being configured to electronically compare the user's bid unique user code and the user's bid personal verification code with a user's registration unique user code and a user's registration personal verification code, and make a matching determination for verifying the user's authority to access the rule-module nexus, said matching determination comprising any one of the following: making a positive matching determination verifying that the user is authorized to access the rule-module nexus, and; failing to make a positive matching determination, wherein making a negative matching determination is automatic, verifying that the user is not authorized to access the rule-module nexus.

(226) In an illustrative embodiment of the method of the invention, the data-storing in the nexus access token further comprises any one of the following: registering the pattern data of a user via a user interface apparatus; transmitting said pattern data to the rule-module nexus; transmitting a unique user code via the rule-module nexus to a dispensing kiosk co-located, but not conjoined, with said user interface apparatus; data-storing of said unique user code into a nexus access token by the dispensing kiosk, and; dispensing said nexus access token to the user via the dispensing kiosk.

(227) In an illustrative embodiment of the system of the invention, a dispensing kiosk is configured to comprise any one of the following: being co-located but not conjoined with said user interface apparatus; receiving a unique user code via the rule-module nexus, upon the rule-module nexus having received the pattern data of a user via the user interface apparatus; data-storing said unique user code into a nexus access token, and; dispensing to the user said nexus access token storing the unique user code.

(228) In another embodiment of the method of the invention, a party having fiduciary duty for a financial account of a user has a responsibility comprising any one of the following: managing financial data within a financial account; adjusting financial data within a financial account, and; reconciling financial data within a financial account.

(229) In another embodiment of the system of the invention, a party having fiduciary duty for a financial account of a user has a responsibility comprising any one of the following: managing financial data within a financial account; adjusting financial data within a financial account, and; reconciling financial data within a financial account.

(230) In another illustrative embodiment of the method of the invention, a financial account further

comprises any one of the following: an account typing-code specifying the type of financial account; an account name, comprising a text name of the financial account; a currency code, comprising an ISO 4217 code specifying the currency of the financial account.

(231) In another illustrative embodiment of the method of the invention, the type of financial account comprises any one of the following: savings; checking; money market; rewards; scrips; brokerage; money market; money fund; credit; debit; prepaid, and; insurance.

(232) In an illustrative embodiment of the method of the invention, accessing a financial account comprises any one of the following: electronically accessing a rule-module associated with a financial account; electronically querying data associated with a financial account; electronically querying data contained within a financial account; electronically invoking data associated with a financial account; electronically invoking data contained within a financial account.

(233) In an illustrative embodiment of the method of the invention, a unique account identifier comprises any one of the following: a unique code which is electronically associated with only one financial account; a unique code which electronically identifies only one financial account.

(234) In another illustrative embodiment of the system of the invention, a financial account further comprises any one of the following: an account typing-code specifying the type of financial account; an account name, comprising a text name of the financial account; a currency code, comprising an ISO 4217 code specifying the currency of the financial account.

(235) In another illustrative embodiment of the system of the invention, the type of financial account comprises any one of the following: savings; checking; money market; rewards; scrips; brokerage; money market; money fund; credit; debit; prepaid, and; insurance.

(236) In an illustrative embodiment of the system of the invention, accessing a financial account comprises any one of the following: accessing a rule-module associated with a financial account; electronically querying data associated with a financial account; electronically querying data contained within a financial account; electronically invoking data associated with a financial account; electronically invoking data contained within a financial account.

(237) In an illustrative embodiment of the system of the invention, a unique account identifier comprises any one of the following: a unique code which is electronically associated with only one financial account; a unique code which electronically identifies only one financial account.

(238) In another embodiment of the method of the invention, the portable nexus access token comprises any one of the following: being dimensionally small enough to be handheld, and; being of a sufficiently light weight as to be handheld.

(239) In another embodiment of the method of the invention, the user interface apparatus comprises any one of the following: being dimensionally small enough to be handheld, and; being of a sufficiently light weight as to be handheld.

(240) In another embodiment of the method of the invention, the visible electronic text comprises any one of the following: an instant message (IM); a short message service (SMS), and; a J2ME-enabled message.

(241) In another embodiment of the method of the invention, the visible display is an electronically readable bar code image.

(242) In another embodiment of the method of the invention, a rule-module is registered to a payee, said registered rule-module comprising a pattern data associated with an execution command.

(243) In another embodiment of the method of the invention, the pattern data comprises any one of the following: the payee's legal name; a user interface apparatus hardware verification code; an employer identification number; a payee's electronic positioning system code; financial account access authorization fields; a unique payee code; a payee verification code; a transaction terminal identification code; an emergency code; a financial account; an email address; a telephone number; a mailing address; authority of at least one employee of the payee; a digital certificate; a network credential; an Internet protocol address; a digital signature; an encryption key; electronic audible signature, and; an electronic visible signature.

(244) In another embodiment of the method of the invention, the execution command comprises any one of the following: accessing a user's financial account; processing a user's financial transaction; presenting selected data from user's rule-module data, and; alerting an emergency authority.

(245) In another embodiment of the method of the invention, processing the financial transaction further comprises verifying the payee, wherein the verification platform electronically compares bid verification data of the payee with registered verification data of a payee, and makes a matching determination for verifying the payee's authority to access the rule-module nexus, said matching determination comprising any one of the following: making a positive matching determination verifying that the payee is authorized to access the rule-module nexus, and; failing to make a positive matching determination, wherein making a negative matching determination is automatic, verifying that the payee is not authorized to access the rule-module nexus.

(246) In another embodiment of the method of the invention, the bid verification data comprises any registered pattern data of the payee enabling the verification platform to verify that the payee is authorized to access the rule-module nexus.

(247) In another embodiment of the method of the invention, upon the verification platform making a positive matching determination of the payee's authority to access the rule-module nexus, an execution command of the payee is invoked, comprising any one of the following: authorizing a field for accessing rule-modules in the rule-module nexus; accepting a subordinated account user; authorizing a field for accessing a third-party computer being accessed by the rule-module nexus, and; invoking a processing preference of the payee.

(248) In another embodiment of the method of the invention, upon a determination by the rule-module nexus that the payee has committed fraud involving the rule-module nexus, data of the payee data is registered with a fraud prevention platform, said data comprising any one of the following: a pattern data; an execution command, and; a rule-module.

(249) In another embodiment of the method of the invention, a platform is a computer-based set of related data subject to electronic processing with software using predetermined criteria associated with the rule-module nexus, said processing comprising any one of the following: data storage; data queries; data retrieval, and; data modification.

(250) In another embodiment of the method of the invention, the rule-module nexus is remotely located from the user, the user interface apparatus, and the nexus access token.

(251) In another embodiment of the method of the invention, processing the financial transaction further comprises accessing the remotely located rule-module nexus via a network comprising any one of the following: a cable network; a wireless network; a land-line phone network; the Internet; an intranet; a local area network, a wide area network; an electronic positioning network; a satellite network, and; an X.25 network.

(252) In another embodiment of the method of the invention, the electronic positioning code, identifying a real-time geographic location and invoking a unique payee code, is provided by a device comprising any one of the following: a nexus access token; a user interface apparatus, and; a transaction terminal.

(253) In another embodiment of the method of the invention, invoking said unique payee code comprises any one of the following: invoking a financial account of the payee; invoking an arbitrage rule-module; invoking predetermined criteria for tagging and ranking, in a certain order, a plurality of proprietary financial accounts of a user.

(254) In another embodiment of the method of the invention, the predetermined criteria for the ranking further comprises any one of the following: improving a transaction benefit for an account issuer; improving a transaction benefit for the user, and; improving a transaction benefit for a payee.

(255) In another embodiment of the method of the invention, improving a transaction benefit further comprises any one of the following: increasing efficiency; increasing speed; increasing

profit; increasing security; decreasing cost; increasing reward incentives, and; invoking a surcharge predetermined by the user.

(256) In another embodiment of the method of the invention, the ranking invokes presenting financial account data to the user, comprising any one of the following: a default financial account; a plurality of financial accounts, with visibly distinct indicators for the respective rankings of the financial accounts, and; a plurality of financial accounts displayed in sequence as a function of their respective ranking.

(257) In another embodiment of the method of the invention, presenting to the user a default financial account further comprises any one of the following: presenting to the user a first default financial account; presenting to the user a second default financial account if the first default financial account is declined, and; presenting to the user an “N.sup.th” default financial account, in the event the “N.sup.th-1” default financial account is declined.

(258) In another embodiment of the method of the invention, the hardware verification code of the user interface apparatus invokes a unique payee code.

(259) In another embodiment of the method of the invention, invoking said unique payee code comprises any one of the following: invoking a financial account of the payee; invoking predetermined criteria for tagging and ranking, in a certain order, a plurality of proprietary financial accounts of a user; invoking an arbitrage rule-module; invoking a financial account routing code of the payee; transmitting payee data to the user interface apparatus, and; transmitting payee data to the nexus access token.

(260) In another embodiment of the method of the invention, the predetermined criteria for the ranking further comprises any one of the following: improving a transaction benefit for an account issuer; improving a transaction benefit for the user, and; improving a transaction benefit for a payee.

(261) In another embodiment of the method of the invention, improving a transaction benefit further comprises any one of the following: increasing efficiency; increasing speed; increasing profit; increasing security; decreasing cost; increasing reward incentives, and; invoking a surcharge predetermined by the user.

(262) In another embodiment of the method of the invention, the ranking invokes presenting financial account data to the user, comprising any one of the following: a default financial account; a plurality of financial accounts, with visibly distinct indicators for the respective rankings of the financial accounts, and; a plurality of financial accounts displayed in sequence as a function of their respective ranking.

(263) In another embodiment of the method of the invention, presenting to the user a default financial account further comprises any one of the following: presenting to the user a first default financial account; presenting to the user a second default financial account if the first default financial account is declined, and; presenting to the user an “N.sup.th” default financial account, in the event the “N.sup.th-1” default financial account is declined.

(264) In another embodiment of the system of the invention, the portable nexus access token comprises any one of the following: being dimensionally small enough to be handheld, and; being of a sufficiently light weight as to be handheld.

(265) In another embodiment of the system of the invention, the user interface apparatus comprises any one of the following: being dimensionally small enough to be handheld, and; being of a sufficiently light weight as to be handheld.

(266) In another embodiment of the system of the invention, the visible electronic text comprises any one of the following: an instant message (IM); a short message service (SMS), and; a J2ME-enabled message.

(267) In another embodiment of the system of the invention, the visible display is an electronically readable bar code image.

(268) In another embodiment of the system of the invention, a rule-module is registered to a payee,

said registered rule-module comprising a pattern data associated with an execution command.

(269) In another embodiment of the system of the invention, the pattern data comprises any one of the following: the payee's legal name; a user interface apparatus hardware verification code; an employer identification number; a payee's electronic positioning system code; financial account access authorization fields; a unique payee code; a payee verification code; a transaction terminal identification code; an emergency code; a financial account;

(270) an email address; a telephone number; a mailing address; authority of at least one employee of the payee; a digital certificate; a network credential; an Internet protocol address; a digital signature; an encryption key; electronic audible signature, and; an electronic visible signature.

(271) In another embodiment of the system of the invention, the execution command comprises any one of the following: accessing a user's financial account; processing a user's financial transaction; presenting selected data from user's rule-module data, and; alerting an emergency authority.

(272) In another embodiment of the system of the invention, the financial transaction further comprises verifying the payee, wherein the verification platform electronically compares bid verification data of the payee with registered verification data of a payee, and makes a matching determination for verifying the payee's authority to access the rule-module nexus, said matching determination comprising any one of the following: making a positive matching determination verifying that the payee is authorized to access the rule-module nexus, and; failing to make a positive matching determination, wherein making a negative matching determination is automatic, verifying that the payee is not authorized to access the rule-module nexus.

(273) In another embodiment of the system of the invention, the bid verification data comprises any registered pattern data of the payee enabling the verification platform to verify that the payee is authorized to access the rule-module nexus.

(274) In another embodiment of the system of the invention, upon the verification platform making a positive matching determination of the payee's authority to access the rule-module nexus, an execution command of the payee is invoked, comprising any one of the following: authorizing a field for accessing rule-modules in the rule-module nexus; accepting a subordinated account user; authorizing a field for accessing a third-party computer being accessed by the rule-module nexus, and; invoking a processing preference of the payee.

(275) In another embodiment of the system of the invention, upon a determination by the rule-module nexus that the payee has committed fraud involving the rule-module nexus, data of the payee data is registered with a fraud prevention platform, said data comprising any one of the following: a pattern data; an execution command, and; a rule-module.

(276) In another embodiment of the system of the invention, a platform is a computer-based set of related data subject to electronic processing with software using predetermined criteria associated with the rule-module nexus, said processing comprising any one of the following: data storage; data queries; data retrieval, and; data modification.

(277) In another embodiment of the system of the invention, the rule-module nexus is remotely located from the user, the user interface apparatus, and the nexus access token.

(278) In another embodiment of the system of the invention, processing the financial transaction further comprises accessing the remotely located rule-module nexus via a network comprising any one of the following: a cable network; a wireless network; a land-line phone network; the Internet; an intranet; a local area network, a wide area network; an electronic positioning network; a satellite network, and; an X.25 network.

(279) In another embodiment of the system of the invention, the electronic positioning code, identifying a real-time geographic location and invoking a unique payee code, is provided by a device comprising any one of the following: a nexus access token; a user interface apparatus, and; a transaction terminal.

(280) In another embodiment of the system of the invention, invoking said unique payee code

comprises any one of the following: invoking a financial account of the payee; invoking an arbitrage rule-module; invoking predetermined criteria for tagging and ranking, in a certain order, a plurality of proprietary financial accounts of a user.

(281) In another embodiment of the system of the invention, the predetermined criteria for the ranking further comprises any one of the following: improving a transaction benefit for an account issuer; improving a transaction benefit for the user, and; improving a transaction benefit for a payee.

(282) In another embodiment of the system of the invention, improving a transaction benefit further comprises any one of the following: increasing efficiency; increasing speed; increasing profit; increasing security; decreasing cost; increasing reward incentives, and; invoking a surcharge predetermined by the user.

(283) In another embodiment of the system of the invention, the ranking invokes presenting financial account data to the user, comprising any one of the following: a default financial account; a plurality of financial accounts, with visibly distinct indicators for the respective rankings of the financial accounts, and; a plurality of financial accounts displayed in sequence as a function of their respective ranking.

(284) In another embodiment of the system of the invention, presenting to the user a default financial account further comprises any one of the following: presenting to the user a first default financial account; presenting to the user a second default financial account if the first default financial account is declined, and; presenting to the user an “N.sup.th” default financial account, in the event the “N.sup.th-1” default financial account is declined.

(285) In another embodiment of the system of the invention, the hardware verification code of the user interface apparatus invokes a unique payee code.

(286) In another embodiment of the system of the invention, invoking said unique payee code comprises any one of the following: invoking a financial account of the payee; invoking predetermined criteria for tagging and ranking, in a certain order, a plurality of proprietary financial accounts of a user; invoking a financial account routing code of the payee; invoking an arbitrage rule-module; transmitting payee data to the user interface apparatus, and; transmitting payee data to the nexus access token.

(287) In another embodiment of the system of the invention, the predetermined criteria for the ranking further comprises any one of the following: improving a transaction benefit for an account issuer; improving a transaction benefit for the user, and; improving a transaction benefit for a payee.

(288) In another embodiment of the system of the invention, improving a transaction benefit further comprises any one of the following: increasing efficiency; increasing speed; increasing profit; increasing security; decreasing cost; increasing reward incentives, and; invoking a surcharge predetermined by the user.

(289) In another embodiment of the system of the invention, the ranking invokes presenting financial account data to the user, comprising any one of the following: a default financial account; a plurality of financial accounts, with visibly distinct indicators for the respective rankings of the financial accounts, and; a plurality of financial accounts displayed in sequence as a function of their respective ranking.

(290) In another embodiment of the system of the invention, presenting to the user a default financial account further comprises any one of the following: presenting to the user a first default financial account; presenting to the user a second default financial account if the first default financial account is declined, and; presenting to the user an “N.sup.th” default financial account, in the event the “N.sup.th-1” default financial account is declined.

(291) In another embodiment of the method of the invention, accessing financial accounts further comprises accessing a financial interchange platform, wherein units of financial data are electronically transmitted, comprising any one of the following: being transmitted between a

plurality of proprietary financial accounts of a user, and; being transmitted between a plurality of proprietary financial accounts of a plurality of users.

(292) In another embodiment of the method of the invention, being transmitted further comprises any one of the following: converting financial data; trading financial data; exchanging financial data, and; transferring financial data.

(293) In another embodiment, the system of the invention further comprises a financial interchange platform configured to comprise: accessing a financial interchange platform, wherein units of financial data are electronically transmitted, comprising any one of the following: being transmitted between a plurality of proprietary financial accounts of a user, and; being transmitted between a plurality of proprietary financial accounts of a plurality of users.

(294) In another embodiment of the system of the invention, being transmitted further comprises any one of the following: converting financial data; trading financial data; exchanging financial data, and; transferring financial data.

(295) In another embodiment, the method of the invention further comprises invoking a rule-module nexus routing code which is electronically stored within, and invoked by, a device comprising any one of the following: a user interface apparatus, and; a transaction terminal.

(296) In another embodiment, the method of the invention further comprises invoking a rule-module nexus routing code by the device occurs via an event comprising any one of the following: scanning of a unique user code from a nexus access token, and; instructing the device to process the financial transaction via the rule-module nexus.

(297) In another embodiment of the method of the invention, instructing the device comprises any one of the following: voice-command; key-punching; and touching a screen.

(298) In another embodiment, the system of the invention further comprises a rule-module nexus routing code platform within a device, said device comprising any one of the following: a user interface apparatus, and; a transaction terminal.

(299) In another embodiment of the system of the invention, the rule-module nexus routing code platform is configured to comprise any one of the following: electronically storing the rule-module nexus routing code, and; electronically invoking the rule-module nexus routing code.

(300) In another embodiment of the system of the invention, invoking a rule-module nexus routing code occurs via an event comprising any one of the following: scanning of a unique user code from a nexus access token, and; data-entering, comprising voice-command, key-punching, and touch-screen.

(301) In another embodiment of the method of the invention, the nexus access token further comprises data-scanning capability, comprising any one of the following: a radio frequency interrogator, and; a bar-code scanner.

(302) In another embodiment, the method of the invention further comprising compiling a virtual shopping platform, wherein electronically scanned data is compiled and stored, said electronically scanned data comprising any one of the following: product name; product brand; product manufacturer; unique package code; unique product inventory/tracking code, and; product price.

(303) In another embodiment of the method of the invention, the virtual shopping platform resides within a device comprising any one of the following: the nexus access token; the rule-module nexus, and; a third-party platform.

(304) In another embodiment of the system of the invention, the nexus access token further comprises data-scanning capability, comprising any one of the following: a radio frequency interrogator, and; a bar-code scanner.

(305) In another embodiment of the system of the invention, a virtual shopping platform is configured to comprise any one of the following: compiling electronically scanned data, and; storing electronically scanned data.

(306) In another embodiment of the system of the invention, the electronically scanned data further comprises any one of the following: product name; product brand; product manufacturer; unique

package code; unique product inventory/tracking code, and; product price.

(307) In another embodiment of the system of the invention, the virtual shopping platform resides within a device comprising any one of the following: the nexus access token; the rule-module nexus, and; a third-party platform.

(308) In another embodiment of the method of the invention, processing the financial transaction requires no live financial account data being provided directly from the nexus access token, said live financial account data comprising any one of the following: data which uniquely and directly identifies a specific financial account of the user, and; payment-enabling data.

(309) In another embodiment of the system of the invention, the financial transaction processing platform requires no live financial account data being provided directly from the nexus access token to process a financial transaction, said live financial account data comprising any one of the following: no data which uniquely and directly identifies a specific financial account of the user, and; no live financial account data.

(310) In another embodiment of the method of the invention, accessing an online user account registry comprises any one of the following: a) Approving access to an online user account registry via a verification platform comparing a bid verification data, comprising a unique user code provided from a nexus access token, with a registered verification data; b) Accessing the online user account registry via a rule-module invoked from a rule-module nexus, said online user account registry comprising a plurality of financial accounts of the user; Whereby an online account registry, comprising a plurality of financial accounts, is accessed via a unique user code provided from a nexus access token.

(311) In another embodiment of the method of the invention, the unique user code is thin-client, comprising any one of the following: no financial account identifier; no live financial data; no payment-enabling data; a rule-module nexus routing code; a user account registry code; an electronic positioning system code, and; a network routing instruction for processing the financial transaction via a rule-module nexus.

(312) In another embodiment of the method of the invention, the online user account registry comprises any one of the following: storing data apart from the nexus access token, and; accessing the stored data via communicating externally to the nexus access token.

(313) In another embodiment of the method of the invention, communicating externally to the nexus access token comprises any one of the following: using an electronically readable magnetic stripe; using a smart card; using a bar code; using a radio signal; using an infrared signal; using an audible signal; using a USB connection; using a global positioning signal; using a local area network; using a wide area network; using an X.25 network; using an internet connection; using a cellular signal; using a near-field communications signal; using an electronically readable alpha-numeric code; using a digital signal; using an electronically visible display; using an acoustic signal; using a microwave signal; using a satellite signal; using an X-ray signal; using a pulsar signal; using a television signal, and; using a photo signal.

(314) In another embodiment of the method of the invention, accessing the online user account registry further comprises any one of the following: displaying a financial account; querying a financial account; displaying financial data of a financial account; processing a financial transaction via selecting a financial account; displaying an electronic audible signature of a financial account; displaying an electronic visible signature of a financial account; displaying a statement of a financial account; aggregating a plurality of invoices of a plurality of financial accounts; aggregating billing of a plurality of financial accounts; invoking a rule-module associated with a financial account; displaying a rule-module associated with a financial account; editing a rule-module associated with a financial account; registering a rule-module of a user; registering a financial account of the user.

(315) In another embodiment of the method of the invention, processing a financial transaction comprises any one of the following: invoking a rule-module via a rule-module nexus; a buyer-seller

exchange transaction wherein a user's financial account balance is adjusted and a payee's financial account is correspondingly adjusted, wherein the user and the payee are different parties; transferring financial data from a financial account of the user to another financial account of the user, wherein there is no buyer-seller exchange; redeeming a pre-paid ticket transaction for venue admittance, wherein there is no buyer-seller exchange; redeeming a pre-paid membership benefit transaction for venue admittance, wherein there is no buyer-seller exchange; an electronic benefits transfer, wherein there is no buyer-seller exchange.

(316) In another embodiment of the method of the invention, the buyer-seller exchange further comprises any one of the following: a credit transaction; a debit transaction; a scrip transaction; a rewards transaction; an electronic check transaction; a private label transaction; a stored value transaction; a brokerage trade transaction.

(317) Another embodiment of the method of the invention comprises registering a rule-module to a user within a rule-module nexus, said registered rule-module further comprising a pattern data associated with an execution command.

(318) In another embodiment of the method of the invention, the pattern data comprises any one of the following: personal legal name; a private code; a driver's license number; a unique user code; a physical POS unique user code; a virtual POS unique user code; a nexus access token identification code; a personal verification code; a secondary personal verification code; an emergency code; a plurality of financial accounts; demographic information; an email address; social security number; a mother's maiden name; a facial photograph; an Internet browsing pattern; a telephone number; a mailing address; a purchasing pattern; an authorized subordinated user; electronic data usage patterns; employee status; job title; data on user behavior patterns; a credit score; a digital certificate; a network credential; an Internet protocol address; a digital signature; an encryption key; an instant messaging address; personal medical records; an electronic audible signature, and; an electronic visible signature.

(319) In another embodiment of the method of the invention, the execution command comprises invoking at least one of the following: accessing the rule-module nexus; accessing a user account registry; accessing a financial account; authorizing a subordinated user to access a financial account of the user; displaying a financial account of the user; completing the financial transaction; authorizing settlement of the online transaction; displaying the pattern data; displaying the execution command; displaying the rule-module; notifying an emergency authority upon rule-module nexus receiving an emergency code of the user; accessing a third-party database; accessing an account issuer database, and; accessing a payee database.

(320) In another embodiment of the method of the invention, selecting a financial account comprises any one of the following: automatically selecting a financial account via invoking a rule-module of a rule-module nexus, and; manually selecting a financial account by the user via a user interface apparatus.

(321) In another embodiment of the method of the invention, manually selecting a financial account, comprises any one of the following: displaying a plurality of financial accounts to the user in parallel; displaying a plurality of financial accounts to the user in sequence; data-entering by the user via a touch-screen; data-entering by the user via a key pad; data-entering by the user via an audio receiver.

(322) In another embodiment of the method of the invention, invoking a rule-module associated with a financial account comprises any one of the following: appending a surcharge to a transaction based on predetermined criteria; preserving a processing preference of a party registered with the rule-module nexus, said party comprising any one of the following: an account issuer; a user, and; a payee.

(323) In another embodiment of the method of the invention, preserving the processing preference of the party comprises any one of the following: invoking criteria predetermined by the party for declining the financial transaction; invoking criteria predetermined by the party for approving the

financial transaction, and; invoking criteria predetermined by the party for a settlement of the financial transaction.

(324) In another embodiment of the method of the invention, accessing the online user account registry further comprises any one of the following: electronically comparing, via the verification platform, the user's bid unique user code with a user's registration unique user code, and making a matching determination for verifying the user's authority to access the rule-module nexus, said matching determination comprising any one of the following: a positive matching determination verifying that the user is authorized to access the user account registry, and; a negative matching determination, verifying that the user is not authorized to access the user account registry.

(325) In another embodiment of the method of the invention, upon a positive matching determination, the verification platform issues a verification approval code invoking a rule-module of a rule-module nexus.

(326) In another embodiment of the method of the invention, the verification approval code invokes a user account registry code identifying a user account registry, said user account registry code comprising any one of the following: no financial account identifier; no data identifying a primary financial account of the user for all financial transactions of the user; no financial data.

(327) In another embodiment of the method of the invention, the verification data further comprises a bid personal verification code provided directly by the user, wherein the verification platform electronically compares the user's bid unique user code and the user's bid personal verification code with a user's registration unique user code and a user's registration personal verification code, and makes a matching determination for verifying the user's authority to access the rule-module nexus, said matching determination comprising any one of the following: making a positive matching determination verifying that the user is authorized to access the rule-module nexus, and; failing to make a positive matching determination, wherein making a negative matching determination is automatic, verifying that the user is not authorized to access the rule-module nexus.

(328) In another embodiment of the method of the invention, ranking the plurality of financial accounts of a user account registry comprises predetermined criteria being used for tagging and ranking the plurality of financial accounts in a certain order.

(329) In another embodiment of the method of the invention, the predetermined criteria for the ranking further comprises any one of the following: improving a transaction benefit for an account issuer; improving a transaction benefit for the user, and; improving a transaction benefit for a payee.

(330) In another embodiment of the method of the invention, an auction rule-module computes the automatic selection of a financial account from among a plurality of financial accounts competing to be offered to the User based up predetermined criteria comprising any one of the following: improving a transaction benefit for an account issuer; improving a transaction benefit for the user, and; improving a transaction benefit for a payee.

(331) In another embodiment of the method of the invention, the nexus access token identification code comprises any one of the following: being associated with an account issuer, and; the account issuer receiving a business benefit via the rule-module nexus.

(332) In another embodiment of the method of the invention, the business benefit comprises any one of the following: a percentage-based fee; a flat fee; a one-time fee; a recurring fee.

(333) In another embodiment of the system of the invention, accessing an online user account registry, comprises any one of the following: a) A nexus access token, comprising a verification data storage means, said verification data comprising a unique user code; b) An online user account registry, comprising a financial account storage means and a plurality of financial accounts of a user; c) A rule-module nexus, comprising a scalable rule-module storage means and a rule-module; d) A verification platform, comprising comparing means; e) Interconnecting communications means between devices, said devices comprising any one of the following: the nexus access token; the online user account registry; the verification platform, and; the rule-module nexus.

(334) In another embodiment of the system of the invention, the unique user code is thin-client, comprising any one of the following: no financial account identifier; no live financial data; no payment-enabling data; a rule-module nexus routing code; a user account registry code; an electronic positioning system code, and; a network routing instruction for processing the financial transaction via a rule-module nexus.

(335) In another embodiment of the system of the invention, the online user account registry comprises any one of the following: storage means apart from the nexus access token, and; accessibility via communicating externally to the nexus access token.

(336) In another embodiment of the system of the invention, communicating means external to the nexus access token comprises any one of the following: an electronically readable magnetic stripe; a smart card; a bar code; a radio signal; an infrared signal; an audible signal; a USB connection; a global positioning signal; a local area network; a wide area network; an X.25 network; an internet connection; a cellular signal; a near-field communications signal; an electronically readable alpha-numeric code; a digital signal; an electronically visible display; an acoustic signal; a microwave signal; a satellite signal; an X-ray signal; a pulsar signal; a television signal, and; a photo signal.

(337) In another embodiment of the system of the invention, the online user account registry further comprises any one of the following: a financial account display means; a financial account querying means; a financial data display means; a financial account selecting means; a financial transaction processing means; a financial account electronic audible signature display means; a financial account electronic visible signature display means; a financial account statement display means; a plurality of financial accounts invoice aggregating means; a plurality of financial accounts billing aggregating means; a rule-module invoking means; a rule-module display means; a rule-module editing means; a rule-module registering means; a financial account registering means.

(338) In another embodiment of the system of the invention, the financial transaction processing means further comprises any one of the following: a rule-module invoking means via a rule-module nexus; a buyer-seller exchange means wherein a user's financial account balance is adjusted and a payee's financial account is correspondingly adjusted, wherein the user and the payee are different parties; a financial data transferring means, comprising means for transferring financial data from a financial account of the user to another financial account of the user, wherein there is no buyer-seller exchange; a pre-paid ticket redeeming means, comprising means for venue admittance, wherein there is no buyer-seller exchange; a pre-paid membership benefit redeeming means, comprising means for venue admittance, wherein there is no buyer-seller exchange; an electronic benefits transfer means, wherein there is no buyer-seller exchange.

(339) In another embodiment of the system of the invention, the buyer-seller exchange further comprises any one of the following: a credit transaction; a debit transaction; a scrip transaction; a rewards transaction; an electronic check transaction; a private label transaction; a stored value transaction; a brokerage trade transaction.

(340) Another embodiment of the system of the invention comprises a rule-module registering means via a rule-module nexus, said registered rule-module further comprising a pattern data associated with an execution command.

(341) In another embodiment of the system of the invention, the pattern data comprises any one of the following: personal legal name; a private code; a driver's license number; a unique user code; a physical POS unique user code; a virtual POS unique user code; a nexus access token identification code; a personal verification code; a secondary personal verification code; an emergency code; a plurality of financial accounts; demographic information; an email address; social security number; a mother's maiden name; a facial photograph; an Internet browsing pattern; a telephone number; a mailing address; a purchasing pattern; an authorized subordinated user; electronic data usage patterns; employee status; job title; data on user behavior patterns; a credit score; a digital certificate; a network credential; an Internet protocol address; a digital signature; an encryption key; an instant messaging address; personal medical records; an electronic audible signature, and;

an electronic visible signature.

(342) In another embodiment of the system of the invention, the execution command comprises any one of the following: rule-module nexus accessing means; an online user account registry accessing means; a financial account accessing means; a financial account subordinated user access authorizing means; a financial account displaying means; a financial transaction completing means; a financial transaction settlement authorizing means; a pattern data displaying means; an execution command displaying means; a rule-module displaying means; an emergency authority notification means via a rule-module nexus receiving an emergency code; a third-party database accessing means; an account issuer database accessing means, and; a payee database accessing means.

(343) In another embodiment of the system of the invention, the financial account selecting means further comprises any one of the following: a financial account automatic selecting means invoking a rule-module of a rule-module nexus, and; a financial account manual selecting means via a user interface apparatus.

(344) In another embodiment of the system of the invention, the financial account manual selecting means further comprises any one of the following: means for displaying a plurality of financial accounts in-parallel; means for displaying a plurality of financial accounts in-sequence; a touch-screen data-entering means; a key pad data-entering means; an audio receiver data-entering means.

(345) In another embodiment of the system of the invention, the rule-module invoking means further comprises any one of the following: a financial transaction surcharge appending means via predetermined criteria; a financial transaction processing preference preserving means.

(346) In another embodiment of the system of the invention, the financial transaction processing preference preserving means further comprises any one of the following: a financial transaction declining criteria invoking means; a financial transaction approving criteria invoking means, and; a financial transaction settlement criteria invoking means.

(347) In another embodiment of the system of the invention, the verification platform comparing means further comprises any one of the following: the user's bid unique user code being electronically compared with a user's registered unique user code; a matching determination means, verifying the user's authority to access the rule-module nexus, said matching determination means further comprising any one of the following: a positive matching determination means, verifying that the user is authorized to access the user account registry, and; a negative matching determination means, verifying that the user is not authorized to access the user account registry.

(348) In another embodiment of the system of the invention, the positive matching determination means further comprises any one of the following: a verification approval code invoked via the verification platform; a rule-module invoked via a rule-module nexus.

(349) Another embodiment of the system of the invention comprises an online user account registry identifying means, wherein a user account registry code invoked via the verification approval code, said user account registry code comprising any one of the following: no financial account identifier; no data identifying a primary financial account of the user for all financial transactions of the user; no financial data.

(350) In another embodiment of the system of the invention, the verification platform comparing means further comprises any one of the following: the user's bid personal verification code and bid unique user code being electronically compared with a user's registration unique user code and a user's registration personal verification code; a matching determination means, verifying the user's authority to access the rule-module nexus, said matching determination further comprising any one of the following: a positive matching determination means, verifying that the user is authorized to access the rule-module nexus, and; a negative matching determination means, verifying that the user is not authorized to access the rule-module nexus.

(351) Another embodiment of the system of the invention comprises a plurality of financial accounts ranking means via the user account registry, comprising predetermined criteria being used for tagging and ranking the plurality of financial accounts in a certain order.

(352) In another embodiment of the system of the invention, the predetermined criteria for the ranking further comprises any one of the following: improving a transaction benefit for an account issuer; improving a transaction benefit for the user, and; improving a transaction benefit for a payee.

(353) Another embodiment of the system of the invention comprises an auction rule-module computing means, wherein automatically selecting a financial account from among a plurality of financial accounts competing to be offered to the User, is based upon predetermined criteria, said criteria comprising any one of the following: improving a transaction benefit for an account issuer; improving a transaction benefit for the user, and; improving a transaction benefit for a payee.

(354) In another embodiment of the system of the invention, the nexus access token identification code comprises any one of the following: being associated with an account issuer, and; the account issuer receiving a business benefit via the rule-module nexus.

(355) In another embodiment of the system of the invention, the business benefit comprises any one of the following: a percentage-based fee; a flat fee; a one-time fee; a recurring fee.

(356) In another embodiment of the method of the invention, accessing an online user account registry, comprises any one of the following: a) Approving access to an online user account registry via a verification platform comparing a bid verification data, comprising a unique user code provided from a nexus access token, with a registered verification data;

(357) b) Accessing the online user account registry via a rule-module invoked from a rule-module nexus, said online user account registry comprising a plurality of financial accounts of the user, wherein accessing further comprises any one of the following: (i) Displaying a default financial account invoked by a rule-module from a rule-module nexus, and; (ii) Displaying a plurality of financial accounts via a rule-module invoked from a rule-module nexus; Whereby an online account registry, comprising a plurality of financial accounts, is accessed via a unique user code provided from a nexus access token.

(358) In another embodiment of the system of the invention, accessing an online user account registry, comprises any one of the following: a) A nexus access token, comprising a verification data storage means, said verification data comprising a unique user code; b) An online user account registry, comprising a financial account storage means and a plurality of financial accounts of a user; c) A verification platform, comprising a comparing means; d) A rule-module nexus, comprising a scalable rule-module storage means and a rule-module; e) A display means, comprising any one of the following: (i) A default financial account display means; (ii) A plurality of financial accounts display means; f) Interconnecting communications means between devices, said devices comprising any one of the following: the nexus access token; the online user account registry; the verification platform; the rule-module nexus, and; the display means.

(359) In another embodiment of the method of the invention, the nexus access token further comprises any one of the following: being of a handheld size; being smaller than of a handheld size.

(360) In another embodiment of the system of the invention, the nexus access token further comprises any one of the following: being of a handheld size; being smaller than of a handheld size.

(361) Another embodiment of the method of the invention comprises any one of the following: a) Approving access to an online user account registry via a verification platform comparing a bid verification data, comprising a personal verification code (and optionally a secondary personal verification code), with a registered verification data; b) Accessing the online user account registry via a rule-module invoked from a rule-module nexus, said online user account registry comprising a plurality of financial accounts of the user; Whereby an online account registry, comprising a plurality of financial accounts, is accessed via a personal verification code.

(362) Another embodiment of the system of the invention comprises any one of the following: a) A user interface apparatus, comprising data entry means, wherein said data comprises a personal

verification code; b) An online user account registry, comprising a financial account storage means and a plurality of financial accounts of a user; c) A rule-module nexus, comprising a scalable rule-module storage means and a rule-module; d) A verification platform, comprising comparing means; e) Interconnecting communications means between devices, said devices comprising any one of the following: the user interface apparatus; the online user account registry; the verification platform, and; the rule-module nexus.

(363) Another embodiment of the method of the invention comprises any one of the following: a) Approving access to an online user account registry via a verification platform comparing a bid verification data, comprising a personal verification code, with a registered verification data; b) Accessing the online user account registry via a rule-module invoked from a rule-module nexus, said online user account registry comprising a plurality of financial accounts of the user, wherein accessing further comprises any one of the following: (i) Displaying a default financial account invoked by a rule-module from a rule-module nexus, and; (ii) Displaying a plurality of financial accounts via a rule-module invoked from a rule-module nexus; Whereby an online account registry, comprising a plurality of financial accounts, is accessed via a personal verification code.

(364) Another embodiment of the system of the invention comprises any one of the following: a) A user interface apparatus, comprising data entry means, wherein said data comprises a personal verification code; b) An online user account registry, comprising a financial account storage means and a plurality of financial accounts of a user; c) A verification platform, comprising a comparing means; d) A rule-module nexus, comprising a scalable rule-module storage means and a rule-module; e) A display means, comprising any one of the following: (i) A default financial account display means; (ii) A plurality of financial accounts display means; f) Interconnecting communications means between devices, said devices comprising any one of the following: the user interface apparatus; the online user account registry; the verification platform; the rule-module nexus, and; the display means.

(365) In another embodiment of the method of the invention, accessing an online financial transaction via a rule-module nexus comprises the steps of: a) registering a user, comprising any one of the following: (i) registering a plurality of rule-modules to a user within a rule-module nexus, said registered rule-module further comprising pattern data associated with an execution command; (ii) registering to the user within an online user account registry, a plurality of proprietary financial accounts, each comprising an account identifier, said account identifier further comprising any one of the following: a code identifying an account issuer, as approved by the International Organization for Standardization [(Bank Identification Code, or BIC Code) or (Bank Identification Number, or BIN)]; a code identifying an account issuer, as approved by the Society for Worldwide Interbank Financial Telecommunications (SWIFT Address or SWIFT Code), and; a code identifying an account issuer, as approved by the American Banker's Association (ABA Routing Code); b) processing pan-portfolio analytics, comprising any one of the following: (i) accessing a plurality of proprietary financial accounts, wherein upon a verification platform verifying a party is authorized to access the online user account registry, the rule-module nexus invokes a rule-module governing the party's access to a plurality of proprietary financial accounts within the online user account registry; (ii) surveying financial data, wherein a pattern in processing a plurality of online financial transactions is detected by an authorized party via surveying financial data residing within the plurality of proprietary financial accounts; (iii) presenting analytics, wherein an analysis of a detected pattern is presented to the authorized party; Whereby, pan-portfolio analytics are presented by an authorized party accessing a plurality of proprietary financial accounts in an online user account registry via a rule-module nexus.

(366) In another embodiment of the method of the invention, the analytics further comprises any one of the following: a pattern of fraud; a pattern of product purchasing; a pattern of preferences for processing a financial transaction; a pattern of financial account selection; a pattern of rewards redemption; a pattern of user geographic locations; a pattern retailer locations; a pattern of costs in

processing a financial transaction.

(367) In another embodiment of the method of the invention, the pattern further comprises any one of the following: a prior pattern; a real-time pattern; a predictive pattern.

(368) In another embodiment of the method of the invention, a personal verification code (PVC) association step comprises any one of the following: matching the personal verification code (PVC) with a live personal identification number (PIN) for processing a debit financial transaction of the user, and; translating the personal verification code (PVC) into a live personal identification number (PIN) for processing a debit financial transaction of the user.

(369) In another embodiment of the method of the invention, a personal verification code (PVC) forwarding step comprises any one of the following: forwarding the PIN to a debit processing network, and authorizing a PIN-enabled debit financial transaction; forwarding the PVC to a debit financial transaction processing network, and authorizing a PIN-enabled debit financial transaction; forwarding the PVC to a financial transaction processing network and authorizing a PIN-enabled financial transaction; forwarding the PVC to a financial transaction processing network and authorizing a non-PIN enabled financial transaction.

(370) In another embodiment of the method of the invention, the debit processing network comprises any one of the following: Star®; NYCE®; PULSE®, and; Interlink®.

(371) In another embodiment of the method of the invention, the debit financial transaction further comprises any one of the following: a demand deposit account; an automated clearinghouse (ACH); a cash-back to the user at a point of sale.

(372) In another embodiment of the method of the invention, a private code presenting step comprises a private code registered to the user, distinct from a personal verification code and not used in verifying the user, being presented to the user via the user interface apparatus for verifying that the user interface apparatus is authentic.

(373) In another embodiment of the method of the invention, compiling pan-portfolio analytics comprises any one of the following: a) Approving access to an online user account registry via a verification platform comparing: a bid verification data, comprising a unique user code provided from a nexus access token, with a registered verification data; b) Accessing the online user account registry via a rule-module invoked from a rule-module nexus, said online user account registry comprising a plurality of financial accounts of the user; c) Accessing a plurality of proprietary financial accounts registered in the user account registry; d) Detecting patterns via analyzing financial data within said proprietary financial accounts, said patterns comprising any one of the following: a pattern of preferences for processing a financial transaction; a pattern of financial account selection; a pattern of rewards redemption; a pattern of user geographic locations; a pattern retailer locations; a pattern of costs in processing a financial transaction; Whereby pan-portfolio analytics are compiled via a plurality of otherwise inaccessible proprietary financial accounts being rendered accessible by the rule-module nexus.

(374) In another embodiment of the method of the invention, the analytics further comprises any one of the following: a pattern of fraud; a pattern of product purchasing; a pattern of preferences for processing a financial transaction; a pattern of financial account selection; a pattern of rewards redemption; a pattern of user geographic locations; a pattern retailer locations; a pattern of costs in processing a financial transaction.

(375) In another embodiment of the method of the invention, the pattern further comprises any one of the following: a prior pattern; a real-time pattern; a predictive pattern.

(376) In another embodiment of the system of the invention, accessing an online user account registry via a rule-module nexus comprises any one of the following: a) a rule-module nexus, configured to register a plurality of rule-modules to a user within a rule-module nexus, said rule-module further comprising pattern data associated with an execution command; b) an online user account registry, configured to register a plurality of proprietary financial accounts, each comprising an account identifier, said account identifier further comprising any one of the

following: a code identifying an account issuer, as approved by the International Organization for Standardization [(Bank Identification Code, or BIC Code) or (Bank Identification Number, or BIN)]; a code identifying an account issuer, as approved by the Society for Worldwide Interbank Financial Telecommunications (SWIFT Address or SWIFT Code), and; a code identifying an account issuer, as approved by the American Banker's Association (ABA Routing Code); c) a verification platform, configured to verify a party is authorized to access the online user account registry; d) a pan-portfolio analytics processing platform, comprising any one of the following: (i) means for accessing a plurality of proprietary financial accounts via the verification platform, the rule-module nexus, and the user account registry; (ii) means for surveying financial data within said proprietary financial accounts, and detecting a pattern in processing a plurality of online financial transactions; e) analytics presenting means, comprising means for presenting to the authorized party an analysis of a detected pattern; Whereby, pan-portfolio analytics are presented by an authorized party accessing a plurality of proprietary financial accounts in an online user account registry via a rule-module nexus.

(377) In another embodiment of the system of the invention, the analytics further comprises any one of the following: a pattern of fraud; a pattern of product purchasing; a pattern of preferences for processing a financial transaction; a pattern of financial account selection; a pattern of rewards redemption; a pattern of user geographic locations; a pattern retailer locations; a pattern of costs in processing a financial transaction.

(378) In another embodiment of the system of the invention, the pattern further comprises any one of the following: a prior pattern; a real-time pattern; a predictive pattern.

(379) In another embodiment of the system of the invention, a personal verification code (PVC) association platform comprises any one of the following: matching means, wherein the personal verification code (PVC) is matched with a live personal identification number (PIN) for processing a debit financial transaction of the user, and; translating means, wherein the personal verification code (PVC) is translated into a live personal identification number (PIN) for processing a debit financial transaction of the user.

(380) In another embodiment of the system of the invention, a personal verification code (PVC) forwarding platform comprises any one of the following: forwarding means, wherein the PIN is forwarded to a debit processing network, and a PIN-enabled debit financial transaction is authorized, and; forwarding means, wherein the PVC is forwarded to a debit financial transaction processing network, and a PIN-enabled debit financial transaction is authorized; forwarding means, wherein the PVC is forwarded to a financial transaction processing network, and a PIN-enabled financial transaction is authorized; forwarding means, wherein the PVC is forwarded to a financial transaction processing network, and a non-PIN enabled financial transaction is authorized.

(381) In another embodiment of the system of the invention, the debit processing network comprises any one of the following: Star®; NYCE®; PULSE®, and; Interlink®.

(382) In another embodiment of the system of the invention, the debit financial transaction further comprises any one of the following: a demand deposit account; an automated clearinghouse (ACH); a cash-back to the user at a point of sale.

(383) In another embodiment of the system of the invention, the user interface apparatus is configured to present to the user, via the rule-module nexus, a private code registered to the user, distinct from a personal verification code and not used in verifying the user, for verifying that the user interface apparatus is authentic.

(384) In another embodiment of the system of the invention, compiling pan-portfolio analytics comprises any one of the following: a) a verification platform, configured to compare a bid verification data, comprising a unique user code provided from a nexus access token, with a registered verification data; b) a rule-module nexus, configured to invoke a rule-module registered to a user, said rule-module comprising a pattern data associated with an execution command; c) an online user account registry, configured to comprise a plurality of proprietary financial accounts of

the user; c) a pan-portfolio analytics platform, configured to analyze financial data within said plurality of proprietary financial accounts, and to detect patterns, said patterns comprising any one of the following: a pattern of preferences for processing a financial transaction; a pattern of financial account selection; a pattern of rewards redemption; a pattern of user geographic locations; a pattern retailer locations; a pattern of costs in processing a financial transaction; Whereby pan-portfolio analytics are compiled via a plurality of otherwise inaccessible proprietary financial accounts being rendered accessible by the rule-module nexus.

(385) In another embodiment of the system of the invention, the analytics further comprises any one of the following: a pattern of fraud; a pattern of product purchasing; a pattern of preferences for processing a financial transaction; a pattern of financial account selection; a pattern of rewards redemption; a pattern of user geographic locations; a pattern retailer locations; a pattern of costs in processing a financial transaction.

(386) In another embodiment of the system of the invention, the pattern further comprises any one of the following: a prior pattern; a real-time pattern; a predictive pattern.

(387) In another embodiment of the method of the invention, auto-populating a user's pattern data with a plurality of financial accounts, comprises any one of the following: electronically pulling a credit report of the user; selecting financial accounts contained in said credit report; automatically registering said selected financial accounts via the rule-module nexus.

(388) In another embodiment of the method of the invention, auto-populating communications occur via a network comprising any one of the following: a cable network; a wireless network; a land-line phone network; the Internet; an intranet; a local area network, a wide area network; an electronic positioning network; a satellite network, and; an X.25 network.

(389) In another embodiment of the system of the invention, the registration platform is configured to auto-populate a user's pattern data with a plurality of financial accounts, comprising any one of the following: electronically pulling a credit report of the user; selecting financial accounts contained in said credit report; automatically registering said selected financial accounts via the rule-module nexus.

(390) In another embodiment of the system of the invention, the registration platform auto-populates financial accounts via a network comprising any one of the following: a cable network; a wireless network; a land-line phone network; the Internet; an intranet; a local area network, a wide area network; an electronic positioning network; a satellite network, and; an X.25 network.

(391) In another embodiment of the method of the invention, an on-line financial transaction comprises the steps of: a) registering a user, wherein a rule-module is registered to a user within a rule-module nexus, said rule-modules further comprising pattern data associated with an execution command; b) data-storing in a nexus access token, wherein a unique user code of the user is stored in a portable nexus access token; c) processing an on-line financial transaction, using the nexus access token and a user interface apparatus located remotely from the rule-module nexus, comprising any one of the following: (i) verifying a user, wherein a user's authority to access the rule-module nexus is verified on-line by a verification platform using verification data provided via the user interface apparatus, said verification data comprising a bid unique user code provided directly from the nexus access token, and; (ii) accessing financial accounts, wherein upon the verification platform verifying the user is authorized to access the rule-module nexus, the rule-module nexus enables concurrent access to a plurality of proprietary financial accounts of the user for completing the on-line financial transaction; Whereby, a nexus access token and a rule-module nexus provide an authorized user concurrent access to a plurality of proprietary financial accounts for processing an on-line financial transaction.

(392) In another embodiment of the method of the invention, registering a user further comprises a plurality of rule-modules being registered to a user.

(393) In another embodiment of the system of the invention, an on-line financial transaction comprises any one of the following: a) a registration platform, configured within a rule-module

nexus to comprise registering a rule-module to a user, said rule-modules further comprising pattern data associated with an execution command; b) a nexus access token, configured to comprise: portability, and; storing a unique user code of the user; c) an on-line financial transaction processing platform, comprising any one of the following: (i) a nexus access token; (ii) a user interface apparatus located remotely from the rule-module nexus, configured to gather bid verification data of the user, said verification data comprising a bid unique user code provided directly from the nexus access token, and; (iii) a verification platform configured to verify a user on-line using the bid verification data, wherein a user's authority to access the rule-module nexus is verified, and; (iv) a user account registry platform configured to access financial accounts via the rule-module nexus, wherein upon the verification platform verifying the user is authorized to access the rule-module nexus, the user is provided on-line concurrent access to a plurality of proprietary financial accounts of the user for completing the on-line financial transaction; Whereby, a nexus access token and a rule-module nexus provide an authorized user concurrent access to a plurality of proprietary financial accounts for processing an on-line financial transaction.

(394) In another embodiment of the system of the invention, the registration platform is further configured to comprise registering a plurality of rule-modules to a user.

(395) In another embodiment of the method of the invention, the verification platform verifies the user interface apparatus hardware verification code, and the rule-module nexus invokes a rule-module, said rule-module comprising any one of the following: accessing a financial account of a payee; a default electronic audible signature; a default electronic visible signature; a default financial account of a user; a private code of a user; predetermined criteria for tagging and ranking, in a certain order, a plurality of proprietary financial accounts of a user; a financial account routing code of a payee; transmitting payee data to the user interface apparatus, and; transmitting payee data to the nexus access token; settlement of the financial transaction wherein a financial account of the user is debited and a financial account of a payee is credited; a rewards incentive; an arbitrage rule-module, and; a financial transaction processing preference.

(396) In another embodiment of the system of the invention, the verification platform is configured to verify the user interface apparatus hardware verification code, and to invoke a rule-module from the rule-module nexus, said rule-module comprising any one of the following: accessing a financial account of a payee; a default electronic audible signature; a default electronic visible signature; a default financial account of a user; a private code of a user; predetermined criteria for tagging and ranking, in a certain order, a plurality of proprietary financial accounts of a user; a financial account routing code of a payee; transmitting payee data to the user interface apparatus, and; transmitting payee data to the nexus access token; settlement of the financial transaction wherein a financial account of the user is debited and a financial account of a payee is credited; a rewards incentive; an arbitrage rule-module, and; a financial transaction processing preference.

(397) In another embodiment of the method of the invention, processing an online financial transaction, comprising the steps of: a. registering a plurality of users within a rule-module nexus, comprising for each user: (i) registering a plurality of rule-modules to a plurality of users within a rule-module nexus, each rule-module comprising pattern data associated with an execution command; (ii) registering to each user an online user account registry, remotely located from the user, comprising: (a) a thin-client user account registry code, and; (b) a plurality of proprietary financial accounts, each having a registry financial account identifier comprising any one of the following: a code identifying an account issuer, as approved by the International Organization for Standardization [(Bank Identification Code, or BIC Code) or (Bank Identification Number, or BIN)]; a code identifying an account issuer, as approved by the Society for Worldwide Interbank Financial Telecommunications (SWIFT Address or SWIFT Code), and; a code identifying an account issuer, as approved by the American Banker's Association (ABA Routing Code); b. gifting to a user, comprising: (i) sending a transmission from a first user to a second user via the rule-module nexus, requesting permission of the second user to accept a gift from the first user, the gift

comprising any one of the following: (a) a credit of financial data to a financial account of the second user, and; (b) a purchase by the first user of a smart-product for the second user; (ii) sending a transmission from the second user to the first user via the rule-module nexus, authorizing permission for the first user to provide the gift to the second user; c. processing an online financial transaction, comprising: (i) verifying the first user, wherein the first user's authority to access the rule-module nexus is verified by an on-line verification platform using verification data provided by the first user via a user interface apparatus, said verification data comprising any one of the following: (a) a thin-client, non-biometric personal verification code; (b) a thin-client unique user code provided directly from a portable nexus access token, and; (c) a thin-client, non-biometric personal verification code and a thin-client unique user code provided directly from a portable nexus access token; (ii) accessing financial accounts, wherein upon the verification platform verifying the first user is authorized to access the rule-module nexus, a rule-module is invoked enabling the first user to access to a plurality of proprietary financial accounts via the first user's online user account registry, for completing the on-line financial transaction, said on-line financial transaction comprising any of the following: (a) debiting financial data from a financial account of the first user, and a corresponding credit of financial data to a financial account of the second user, and; (b) purchasing by the first user of a smart-product for the second user; Whereby, an on-line financial transaction between a plurality of socially networked users is processed via a rule-module nexus and an online user account registry.

(398) Another embodiment of the method of the invention further comprises processing pan-portfolio analytics for the smart-product, comprising any one of the following: a. accessing a plurality of proprietary financial accounts of the second user, wherein upon the second user's authorizing permission for the first user to purchase a smart-product for the second user, the rule-module nexus invokes a rule-module for surveying financial data residing within the plurality of proprietary financial accounts of the second user, wherein a pattern from a plurality of online financial transactions is detected by the rule-module nexus; b. presenting analytics, wherein an analysis of the detected pattern is presented to the first user, such that a smart-product may be purchased by the first user for the second user.

(399) In another embodiment of the method of the invention, the pan-portfolio analytics further comprise any one of the following: a pattern of product purchasing; a pattern of financial account selection; a pattern of rewards redemption; a pattern of user geographic locations; a pattern of retailer locations; Whereby pan-portfolio analytics are compiled via a plurality of otherwise inaccessible proprietary financial accounts being rendered accessible by virtue of the rule-module nexus.

(400) In another embodiment of the method of the invention, the pattern further comprises any one of the following: a prior pattern; a real-time pattern; a predictive pattern.

(401) In another embodiment of the method of the invention, a rule-module is invoked by the second user to select at least one pattern to be shared via the rule-module nexus with the first user.

(402) Another embodiment of the method of the invention further comprises gift branding, wherein the gift is branded in the name of the first user, comprising any one of the following: the credit of financial data to a financial account of the second user is branded in the name of the first user; and; the smart-product is branded in the name of the first user.

(403) In another embodiment of the method of the invention, gift branding further comprises any one of the following: creating a new financial account within the second user's user account registry, wherein said new financial account is branded in the name of the first user; crediting financial data to an existing account within the second user's user account registry, wherein said credited financial data is branded in the name of the first user; labelling the smart-product with the name of the first user, such that the second user is notified that the smart-product was given by the first user.

(404) Another embodiment of the method of the invention further comprises displaying the gift

branding, comprising any one of the following: visibly or audibly displaying by a user interface apparatus of the gift branded financial account via an electronic signature in the name of the first user; visibly or audibly displaying by a user interface apparatus of the gift branded financial data associated with an existing financial account via an electronic signature in the name of the first user; visibly or audibly displaying by a smart-product of the name of the first user.

(405) Another embodiment of the method of the invention further comprises processing an online financial transaction, comprising: a. verifying the second user, wherein the second user's authority to access the rule-module nexus is verified by an on-line verification platform using verification data provided by the second user via a user interface apparatus, said verification data comprising any one of the following: (a) a thin-client, non-biometric personal verification code; (b) a thin-client unique user code provided directly from a portable nexus access token, and; (c) a thin-client, non-biometric personal verification code and a thin-client unique user code provided directly from a portable nexus access token; b. accessing financial accounts, wherein upon the verification platform verifying the second user is authorized to access the rule-module nexus, a rule-module is invoked enabling the second user to access to a plurality of proprietary financial accounts via the second user's online user account registry, for completing the on-line financial transaction, whereby the second user can select via a user interface apparatus from any one of the following: the gift branded financial account; the gift branded financial data; Whereby, an online user account registry and a rule-module nexus provide an authorized user access to a plurality of proprietary financial accounts for processing an on-line financial transaction.

(406) Another embodiment of the method of the invention further comprises gift notification, wherein the first user receives an electronic transmission via the rule-module nexus, notifying the first user that any one of the following has occurred: the receipt by the second user of the gift; the receipt by the second user of the credit of financial data; the usage by the second user of the gift-branded financial data; the usage by the second user of the gift-branded financial account.

(407) Another embodiment of the system of the invention comprises: a. a rule-module nexus, comprising means for registering a plurality of rule-modules to a plurality of users within a rule-module nexus, each rule-module comprising pattern data associated with an execution command; b. an online user account registry for each user, comprising means for registering to each user account registry: (i) a thin-client user account registry code, and; (ii) a plurality of proprietary financial accounts, each having a registry financial account identifier comprising any one of the following: a code identifying an account issuer, as approved by the International Organization for Standardization [(Bank Identification Code, or BIC Code) or (Bank Identification Number, or BIN)]; a code identifying an account issuer, as approved by the Society for Worldwide Interbank Financial Telecommunications (SWIFT Address or SWIFT Code), and; a code identifying an account issuer, as approved by the American Banker's Association (ABA Routing Code); c. means for gifting to a user, comprising: (i) means for sending a transmission from a first user to a second user via the rule-module nexus, requesting permission of the second user to accept a gift from the first user, the gift comprising any one of the following: (a) a credit of financial data to a financial account of the second user, and; (b) a purchase by the first user of a smart-product for the second user; (ii) means for sending a transmission from the second user to the first user via the rule-module nexus, authorizing permission for the first user to provide the gift to the second user; d. means for processing an online financial transaction, comprising: (i) an on-line verification platform using verification data provided by the first user via a user interface apparatus, said verification data comprising any one of the following: (a) a thin-client, non-biometric personal verification code; (b) a thin-client unique user code provided directly from a portable nexus access token, and; (c) a thin-client, non-biometric personal verification code and a thin-client unique user code provided directly from a portable nexus access token; (ii) means for accessing financial accounts, wherein upon the verification platform verifying the first user is authorized to access the rule-module nexus, a rule-module is invoked enabling the first user to access to a plurality of

proprietary financial accounts via the first user's online user account registry, for completing the on-line financial transaction, said on-line financial transaction comprising any of the following: (a) debiting financial data from a financial account of the first user, and a corresponding credit of financial data to a financial account of the second user, and; (b) purchasing by the first user of a smart-product for the second user; Whereby, an on-line financial transaction between a plurality of socially networked users is processed via a rule-module nexus and an online user account registry. (408) Another embodiment of the system of the invention further comprises means for processing pan-portfolio analytics for the smart-product, comprising any one of the following: a. means for accessing a plurality of proprietary financial accounts of the second user, wherein upon the second user's authorizing permission for the first user to purchase a smart-product for the second user, the rule-module nexus invokes a rule-module for surveying financial data residing within the plurality of proprietary financial accounts of the second user, wherein a pattern from a plurality of online financial transactions is detected by the rule-module nexus; b. means for presenting analytics, wherein an analysis of the detected pattern is presented to the first user, such that a smart-product may be purchased by the first user for the second user.

(409) In another embodiment of the system of the invention, the pan-portfolio analytics further comprise any one of the following: a pattern of product purchasing; a pattern of financial account selection; a pattern of rewards redemption; a pattern of user geographic locations; a pattern of retailer locations; Whereby pan-portfolio analytics are compiled via a plurality of otherwise inaccessible proprietary financial accounts being rendered accessible by virtue of the rule-module nexus.

(410) In another embodiment of the system of the invention, the pattern further comprises any one of the following: a prior pattern; a real-time pattern; a predictive pattern.

(411) Another embodiment of the system of the invention comprises means for invoking a rule-module, whereby the second user selects at least one pattern to be shared via the rule-module nexus with the first user.

(412) Another embodiment of the system of the invention, further comprises means for gift branding, wherein the gift is branded in the name of the first user, comprising any one of the following: the credit of financial data to a financial account of the second user is branded in the name of the first user; and; the smart-product is branded in the name of the first user.

(413) In another embodiment of the system of the invention, the means for gift branding further comprises any one of the following: creating a new financial account within the second user's user account registry, wherein said new financial account is branded in the name of the first user; crediting financial data to an existing account within the second user's user account registry, wherein said credited financial data is branded in the name of the first user; labelling the smart-product with the name of the first user, such that the second user is notified that the smart-product was given by the first user.

(414) Another embodiment of the system of the invention further comprising means for displaying the gift branding, comprising any one of the following: visibly or audibly displaying by a user interface apparatus of the gift branded financial account via an electronic signature in the name of the first user; visibly or audibly displaying by a user interface apparatus of the gift branded financial data associated with an existing financial account via an electronic signature in the name of the first user; visibly or audibly displaying by a smart-product of the name of the first user.

(415) In another embodiment of the system of the invention, processing an online financial transaction comprises: a. an on-line verification platform, comprising means for verifying the second user's authority to access the rule-module nexus by using verification data provided by the second user via a user interface apparatus, said verification data comprising any one of the following: (i) a thin-client, non-biometric personal verification code; (ii) a thin-client unique user code provided directly from a portable nexus access token, and; (iii) a thin-client, non-biometric personal verification code and a thin-client unique user code provided directly from a portable

nexus access token; b. an online user account registry comprising means for accessing financial accounts, wherein upon the verification platform verifying the second user is authorized to access the rule-module nexus, a rule-module is invoked enabling the second user to access to a plurality of proprietary financial accounts via the second user's online user account registry, for completing the on-line financial transaction, whereby the second user can select via a user interface apparatus from any one of the following: the gift branded financial account; the gift branded financial data; Whereby, an online user account registry and a rule-module nexus comprise means for authorizing user access to a plurality of proprietary financial accounts for processing an on-line financial transaction.

(416) Another embodiment of the system of the invention further comprises means for gift notification, wherein the first user receives an electronic transmission via the rule-module nexus, notifying the first user that any one of the following has occurred: the receipt by the second user of the gift; the receipt by the second user of the credit of financial data; the usage by the second user of the gift-branded financial data; the usage by the second user of the gift-branded financial account.

(417) Another embodiment of the method of the invention further comprises processing an online financial transaction executed at least partially by a computing device coupled to a communications network, comprising the steps of: a. registering a rule-module to a user within a rule-module nexus comprising computer executable programs tangibly embodied on a computer readable medium, each rule-module comprising pattern data associated with an execution command, wherein said pattern data comprises at least four of the following: (i) a plurality of proprietary financial accounts of the user; (ii) a thin-client user account registry code; (iii) a thin-client primary personal verification code, comprising a non-biometric data string formed from the input of alpha-numeric-symbolic characters; (iv) a thin-client secondary personal verification code, comprising a non-biometric data string formed from the input of alpha-numeric-symbolic characters; (v) a thin-client unique user code comprising a binary data string and encoded in a nexus access token, and; (vi) a thin-client private code, comprising a non-biometric data string, distinct from a personal verification code of the user and not used in verifying the user; presented to the user in the form of visible alpha-numeric-symbolic characters or audible sound; b. processing an online payor-payee financial transaction, wherein the user is the payor and a merchant is the payee, comprising: (i) Connecting to the rule-module nexus, comprising at least one of the following: (a) the user and the payee are each directly and independently connected to the rule-module nexus during simultaneous logon sessions via a communication network, and wherein upon the user clicking on a website widget of the payee, the rule-module nexus transmits secure socket layer data directly to the user interface apparatus for display to the user via a pop-up window, without requiring application programming interface integration with a website of the payee and without requiring a form re-direct of the user away from the website of the payee; (b) the buyer, the seller, and the acquirer are each directly and independently connected to the rule-module nexus via a communication network, not necessarily during simultaneous logon sessions, and wherein the rule-module nexus hibernates the financial transaction pending at least one of the following: verification of the user; invocation of a rule-module; selection of the financial account, and; submission of the selected financial account to the acquirer; (i) verifying the user, wherein the user's authority to access the rule-module nexus is verified by an on-line verification platform using verification data comprising at least two of the following: the primary personal verification code provided directly by the user via a user interface apparatus (UIA); the secondary personal verification code provided directly by the user via the UIA, and; the unique user code provided via the nexus access token; (ii) accessing financial accounts, wherein upon the verification platform verifying the user is authorized to access the rule-module nexus, at least one rule-module is invoked enabling the user to access a plurality of proprietary financial accounts via the user's online user account registry, for account selection and completion of the on-line financial transaction; Whereby, an on-line financial transaction is

processed via a rule-module nexus and an online user account registry comprising a plurality of proprietary financial accounts of the user.

(418) In another embodiment of the method of the invention, the unique user code comprises any of the following: a telephone number; a browser cookie; an email address; a subscriber identification module card number, and; an internet protocol address.

(419) In another embodiment of the method of the invention, the nexus access token, conjoined with the user's UTA, comprises any of the following: a cell phone; a personal computer; a USB thumb drive; a car's dashboard computer with interactive graphical user interface.

(420) Another embodiment of the method of the invention further comprises third-party transmittal of the unique user code, wherein the unique user code is transmitted to the verification platform from a trusted third-party with which the user has initiated a secure logon session using the nexus access token and the user interface apparatus.

(421) In another embodiment of the method of the invention, the secure socket layer data comprises at least one of the following: the private code; a data entry line and a prompt for user input of the primary personal verification code; a data entry line and a prompt for user input of the secondary personal verification code; HTTPS data; S-HTTP data.

(422) In another embodiment of the method of the invention, upon display of the private code, the user inputs verification data via the pop-up window, comprising at least one of the following: the primary personal verification code, and; the secondary personal verification.

(423) In another embodiment of the method of the invention, upon verification of the user's primary and/or secondary personal verification code(s), the rule-module nexus invokes a rule-module comprising at least one of the following: pre-determined instructions for shipping of a product being purchased by the user from the payee; a mailing address for the user; a user-customized rewards incentive; a user-customized advertisement; a user-customized default financial account; the invoking and forwarding to an acquirer by a PVC association platform, of a pre-registered personal identification number associated with a debit financial account.

(424) In another embodiment of the method of the invention, the rule-module invokes at least one audible/visible account signature for display to the user via the user interface apparatus.

(425) Another embodiment of the method of the invention further comprises registering a payee, wherein the payee registers a rule-module within the rule-module nexus, said rule-module comprising pattern data associated with an execution command, and said pattern data comprising at least one of the following: a unique payee code; a payee account registry code; a plurality of proprietary financial accounts; payee positioning system coordinates; a payee's roster of goods and/or services including associated pricing and universal product code(s); a payee's UTA-verification code.

(426) Another embodiment of the method of the invention, further comprises verifying the payee, wherein the payee provides a unique payee code to the verification platform for verifying the payee's authority to access the rule-module nexus.

(427) Another embodiment of the method of the invention further comprises invoking a payee rule-module, wherein upon the payee being verified by the verification platform, a rule-module of the payee is invoked comprising any of the following: instructions enabling the payee to access a plurality of proprietary financial accounts via the payee account registry; instructions authorizing the payee to provide the unique user code of the user to the rule-module nexus; instructions authorizing the payee to provide a user-customized payment transaction request for transmittal to the user.

(428) In another embodiment of the method of the invention, payee access to a plurality of proprietary financial accounts comprises invoking a payee depository account for receipt of funds from the user pursuant to the user-customized payment transaction request.

(429) Another embodiment of the method of the invention further comprises displaying a confirmation code, wherein upon the user's completion of the financial transaction and transmittal

of a financial transaction response packet to the payee, the user's UIA displays a confirmation code for presentation to a UIA of the payee, said confirmation code comprising any of the following: a computer-readable code, and; a visually readable alpha-numeric-symbolic code.

(430) Another embodiment of the system of the invention further comprises processing an online financial transaction, comprising a computing device coupled to a communications network: a. a rule-module nexus, comprising computer executable programs tangibly embodied on a computer readable medium, configured for registering a rule-module comprising pattern data associated with an execution command, wherein said pattern data comprises at least four of the following: (i) a plurality of proprietary financial accounts of the user; (ii) a thin-client user account registry code; (iii) a thin-client primary personal verification code, comprising a non-biometric data string formed from the input of alpha-numeric-symbolic characters; (iv) a thin-client secondary personal verification code, comprising a non-biometric data string formed from the input of alpha-numeric-symbolic characters; (v) a thin-client unique user code comprising a binary data string and encoded in a nexus access token, and; (vi) a thin-client private code, comprising a non-biometric data string, distinct from a personal verification code of the user and not used in verifying the user; presented to the user in the form of visible alpha-numeric-symbolic characters or audible sound; b. means for processing an online payor-payee financial transaction, wherein the user is the payor and a merchant is the payee, comprising: (i) Means for connecting to the rule-module nexus, comprising at least one of the following: (a) the user and the payee are each directly and independently connected to the rule-module nexus during simultaneous logon sessions via a communication network, and wherein upon the user clicking on a website widget of the payee, the rule-module nexus transmits secure socket layer data directly to the user interface apparatus for display to the user via a pop-up window, without requiring application programming interface integration with a website of the payee and without requiring a form re-direct of the user away from the website of the payee; (b) the buyer, the seller, and the acquirer are each directly and independently connected to the rule-module nexus via a communication network, not necessarily during simultaneous logon sessions, and wherein the rule-module nexus hibernates the financial transaction pending at least one of the following: verification the user; invocation of a rule-module; selection of the financial account, and; submission of the selected financial account to the acquirer; (ii) An on-line verification platform, comprising means for verifying the user, wherein the user's authority to access the rule-module nexus is verified by an on-line verification platform using verification data comprising at least two of the following: the primary personal verification code provided directly by the user via a user interface apparatus (UIA); the secondary personal verification code provided directly by the user via the UIA, and; the unique user code provided via the nexus access token; (iii) An online user account registry, comprising means for accessing financial accounts, wherein upon the verification platform verifying the user is authorized to access the rule-module nexus, at least one rule-module is invoked enabling the user to access a plurality of proprietary financial accounts via the user's online user account registry, for account selection and completion of the on-line financial transaction, said on-line financial transaction comprising a payor-payee transaction, wherein the user is the payor and a merchant is the payee, and; Whereby, an on-line financial transaction is processed via a rule-module nexus and an online user account registry comprising a plurality of proprietary financial accounts of the user.

(431) Another embodiment of the system of the invention further comprises the unique user code comprises any of the following: a telephone number; a browser cookie; an email address; a subscriber identification module card number, and; an internet protocol address.

(432) In another embodiment of the system of the invention, the nexus access token, conjoined with the user's UTA, comprises any of the following: a cell phone; a personal computer; a USB thumb drive; a car's dashboard computer with interactive graphical user interface.

(433) Another embodiment of the system of the invention further comprises means for third-party transmittal of the unique user code, wherein the unique user code is transmitted to the verification

platform from a trusted third-party with which the user has initiated a secure logon session using the nexus access token and the user interface apparatus.

(434) In another embodiment of the system of the invention, the secure socket layer data comprises at least one of the following: the private code; a data entry line and a prompt for user input of the primary personal verification code; a data entry line and a prompt for user input of the secondary personal verification code; HTTPS data; S-HTTP data.

(435) In another embodiment of the system of the invention, upon display of the private code, the user inputs verification data via the pop-up window, comprising at least one of the following: the primary personal verification code, and; the secondary personal verification.

(436) In another embodiment of the system of the invention, upon verification of the user's primary and/or secondary personal verification code(s), the rule-module nexus invokes a rule-module comprising at least one of the following: pre-determined instructions for shipping of a product being purchased by the user from the payee; a mailing address for the user; a user-customized rewards incentive; a user-customized advertisement; a user-customized default financial account; the invoking and forwarding to an acquirer by a PVC association platform, of a pre-registered personal identification number associated with a debit financial account.

(437) In another embodiment of the system of the invention, the rule-module invokes at least one audible/visible account signature for display to the user via the user interface apparatus.

(438) Another embodiment of the system of the invention further comprises means for registering a payee, wherein the payee registers a rule-module within the rule-module nexus, said rule-module comprising pattern data associated with an execution command, and said pattern data comprising at least one of the following: a unique payee code; a payee account registry code; a plurality of proprietary financial accounts; payee positioning system coordinates; a payee's roster of goods and/or services including associated pricing and universal product code(s); a payee's UTA-verification code.

(439) Another embodiment of the system of the invention further comprises means for verifying the payee, wherein the payee provides a unique payee code to the verification platform for verifying the payee's authority to access the rule-module nexus.

(440) Another embodiment of the system of the invention further comprises means for invoking a payee rule-module, wherein upon the payee being verified by the verification platform, a rule-module of the payee is invoked comprising any of the following: instructions enabling the payee to access a plurality of proprietary financial accounts via the payee account registry; instructions authorizing the payee to provide the unique user code of the user to the rule-module nexus; instructions authorizing the payee to provide a user-customized payment transaction request for transmittal to the user.

(441) In another embodiment of the system of the invention, payee access to a plurality of proprietary financial accounts comprises invoking a payee depository account for receipt of funds from the user pursuant to the user-customized payment transaction request.

(442) Another embodiment of the system of the invention further comprises means for displaying a confirmation code, wherein upon the user's completion of the financial transaction and transmittal of a financial transaction response packet to the payee, the user's UIA displays a confirmation code for presentation to a UIA of the payee, said confirmation code comprising any of the following: a computer-readable code, and; a visually readable alpha-numeric-symbolic code.

(443) In another embodiment of the method of the invention, accessing an online user account registry, comprises: a) approving access authorization to an online user account registry via a verification platform comparing a bid verification data, comprising a thin-client unique user code provided by a nexus access token of a user, with a registered verification data, comprising a previously stored thin-client unique user code; b) accessing the online user account registry via a rule-module invoked from a rule-module nexus, said online user account registry comprising a plurality of proprietary financial accounts of the user; whereby an online account registry is

accessed via a thin-client unique user code provided from a nexus access token.

(444) In another embodiment of the invention, a computer implemented method for processing an online financial transaction in a computer system is executed at least partially by at least one computing device coupled to a communications network, comprising the steps of: a) registering a rule-module to a user within a rule-module nexus associated with the at least one computing device, each rule-module comprising pattern data associated with an execution command, wherein said pattern data comprises at least two of the following: (i) a plurality of audible/visible account signatures of the user; (ii) a thin-client user account registry code; (iii) a thin-client primary personal verification code, comprising a non-biometric data string formed from the input of alpha-numeric-symbolic characters; (iv) a thin-client secondary personal verification code, comprising a non-biometric data string formed from the input of alpha-numeric-symbolic characters; (v) a thin-client unique user code comprising a binary data string and encoded in a nexus access token, and; (vi) a thin-client private code, comprising a non-biometric data string, distinct from a personal verification code of the user and not used in verifying the user; presented to the user in the form of visible alpha-numeric-symbolic characters or audible sound; b) processing an online payor-payee financial transaction, wherein the user is the payor and a merchant is the payee, comprising: (i) Connecting to the rule-module nexus, comprising the nexus access token (NAT) of the payor being in direct communication with the rule-module nexus (RMN), wherein the NAT submits a payment request to the RMN using any of the following: a mobile application, an SMS message, and a web browser; (ii) verifying the user, wherein the user's authority to access the RMN is verified by an on-line verification platform using verification data comprising any of the following: the primary personal verification code provided directly by the user via a user interface apparatus (UIA); the secondary personal verification code provided directly by the user via the UIA, and; the unique user code provided via the NAT; (iii) Transmitting via the RMN of an electronic notification of the payment request to a device of the payee using any of the following: a mobile application, an SMS message, and a web browser; (iv) Transmitting to the RMN, via the device of the Payee, of a message approving the payment request, whereby the RMN: (a) invokes the rule-module selecting at least one financial account of the payee, and; (b) electronically transmits to the NAT a displayable message comprising both the payee's payment request approval and the at least one financial account signature; (v) Transmitting an electronic message by the user from the NAT to the RMN approving selection of the at least one audible/visible account signature, whereby the RMN electronically transmits the payment transaction to a merchant acquirer of the payee; (vi) Transmitting from the merchant acquirer to the RMN of a transaction authorization message, whereby the RMN electronically transmits a confirmation message to both the NAT and the device of the payee via any of the following: a mobile application, an SMS message, and a web browser; Whereby, the online financial transaction is processed via the rule-module nexus.

(445) In another embodiment of the method of the invention, the unique user code comprises any of the following: a telephone number; a browser cookie; an email address; a subscriber identification module card number, and; an internet protocol address.

(446) In another embodiment of the method of the invention, the NAT, conjoined with the user's UIA, comprises any of the following: a cell phone; a personal computer; a USB thumb drive; a car's dashboard computer with interactive graphical user interface.

(447) In another embodiment of the method of the invention, the unique user code is transmitted to the verification platform from a third-party with which the user has initiated a secure logon session using the NAT and the UIA.

(448) In another embodiment of the method of the invention, the rule-module nexus transmits secure socket layer data directly to the user interface apparatus for display to the user via a pop-up window, and wherein the secure socket layer data comprises at least one of the following: the private code; a data entry line and a prompt for user input of the primary personal verification code; a data entry line and a prompt for user input of the secondary personal verification code; HTTPS

data; S-HTTP data.

(449) In another embodiment of the method of the invention, the private code is displayed by the device of the user, whereupon the user inputs verification data via the pop-up window, comprising at least one of the following: the primary personal verification code, and; the secondary personal verification.

(450) In another embodiment of the method of the invention, upon verification of the user's primary and/or secondary personal verification code(s), the rule-module nexus invokes a rule-module comprising at least one of the following: pre-determined instructions for shipping of a product being purchased by the user from the payee; a mailing address for the user; a user-customized rewards incentive; a user-customized advertisement; a user-customized default financial account; the invoking and forwarding to an acquirer by a personal verification code association platform, of a pre-registered personal identification number associated with a debit financial account.

(451) In another embodiment of the method of the invention, the rule-module invokes a plurality of audible/visible account signatures for display to the user via the UTA of the NAT.

(452) In another embodiment of the method of the invention, the payee is registered, wherein the payee registers a rule-module within the rule-module nexus, said rule-module comprising pattern data associated with an execution command, and said pattern data comprising at least one of the following: a unique payee code; a payee account registry code; a plurality of financial accounts; payee positioning system coordinates; a payee's roster of goods and/or services including associated pricing and universal product code(s); a payee's UTA-verification code.

(453) In another embodiment of the method of the invention, the payee is verified, wherein the payee provides a unique payee code to the verification platform for verifying the payee's authority to access the rule-module nexus.

(454) In another embodiment of the method of the invention, a payee rule-module is invoked, wherein upon the payee being verified by the verification platform, a rule-module of the payee is invoked comprising any of the following: instructions enabling the payee to access a plurality of financial accounts via the payee account registry; instructions authorizing the payee to provide the unique user code of the user to the rule-module nexus; instructions authorizing the payee to provide a user-customized payment transaction request for transmittal to the user.

(455) In another embodiment of the method of the invention, payee access to a plurality of financial accounts comprises invoking a payee depository account for receipt of funds from the user pursuant to the user-customized payment transaction request.

(456) In another embodiment of the method of the invention, a confirmation code is displayed, wherein upon the user's completion of the financial transaction and transmittal of a financial transaction response packet to the payee, the user's UIA displays a confirmation code for presentation to a UIA of the payee, said confirmation code comprising any of the following: a computer-readable code, and; a visually readable alpha-numeric-symbolic code.

(457) Another embodiment of the invention comprises a computer system for processing an online financial transaction, comprising at least one computing device coupled to a communications network, the at least one computing device operable to perform a method comprising the steps of: a) registering a rule-module to a user within a rule-module nexus associated with the at least one computing device, each rule-module comprising pattern data associated with an execution command, wherein said pattern data comprises at least two of the following: (i) a plurality of audible/visible account signatures of the user; (ii) a thin-client user account registry code; (iii) a thin-client primary personal verification code, comprising a non-biometric data string formed from the input of alpha-numeric-symbolic characters; (iv) a thin-client secondary personal verification code, comprising a non-biometric data string formed from the input of alpha-numeric-symbolic characters; (v) a thin-client unique user code comprising a binary data string and encoded in a nexus access token, and; (vi) a thin-client private code, comprising a non-biometric data string,

distinct from a personal verification code of the user and not used in verifying the user; presented to the user in the form of visible alpha-numeric-symbolic characters or audible sound; b) Processing an online payor-payee financial transaction, wherein the user is the payor and a merchant is the payee, comprising: (i) A communications network connected to the rule-module nexus, comprising the nexus access token (NAT) of the payor being in direct communication with the rule-module nexus (RMN), wherein the NAT submits a payment request to the RMN using any of the following: a mobile application, an SMS message, and a web browser; (ii) An on-line verification platform, comprising operability to verify the user, wherein the user's authority to access the rule-module nexus is verified by the on-line verification platform using verification data comprising any of the following: the primary personal verification code provided directly by the user via a user interface apparatus (UIA); the secondary personal verification code provided directly by the user via the UIA, and; the unique user code provided via the nexus access token; (iii) An electronic notification of the payment request, transmitting via the RMN to a device of the payee using any of the following: a mobile application, an SMS message, and a web browser; (iv) An electronic message approving the payment request, transmitted to the RMN via the device of the Payee, whereby the RMN: (a) invokes the rule-module selecting at least one financial account of the payee, and; (b) electronically transmits to the NAT a displayable message comprising both the payee's payment request approval and the at least one audible/visible account signature; (v) An electronic message by the user transmitted from the NAT to the RMN approving selection of the at least one audible/visible account signature, whereby the RMN electronically transmits the payment transaction to a merchant acquirer of the payee; (vi) A transaction authorization message, transmitted from the merchant acquirer to the RMN, whereby the RMN electronically transmits a confirmation message to both the NAT and the device of the payee via any of the following: a mobile application, an SMS message, and a web browser; Whereby, the online financial transaction is processed via the rule-module nexus.

(458) In another embodiment of the system of invention, the unique user code comprises any of the following: a telephone number; a browser cookie; an email address; a subscriber identification module card number, and; an internet protocol address.

(459) In another embodiment of the system of invention, the nexus access token, conjoined with the user's UTA, comprises any of the following: a cell phone; a personal computer; a USB thumb drive; a car's dashboard computer with interactive graphical user interface.

(460) In another embodiment of the system of invention, the method further comprises transmitting the unique user code by a third-party, wherein the unique user code is transmitted to the verification platform from a third-party with which the user has initiated a secure logon session using the nexus access token and the user interface apparatus.

(461) In another embodiment of the system of invention, the rule-module nexus transmits secure socket layer data directly to the user interface apparatus for display to the user via a pop-up window, and wherein the secure socket layer data comprises at least one of the following: the private code; a data entry line and a prompt for user input of the primary personal verification code; a data entry line and a prompt for user input of the secondary personal verification code; HTTPS data; S-HTTP data.

(462) In another embodiment of the system of invention, the method further comprises displaying the private code by the device of the user, whereupon the user inputs verification data via the pop-up window, comprising at least one of the following: the primary personal verification code, and; the secondary personal verification.

(463) In another embodiment of the system of invention, the method further comprises, upon verification of the user's primary and/or secondary personal verification code(s), invoking a rule-module, said rule-module comprising at least one of the following: pre-determined instructions for shipping of a product being purchased by the user from the payee; a mailing address for the user; a user-customized rewards incentive; a user-customized advertisement; a user-customized default

financial account; the invoking and forwarding to an acquirer by a personal verification code association platform, of a pre-registered personal identification number associated with a debit financial account.

(464) In another embodiment of the system of invention, the rule-module invokes at least one audible/visible account signature for display to the user via the user interface apparatus of the nexus access token.

(465) In another embodiment of the system of invention, the method further comprises registering the payee, wherein the payee registers a rule-module within the rule-module nexus, said rule-module comprising pattern data associated with an execution command, and said pattern data comprising at least one of the following: a unique payee code; a payee account registry code; a plurality of financial accounts; payee positioning system coordinates; a payee's roster of goods and/or services including associated pricing and universal product code(s); a payee's UIA-verification code.

(466) In another embodiment of the system of invention, the method further comprises verifying the payee, wherein the payee provides a unique payee code to the verification platform for verifying the payee's authority to access the rule-module nexus.

(467) In another embodiment of the system of invention, the method further comprises invoking a payee rule-module, whereupon the payee being verified by the verification platform, a rule-module of the payee is invoked comprising any of the following: instructions enabling the payee to access a plurality of financial accounts via the payee account registry; instructions authorizing the payee to provide the unique user code of the user to the rule-module nexus; instructions authorizing the payee to provide a user-customized payment transaction request for transmittal to the user.

(468) In another embodiment of the system of invention, payee access to a plurality of financial accounts comprises invoking a payee depository account for receipt of funds from the user pursuant to the user-customized payment transaction request.

(469) In another embodiment of the system of invention, the method further comprises displaying a confirmation code, whereupon the user's completion of the financial transaction and transmittal of a financial transaction response packet to the payee, the user's UIA displays a confirmation code for presentation to a UIA of the payee, said confirmation code comprising any of the following: a computer-readable code, and; a visually readable alpha-numeric-symbolic code.

(470) In another embodiment of the method of the invention, a computer implemented method for enabling a payment transaction in a computer system is executed at least partially by a computer of a user connected via a communication network to a remote computer comprising a database of payment-related data, said method comprising the steps of: a) accessing the payment-related data of a user from within the database, wherein a web browser extension on the user's computer is invoked by the user clicking on a web browser extension graphical user interface (GUI) widget displayed on the user's computer, wherein secure network access is established to access the user's payment-related data from within the database; b) presenting the user's payment-related data via a pop-up window displayed on the user's computer, without requiring application programming interface integration with a website or a web application of a payee, and without requiring a form re-direct away from the website or the web application of the payee; c) enabling the payment transaction, wherein the web browser extension invokes an auto-complete function to automatically populate fields of a checkout form presented from the website or the web application of the payee with the user's payment-related data.

(471) In another embodiment of the method of the invention, the payment-related data comprises any of the following: an account number; a visible signature; an audible signature; the user's first and/or last name; an expiration date of the account number; a mailing address associated with the account number; a card verification value (CVV) number; a shipping address associated with the user; browsing data and/or purchasing data of the user; an advertisement customized for the user based on the browsing data and/or the purchasing data of the user; a rewards incentive customized

for the user based on browsing data and/or purchasing data of the user.

(472) An illustrative embodiment of the method of the invention, further comprises verifying the user, wherein to access the user's payment-related data in the database, any of the following data is provided via the web browser extension: a telephone number; an email address; a subscriber identification module (SIM) card number; an internet protocol address; a media access control (MAC) address; a globally unique identifier (GUID).

(473) An illustrative embodiment of the method of the invention, further comprises automatically populating the user's payment-related data on the database, wherein automatically populating comprises any of the following: a) electronically accessing a credit report of the user, automatically detecting at least one account number and other data of the user from within said credit report, and storing in the database said account number and said other data of the user; b) taking a picture of a payment or rewards card, invoking an optical character reading software program from within the database, and extracting payment-related data from the picture, said payment-related data comprising at least an account number.

(474) An illustrative embodiment of the method of the invention, further comprises transmitting the payment-related data via the communication network as secure socket layer data from the database to the user's computer.

(475) In another embodiment of the method of the invention, the communication network comprises any of the following: a cable network; a wireless network; a land-line phone network; the Internet; an intranet; a local area network, a wide area network; an electronic positioning network; a satellite network; an X.25 network.

(476) An illustrative embodiment of the method of the invention, further comprises verifying the database to the user, wherein a private code, previously registered with the database and associated with the payment-related data of the user, is transmitted from the database and displayed to the user via the user's computer.

(477) An illustrative embodiment of the method of the invention, further comprises processing browsing and/or purchasing analytics by the database, comprising: a) collecting the user's browsing data and/or purchasing data via the web browser extension and storing said browsing and/or said purchasing data in the database; b) the database surveying said browsing data and/or said purchasing data and detecting a pattern; c) presenting analytics, wherein an analysis by the database of the pattern detected in said browsing data and/or said purchasing data is presented to the user via the pop-up window on the display of the user's computer.

(478) In another embodiment of the method of the invention, the web browser extension is a computer program using an application programming interface from the database to integrate with any of the following web browsers: Google Chrome; Opera; Microsoft Explorer; Apple Safari; Mozilla Firefox.

(479) In another embodiment of the method of the invention, the pattern further comprises any of the following: a prior pattern; a real-time pattern; a predictive pattern.

(480) In another embodiment of the method of the invention, the analytics further comprises any of the following: a pattern of fraudulent use of the user's payment-related data; a pattern of product purchasing by the user; a pattern of the user's preferences for enabling a payment transaction; a pattern of payment account selection by the user; a pattern of rewards redemption by the user; a pattern of user geographic locations of the user; a pattern retailer locations; a pattern of costs in processing a payment transaction; a pattern by the user of viewing products or services on the web; a pattern by the user of viewing website content on the web.

(481) In another embodiment of the method of the invention, the analysis comprises any of the following: an advertisement customized by the database for the user, wherein said advertisement comprises a clickable; a reward incentive customized by the database for the user; an informational notification customized by the database for the user.

(482) In another embodiment of the system of the invention, at least one computer executable

program is tangibly embodied on a computer readable medium, comprising: a) a database of payment-related data stored on a remote computer; b) a communication network connecting the remote computer with a user's computer, wherein a web browser extension on the user's computer is invoked by the user clicking on a web browser extension GUI widget displayed on the user's computer, wherein secure network access is established to access the user's payment-related data from within the database; c) a display on the user's computer, wherein: (i) upon accessing the database, the payment-related data is presented to the user via a pop-up window on the display, without requiring application programming interface integration with a website or a web application of a payee, and without requiring a form re-direct away from the website or the web application; (ii) upon a checkout form from the website or web application of the payee being presented on the display, the web browser extension invokes an auto-complete function to automatically populate fields of the checkout form with the user's payment-related data.

(483) In another embodiment of the system of the invention, the payment-related data comprises any of the following: an account number; the user's first and last name; an expiration date of the account number; a mailing address associated with the account number; a card verification value (CVV) number; a shipping address associated with the user; browsing data and/or purchasing data of the user; an advertisement customized for the user based on the browsing data and/or the purchasing data of the user; a rewards incentive customized for the user based on browsing data and/or purchasing data of the user.

(484) Another embodiment of the system of the invention, further comprises operability to verify the user, wherein to access the user's payment-related data in the database, any of the following data is provided to the database via the web browser extension: a telephone number; a browser cookie; an email address; a subscriber identification module (SIM) card number; a media access control (MAC) address; a globally unique identifier (GUID).

(485) Another embodiment of the system of the invention, further comprises operability to automatically populate the user's payment related data on the database, wherein automatically populating comprises any of the following: a) electronically accessing a credit report of the user, automatically detecting at least one account number and other data of the user from within said credit report, and storing in the database said account number and said other data of the user; b) taking a picture of a payment or rewards card, invoking an optical character reading software program from within the database, and extracting payment-related data from the picture, said payment-related data comprising at least an account number.

(486) Another embodiment of the system of the invention, further comprises operability to transmit the payment-related data via the communication network as secure socket layer data from the database to the user's computer.

(487) In another embodiment of the system of the invention, the network comprises any of the following: a cable network; a wireless network; a land-line phone network; the Internet; an intranet; a local area network, a wide area network; an electronic positioning network; a satellite network; an X.25 network.

(488) Another embodiment of the system of the invention, further comprises operability to verify the database to the user, wherein a private code, previously registered with the database and associated with the payment-related data of the user, is transmitted from the database and displayed to the user via the user's computer.

(489) Another embodiment of the system of the invention, further comprises operability to process browsing and/or purchasing analytics by the database, comprising: a) collecting the user's browsing data and/or purchasing data via the web browser extension and storing said browsing and/or said purchasing data in the database; b) the database surveying said browsing data and/or said purchasing data and detecting a pattern; c) presenting analytics, wherein an analysis by the database of the pattern detected in said browsing data and/or said purchasing data is presented to the user via the pop-up window on the display of the user's computer.

(490) In another embodiment of the system of the invention, the web browser extension is a computer program using an application programming interface from the database to integrate with any of the following web browsers: Google Chrome; Opera; Microsoft Explorer; Apple Safari; Mozilla Firefox.

(491) In another embodiment of the system of the invention, the pattern further comprises any of the following: a prior pattern; a real-time pattern; a predictive pattern.

(492) In another embodiment of the system of the invention, the analytics further comprises any of the following: a pattern of fraudulent use of the user's payment-related data; a pattern of product purchasing by the user; a pattern of the user's preferences for enabling a payment transaction; a pattern of payment account selection by the user; a pattern of rewards redemption by the user; a pattern of user geographic locations of the user; a pattern retailer locations; a pattern of costs in processing a payment transaction; a pattern by the user of viewing products or services on the web; a pattern by the user of viewing website content on the web.

(493) In another embodiment of the system of the invention, the analysis comprises any of the following: an advertisement customized by the database for the user; a reward incentive customized by the database for the user; an informational notification customized by the database for the user.

(494) In another embodiment of the method of the invention, a computer implemented method for enabling a payment transaction in a computer system is executed at least partially by at least one computer connected via at least one communication network to a remote computer comprising a database storing payment-related data of a user, comprising the following steps: a) integrating the remote computer's payment-related data of the user with a database residing on a payee's computer using an application programming interface provided from the remote computer's database; b) presenting to the user, via a display on the user's computer, a graphical user interface (GUI) widget on a website or a web application of the payee; c) the user clicking on the GUI widget, wherein any of the following occurs without a form re-direct away from the website or the web application of the payee: (i) the user's payment-related data is presented to the user via a pop-up window displayed on the user's computer; (ii) an auto-complete function is invoked to automatically populate fields of a checkout form from the website or the web application with the user's payment-related data, said checkout form being presented to the user on the user's computer.

(495) In another embodiment of the method of the invention, the user's payment-related data comprises any of the following: an account number; a visible signature; an audible signature; the user's first and last name; an expiration date of the account number; a mailing address associated with the account number; a card verification value (CVV) number; a shipping address associated with the user; a private code, previously registered with the database and associated with the payment-related data of the user, which when transmitted from the database and displayed to the user via the pop-up window, serves to verify the database to the user; a data entry line and a prompt for the user to input at least one code comprising alpha, numeric, and/or symbolic data previously registered by the user with the database; a data entry line and a prompt for the user to input biometric data previously registered by the user with the database; HTTPS data; S-HTTP data; browsing data and/or purchasing data of the user collected via the web browser extension and stored in the database; an advertisement customized for the user by the database based on the browsing data and/or the purchasing data of the user collected via the web browser extension; a rewards incentive customized for the user by the database based on browsing data and/or purchasing data of the user collected via the web browser extension.

(496) Another embodiment of the method of the invention, further comprises verifying the user, wherein to access the user's payment-related data in the database, any of the following data is provided to the database via the user's computer: a telephone number; an email address; a subscriber identification module (SIM) card number; a media access control (MAC) address; a globally unique identifier (GUID).

(497) Another embodiment of the method of the invention, further comprises automatically populating the user's payment-related data on the database, wherein automatically populating comprises any of the following: a) electronically accessing a credit report of the user, automatically detecting at least one account number and other data of the user from within said credit report, and storing in the database said account number and said other data of the user; b) taking a picture of a payment or rewards card, invoking an optical character reading software program from within the database, and extracting payment-related data from the picture, said payment-related data comprising at least an account number.

(498) In another embodiment of the method of the invention, further comprises installing an application in the web browser of the user's computer as an extension of the web browser to provide any of the following: secure network access to the database; tracking the user's browsing patterns; tracking the user's purchasing patterns.

(499) In another embodiment of the method of the invention, the network comprises any of the following: a cable network; a wireless network; a land-line phone network; the Internet; an intranet; a local area network, a wide area network; an electronic positioning network; a satellite network; an X.25 network.

(500) Another embodiment of the method of the invention, further comprises verifying the database, wherein the private code is transmitted from the database and displayed to the user via the pop-up window, to verify the database to the user.

(501) Another embodiment of the method of the invention, further comprises processing browsing and/or purchasing analytics by the database, comprising: a) collecting the user's browsing data and/or purchasing data via the web browser extension and storing said browsing and/or said purchasing data in the database; b) the database surveying said browsing data and/or said purchasing data and detecting a pattern; c) presenting analytics, wherein an analysis by the database of the pattern detected in said browsing data and/or said purchasing data is presented to the user via the pop-up window on the display of the user's computer.

(502) In another embodiment of the method of the invention, the web browser extension is a computer program using an application programming interface from the database to integrate with any of the following web browsers: Google Chrome; Opera; Microsoft Explorer; Apple Safari; Mozilla Firefox.

(503) In another embodiment of the method of the invention, the pattern further comprises any of the following: a prior pattern; a real-time pattern; a predictive pattern.

(504) In another embodiment of the method of the invention, the analytics further comprises any of the following: a pattern of fraudulent use of the user's payment-related data; a pattern of product purchasing by the user; a pattern of the user's preferences for enabling a payment transaction; a pattern of payment account selection by the user; a pattern of rewards redemption by the user; a pattern of user geographic locations of the user; a pattern retailer locations; a pattern of costs in processing a payment transaction; a pattern by the user of viewing products or services on the web; a pattern by the user of viewing website content on the web.

(505) In another embodiment of the method of the invention, the analysis comprises any of the following: an advertisement customized by the database for the user; a reward incentive customized by the database for the user; an informational notification customized by the database for the user.

(506) In another embodiment of the system of the invention, at least one computer executable program is tangibly embodied on a computer readable medium, comprising: a) a database of payment-related data of a user configured on a remote computer; b) a computer of a user and a computer of a payee connected via at least one communication network to the remote computer; c) operability to integrate the remote computer's payment-related data of the user with a database residing on the payee's computer using an application programming interface provided from the remote computer's database; d) operability to present to the user, via a display on the user's

computer, a graphical user interface (GUI) widget on a website or web application of the payee; c) operability enabling the user to click on the GUI widget, wherein any of the following occurs without a form re-direct away from the website or the web application of the payee: (i) the user's payment-related data is presented to the user via a pop-up window displayed on the user's computer; (ii) an auto-complete function is invoked to automatically populate fields of a checkout form from the website or the web application with the user's payment-related data, said checkout form being presented to the user on the user's computer.

(507) In another embodiment of the system of the invention, the user's payment-related data comprises any of the following: an account number; a visible signature; an audible signature; the user's first and last name; an expiration date of the account number; a mailing address associated with the account number; a card verification value (CVV) number; a shipping address associated with the user; a private code, previously registered with the database and associated with the payment-related data of the user, which when transmitted from the database and displayed to the user via the pop-up window, serves to verify the database to the user; a data entry line and a prompt for the user to input at least one code comprising alpha, numeric, and/or symbolic data previously registered by the user with the database; a data entry line and a prompt for the user to input biometric data previously registered by the user with the database; HTTPS data; S-HTTP data; browsing data and/or purchasing data of the user collected via the web browser extension and stored in the database; an advertisement customized for the user by the database based on the browsing data and/or the purchasing data of the user collected via the web browser extension; a rewards incentive customized for the user by the database based on browsing data and/or purchasing data of the user collected via the web browser extension.

(508) Another embodiment of the system of the invention, further comprises operability to verify the user, wherein to access the user's payment-related data in the database, any of the following data is provided to the database via the user's computer: a telephone number; an email address; a subscriber identification module (SIM) card number; a media access control (MAC) address; a globally unique identifier (GUID).

(509) Another embodiment of the system of the invention, further comprises operability to automatically populate the user's payment-related data on the database, comprising any of the following: a) electronically accessing a credit report of the user, automatically detecting at least one account number and other data of the user from within said credit report, and storing in the database said account number and said other data of the user; b) taking a picture of a payment or rewards card, invoking an optical character reading software program from within the database, and extracting payment-related data from the picture, said payment-related data comprising at least an account number.

(510) Another embodiment of the system of the invention, further comprises operability to install an application into the web browser as an extension of the web browser to provide any of the following: secure network access to the database; tracking the user's browsing patterns; tracking the user's purchasing patterns.

(511) In another embodiment of the system of the invention, the network comprises any of the following: a cable network; a wireless network; a land-line phone network; the Internet; an intranet; a local area network, a wide area network; an electronic positioning network; a satellite network; an X.25 network.

(512) Another embodiment of the system of the invention, further comprises operability to verify the database, wherein the private code is transmitted from the database and displayed to the user via the pop-up window, to verify the database to the user.

(513) Another embodiment of the system of the invention, further comprises operability to process browsing and/or purchasing analytics, comprising: a) collecting the user's browsing data and/or purchasing data via the web browser extension and storing said browsing and/or said purchasing data in the database; b) the database surveying said browsing data and/or said purchasing data and

detecting a pattern; c) presenting analytics, wherein an analysis by the database of the pattern detected in said browsing data and/or said purchasing data is presented to the user via the pop-up window on the display of the user's computer.

(514) In another embodiment of the system of the invention, the web browser extension is a computer program using an application programming interface from the database to integrate with any of the following web browsers: Google Chrome; Opera; Microsoft Explorer; Apple Safari; Mozilla Firefox.

(515) In another embodiment of the system of the invention, the pattern further comprises any of the following: a prior pattern; a real-time pattern; a predictive pattern.

(516) In another embodiment of the system of the invention, the analytics further comprises any of the following: a pattern of fraudulent use of the user's payment-related data; a pattern of product purchasing by the user; a pattern of the user's preferences for enabling a payment transaction; a pattern of payment account selection by the user; a pattern of rewards redemption by the user; a pattern of user geographic locations of the user; a pattern retailer locations; a pattern of costs in processing a payment transaction; a pattern by the user of viewing products or services on the web; a pattern by the user of viewing website content on the web.

(517) In another embodiment of the system of the invention, the analysis comprises any of the following: an advertisement customized by the database for the user; a reward incentive customized by the database for the user; an informational notification customized by the database for the user.

(518) In another embodiment of the method of the invention, the auto-complete function is invoked via any of the following: by the user clicking on the payment-related data presented via the pop-up window; by the web browser extension automatically detecting the presentation of the checkout form on the display of the computer.

(519) In another embodiment of the system of the invention, the auto-complete function is invoked via any of the following: by the user clicking on the payment-related data presented via the pop-up window; by the web browser extension automatically detecting the presentation of the checkout form on the display of the computer.

(520) In another embodiment of the method of the invention, the auto-complete function is invoked via any of the following: a program residing on the user's computer; a program residing on the payee's computer; a program residing within the database on the remote computer.

(521) Another embodiment of the method of the invention, furthering comprises verifying the user, wherein the user submits verification data to the database via the pop-up window, said verification data comprising any of the following: at least one code comprising alpha, numeric, and/or symbolic data previously registered by the user with the database; biometric data previously registered by the user with the database.

(522) Another embodiment of the system of the invention, furthering comprises operability to verify the user, wherein the user submits verification data to the database via the pop-up window, said verification data comprising any of the following: a code comprising alpha, numeric, and/or symbolic data previously registered by the user with the database; biometric data previously registered by the user with the database.

(523) Another embodiment of the method of the invention, furthering comprises verifying the user, wherein the user submits verification data to the database via the pop-up window, said verification data comprising any of the following: a code comprising alpha, numeric, and/or symbolic data previously registered by the user with the database; biometric data previously registered by the user with the database.

(524) Another embodiment of the system of the invention, furthering comprises operability to verify the user, wherein the user submits verification data to the database via the pop-up window, said verification data comprising any of the following: a code comprising alpha, numeric, and/or symbolic data previously registered by the user with the database; biometric data previously

registered by the user with the database.

(525) In another embodiment of the method of the invention, the auto-complete function is invoked via any of the following: by the user clicking on the payment-related data presented via the pop-up window; by the web browser extension automatically detecting the presentation of the checkout form on the display of the computer.

(526) In another embodiment of the method of the invention, the computer comprises any of the following: a desktop personal computer; a smartphone; a tablet computer; a laptop personal computer; a mainframe.

(527) In another embodiment of the system of the invention, the computer comprises any of the following: a desktop personal computer; a smartphone; a tablet computer; a laptop personal computer; a mainframe.

(528) In another embodiment of the method of the invention, the account number comprises any of the following: a payment account number; a proxy number for a payment account number.

(529) In another embodiment of the system of the invention, the account number comprises any of the following: a payment account number; a proxy number for a payment account number.

(530) In another embodiment of the method of the invention, the payment-related data comprises at least one payment account being preferred, wherein the at least one payment account is visually indicated as being preferred for enabling the payment transaction based on having been ranked in the database as a function of any of the following criteria: lowest interest rate; highest available balance; lowest transaction processing fee; longest float of funds; highest rewards incentive; highest competitive bid between at least two third-parties as tendered to the database.

(531) In another embodiment of the method of the invention, the ranking is determined by any of the following: the user; the payee; at least one third-party.

(532) In another embodiment of the method of the invention, upon the payee determining the ranking, the user is provided a financial incentive for selecting the payment account which is preferred, and the financial incentive is provided by any of the following: the payee; the database.

(533) In another embodiment of the method of the invention, the database updates the criteria periodically via application programming interface integration with at least one financial institution associated with at least one payment account.

(534) In another embodiment of the system of the invention, the payment-related data comprises at least one payment account being preferred, wherein the at least one payment account is visually indicated as being preferred for enabling the payment transaction based on having been ranked in the database as a function of any of the following criteria: lowest interest rate; highest available balance; lowest transaction processing fee; longest float of funds; highest rewards incentive; highest competitive bid between at least two third-parties as tendered to the database.

(535) In another embodiment of the system of the invention, the ranking is determined by any of the following: the user; the payee; at least one third-party.

(536) In another embodiment of the system of the invention, upon the payee determining the ranking, the user is provided a financial incentive for selecting the payment account which is preferred, and the financial incentive is provided by any of the following: the payee; the database.

(537) In another embodiment of the system of the invention, the database updates the criteria periodically via application programming interface integration with at least one financial institution associated with at least one payment account.

(538) In another embodiment of the method of the invention, upon the payment-related data comprising the advertisement, the user is provided a financial incentive via the database based on any of the following: displaying the advertisement to the user; the user clicking on the advertisement; the user making a purchase pursuant to clicking on the advertisement.

(539) In another embodiment of the system of the invention, upon the payment-related data comprising the advertisement, the user is provided a financial incentive via the database based on any of the following: displaying the advertisement to the user; the user clicking on the

advertisement; the user making a purchase pursuant to clicking on the advertisement.

(540) It will be appreciated that the invention illustratively disclosed herein through illustrative embodiments may suitably be practiced in the absence of any element which is not specifically disclosed herein, particularly in an illustrative embodiment.

(541) These and other advantages of the invention will become more fully apparent when the following detailed description of the invention is read in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Additional aspects of the present invention will become evident upon reviewing the non-limiting, illustrative embodiments described in the specification and the claims taken in conjunction with the accompanying figures, wherein like reference numerals denote like elements.
- (2) FIG. 1 shows an illustrative embodiment of the Rule-Module Nexus (RMN or Nexus), with a Nexus access token (NAT) and a User interface apparatus (UIA).
- (3) FIG. 2 shows an illustrative embodiment of the Rule-Module Nexus, showing the network connections with various Account Issuers (Authorized Issuer).
- (4) FIG. 3 shows an embodiment of the invention depicting various configurations with an User Account Registry 15, internal Execution Platforms and external Execution Platforms, and Third-Party Platforms.
- (5) FIG. 3A shows a schematic illustration of an illustrative User Interface Apparatus in accordance with the present invention.
- (6) FIG. 3B illustrates an embodiment of the invention, wherein upon associating processing preferences with Financial Account(s), an illustrative embodiment of Rankings is visibly indicated with each associated to their respective Financial Accounts.
- (7) FIG. 3C is a diagram of an Account Issuer or Rule-Module Nexus administrator access key creation process
- (8) FIG. 3D is a diagram of a User key creation process.
- (9) FIG. 3E is illustrative of embodiments of a NAT with a Virtual POS UUC and a Physical POS UUC, further depicting: Front and Back views of a card with Risk Key data, magnetic stripe, bar code, magnetic ink characters, embedded RFID chip, and; Front and Back views of a USB thumb drive with Risk Key data, bar code, magnetic ink characters, embedded RFID chip.
- (10) FIG. 3F illustrates an embodiment of a NAT depicting Front and Back views of a card with a plurality of magnetic strips and associated UUC's on the Back.
- (11) FIG. 3G illustrates embodiments of displays of UIA's.
- (12) FIGS. 4 and 4A-1 show embodiments of the invention wherein the Rule-Module Nexus's network connectivity maintains dedicated frame relay lines to RMN Platforms co-located with Account Issuers' Acquirer sites and Account Issuers' Merchant Network Operation Centers (NOCs) in order to update and backup the RMN's data network. In turn, these RMN Platforms are linked to UIAs in Account Issuers' merchant locations through either the Merchant's in-store dedicated link to the Network Operations Center (NOC), or dial-up connectivity to an Acquirer/Processor.
- (13) FIG. 4A shows a flow chart of an embodiment of a verification (or authentication) process.
- (14) FIG. 4B, FIG. C, and FIG. 4D show, respectively: an embodiment of a "thin-client" financial transaction request packet (or message); an embodiment of a financial transaction response packet (or message); an embodiment of the construction of a financial transaction response packet (or message).
- (15) FIG. 4E shows a flow chart of the operation of the User Interface Apparatus and the Transaction Terminal (or Terminal) for generating a request packet.

- (16) FIG. 4F shows a flow chart depicting the data encryption and sealing process at the User Interface Apparatus.
- (17) FIG. 4G shows a flow chart depicting the data decryption and counter party identification process at the Rule-Module Nexus.
- (18) FIG. 4H shows a flow chart depicting the data encryption and sealing process at the Rule-Module Nexus.
- (19) FIG. 4I shows a representational diagram of the request packet and the mandatory and optional data it contains.
- (20) FIG. 4J shows a representational diagram of the response packet and the mandatory and optional data it contains.
- (21) FIG. 5 shows an embodiment of the Rule-Modules for a plurality of Users within the Rule-Module Nexus.
- (22) FIG. 5A shows an embodiment of the Verification Platform.
- (23) FIG. 6, FIG. 7A, FIG. 7B, and FIG. 7C show various embodiments of Rule-Modules, with various associations between Pattern Data and Execution Commands, including Global Queries and Global Execution Commands.
- (24) FIG. 8 and FIG. 8A show embodiments of the invention, depicting various NAT's, wireless modem and networked connections between various UIA's and the RMN, including Subset RMN's, a Master RMN, Redundant Master RMN, and Third-Party Platforms.
- (25) FIG. 8A-1 shows an embodiment of the invention, depicting wireless modem and networked connections between various UIA's and the RMN, including Subset RMN's, a Master RMN, Redundant Master RMN, and Third-Party Platforms.
- (26) FIG. 9, FIG. 10, and FIG. 11 shows a system block diagram and system architecture for a Merchant Point-of-Sale connected to a remote RMN.
- (27) FIG. 11A shows a block diagram of an illustrative data structure of the UAR platform in accordance with the present invention.
- (28) FIG. 11B illustrates an embodiment wherein an electronic Audible/Visible Signature is matched via the RMN with a Financial Account of a UAR.
- (29) FIG. 11C illustrates an embodiment wherein a User's credit report enables a UAR to be auto-populated with Financial Account(s) via Internet registration.
- (30) FIG. 11D illustrates an embodiment wherein a User activates a NAT using magnetic strip at a physical-world POS.
- (31) FIG. 11E illustrates an embodiment wherein mail order and telephone order transactions are conducted.
- (32) FIG. 12 shows a high performance, semi-redundant configuration for the RMN
- (33) FIGS. 13 and 13A-1 show RMN financial transactions for rewards.
- (34) FIG. 13A illustrates an embodiment wherein the RMN 14 and UAR 15 may convert, translate and/or aggregate financial data via rewards or loyalty programs.
- (35) FIGS. 14 and 14A show RMN financial transactions for an electronic check (eCheck).
- (36) FIG. 15 and FIG. 16 show RMN financial transactions for Credit and Debit.
- (37) FIGS. 15A and 15B illustrate illustrative embodiments of a registration process including a self-serve kiosk.
- (38) FIGS. 15C and 15D illustrate illustrative embodiments of a settlement process when converting an ATM/POS transaction to an ACH System, using verification data comprising a personal verification code (or primary verification code, or 1° PVC), and optionally, a secondary personal verification code (or 2° PVC).
- (39) FIG. 16A, FIG. 16A-1, FIG. 16A-2, FIG. 16A-3, FIG. 16B, FIG. 16B-1, FIG. 16C, FIG. 16C-1, and FIG. 16D show a flow charts of embodiments for accessing a User Account Registry via a Verification Platform.
- (40) FIGS. 17 and 17A show flow charts of registration and transaction processing with the Rule-

Module Nexus.

- (41) FIG. 18 and FIG. 18A-1 show flow charts of registration and transaction processing with the Rule-Module Nexus, User Account Registry, and Third-Party Platforms.
- (42) FIG. 18A, FIG. 18B, FIG. 18C, FIG. 18D, FIG. 18E, FIG. 18F, FIG. 18G, FIG. 18H, and FIG. 18I show embodiments of the Rule-Module Nexus, the User Account Registry, the Payee Account Registry, and Rule-Modules for financial account selection.
- (43) FIG. 19, FIG. 20, and FIG. 21 show use-sensitive embodiments, with Master, Intermediary, and Local (or Subset) configurations.
- (44) FIG. 22, FIG. 23, FIG. 24, FIG. 25, FIG. 26, FIG. 27, and FIG. 28 show process flow embodiments of Rule-Module Nexus processing of financial transactions.
- (45) FIG. 29A, FIG. 29A-1, FIG. 29A-2, FIG. 29A-3, FIG. 29B, FIG. 29C, FIG. 29D, FIG. 29E, FIG. 29F, FIG. 29G, FIG. 29G-1, FIG. 29G-2, FIG. 29G-3, FIG. 29G-4, FIG. 29G-5, FIG. 29G-6, FIG. 29G-7, FIG. 29G-8, and FIG. 29H show embodiments of financial transactions using mobile payments, the Nexus Access Token, and the Rule-Module Nexus.
- (46) FIG. 30, FIG. 30A, FIG. 30B, FIG. 30C, and FIG. 30D show further embodiments of financial transactions using mobile payments, the Nexus Access Token, and the Rule-Module Nexus.
- (47) FIG. 31 shows an embodiment of an aftermarket retrofit of the Nexus Access Token onto a legacy token.
- (48) FIGS. 32A, 32B-1, 32B-2, 32C-1 and 32C-2, shows embodiments of financial transactions which are processed via a centralized National/International (or Global) Gateway Platform and co-located Master Rule-Module Nexus and/or Master User Account Registry.
- (49) FIG. 33A illustrates an embodiment of a distributed system architecture for processing financial transactions via a Subset Rule-Module Nexus (and/or a Subset User Account Registry, with use of a National/International (or Global) Gateway Platform, optionally with co-located Master Rule-Module Nexus (and/or Master User Account Registry).
- (50) FIG. 34 illustrates an embodiment of the invention for accessing an online user account registry.
- (51) FIG. 35 illustrates an embodiment of the invention for processing pan-portfolio analytics.
- (52) FIG. 36 illustrates an embodiment of the invention for processing a financial transaction via associating a personal verification code with a personal identification number.
- (53) FIG. 36A illustrates an embodiment of the User Account Registry and a PVC Association Platform.
- (54) FIG. 37 illustrates an overview of an embodiment of financial transactions via the Rule-Module Nexus and the User Account Registry.
- (55) FIG. 38A and FIG. 38B illustrate functions and components of the pan-portfolio Neural Subsection of the Rule-Module Nexus.
- (56) FIG. 38C illustrates a multiple-output predictive model of the pan-portfolio Neural Subsection of the Rule-Module Nexus.
- (57) FIG. 39, FIG. 40A, FIG. 40B, FIG. 41, and FIG. 42 illustrate embodiments of smart-gifting via the Rule-Module Nexus.
- (58) FIG. 43, FIG. 43A, and FIG. 43B illustrate embodiment of registration and financial transactions using the Rule-Module Nexus.
- (59) FIG. 44A, FIG. 44B, FIG. 44C, FIG. 44D, FIG. 44E, FIG. 44F, FIG. 44G, FIG. 44H, FIG. 44I, FIG. 44J, FIG. 44K, FIG. 44L, and FIG. 44M are illustrative of embodiments of financial transactions via the Rule-Module Nexus using website widgets.
- (60) FIG. 45, FIG. 46, FIG. 46A, FIG. 46B, FIG. 46C, FIG. 46D, FIG. 46E, FIG. 46F, FIG. 46G, FIG. 46H, FIG. 46I, FIG. 46J, FIG. 46K, FIG. 46L, FIG. 46M, FIG. 46N, FIG. 46O, FIG. 47, and FIG. 48 are illustrative of mobile financial transactions via the Rule-Module Nexus.
- (61) FIG. 49 is illustrative of an embodiment of the web browser extension being used to facilitate a payment transaction.

(62) FIG. 50 is illustrative of an embodiment of the remote computer/database being integrated with a payee's computer/database using an API from the remote computer.

(63) FIG. 51 is illustrative of an embodiment of the invention wherein while a user is visiting a payee or merchant website, the web cross-browser extension GUI widget is clicked on to invoke the pop-up window for displaying the user's payment-related data and for selecting a financial account of the user, with a preferred account being indicated.

(64) FIG. 51A is illustrative of an embodiment of the invention wherein upon the user selecting a financial account from the pop-up window, and upon a checkout form from the website or web application of the payee being presented on the display, the web cross-browser extension invokes an auto-complete function to automatically populate fields of the payee's checkout form with the user's payment-related data.

(65) FIG. 51B is illustrative of an embodiment of the invention wherein an intelligent software agent is operable to track the user's Internet browsing via the cross-browser extension and operable to automatically conduct on-line data retrievals reflecting user-customized Web browsing priorities and/or preferences for competitive product and/or service prices across a plurality of on-line merchants, reward incentives providers, banks, internet payment providers, barter providers, insurance companies, credit account organizations, and/or brokerage firms.

(66) FIG. 51C is illustrative of an embodiment of the invention wherein an intelligent software agent is operable to track the user's Internet browsing via the cross-browser extension and operable to automatically conduct on-line data retrievals reflecting user-customized Web browsing priorities and/or preferences for extrapolated suggestions for new products and/or services.

(67) FIG. 52 is illustrative of an embodiment of the invention wherein electronically pulling a credit report of the user invokes auto-populating the user's pattern data with a plurality of financial accounts.

GLOSSARY

(68) Authorized Issuer (ACCOUNT ISSUER) [or FA (Financial Administrator) or AA (Account Administrator)] comprises any one of the following: an owner, comprising a party that owns a Financial Account; a servicer, comprising a party that services a Financial Account on behalf of an account owner; an agent, comprising a party that acts on behalf of another party associated with a Financial Account; a change-agent, comprising a party authorized to influence or change the selection of a Financial Account, and/or a party authorized to influence or change the use of Financial Data of a Financial Account or of a Financial Account itself, and; a fiduciary, comprising a party with a fiduciary duty to the Financial Account. An Account Issuer may comprise any one of the following: a party administering Financial Data within the Financial Account, and; a party reconciling the Financial Data within the Financial Account upon settlement of a financial transaction. An Account Issuer may comprise any one of the following: a bank; a retailer or merchant; a payee; an educational institution; an affinity group such as The Johns Hopkins University and The Sundance Institute®; a branding and marketing agent; an insurance company; and a processor such as First Data Corporation®, General Electric Capital (GE Capital®), and, Total® Processing.

(69) ACCOUNT ISSUER BATCH: A collection of “add” and “delete” instructions complete with UUC-IDs, financial asset accounts, and account index codes verified and submitted by an account issuer to the RMN.

(70) Authorized Issuer (Account Issuer Platform): Platform comprising data for account issuers that are allowed to add and delete financial asset account numbers with the RMN or RMN-authorized Third-Party platforms.

(71) Authorized Issuer (Account Issuer Transaction Terminals): Provides a batch connection to the RMN or third-party platforms for account issuers to add and remove (their own) financial asset account numbers from specific individual's UUC records.

(72) AOD (or AOP): Apparatus Owner Database (or Apparatus Owner Platform): stores

information about the owners of UTA devices.

(73) ASCII: American Standard Code for Information Interchange

(74) ATM (Automated Teller Machinery): Uses encoded UUC packet (or optionally UUC-PVC packet) verification information to obtain access to an account issuer or third-party platform, including authorizing cash dispensing and account management.

(75) BRT (Banking Retail Transaction Terminal-UTA): UUC Registration and Re-Coding UTA's, located at retail banking outlets, BRT's utilizes UUC registration information with personal verification data to register individuals with the RMN or third-party platforms.

(76) BRT-SP: Banking Retail Transaction Terminal-Subset Platform

(77) CBC (Cipher Block Chaining): an encryption mode for the DES.

(78) CCD: Charged-Coupled Device

(79) CPT (Cable-TV Point-of-Sale UIA): Transaction terminal combining an onscreen display simulcast digital signal informing TV-top cable box of product information with product video, and an UIA controller remote which performs the UUC (or optionally UUC-PVC) validation using the CATV communications network. Order/autho/mailling-address/item-id forwarded to merchant. Results of authorization are displayed on the TV.

(80) CST (Customer Service Transaction Terminals): Provide RMN or third-party platforms User service personnel with varying degrees of access (based on access privilege) the ability to retrieve and modify information on individuals in order to help people with account problems.

(81) DUAL SEALING STEP: The conversion of plain text to cipher text (known as "encryption") in combination with the encrypted check-summing of a message that allows information to remain in plain text while at the same time providing a means for detecting any subsequent modification of the message.

(82) DES (Digital Encryption Standard): A standard for the cryptographic protection of digital data. See standard ANSI X3.92-1981.

(83) DETERMINATION: the status of the command processed during the execution step.

(84) DSP (Digital Signal Processor): a class of integrated circuits that specialize in the mathematical operations required by the signal processing applications.

(85) DUKPT (Derived Unique Key Per Transaction): See standard ANSI/ABA X9.24-1992.

(86) EMERGENCY CODE: The alpha-numeric sequence, visible image or audible signal selected by an individual which, when accessed, will result in a transaction being associated by the RMN or third-party platforms as an emergency alert, potentially causing the display of false screens and/or the notification of emergency authorities that said individual has been coerced into performing a transmission or transaction. An emergency authority may comprises any one of the following: the RMN; a governmental agency (e.g., fire, medical, police, sheriff), and; a private third-party company (e.g., Brinks™, Bay Alarm™).

(87) EP: Execution Platform (or ECP: Execution Command Platform).

(88) ESP (Electronic Signature Platform): Platform comprising all MD5 and electronic signatures of all documents signed by anybody, referenced by authorization number.

(89) EST (Electronic Signature Transaction Terminal): Uses UTA to verify, computer calculates checksum on document, sends checksum to RMN or third-party platforms, RMN or third-party platforms validates, timestamps, saves checksum, and returns with sig-code. Uses Internet as transport. EST also verifies signatures given a sig code and an MD5 calculation.

(90) EXECUTION COMMANDS: A program or subroutine residing in Rule-Modules of the RMN that performs a specific task, activated by a request message sent from a UIA-conjoined Transaction Terminal.

(91) FAR (False Accept Rate): The statistical likelihood that one user's UUC (or optionally UUC-PVC) algorithmic verification will be incorrectly verified as the UUC-PVC of another user.

(92) FALSE SCREENS: Displays of information which has been intentionally pre-determined to be subtly inaccurate such that a coercing party will not illegally obtain accurate data about an

individual's financial assets, all the while remaining unaware of the alteration of the information.
(93) FDDI (Fiber Digital Device Interface): a networking device that utilizes a fiber optic token ring.

(94) FS: Field Separator.

(95) FW (Firewall Platform): The network—Local or Subset net router that regulates traffic into and out of the RMN.

(96) FINANCIAL ACCOUNT (or online FINANCIAL ACCOUNT or ACCOUNT or ASSET ACCOUNT), comprises any one of the following: a plurality of related online Financial Data electronically stored apart from a Nexus Access Token (NAT); a unique identifier (“Financial Account-Identifier” or “Financial Account-ID”) distinguishing the plurality of related Financial Data from all other pluralities of related Financial Data, and; storage means apart from the NAT and electronically accessible from the NAT via communications means (or interconnecting communications means) external to the NAT. A Financial Account further comprises any one of the following: a credit account; a micropayment account; a pre-paid account; a debit account; a rewards/loyalty account; a scrips account; a checking account; a savings account; an investment account; a brokerage account; and an insurance account. However, it should be noted that the invention may not be so limited, and other Financial Accounts may be contemplated to be within the scope of the present invention, including a Financial Account enabling a review of Financial Data, a transfer of Financial Data, and a financial transaction comprising an exchange of Financial Data, and an exchange, purchase or sale of goods and/or services.

(97) FINANCIAL ACCOUNT-ID (Financial Account Identifier or Account Identifier or Account-ID): Financial Account-ID, wherein a unique code identifies a specific Financial Account and is associated with only said specific Financial Account.

(98) FINANCIAL DATA (or LIVE FINANCIAL ACCOUNT DATA or ACTIONABLE-ACCOUNT DATA or FUNDING-ACCOUNT DATA or PAYMENT-ENABLING DATA): Any unit of electronic data comprising any one of the following: monetary value as currency or currency-equivalent; legal tender or legal tender-equivalent. As such, a display of these units may provide a User with: a balance in a Financial Account of the User; a history of financial transactions involving a Financial Account of the User; a history of activity of a Financial Account of the User. Further, a User's purchase, expenditure, transfer, exchange or usage of these units may comprise any one of the following: a credit of said units to a Financial Account; a debit of said units from a Financial Account; a purchase, sale, transfer, discounted cost, or exchange of goods, services or currency involving an Account Issuer. Financial data further comprises units of: currency; rewards; loyalty; incentive; discount; and scrips points. Examples include: dollars; discount points; insurance credits; minutes of telephone calling time; miles towards earning a free airplane flight, points towards a gallon of gas, and the like. However, it should be noted that the invention may not be so limited, and other Financial Data may be contemplated to be within the scope of the present invention.

(99) FINANCIAL INTERCHANGE PLATFORM (FIP): A computing platform, optionally within the Rule-Module Nexus, comprising means for converting, trading, exchanging, transferring and/or interchanging Financial Data between or among a plurality of proprietary Financial Accounts of a User or of a plurality of Users.

(100) FINANCIAL TRANSACTION: comprises any electronic transfer, exchange or use of Financial Data in a Financial Account accessed via an online User Account Registry, wherein Financial Data has a predetermined monetary or monetary-equivalent unit value which is legal tender or a legal tender-equivalent. As such, a Financial Transaction further comprises any one of the following: debiting Financial Data from a Financial Account; crediting Financial Data to a Financial Account; a transfer of Financial Data between a plurality of Financial Accounts from a User Account Registry; a transfer of Financial Data between a Financial Account from a User Account Registry and a Financial Account of an Account Issuer. In the latter instance, a financial

transaction further comprises a User's purchase, expenditure or usage of said Financial Data results in the User's purchase or sale of goods, services or currency involving an Account Issuer. Note that a user's approving a Financial transaction comprises any one of the following: a user signalling consent via selecting a financial account from a plurality of financial accounts displayed to the user in parallel; a user signalling consent via selecting a financial account from a plurality of financial accounts displayed to the user in sequence, and; a user signalling consent via accepting a financial account automatically selected by the rule-module nexus and displayed to the user based upon predetermined criteria.

(101) GEC (Global Execution Command): In an illustrative embodiment customized to the user. Note that in an illustrative embodiment of this invention, a GEC does not require all financial transactions of all users to automatically comprise any of the following: being linked to any particular account issuer; invoking a specific online transaction processing preference for all Financial Accounts for all users; being appended to any particular product or service, and; being diverted from any predetermined processing preferences of an account issuer which would otherwise apply to a Financial Account selected by a user during a financial transaction.

(102) GP (Gateway Platform): the main communications directing platforms in the RMN; which direct the flow of electronic messages to other platforms within the RMN and with third-party platforms.

(103) INTERMEDIARY PLATFORM (or Intermediary Computer): Both defined the same, as computer hardware and software storing a more complete set of data than the Subset Platform, but storing less data than the Master Platform.

(104) IPT (Internet Point-of-Sale Transaction Terminal): Communicates product/services data and merchant code from the internet, appends UUC (or optionally UUC-PVC) via UTA for validation and sends to RMN using Internet, wherein autho/order/PO # forwarded to merchant. RMN response using internet as well, displaying results on screen of the Transaction Terminal.

(105) ITT (Internet Teller UIA): Authorizes network UIA session using encrypted credential obtained from RMN using UUC-ID.

(106) LCD (Liquid Crystal Display): A technology used for displaying text and visible images.

(107) MAC (Message Authentication Code): an encrypted checksum algorithm, the MAC provides assurance that the contents of a message have not been altered subsequent to the MAC calculation. See standard ANSI X9.9-1986

(108) MACP (or MACM): Message Authentication Code Platform (or Message Authentication Code Module): a software platform in the RMN that handles MAC validation and generation for inbound and outbound packets.

(109) MDP (or MDM): Message Decrypt Platform (or Message Decrypt Module): a software module in the RMN that encrypts and decrypts packets from or destined to an UIA device.

(110) MPP (or MDM): Message Processing Platform (or Message Processing Module): a software module in the RMN that performs the processing of request packets.

(111) MASTER PLATFORM (or Master Computer): Both defined the same, as computing hardware and software storing a complete set of all data being used in the invention.

(112) MERCHANT (or Retailer or Payee): A party retailing or selling services or goods via financial transactions to other parties or entities by means of the Internet or a physical storefront.

(113) MPU (Merchant Point-of-Sale UIA) or (RPU: Retailer Point-of-Sale UIA): a combines encoded UUC verification information with retail transaction information (possibly from an electronic cash register) and formulates authorization requests of the RMN or third-party platforms using X.25 networks, modems, etc.

(114) MSP (Merchant Subset Platform) or (RSP: Retailer Subset Platform): A Platform residing with a Merchant, comprising a subset of all data in the RMN pertaining to a user

(115) MDP (Message Decrypt Platform): A software platform in the RMN that encrypts and decrypts packets from or destined to an UIA device.

(116) MPP (Message Processing Platform): A software platform, optionally in the RMN, that performs the processing of request packets.

(117) NAT (Nexus Access Token): A token comprising: data storage means with a UUC encoded therein; means for communicating with networks external to the NAT. Illustrative embodiments of NAT storage means comprise any one of the following: magnetic strip; bar code; imprinting; printing; encoding; embossing; an integrated chip (IC); a memory chip. Illustrative embodiments of NAT communications means (or interconnecting communications means) comprise any one of the following: contact-based scannable data; contactless scannable data; radio frequency (e.g., RFID); infrared beam; audible signal; USB connection; global positioning signal; cellular phone signal; DSL connection; dial-up connection; near-field communications comprising short-range high frequency wireless communication technologies; open wireless technology standard for exchanging data over short distances (using short length radio waves) from fixed and mobile devices, creating personal area networks (PANs) with high levels of security (e.g., Bluetooth). Embodiment of NAT tokens comprise any one of the following: a plastic card; a key fob; a cell phone; USB “thumb” drive; a personal digital assistant; a pager; a personal computer. Optionally, the NAT may also comprise means for being conjoined with a UTA, either temporarily or permanently integrated therewith. Also, the NAT may optionally be a “thin-client” token, wherein the NAT stores only “thin-client” data.

(118) NETWORK CREDENTIAL: Both the user and the account issuer are identified by the RMN to create the network credential. The credential includes the individual's identification as well as the context of the connection (i.e., the TCP/IP source and destination ports). RMN creates a network credential using the individual's account id, the time of day, and the bank code. The RMN signs this credential using Public Key Encryption and the RMN's Private Key.

(119) ONLINE (or ON-LINE) FINANCIAL TRANSACTION: A financial transaction starting in real- (or actual-) time elapsed from verification of the user through until declining the financial transaction or successful selection of the financial account, during which the NAT and/or the UTA remain(s) remotely located from, and remain(s) in communication with, at least one of the following: the Rule-Module Nexus; Verification Platform, and; the User Account Registry. Said communication preferably occurs via a communications network comprising at least one of the following: cable network, wireless network, telephone network, the Internet, an intranet, a local area network (LAN), a wide area network (WAN), and an X.25 network.

(120) In an illustrative embodiment of the method of the invention, completing the financial transaction comprises any one of the following: declining the financial transaction, and; settlement of the financial transaction.

(121) PAN-PORTFOLIO ANALYTICS: Analytics using a predictive patterns and/or a neural network for fraud-detection, rewards programs, micro-merchandising, data-mining, and the like, comprising financial data compiled via the RMN accessing a plurality of Proprietary Financial Accounts. As such, pan-portfolio analytics are enabled via a plurality of otherwise inaccessible Authorized Issuer-PFAs being rendered accessible as a function of the RMN.

(122) PAYEE: A party receiving an electronic payment via the RMN from a User, wherein a payee comprises any one of the following: an account issuer; a merchant (internet, point-of-sale), and; a user.

(123) PAR (Payee Account Registry): a platform, in an illustrative embodiment, within the rule-module nexus, comprising Financial Accounts of a payee, identified by a PAR-Code, said PAR-Code, in an illustrative embodiment, comprising any one of the following: no “live” financial account data; no “funding” account data; no payment-enabling data; no data directly identifying any specific or primary Financial Account of a payee. The PAR may also comprise rule-modules invoking preferences, tagging, rankings, and/or default Financial Account(s) of a payee. The PAR may also comprise a rule-module of a payee, said payee rule-module further comprising any one of the following: a payee name; a payee financial depository account; a payee Financial Account; a

payee financial depository account routing code, and; payee geographic positioning data.

(124) PARTY: An entity or individual comprising any one of the following: a user; an account issuer; the rule-module nexus; a payee, and; a third-party.

(125) PFP (Prior Fraud Platform): A platform for UUCP records which have had prior fraud associated with them. Every new User's UUCs are checked against all PFP records with the intent of reducing recidivism.

(126) PGL: Personal Verification Code—Group List

(127) PVC (Personal Verification Code): an optional code or a data sequence or data packet for protecting access to an individual's account through secret knowledge, formed from either numbers, symbols, or alphabetic characters (or alpha-numeric-symbolic characters), which can be provided via keypad, touch screen, audio commands or instructions, and the like. A Personal Verification Code may be selected by the User, the Rule-Module Nexus, or an authorized third-party.

(128) POS (Point-Of-Sale): A physical place where goods or services are sold.

(129) PPU (Phone Point-of-Sale UTA): Transaction terminal combining phone number with merchant price and product information to authorize a transaction over a UTA-equipped telephone. Order/authorization/mailling-address/PO forwarded to merchant. Resulting authorization is displayed on phone LCD, or “spoken”, along with the individual's private code.

(130) POSITIONING SYSTEM (or GLOBAL POSITIONING or POSITIONING): electronic positioning communications and systems, including electronic navigating communications and systems, electronic tracking communications and systems, electronic mapping communications and systems, and electronic guiding communications and systems. Such systems and their related software and hardware components, known in the art, comprise any one of the following: Global Positioning System (GPS); Standard Positioning Service (SPS); Differential GPS (DGPS); basic local area DGPS (LADGPS), wide area DGPS (WADGPS) such as WAAS, ARGOS satellite system, Cell phone Networks 18 systems such as Cell-ID, AOA, TDOA, TOA, E-OTD, A-FLT and OTDOA, E-CID, U-TDOA, A-GPS; Matrix, Enhanced-GPS (E-GPS); Polaris Wireless Location Signatures (PWLS); TV-GP; Precision Location Technology (PLT), based on time of arrival (TOA); Ekahau Positioning Engine (EPE); AeroScout; PanGo Locator; AXCESS tracking; InnerWireless RF; Ultra Band Width (UWB)-based tracking; Laser Positioning System; Microwave-based tracking systems; Radar/Sonar positioning; Universal Time Coordinated (UTC), and; Acoustic Positioning Systems (APS).

(131) PROPRIETARY Financial Account: a Financial Account, accessed via the User Account Registry, wherein an Account Identifier comprises any one of the following: a code identifying an account issuer, as approved by the International Organization for Standardization [(Bank Identification Code, or BIC Code) or (Bank Identification Number, or BIN)]; a code identifying an account issuer, as approved by the Society for Worldwide Interbank Financial Telecommunications (SWIFT Address or SWIFT Code), and; a code identifying an account issuer, as approved by the American Banker's Association (ABA Routing Code). The Account Identifier of a Proprietary Financial Account further comprises an 8 or 11 alphanumeric characters long code, comprising any one of the following: Four characters—Account Issuer code; Two characters—ISO 3166-1 alpha-2 country code; Two characters—location code; Three characters—Account Issuer branch code, optional (‘XXX’ for Account Issuer's primary office). Where an 8-digit code is given, it may refer to the primary office. The ABA Routing Number, (a.k.a. ABA number; Routing Transit number) is devised by the American Bankers Association (ABA), serves to identify the specific account issuer responsible for the payment of a negotiable instrument, and may also designate participants in automated clearinghouses, electronic funds transfer, and on-line banking. In other embodiments, a Financial Account is Proprietary to a specific Account Issuer (Account Issuer-Proprietary Financial Account, or Authorized Issuer-PFA), such that accessing a PFA of a first Account Issuer (1° Authorized Issuer-PFA) with accessing a PFA of a second Account Issuer (2° Authorized Issuer-

PFA) is enabled as a function of the RMN and/or the UAR. Alternatively, absent interconnectivity via the RMN and/or the UAR, accessing the 1° Authorized Issuer-PFA in concurrently with accessing the 2° Authorized Issuer-PFA may not be possible. As used herein, “concurrently” refers to proprietary Financial Accounts intersecting at the same point, said same point being a User Account Registry and/or the Rule-Module Nexus. Alternatively, a plurality of otherwise concurrently inaccessible Authorized Issuer-PFAs may be rendered concurrently accessible via the RMN and/or the UAR. In an exemplary embodiment, a 1° Authorized Issuer is a bank named Wells Fargo® with which a User has a Proprietary Financial Account (e.g., a credit account), whereby this 1° Authorized Issuer-PFA is registered in said User's UAR, and; a 2° Authorized Issuer is a bank named Chase® with which said User has a Proprietary Financial Account (e.g., a checking account), whereby this 2° Authorized Issuer-PFA is also registered in said User's UAR. As such, accessing the 1° Authorized Issuer-PFA concurrently with accessing 2° Authorized Issuer-PFA is enabled via the RMN. Absent the RMN, said 1° Authorized Issuer-PFA and said 2° Authorized Issuer-PFA may not be concurrently accessible.

(132) RAM: Random Access Memory

(133) RC: Routing Code wherein the UUC is electronically transmitted to either the online VP or the online UAR for accessing a Financial Account from the UAR.

(134) RF (Radio Frequency): Generally refers to radio frequency energy emitted during the normal operation of electrical devices.

(135) REGISTERS: Memory reserved for a specific purpose, data set aside on chips and stored operands to instructions

(136) REGISTRY FINANCIAL ACCOUNT comprises any one of the following: an online Financial Account of a User stored within an online User Account Registry of the User, and; a Registry Financial Account-ID, unique to each Registry Financial Account. A plurality of Registry Financial Accounts, are two or more Registry Financial Accounts have in common at least their association with the same designated User of the User Account Registry.

(137) REGISTRY FINANCIAL ACCOUNT IDENTIFIER (or Registry Financial Account-ID or Financial Account-ID): Computer-readable data, preferably in binary code and comprising a complete financial account number, which directly identifies a unique and specific online financial account of a user, and which is singular to said financial account.

(138) REQUESTS: Electronic instructions from the UTA to RMN instructing the RMN to verify the individual and thereby process the individual's command in the event the identification or verification is successful.

(139) RM (Rule-Module): Comprising software an association between a Pattern Data and an Execution Command, invoked by a Positive Matching Determination, and governing access to a User Account Registry.

(140) RMN (Rule-Module Nexus): A subset or master Rule-Module Nexus is a platform comprising any one of the following: a rule-module, and; capability of comprising a plurality of rule-modules. A Rule-Module Nexus, in an illustrative embodiment, communicates with, and may be integrated with, a User Account Registry and/or a Verification Platform, and optionally is capable of communicating with a plurality of Account Issuers and their associated networks, and communicates with a transaction terminal and a User Interface Apparatus. The RMN is optionally integrated with a UUC (or optionally UUC-PVC) verification platform, and invokes a Rule-Module to access and process financial transactions. During a financial transaction, the RMN may be stored directly on or within a NAT, or the RMN may be temporarily and removeably conjoined with a NAT. In an illustrative embodiment, the Rule-Module Nexus is online and stored apart from the NAT (“Online-RMN”), wherein the UUC is transmitted from the NAT to the Online-RMN via communications means (or interconnecting communications means) external to the NAT. In one embodiment, during a financial transaction, the RMN is remotely located such that it is not physically accessible to the user, either by physical distance or by physical barrier. In an illustrative

embodiment, to commence processing a financial transaction, the RMN initially relies only on a thin-client data packet, or electronic message, transmitted from a NAT, a NAT-UTA, a UTA, and/or a UIA-Terminal. In an illustrative embodiment, this thin-client packet comprises any one of the following: no Financial Account-ID; no user “live” financial account data; no user payment-enabling data; no user funding account data; a unique user code; a rule-module nexus routing code; positioning data, and; over-the-air provisioning data.

(141) RMN-RC (Rule-Module Nexus Routing Code): Data comprising network instructions for routing a financial transaction via the Rule-Module Nexus for processing. Electronic storage of a RMN-RC may comprise any one of the following: storage on a NAT; storage in a UIA, and; storage on a third-party platform. Invoking a RMN-RC may comprise any one of the following: manually invoking via a user's data entry into a UIA; automatic invoking via a scanning of a user's nexus access token comprising stored RMN-RC; manually invoking a third-party's data entry of the RMN-RC into a UIA or a Terminal.

(142) RMP (Remote Merchant Platform): Comprises all merchant identification codes for merchant telephone and Cable TV order shops; indexed by merchant ID. Comprises per-merchant encryption codes as well.

(143) SMART-PRODUCT: A product or service which is purchased by a user using pan-portfolio analytics from the RMN and UAR of the user or of another user.

(144) SNP (Sequence Number Platform): A software platform in the RMN that handles the DUKPT sequence number processing for inbound request packets. Sequence number processing protects against replay attacks.

(145) SUBSET PLATFORM (or Local Computer): Both defined the same, as computing hardware and software storing a set of related data which represents only a portion of all data stored in the Master Computer or Master Platform.

(146) THIN-CLIENT (or THIN-CLIENT DATA): Computer readable data, which: (a) is not an account number nor does it contain “live” account data [“Live” account data means any data that enables a financial account to be charged or debited “off-line”, wherein there is not a real-time verification of the user via the Verification Platform of the RMN, whether or not subsequently in a batched settlement]; (b) does not directly identify, directly correspond to, and/or directly access a specific online financial account of the user, and; (c) comprises at least one of the following: (i) a binary string having a sequence of data values, usually bytes, and represented in the form of zeros and ones (e.g., 01101000); (ii) a sequence of alpha-numeric-symbolic characters (e.g., meaning any or all of the following: alphabetical characters; numeric characters, and/or; symbols), and; (iii) American Standard Code for Information Interchange (ASCII) encoded characters.

(147) TRANSMISSION: An transmission is the online electronic communication of content which is non-Financial Data and is not a financial transaction.

(148) TRANSACTION TERMINAL (or TERMINAL): A transaction platform, usually associated with a merchant, which is remotely located from, and electronically communicating with, a NAT and/or a UTA, and which participates in forming request messages being sent to the RMN for executing financial transactions. Transaction Terminals optionally appends ancillary transaction information to request messages, such as purchasing data, verifying counterparties and the like. In the embodiment where the User Interface Apparatus and Transaction Terminal are conjoined without being fully integrated, the two devices each remain operationally and functionally separate from each other, but can communicate to exchange data. In the embodiment where the UIA and the Transaction Terminal are fully integrated, the two devices are operationally and functionally united as one device. Optionally, the Transaction Terminal may also electronically store the RMN-RC for invocation upon the Transaction Terminal scanning a UUC from a NAT, or upon the Transaction Terminal being instructed by User or an attendant via data-entry comprising voice-command, key-pad, touch screen, and the like.

(149) TITLE INDEX CODE: Alpha-numeric sequence uniquely verifying an individual's

authorized role or capacity within the context of his employment

(150) TRACKING CODE: An alpha-numeric sequence assigned to data stored in or transmitted by the RMN, such that said sequence may be used to recall the data or obtain a report on the status of the transmission of the data.

(151) UAP (UIA Authorization Platform): Comprises the list of parties, whether users, individuals or account issuers, authorized to use modify and issue UIA devices.

(152) UAR (User Account Registry or Online User Account Registry): an online platform comprising any one of the following: a UAR-Code; a plurality of Registry Financial Accounts of a User; a plurality of Registry Financial Accounts associated with a User, and; storage means apart from the NAT and electronically accessible from the NAT via communications means (or interconnecting communications means) external to the NAT. The UAR may comprise any one of the following: a plurality of co-located Registry Financial Accounts; a plurality of Registry Financial Accounts conjoined on a distributed basis. In an illustrative embodiment, the UAR is accessible to the User via a UUC stored on a NAT, wherein the UUC comprises no Registry Financial Account-ID. In an illustrative embodiment a UAR is not dependant on a primary Registry Financial Account. The UAR may also comprise any one of the following: being wholly integrated with a Rule-Module Nexus; being conjoined and co-located with a Rule-Module Nexus; being conjoined but not co-located with a Rule-Module Nexus. While a UAR may be stored apart from, or external to, the NAT, the UAR may be stored within a UTA or UIA-Terminal temporarily and removeably conjoined with a NAT. The UAR is, in an illustrative embodiment, not stored directly or permanently on a NAT.

(153) UAR-CODE (User Account Registry Code): Uniquely identifies a User Account Registry, said User Account Registry comprising a plurality of Financial Accounts of the User. The UAR-Code is preferably "thin-client", meaning that it comprises no data which directly and uniquely identifies an online financial account of the user (i.e., the UAR-Code comprises no Registry Financial Account-ID). In an illustrative embodiment said User Account Registry Code is unique for each user, and, in an illustrative embodiment, does not directly identify or depend upon any specific Financial Account of the User or require designating a specific primary Financial Account of a User. While the UAR-Code is unique to each UAR, a plurality of Users may be registered with the RMN to be authorized to access a single UAR.

(154) UIA (User Interface Apparatus): A device which is remotely located from the RMN and communicates with RMN during a financial transaction, including transmitting the User's verification data (comprising any of the following: UUC; primary PVC; and/or secondary PVC), optionally encrypted, to the RMN. Optionally, the UIA may also be conjoined with a NAT, either temporarily or permanently integrated therewith. The UIA may also communicate with a co-located or remotely located Transaction Terminal. The UIA preferably also comprises means for gathering the User's UUC, primary PVC and/or secondary PVC. Said means for gathering the User's primary PVC and/or secondary PVC comprise at least one of the following: a microphone for receiving voice-commands; a key-pad for data-entry; a touch screen for data-entry. Optionally, the UIA may comprise means for being temporarily conjoined with a UAR, without directly store or permanently connecting to a UAR. Illustrative embodiments of UIA communications means (or interconnecting communications means) comprise any one of the following: contact-based scanning capability; contactless scanning capability; radio frequency (e.g., RFID) interrogator; infrared scanning capability; bar-code scanning capability; USB connection; global positioning signal receiving and transmitting capability; cellular or landline phone network connectivity (e.g., DSL connection, cable connection, etc.); near-field communications comprising short-range high frequency wireless communication connectivity; open wireless technology standard for exchanging data over short distances (using short length radio waves) from fixed and mobile devices, creating personal area networks (PANs) with high levels of security (e.g., Bluetooth). Embodiments of UIAs comprise any one of the following: a cell phone; a landline phone; a personal digital assistant; a pager; a

personal computer, a PIN-pad with encrypting software and optionally a magnetic stripe card reader. Optionally, the UIA may also comprise: means for being conjoined with a NAT, either temporarily or permanently integrated therewith; means for being conjoined with a Terminal, either temporarily or permanently integrated therewith. Also, the UIA may optionally be a “thin-client”, wherein the UIA stores only “thin-client” data. The UIA may be owned by an acquirer, an account issuer, a merchant (e.g., a payee), or a User (e.g., a payor).

(155) UIA-SP: User Interface Apparatus-Subset Platform

(156) UIA-VC (User Interface Apparatus Verification Code, or User Input Apparatus hardware Verification Code): a unique code which is assigned to a UIA unit, which is appended to a financial transaction and transmitted to the Rule-Module Nexus in order to verify the authenticity of the UIA to the Rule-Module Nexus. The RMN may associate the UIA-VC with additional verifying data, such as a User's UUC, a merchant location, a merchant identity, a Unique Payee Code, and the like.

(157) UOP (User Interface Apparatus Owner Platform): A platform comprising the geographic and contact information on the owner of each UIA.

(158) UPC (A unique payee code): Stored in a PAR (payee account registry), in an illustrative embodiment, within the rule-module nexus, the UPC is distinct for each payee of the RMN or Third-Party Platforms, said UPC an illustrative embodiment, comprising any one of the following: no UPC-identifier (UPC-ID) which uniquely identifies and directly accesses a UPC; no “live” financial account data of a payee; no “funding” Financial Account data of a payee; no payment-enabling Financial Account data of a payee; no stored data which uniquely and directly identifies a specific online Financial Account of the payee; a rule-module nexus routing code, and; stored data which is unique to the payee.

(159) UPP (UUC-PVC Platform): a software platform in the RMN maintains a registry of which UUCs are assigned to which PVC's, including a registry of algorithmically dissimilar UUCs linked to the same PVC.

(160) UST: User Support Terminal

(161) UUC (or THIN-CLIENT UUC or UNIQUE USER CODE): The UUC comprises “Thin-client” data only, and is a code unique to a User, which is stored in a nexus access token (“NAT”) and submitted to the Verification Platform for verifying the User's authority to access the RMN and to thereby invoke at least one rule-module to access financial account(s) via the user account registry. The UUC comprising any one of the following in computer-readable form: message; indicia; number; alpha-numeric sequence; electronic signal; symbol; digital certificate; digital signal, and; analog signal. In an illustrative embodiment, the UUC may comprise a RMN-RC with instructions for routing a financial transaction via the RMN. Any one user may have, or be associated with, more than one UUC, but any one UUC should not be associated with more than one user. Thus a user may have a different UUC stored on each of two different NATs, but both of said two different UUC's are associated only with said one user. Further, optionally the UUC itself may comprise a UAR-Code. Examples of UUC's include: a phone number (land line or mobile); SIM card data; or a persistent cookie stored on the hard drive of a User's personal computer NAT **62-UIA 16**, which RMN may automatically detect as the User's pattern data from a previously authenticated session between User and RMN.

(162) UUC-ID (A unique user code identification): An identifier used by the RMN to uniquely verify an individual's UUC record (IRID—Individual Record ID), which, in an illustrative embodiment, comprises no data directly identifying any specific online Financial Account of a user.

(163) UUCP (Unique User Code Platform): A platform for unique user codes. Queries against the UUCP are used to verify authority for financial transactions.

(164) VAP (or VAD): Valid Apparatus Platform (or Valid Apparatus Database): A platform in which each UIA (with associated unique encryption codes) is identified, along with the owner of the UIA.

(165) VAC (Verification Approval Code): A message, indicia, number or alpha-numeric sequence, electronic signal (and the like) invoked as Pattern Date from a Rule-Module Nexus upon the

Verification Platform making a Positive Matching Determination. The VAC invokes access to the UAR. The VAC may comprise a UAR-Code to identify a user's User Account Registry while, in an illustrative embodiment, not directly identifying or directly accessing any specific Financial Account of the user. Further, the VAC may be unique to each Positive Matching Determination for a User; may be common to all Positive Matching Determinations made for a User; may be common among a plurality of Positive Matching Determination made for a plurality of Users.

(166) VP (Verification Platform): a platform comprising any one of the following: a registered Unique User Code; a registered user personal verification code (PVC), or primary PVC; a registered user private code; a unique payee code; a user interface apparatus verification code; a registered user secondary PVC. The VP, in an illustrative embodiment, communicates with, and may be integrated with, a User Account Registry and/or a RMN, and optionally is capable of communicating with a plurality of Account Issuers and their associated networks, and communicates with a transaction terminal and a User Interface Apparatus. The VP invokes a Rule-Module from the RMN to access and process financial transactions via the UAR. During a financial transaction, the VP may be stored directly on or within a NAT, or the VP may be temporarily and removeably conjoined with a NAT. In an illustrative embodiment, the VP is online and stored apart from the NAT ("Online-VP"), wherein the UUC is transmitted from the NAT to the Online-VP via communications means (or interconnecting communications means) external to the NAT. In an illustrative embodiment, the VP may also comprise any one of the following: being wholly integrated with a Rule-Module Nexus; being conjoined and co-located with a Rule-Module Nexus; being conjoined but not co-located with a Rule-Module Nexus. While a VP may be stored apart from, or external to, the NAT, the VP may be stored within a UTA or UTA-Terminal temporarily and removeably conjoined with a NAT. The VP is, in an illustrative embodiment, not stored directly or permanently on a NAT. The VP, being online, requires communications means (or interconnecting communications means) external to the NAT for receiving the UUC from the NAT. Receives incoming bid verification data, compares said bid verification data to registered verification data, making a matching determination and, if successful, in an illustrative embodiment, outputs or invokes a VAC enabling access to a UAR.

DETAILED DESCRIPTION OF THE INVENTION

(167) It should be noted that as used herein, the article "a", or "an", means "one or more" of anything to which it refers, such as a platform, component, feature, element, process, step, method, system or the like of the invention. For example, "a platform" means "one or more platforms". It should also be noted that as used herein, the term "plurality" means "two or more" of anything to which it refers. For example, "plurality of Financial Accounts" means "two or more Financial Accounts". It should also be noted that the term "comprising" (or "comprises") or "comprising any of the following", means "including, but not limited to" at least one item to which said "comprising" refers, inclusive of meaning singularly, in combination, in any combination, and/or in any order, and to any degree to which said items can be made operatively or functionally compatible with each other. For example, "Financial Accounts comprising: debt; credit; and; stored value" means "including, but not limited to, at least one of the Financial Accounts of: debit; credit, and/or; stored value". It should be further noted that the phrase "of a Financial Account" means "relating to, or associated with a Financial Account". It should also be noted that the term "communication" includes an electronic message, whether presented, transmitted, received, or a combination thereof.

(168) It should also be noted that "via", as used herein, means "by the use of", "by the agency of" or "by the instrumentality of". Further, it should be noted that "conjoin", as used herein, means to join, connect, associate and/or combine, whether in physical proximity or remotely. It should also be noted that "presenting", as used herein, means "displaying", and vice-versa. Further, it should be noted that "scalable" comprises the capability of storing, using and/or invoking a plurality of data, such as a plurality of rule-modules, or a plurality of software programs.

(169) As used herein, “invoke” comprises activating or calling up a computer or communications code, function or method, any of which may be static or dynamic. In one embodiment, an existing, reusable code may be repeatedly invoked, or a dynamic code, instantly formulated and possibly non-reusable, may be invoked.

(170) It should further be noted that, as used herein, unless otherwise specified, the terms “navigating system”, “mapping system”, “guiding system”, and “positioning system” are deemed to have similar and at times identical meanings, for the purposes of describing systems which electronically identify the geographic location of a mobile device. Other definitions and meanings shall be referred to in the glossary and elsewhere herein.

(171) The method and system of the present invention may be described herein in terms of functional block components, flow charts, screen shots, optional selections and various processing steps. It should be appreciated that such functional blocks may be realized by any number of hardware and/or software components configured to perform the specified functions. For example, the present invention may employ various platforms comprising software and hardware components (e.g., memory elements, processing elements, logic elements, look-up tables, and the like), which may carry out a variety of functions under the control of one or more microprocessors or other control devices. Accordingly, functional blocks of the block diagrams and flowchart illustrations support combinations of means for performing the specified functions, combinations of steps for performing the specified functions, and program instruction means for performing the specified functions. It will also be understood that each functional block of the block diagrams and flowchart illustrations, and combinations of functional blocks in the block diagrams and flowchart illustrations, can be implemented by either special purpose hardware-based computer systems which perform the specified functions or steps, or suitable combinations of special purpose hardware and computer instructions.

(172) Similarly, the platforms of the present invention may be implemented with any programming or scripting language such as C, C++, Java, COBOL, assembler, PERL, or the like, with the various algorithms being implemented with any combination of data structures, objects, processes, routines or other programming elements. Further, it should be noted that the platforms of present invention may employ any number of conventional techniques for data transmission, signaling, data processing, network control, and the like. For a basic introduction of cryptography, please review a text written by Bruce Schneier which is entitled “Applied Cryptography: Protocols, Algorithms, And Source Code In C,” published by John Wiley & Sons (second edition, 1995), which is hereby incorporated by reference. Protocols, as known in the art, are computing languages or computing instructions.

(173) These computer program instructions may also be stored in a computer-readable memory that can direct a computer or other programmable data processing apparatus to function in a particular manner, such that the instructions stored in the computer-readable memory produce an article of manufacture including instruction means which implement the function specified in the flowchart block or blocks. The computer program instructions may also be loaded onto a computer or other programmable data processing apparatus to cause a series of operational steps to be performed on the computer or other programmable apparatus to produce a computer-implemented process such that the instructions which execute on the computer or other programmable apparatus provide steps for implementing the functions specified in the flowchart block or blocks.

(174) It should be appreciated that the particular implementations shown and described herein are illustrative of the invention and its best mode and are not intended to otherwise limit the scope of the present invention in any way. Indeed, for the sake of brevity, conventional data networking, application development, methods of populating data onto platforms, including databases and other functional aspects of the methods and systems herein (and components of the individual operating components of the systems) may not necessarily be described in detail herein. Furthermore, the connecting lines shown in the various figures contained herein are intended to represent illustrative

functional relationships and/or physical couplings between the various elements. It should be noted that many alternative or additional functional relationships or physical connections may be present in a practical electronic transaction system.

(175) It should be appreciated that the particular implementations shown and described herein are illustrative of the invention and its best mode and are not intended to otherwise limit the scope of the present invention in any way. Indeed, for the sake of brevity, conventional data networking, application development, and other functional aspects of the systems (and components of the individual operating components of the systems) may not be described in detail herein.

Furthermore, the connecting lines shown in the various figures contained herein are intended to represent illustrative functional relationships and/or physical couplings between the various elements. It should be noted that many alternative, equivalent, or additional functional relationships or physical connections may be present in a practical electronic transaction system. It should further be noted that the order of the steps denoted in the attached drawings are not intended as limitations and the steps may be accomplished in other orders without deviating from the scope of the present invention. Still further, the actors denoted as performing individual steps in the disclosed process should not be interpreted as limiting in any way as one with ordinary skill in the art appreciates that the steps may be performed by actors different from those disclosed herein without deviating from the spirit and scope of the present invention.

(176) The method and system of the present invention may be described herein in terms of various functional elements as depicted in the attached drawings, configurations and described embodiments. It should be appreciated that such functional elements may be realized by any number of similar elements configured according to this invention to perform the specified functions. Thus, it should be appreciated that the particular implementations shown and described herein are illustrative of the invention and its preferred mode and are not intended to otherwise limit the scope of the present invention in any way. Indeed, for the sake of brevity, conventional elements of underpants may not be described in detail herein. Furthermore, the various figures contained herein are intended to represent illustrative functional embodiments of the invention, and that many alternative or equivalent configurations, elements, and structures are intended to be within the scope of the present invention. It should further be noted that the order of the elements in the attached drawings, specification and claims are not intended as limitations and the drawings, specification and claims may be configured in other orders without deviating from the scope and spirit of the present invention. It should be considered that various components, embodiments, and definitions of the invention discussed herein may at times consist essentially of their respective elements as described herein.

(177) In the following detailed description of an illustrative embodiments, reference may be made to the accompanying drawings, which form a part hereof, and within which are shown by way of illustration specific embodiments by which the invention may be practiced. It is to be understood that other embodiments may be utilized and structural changes may be made without departing from the scope of the invention.

(178) It will be appreciated that the methods, systems, and their comprising elements of this invention, may be practiced in any order, in any timeframe or time delay with respect to other steps, other systems and their comprising elements. It will further be appreciated that the invention illustratively disclosed herein suitably may be practiced in the absence of any element which is not specifically disclosed or required herein.

(179) It will be appreciated, that many applications and embodiments of the present invention could be formulated. One skilled in the art will appreciate that a network may include any system for exchanging data or transacting business, such as the Internet, an intranet, an extranet, WAN, LAN, satellite or wireless communications, and/or the like. Moreover, although the invention and its platforms use protocols such as TCP/IP to facilitate network communications, it will be readily understood that the invention could also be implemented using IPX, Appletalk, IP-6, NetBIOS, OSI

or any number of existing or future protocols. Moreover, the method and the system contemplate the use, sale, exchange, transfer, or any other distribution of any goods, services or information over any network having similar functionality described herein.

(180) Further still, the terms “Internet” or “network” may refer to the Internet, any replacement, competitor or successor to the Internet, or any public or private network, intranet or extranet that is based upon open or proprietary protocols. Specific information related to the protocols, standards, and application software utilized in connection with the Internet may not be discussed herein. For further information regarding such details, see, for example, DILIP N Authorized IssuerK, INTERNET STANDARDS AND PROTOCOLS (1998); JAVA 2 COMPLETE, various authors, (Sybex 1999); DEBORAH RAY AND ERIC RAY, MASTERING HTML 4.0 (1997); LOSHIN, TCP/IP CLEARLY EXPLAINED (1997). All of these texts are hereby incorporated by reference.

(181) As used herein, a Platform may include local or remote database(s) for storing, associating, and retrieving information for accessing a User Account Registry, Rule Module Nexus, and/or Verification Platform. Any databases discussed herein may be any type of database, such as relational, graphical, hierarchical, object-oriented, and/or the like. Common database products that may be used to enable the Platforms include DB2 by IBM (White Plains, N.Y.), any of the database products available from Oracle Corporation (Redwood Shores, Calif.), Microsoft Access or MSSQL by Microsoft Corporation (Redmond, Wash.), or any other database product. The database may be organized in any suitable manner, including as data tables or lookup tables. Association of certain data may be accomplished through any data association technique known and practiced in the art. For example, the association may be accomplished either manually or automatically. Automatic association techniques may include, for example, a database search, a database merge, GREP, AGREP, SQL, and/or the like. The association step may be accomplished by a database merge function, for example, using a “key field” in each of the manufacturer and retailer data tables. A “key field” partitions the database according to the high-level class of objects defined by the key field. For example, a certain class may be designated as a key field in both the first data table and the second data table, and the two data tables may then be merged on the basis of the class data in the key field. In this embodiment, the data corresponding to the key field in each of the merged data tables is, in an illustrative embodiment, the same. However, data tables having similar, though not identical, data in the key fields may also be merged by using AGREP, for example.

(182) As used herein, a web browser extension browser extension comprises a computer program that extends the functionality of a web browser in some way. Depending on the browser and the version, the term may be distinct from similar terms such as plug-in or add-on. Some extensions are authored using web technologies such as HTML, JavaScript, and CSS. Browser extensions can change the user interface of the web browser without directly affecting viewable content of a web page; for example, by adding a “widget”, which is a clickable graphical user interface object. Each browser type has its own architecture and APIs (application programming interfaces) to build the extensions which requires different code and skills for each extension. Nowadays there are development frameworks which allows developers to build cross-browser extensions with only one code base and one API, eliminating the need to develop a different extension version for each one of the Browsers. Examples of those frameworks are Add-ons Framework and Extension Maker, which allow developer to build cross browser extensions for Internet Explorer, Firefox, Chrome, Safari and Opera, and the Crossrider development framework which allows developers to build cross browser extensions for Internet Explorer, Firefox, Chrome and Safari.

(183) Communication between the parties (e.g., an Account Issuer, a User, merchant, and/or third-party computer) and the rule-module nexus of the present invention may be accomplished through any suitable communication means, such as, for example, a telephone network, intranet, Internet, point-of-interaction device (point-of-sale device, personal digital assistant, cellular phone, kiosk, etc.), online communications, offline communications, wireless communications, and/or the like.

One skilled in the art will also appreciate that, for security reasons, any databases, systems, or components of the present invention may consist of any combination of databases or components at a single location or at a plurality of locations, wherein each database or system includes any of various suitable security features, such as firewalls, access codes, encryption, decryption, compression, decompression, and/or the like.

(184) Referencing the computer networked aspect of an illustrative embodiment of this invention, each party is equipped with a computing system to facilitate online commerce transactions. The computing units may be connected with each other via a data communication network. The network is a public network and assumed to be insecure and open to eavesdroppers. In illustrative embodiment, the network is embodied as the Internet. In this context, the computers may or may not be connected to the Internet at all times. For instance, the computer may employ a modem to occasionally connect to the Internet, whereas the rule-module nexus might maintain a permanent connection to the Internet. It is noted that the network may be implemented as other types of networks, such as an interactive television (ITV) network.

(185) The merchant computer and the transaction Account Issuer or provider computing systems may be interconnected via a second network, referred to as a payment network. The payment network represents existing proprietary networks that presently accommodate transactions (or "Xact's") for credit, debit, loyalty/rewards, and other types of financial/banking transactions. The payment network is a closed network that is assumed to be secure from eavesdroppers. Examples of the payment network include the American Express™, VisaNet™ and the Verifone™ network.

(186) The User may interact via the rule-module nexus with a transaction system or a merchant using any transaction terminal such as a telephone, magnetic stripe card reader, radio frequency interrogator, keyboard, mouse, kiosk, personal digital assistant, touch screen, voice recognition device, transponder, handheld computer (e.g., Palm Pilot™), cellular phone, web TV, web phone, blue tooth/beaming device and/or the like. Similarly, the invention could be used in conjunction with any type of personal computer, network, computer, workstation, minicomputer, mainframe, or the like, running any operating system such as any version of Windows, Windows NT, Windows2000, Windows 98, Windows 95, MacOS, OS/2, BeOS, Linux, UNIX, MVS, OS, or the like.

(187) As will be appreciated by one of ordinary skill in the art, the present invention may be embodied as a method, a data processing system, a platform for data processing, and/or a computer program product, wherein the steps and/or processes may be performed in a variety of sequences and/or orders, without restricting the scope of this invention. Accordingly, the present invention may take the form of an entirely software embodiment, an entirely hardware embodiment, or an embodiment combining aspects of both software and hardware. Furthermore, the present invention may take the form of a computer program product on a computer-readable storage medium having computer-readable program code means embodied in the storage medium. Any suitable computer-readable storage medium may be utilized, including hard disks, CD-ROM, optical storage devices, magnetic storage devices, flash card memory and/or the like.

(188) The transaction Account Issuer (Authorized Issuer) (or Account Provider) platform **28** includes any provider of products and/or services that facilitates any type of transaction. As contemplated by an illustrative embodiment of the present invention, the Account Issuer platform **28** establishes and maintains account and/or transaction information for the User. The Account Issuer platform **28** may issue products to the User and may also provide both the User and the Merchant platform **28** with the processes to facilitate the transaction system of the present invention. The Account Issuer platform **28** includes banks, credit unions, credit, debit or other transaction-related companies, telephone companies, or any other type of card or account issuing entities, such as card-sponsoring companies, incentive rewards companies, or third-party providers under contract with financial entities. Unless otherwise specifically set forth herein, although referred to as "Account Issuer," this term should be understood to mean any entity issuing any type

of account to facilitate any transaction, exchange or service, and should not be limited to companies possessing or issuing physical cards. In an illustrative system, the Account Issuer platform **28** may be any transaction facilitating company such as a credit account providers like American Express™, VISA™, Mastercard™, Discover™, and the like. In another embodiment, the Account Issuer platform **28** could be any membership organization or union. In some instances, the Account Issuer platform **28** and the Merchant platform **28** may be the same, for example, where the credit account is issued by the same entity that provides the product or service. A phone card using a credit account issued by a telephone company, where the credit account is associated to the User's home telephone account is one such occasion.

(189) Although the present system for facilitating transactions may exist within one transaction Account Issuer system, illustrative embodiments contemplate use with other third-party authorization and settlement systems and networks.

(190) The authorization and settlement processes may occur as separate steps or as a single step. In one embodiment, referred to herein as an electronic data capture (EDC) system, the Merchant platform **28** sends an authorization request to an Account Issuer platform **28** and if the authorization request is approved, a receipt of charges is created and submitted for the Merchant platform **28**. Separate sequences of file transmissions or messages are therefore not required. Various embodiments, hybrids, and modifications of these processes should be apparent to one skilled in this art.

(191) It will be appreciated that the invention illustratively disclosed herein suitably may be practiced in the absence of any element which is not specifically disclosed herein, particularly in an illustrative embodiment.

(192) The invention is a financial transaction system which uniquely enables a User to, in an illustrative embodiment, present a single “thin-client” token and benefit from centralized Financial Account access, Financial Account aggregation and Financial Account presentation by means of a secure, remotely located Rule-Module Nexus **14**.

(193) A financial transaction is any electronic transfer or exchange of Financial Data having a predetermined monetary or monetary-equivalent unit value which is legal tender or a legal tender-equivalent, and which is honored by an Account Issuer, such that a User's purchase, expenditure or usage of these units results in the User's purchase or sale of goods, services or currency involving an Account Issuer. Financial data can also include units of currency, minutes of telephone calling time, miles towards earning a free airplane flight, points towards a gallon of gas, and the like.

(194) A “Financial Account number” or “transaction number” as used herein, may include Financial Account-ID, comprising any identifier for a Financial Account **65** (e.g., credit, charge debit, checking, savings, reward, loyalty, travel or the like) which may be maintained by a transaction Account Issuer (e.g., payment authorization center) and which may be used to complete a financial transaction. A typical account number (e.g., account data) may be correlated to a credit or debit account, loyalty account, travel or rewards account maintained and serviced by such entities as American Express, Visa and/or MasterCard, such as reward points as currency wherein a User can use reward or scrip points (e.g., Star Alliance™, eScrip™) as currency to pay for purchases). For ease in understanding, the present invention may be described with respect to a credit card account. However, it should be noted that the invention may not be so limited and other accounts permitting an exchange of goods and services for an account data value may be contemplated to be within the scope of the present invention.

(195) In an illustrative embodiment, RMN may refer to Master and/or Subset platforms, e.g., RMN **14** and Subset RMN **17**, unless otherwise specified. Similarly, a Payee may refer to an Account Issuer, unless otherwise specified.

(196) An Account Issuer as defined herein is the named entity with primary fiduciary duty for administering a Financial Account **65** of a User, said fiduciary duty comprising any one of the following: managing the Financial Data within the Financial Account **65**, and; reconciling the

Financial Data within the Financial Account **65** upon settlement of a financial transaction. An Account Issuer comprises any of the following: a bank, a merchant, a scrips provider, credit account organization, a government agency, an insurance company, a brokerage firm, a reward incentives provider, a services barter provider, a product barter provider, an Internet payment provider, and the rule-module nexus of this invention. A Financial Account **65**, storing related Financial Data, resides with an Account Issuer if said Account Issuer has the primary responsibility to store, administer and reconcile the Financial Data within the User's Financial Account **65**.

(197) The Account Issuer may be within the RMN **14** or may be within a RMN-authorized Third-Party Platform **28**.

(198) In an illustrative embodiment, once the User makes an account selection from a plurality of proprietary accounts presented, the RMN **14** does not universally “stand-in” for Account Issuers, nor automatically invoke a default program or Global Execution Command (GEC) **55**, to cause financial transactions of all RMN **14** Users to bypass, supersede, divert or switch an already-existing interchange fee, discount rate, or other settlement process which would otherwise apply to a Financial Account **65** selected by the User, or which would otherwise apply to the selected Financial Account's **65** associated Account Issuer (Authorized Issuer). Alternatively put, the RMN **14** is, in an illustrative embodiment, designed wherever possible to compliment, not contravene, the already-existing interchange fee, discount rate, or settlement process of the User's selected Financial Account **65** and its associated Account Issuer(s). In an illustrative embodiment, the RMN **14** maintains, wherever possible, a Financial Account's **65** already-associated transaction processors (issuing banks, credit associations, national automated clearinghouse association (“NACHA®”), merchant banks), their respective proprietary networks (VisaNet®, Pulse®, Nova®, Interlink® and the like), and their respective processing fees (interchange fees, discount rates), protocols, and procedures. Similarly, in an illustrative embodiment, the RMN **14** does not act to “stand-in” or “mirror” the transaction processing of an Account Issuer (Authorized Issuer) which has a pre-existing association with a Financial Account selected by a User.

(199) In an illustrative embodiment, it is an objective of the present invention to compliment, not contravene, the already-existing interchange fee, discount rate, or settlement process of the User's selected Financial Account **65** and its associated Account Issuer(s). Alternatively put, the invention herein is, in an illustrative embodiment, designed not to universally “stand-in” for an existing Account Issuer, nor to cause all financial transactions of the invention herein to bypass, divert or switch an already-existing interchange fee, discount rate, or settlement process which would otherwise be applied by the Account Issuer(s) of the User's selected Financial Account **65** and its associated transaction processing.

(200) It will be appreciated that an illustrative embodiment of this invention maximizes security of the Rule-Module Nexus (RMN) **14** transactions and minimize the size and cost of the nexus access token, by employing a “thin-client”, passive RFID-based Nexus Access Token (NAT) **62**. As used herein, “thin-client”, in an illustrative embodiment, means that to complete a financial transaction using the RMN **14**, the NAT **62** transmits a “thin-client” packet which is not required to be transaction-enabled by containing any “live” financial account data or “funding” account data which can directly identify or directly access a specific online Financial Account **65** of a user. Thus, during processing of a financial transaction, the NAT **62** does not require use of any stored data on a User's Nexus Access Token **62** which may directly identify or directly access an online Financial Account **65** of the User.

(201) Further, the NAT **62** is designed, in an illustrative embodiment, to be of a sufficiently convenient size and weight as to be hand-holdable (e.g., hand-held) or smaller: small, light-weight and readily mobile for daily use, or having physical dimensions sufficiently small to be hand-held or at most, palm-sized. In an illustrative embodiment, the NAT **62** is compact enough also to be used and/or operated while being held in the hand or hands.

(202) Further, as a passive RFID-based token, the “thin-client” NAT, in an illustrative embodiment,

does not require a battery power supply, nor does it require an integrated circuit chip. In one embodiment, a purely passive or “reflective” NAT **62** may rely upon the electromagnetic energy radiated by a UTA **16** reader to power the RF integrated circuit that makes up the RFID tag within the NAT **62**. As such, in this embodiment, the NAT **62** may be said to be “beam powered”. This “thin-client” NAT may be low-cost, high-security, and miniaturized for conjoining with other conveniently portable tokens, such as a wearable finger ring or a house key. For high security, it is preferred that stored data on a User's NAT **62** comprises a User's Unique User Code (UUC) **200**, which may also comprise a RMN-Routing Code (RMN-RC) **61**. In one embodiment, the NAT **62** does not store the User's User Account Registry Code (UAR-Code) **59**. The UAR-Code **59** identifies a User Account Registry **15** of the RMN **14** which comprises the Financial Accounts **65** of a User, even though, in an illustrative embodiment, said UAR-Code **59** does not itself directly identify any specific Financial Account **65** of the User, nor, in an illustrative embodiment, does the UAR-Code **59** depend upon the designation of any Financial Account **65** of the User as a primary account. In this embodiment, in an illustrative embodiment, no stored data on the NAT **62** can be read or copied off-line which could provide direct access to a specific Financial Account **65** of the User. As such, if a User's NAT **62** is stolen, lost or copied, the NAT **62** and its stored data would not provide an unauthorized User with direct access to a Financial Account **65** of an authorized User of the RMN **14**. In an illustrative embodiment, there are a plurality of different UAR-Codes **59**, each identifying a User Account Registry (UAR) **15** of the User, comprising any one of the following: a Master UAR **15** within the RMN **14**; a Master UAR **15** within a Third-Party Platform **28**; a Subset UAR **19** within a Subset RMN **17**; and the like.

(203) Therefore, an illustrative NAT **62** or UUC **200** in this embodiment contrasts with a standard “fat-client” token requiring stored data which directly identifies, or enables direct access to, a specific Financial Account **65** of a User. For example, a standard “fat-client” magnetic stripe card stores a credit card number in a sixteen-digit format as four spaced sets of numbers, represented by a number like “1234 5678 9101 1213”. The first five to seven digits are reserved for processing purposes and identify the issuing bank, card type, etc. In this example, the last (sixteenth) digit is used as a sum check for the sixteen-digit account number. The intermediary eight-to-ten digits are used to uniquely identify the consumer and the account. A merchant account number may be, for example, any number or alpha-numeric characters that identifies a particular merchant for purposes of card acceptance, account reconciliation, reporting, or the like. In an illustrative embodiment, a NAT **62** embodiment uses a magnetic stripe or a bar-code imprint, optionally in combination with another connectivity or communications capability like RFID, wherein the data stored on the NAT **62** may only include a UUC **200**, optionally including a RMN-RC **61**.

(204) The invention provides a Rule-Module Nexus (RMN) **14** method and system for a financial User to authorize a financial transaction using Financial Accounts **65** either at the merchant point of sale or over the Internet. A Rule-Module Nexus **14** and a Verification Platform **12** are used to accomplish these goals.

(205) A Unique User Code (UUC) **200** is defined as any data string, unique to a User, which is stored into a User's Nexus access token (NAT) **62**, and comprises any of the following: a dynamic code which changes periodically based on predetermined criteria synchronized with the verification platform, and; a static code which remains constant based on a predetermined code synchronized with the verification platform. The UUC **200**, in an illustrative embodiment, does not store any data which directly identifies a specific online Financial Account **65** of a User. The UUC **200** may also comprise a RMN-RC **61**. One such UUC is registered for the User with the Rule-Module Nexus **14**.

(206) A Personal Verification Code (PVC) **202**, distinct from the Unique User Code **200**, comprises any alpha-numeric sequence or data string, not necessarily unique to a User, which is optionally input by the User during a financial transaction. Optionally, at least one such PVC **202** is registered for the User with the Rule-Module Nexus **14**.

(207) A Rule-Module **50** is defined as any Pattern Data **54** associated with an Execution Command **52**. At least one such Rule-Module **50** is registered for the User with the Rule-Module Nexus **14**.

(208) The components of the computer system of this invention comprises any of the following: Nexus access token (NAT); User Interface Apparatus (UTA); Communication lines for a Network; Rule-Module Nexus (RMN);

(209) These components together allow a financial transaction to occur without requiring the User to use a plurality of financial cards, paper coupons, credit cards, debit cards, or other physical objects.

(210) Nexus Access Token (NAT)

(211) The NAT **62** is, in an illustrative embodiment, capable of communicating and/or connectivity with a UIA **16**-Terminal **2** and/or the RMN **14** via at least one mode, including magnetic stripe (Magstripe), bar code imprint, radio frequency identification (RFID), near-field communications (NFC), cell phone network, and the like. In an illustrative embodiment of the invention, the UIA **16** electronically scans an RFID signal, a bar-code imprint, or a magnetic stripe from a NAT **62**, via the data reader attached to the UIA **16**. In one embodiment, NAT **62** is encoded with a UUC **200**, optionally further comprising a RMN-RC **61**.

(212) The UUC **200** may be stored with a NAT **62** via processing comprising; embossing, printing, labelling, encoding or encrypting in an electronic memory unit; storing in an unencrypted memory unit.

(213) In an illustrative embodiment of a NAT **62** using Magstripe (or Magnetic Strip, or Magnetic Swipe, or Magnetic Stripe), the NAT **62** may use a unique DCVC3 (Dynamic card validation code); a unique Static CVC (card validation code); a unique DCVV (Dynamic card validation value), and/or; a unique Static CVV (card validation value).

(214) In an illustrative embodiment of a NAT **62** using a bar code imprint, the NAT **62** may be imprinted with an electronically readable bar code, and optionally a visibly readable imprint, and/or a raised imprint, of alpha-numeric code.

(215) In an illustrative embodiment of a NAT **62** using RFID, would, in an illustrative embodiment, include a portable transponder using passive contactless proximity technologies, in an illustrative embodiment, compatible with close proximity standards ISO 14443 and 13.56 Mhz, wherein either: the NAT **62** reflects energy from the User Interface Apparatus **16**, or; the NAT **62** absorbs and temporarily stores a very small amount of energy from the UIA **16** signal to generate its own quick response. In either case, in this preferred embodiment, the NAT **62** requires strong signals from the User Interface Apparatus **16**, and the signal strength returned from the NAT **62** is constrained to very low levels by the limited energy.

(216) In an illustrative embodiment of a cell phone NAT **62** embodiment, a plasma display may display a bar-code image, readable by a UIA **16** (or UIA **16**-Terminal **2**) bar code reader, in an illustrative embodiment, displaying a thin-client data packet comprising any one of the following: no user "live" financial account data; no user payment-enabling data; no user funding account data; a unique user code; a rule-module nexus routing code; positioning data, and/or; over-the-air provisioning data.

(217) In illustrative embodiments of a NAT **62** using NFC, the NAT **62** is capable of wireless internet detection (WiFi), Bluetooth, and IEEE 802.11 communications.

(218) It will also be appreciated that in an illustrative embodiment of the invention, the "thin-client" NAT **62** of the invention permits optimal miniaturization, in an illustrative embodiment, enabling the NAT **62** to be embedded in a User's most-conveniently portable token, such as a watch, a ring, a door key, a PDA, a bracelet, a cell phone, or the like. In an illustrative embodiment, the NAT **62** does not require a battery; rather, the power is supplied by the UTA **16** conjoined with a radio frequency identification (RFID) scanner having read/write capabilities. When a NAT **62** encounters radio waves from a UTA **16**, a coiled antenna within the NAT **62** forms a magnetic field. The NAT **62** draws power from the magnetic field, energizing the circuits in the NAT **62**, enabling

the UTA **16** to read the stored data on the NAT **62**. In this embodiment of the invention, the thin-client NAT **62** has several advantages, including: not requiring an embedded battery; sustaining its function for twenty years or more; reduced cost; miniaturization to smaller than a grain of rice. (219) In an illustrative embodiment, a NAT **62** may be single-mode, dual-mode or multi-mode connectivity enabled for communicating with a UTA **16**-Terminal **2** and/or with the RMN **14**, comprising embodiments such as: Magnetic Stripe (or Magstripe); a bar-code imprint; a bar-code imprint and a Magstripe; radio frequency identification (RFID) and Magnetic Stripe; RFID and near-field communications (NFC); RFID and Bluetooth®; RFID and Cellular phone network; RFID, NFC and Cellular phone network; instant messaging (IM), short message system (SMS), and RFID. Optionally, the bar-code may be imprinted on a label may be adhesively or removeably attached to the NAT **62**. In another embodiment of the invention, the RFID chip may be adhesively or removeably attached to the NAT **62**.

(220) In one card embodiment of the NAT **62**, the front of the NAT **62** may be reserved for branding by an account issuer; the back may comprise acceptance mark(s), appropriate token numbers (see below), risk items, magnetic stripe, and signature panel for use when a UTA **16** does not have a keypad **70** for PVC **202** entry to validate or authorize the financial transaction. Regarding the data strings, no embossed numbers may be required, since the financial transactions would be authorized online. As such, the token number for the NAT **62**, comprising a portion of the UUC **200**, can be laser printed. There may also be a “virtual POS” number printed on the back, in full, along with a third number—a risk key of three or four digits. In this way, the NAT **62** can effectively work in all transaction environments, without the need for a separate “Virtual card” NAT **62** which might otherwise be used for internet purchases or physical merchant purchases in the event that a magnetic swipe reader is not available.

(221) In one embodiment, the UUC **200** associated with a NAT **62**, is an alphanumeric data string further comprising any one of the following: First #'s (is Nexus Xact Flag) Next ###'s (are the RMN-RC or UAR-RC) Next #####'s (Unique User data portion) Next ###'s (NAT originator/distributor identifier: This enables: a) a third-party having originating or distributing a NAT, to be identified with said NAT for purposes of said third-party receiving a royalty or fee from the RMN, potentially as a function of tracking the activation and usage frequency of said NAT; b) the NAT to present a third-party's BIC Code or BIN, for appending to a RMN-bound transaction such that the Terminal **2** and the Network **18** more readily recognize the data packet being transmitted in IS O/IEC 7813 format Next ###'s (UAR/UUC Expiration date)

(222) FIG. **11D** illustrates an embodiment for activating a magnetic strip NAT **62**, optionally including registration of an initial Financial Account **65**, at POS in a physical-world retailer using an in-market Verifone® Trans330 unit, comprising any one of the following: the User swipes the NAT **62** (Step **513**); the User swipes their driver's license (Step **514**); the User swipes an initial Financial Account **65** card. The data from these cards is transmitted to the RMN **14** to be activated, linked, and stored.

(223) User Interface Apparatus (UIA)

(224) The UIA **16** is a device that gathers or comprises verification information for use in authorizing financial transactions. In one embodiment, the UIA **16** or Transaction Terminal **2** may also store the RMN-RC **61**, wherein the RMN-RC **61** may be invoked upon an event comprising any one of the following: the UIA **16** scanning a UUC **200** from a NAT **62**, wherein the NAT **62** need not necessarily store or transmit a RMN-RC **61**, and; the UIA **16** or Transaction Terminal **2** being instructed by the User or an attendant to invoke the RMN-RC **61**, wherein being instructed comprises key-punching, voice-commands, or touch-screen options.

(225) Each UIA **16** conducts one or more of the following operations:

(226) Gathers UUC **200** input from scanning a NAT **62** of a User, via any of the following: bar-code reader; contactless communications scanner; near-field communications (NFC) scanner; magnetic stripe scanner; magnetic ink character reader (MICR) scanner, and; automated

transponder interrogation or scanning for dedicated contactless proximity communications from a NAT **62**, using an RFID sensor or scanner, an infrared sensor, or an audio frequency sensor, and the like;

(227) Optionally gathers a PVC **202** code from a User via a keypad, touch screen, or audio microphone;

(228) Secure communications between UIA **16** and its conjoined Transaction Terminal **2**, in an illustrative embodiment, using encryption;

(229) Secure storage of secret encryption keys;

(230) Store and retrieve a unique Account Issuer UIA-VC **204**;

(231) Secure enclosure & components from unauthorized tampering;

(232) Display information, allow parties to approve or cancel a financial transaction;

(233) Store, verify, and retrieve an Account Issuer's digital verification code;

(234) Allow parties to select, via a conjoined Transaction Terminal, from among choices of Account Issuer and a plurality of a User's Proprietary Financial Accounts.

(235) In an illustrative embodiment, the UUC **200** input is gathered using a UUC **200** sensor or scanner, located within the UIA **16**. UUC **200** sensor is a dedicated short range contactless communications sensor, however it is understood that other types of UUC **200** sensors for wireless tokens can be used, such as infrared and the like.

(236) For the optional gathering a PVC **202**, the UIA **16** optionally has PVC **202** input means comprising a keypad **70**, touch screen or an audio microphone also located securely inside the UIA **16**.

(237) FIG. **3A** illustrates an illustrative embodiment of a UIA **16** in accordance with the present invention, with illustrative components for use in gathering a User's bid verification data for a financial transaction via the RMN **14**. In general, the operation of UIA **16** may begin when NAT **62** may be presented for payment, and may be interrogated by UUC-scanner within the UIA **16**. NAT **62** and UIA **16** may then engage in mutual authentication after which the NAT **62** may provide the UUC **200** and/or RMN-RC **61** to the UIA **16**, which may further provide the information to the Transaction Terminal **2** conjoined with the UIA **16**.

(238) FIG. **3A** also illustrates an embodiment, wherein the operation of UIA **16** may begin when a User registers a Financial Account **65** with the RMN **14**, which may include the User swiping a magnetic stripe card (e.g., credit, debit, rewards, stored value, or private label) through a magnetic stripe reader (MSR) **89** or running a paper bank check through a magnetic ink character reader (MICR) **89**. The User may register a Primary PVC **200** (not shown) via a keypad **70** or a touch screen Display **6**, optionally using a stylus **93** tethered to the UIA **16**.

(239) FIG. **3A** also illustrates an illustrative embodiment of the invention, wherein a Prompt **91**, in an illustrative embodiment, via a visible screen display, a keypad indicator, or audio signal, is provided to the User via a UIA **16** or a Transaction Terminal **2**. Said Prompt **91** enables the User, a retailer's live attendant, the Rule-Module Nexus, or an Account Issuer to signal intent to process or route a financial transaction via the Rule-Module Nexus **14**. In an illustrative embodiment, employing the Prompt **91** signals intent to process the financial transaction via the RMN **14** invokes a RMN-Routing Code (or RMN-RC) **61** optionally stored with the UIA **16**, the Transaction Terminal **2**, or on a remote computer, and; an additional Prompt **91** for the User to input their Primary personal verification code (Primary PVC or 1° PVC) **200**. In an illustrative embodiment, the RMN-RC **61** contains electronic instructions to a network to process the financial transaction in a manner that routes at least one element or step of the processing to or through the RMN **14**.

(240) FIG. **3B** illustrates an embodiment of the invention, wherein upon associating processing preferences with Financial Account(s) **65**, an illustrative embodiment of Rankings **96** is visibly indicated with each associated to their respective Financial Accounts **65**.

(241) FIG. **3C** is a diagram of an administrator (wherein administrator comprises any one of the following: Account Issuer; User; Rule-Module Nexus **14**) access key creation process. By way of

example and not of limitation, administration security profile input **301** may comprise various data including name **306-A**, physical address **305**, email address **304-A**, client hardware identification signature **303**, and internet protocol (“IP”) address **302-A**. All data may be entered via system graphical user interface (“GUI”). After data is entered **301**, internal software creates administrator access key **300**.

(242) FIG. 3D is a diagram of a User key creation process. Data may be entered **401** into a GUI interface. By way of example and not of limitation, data entry points may comprise data such as name **404**, physical mailing address **406**, email address **408**, social security number **410**, date of birth **411**, IP address **414**, hardware identification signature **415**, user photo **413**, and/or government issued I.D. **402** which could be swiped as a means of input. FIG. 3D also shows optional information that may be entered such as debit card information **403**, credit card information **405**, bank account information **407**, and/or system based credit limit **412**. After data is entered **401** into the user profile, initial user key is created **400**. The data entry depicted in FIG. 3D may also be used to create subsequent user access keys for enrollment agents, financial agents, merchants and users. In some instances, not all of the inputs are used, whereas in some instances, additional inputs may be desired.

(243) Upon having registered a Financial Account **65**, the User may: provide a Primary PVC **200** via a keypad **70** on the UIA **16**, and Signal Intent to use (or to process the financial transaction via) the RMN **14** via pressing a preset a Prompt Button **91**, or alternatively, tapping a touch screen visible Prompt **91**. Signaling Intent to use the RMN **14** may invoke a RMN-RC **61** (not shown) wherein the UIA **16** may encrypt the Primary PVC **200**-RMN-Code **61** packet, which may be transmitted to the Transaction Terminal **2** (not shown) conjoined with the UIA **16**. In one embodiment, upon the VP **12** (not shown) having made a positive determination that the User is authorized to access the RMN **14**, the User may be presented via a touch screen Display **6** with Visible Signatures **81** for a plurality of Financial Accounts **65**, from which the User may select a Financial Account **65** for processing the financial transaction.

(244) Although the present invention may be described with respect to a NAT **62**, the invention may use a NAT **62** conjoined with a key ring, tag, fob, card, cell phone, hat, shirt, audio entertainment device, wristwatch, clothes (e.g., jackets, raincoats, shoes), or any such form capable of being presented for interrogation.

(245) The UIA **16** may be configured to communicate using a RFID internal antenna **106**. Alternatively, UIA **16** may include an external antenna **108** for communications with NAT **62**, where the external antenna may be made remote to the UIA **16** using a suitable cable and/or data link **120**. UIA **16**-Transaction Terminal **2** (conjoined) may be in communication with the RMN **14** via a Network **18**. The UIA **16** may be conjoined or wholly integrated with a Transaction Terminal **2**, including a point-of-interaction device such as, for example, a merchant point-of-sale (POS) electronic cash register or a computer interface (e.g., User interface). In one illustrative embodiment the UIA **16** is conjoined with the Transaction Terminal **2** via a USB connector. As described more fully below, the UIA **16** may optionally include the keypad **70** or touch screen for data-entry of a bid PVC **202** by the User.

(246) In one embodiment, the UIA **16** may have a serial (RS232) and USB1.1 interface, wherein the device application programming interface (“API”) allows the RF field to be turned on/off and provide status and version information. The RFID command-set includes block read, block write and NAT **62** inventory (enumerate UUCs **200** of all NATs **62** in range) commands. The UIA **16** can issue addressed commands (affect only one NAT **62**) and non-addressed commands (obeyed by all NATs **62** in range). C++ class libraries in the UIA **16** may also support digital signatures, based on strong encryption, to detect tampering with (and corruption of) data on the UIA **16**.

(247) Although the UIA **16** may be described herein with respect to being conjoined with a merchant point-of-sale (POS) Transaction Terminal **2**, the invention may not be so limited. Indeed, a Transaction Terminal **2** may be used herein by way of example, and the point-of-interaction

device may be any device capable of being conjoined with a **UIA 16**. The **UIA 16** conjoined with the Transaction Terminal **2** may also be provided with additional or ancillary transaction data to append to the encrypted packet for transmittal to the **RMN 14**. Said ancillary transaction data may include the cost of goods, type of goods, **UIA-VC 204**, and the like. In addition, **UIA 16** conjoined with the Transaction Terminal **2** may be in communication with a **RSP 130**, an Account Issuer host or proprietary Network **18**, and/or any other access point for processing any Transaction Request **251**. In this arrangement, information may be provided via the **UIA 16**-Transaction Terminal **2** to the **RMN 14** using Network **18**.

(248) In one embodiment, the **UIA-VC 204** is appended to additional data for verifying the authenticity of the **UIA 16**, such as its merchant location, merchant identity, **UPC 67**, payee verification code, and the like. Said data may be transmitted to the **VP 12**, whereby the bid verification data of the **UIA 16** is compared with the registered verification data for the **UIA 16**. For example, if the bid **UIA-VC 204** is appended to a merchant location or **UPC 67** which is not registered to that **UIA 16**, the **RMN 14** may invoke a Rule-Module **50** comprising an alert message for presentation via the **UIA 16** Display **7** to alert the User or other parties that a potentially fraudulent transaction may be underway. Further, the **RMN 14** may decline to process the Financial Transaction Request Packet **251**, and not respond with an Audible or Visible Account Signature **81**.

(249) FIG. 3G is illustrative of several embodiments of **UIA 16** Displays **6**, wherein a merchant's **UIA-VC 204** is transmitted to the **RMN 14** with a Financial Transaction Request Packet **251** provided to **RMN 14**, wherein the **UIA-VC 204** invokes Rule-Module(s) **50** which determine or influence whether the User's **UAR 15** selects: a default Account Signature **81** for presentation to the User via Display **6** of the **UIA 16**; a plurality of Account Signatures **81** for presentation in sequence to the User via Display **6** of the **UIA 16** (e.g., one at a time), or; a plurality of Account Signatures **81** for presentation in parallel to the User via Display **6** of the **UIA 16** (e.g., more than one at a time). For, this "randomization" of Account Signature **81** presentations may be based on whether the **UIA 16** has: a single account display capability (e.g., a Verifone® Trans330), or a with, or; a multi-account display capability (e.g., a Hypercom® dynamic plasma screen display), wherein such capabilities are pre-registered with the **RMN 14** in association with the **UIA-VC 204** of the **UIA 16** in question.

(250) FIG. 3A also illustrates a block diagram of an illustrative embodiment of a **UIA 16** in accordance with the present invention. **UIA 16** includes, for example, an antenna **106, 108** coupled to a RF module **302**, which may be further coupled to a control module **304**. In addition, **UIA 16** may include an antenna **108** positioned remotely from the **UIA 16** and coupled to **UIA 16** via a suitable cable **120**, or other wire or wireless connection.

(251) RF module **302** and antenna **106, 108** may be suitably configured to facilitate communication with **NAT 62**. Where **NAT 62** may be formatted to receive a signal at a particular RF frequency, RF module **302** may be configured to provide an interrogation signal at that same frequency. For example, in one illustrative embodiment, **NAT 62** may be configured to respond to an interrogation signal of about 13.56 MHz. In this case, RFID antenna **106, 108** may be 13 MHz and may be configured to transmit an interrogation signal of about 13.56 MHz. That is, **NAT 62** may be configured to include a first and second RF module (e.g., transponder) where the first module may operate using a 134 kHz frequency and the second RF module may operate using a 13.56 MHz frequency. The **UIA 16** may include two receivers which may operate using the 134 kHz frequency, the 13.56 MHz frequency or both. When the **UIA 16** may be operating at 134 kHz frequency, only operation with the 134 kHz module on the **NAT 62** may be possible. When the **UIA 16** may be operating at the 13.56 MHz frequency, only operation with the 13.56 MHz module on the **NAT 62** may be possible. Where the **UIA 16** supports both a 134 kHz frequency and a 13.56 MHz RF module, the **NAT 62** may receive both signals from the **UIA 16**. In this case, the **NAT 62** may be configured to prioritize selection of the one or the other frequency and reject the remaining frequency. Alternatively, the **UIA 16** may receive signals at both frequencies from the **NAT 62** upon

interrogation. In this case, the **UIA 16** may be configured to prioritize selection of one or the other frequency and reject the remaining frequency.

(252) Further, protocol/sequence controller **314** may include an optional feedback function for notifying the User via a Display **6** conjoined with the Transaction Terminal **2** of the status of a particular transaction. For example, the optional feedback may be in the form of a Display **6**, such as an audio transmitter of audible signatures, an LED screen, an LCD screen and/or other visual display which may be configured to light up or display a static, scrolling, flashing and/or other message and/or signal to inform the User that the transaction may be initiated (e.g., **NAT 62** may be being interrogated), the **UUC 200** (or optionally **UUC 200-PVC 202**) may be valid (e.g., User may be verified for accessing the **RMN 14**), transaction may be being processed (e.g., **UUC 200** may be being read by **UIA 16**), and/or the transaction may be completed (e.g., transaction approved or disapproved/denied). Such an optional feedback may or may not be accompanied by an audible indicator (or may present the audible indicator singly) for informing the User of the transaction status. The audible feedback may be a simple tone, multiple tones, musical indicator, and/or voice indicator configured to signify when the **NAT 62** may be being interrogated, the transaction status, or the like.

(253) **RFID antenna 106** may be in communication with a **NAT 62** for transmitting an interrogation signal and receiving at least one of an authentication request signal and/or a **UUC 200** from **NAT 62**. **RFID communicator 306** may be configured to send and/or receive RF signals in a format compatible with antenna **106, 108** in similar manner as with respect to **NAT 62** transponder. For example, where **RFID communicator 306** may be 13.56 MHz RF rated antenna **106, 108** may be 13.56 MHz compatible. Similarly, where **RFID communicator 306** may be ISO/IEC 14443 rated, antenna **106** may be ISO/IEC 14443 compatible.

(254) **RF module 302** may include, for example, **RFID communicator 306** in communication with authentication circuitry **308** which may be in communication with a secure internal platform **310**. In one embodiment, internal platform **310** may store data corresponding to the **NAT 62** being authorized to transact business over the **UIA 16**. Internal platform **310** may additionally store **UIA-VC 204** identifying information for providing to **NAT 62** for use in authenticating whether **UIA 16** may be authorized to be provided the **UUC 200** data stored on **NAT 62** to the Transaction Terminal **2**.

(255) As may be described more fully below, in one embodiment, **NAT 62** and **UIA 16** optionally engage in mutual authentication. In this context, “mutual authentication” may mean that operation of the **UIA 16** may not take place until **NAT 62** authenticates the signal from **UIA 16**, and **UIA 16** authenticates the signal from **NAT 62**.

(256) **FIG. 4A** is a flowchart of an illustrative authentication process in accordance with the present invention. The authentication process of this embodiment may be depicted as one-sided. That is, the flowchart depicts the process of the **UIA 16** interrogating the **NAT 62**, although certain similar steps may be followed in an embodiment wherein the **NAT 62** authenticates **UIA 16**.

(257) As noted, in this embodiment, internal platform **310** may store security keys for encrypting or decrypting signals received from **NAT 62**. In an illustrative authentication process, the **UIA 16** may provide an interrogation signal to **NAT 62** (step **402-A**). The interrogation signal may include a random code generated by the **UIA 16** authentication circuit, which may be provided to the **NAT 62** and which may be encrypted using a unique encryption key corresponding to the **NAT 62**. For example, the protocol/sequence controller **314** may provide a command to activate the authentication circuitry **308**. Authentication circuitry **308** may provide from internal platform **310** an interrogation signal including a random number as a part of the authentication code generated for each authentication signal. The authentication code may be an alphanumeric code which may be recognizable (e.g., readable) by the **UIA 16** and the **NAT 62**. The authentication code may be provided to the **NAT 62** via the **RFID-RF interface 306** and antenna **106** (or alternatively antenna **108**).

(258) NAT **62** receives the interrogation signal (step **404-A**), optionally including the authorization code. Once the NAT **62** is activated, the interrogation signal (step **404-A**), optionally including the authorization code, may be provided to a modulator/demodulator circuit within the NAT **62**, where the signal may be demodulated prior to providing the signal to protocol/sequence controller **314**. Protocol/sequence controller **314** may recognize the interrogation signal as a request for authentication of the NAT **62**, and provide the authentication code to authentication circuit **308**. The NAT **62** may then encrypt the authentication code. In particular, encryption may be done by authentication circuit **308**, which may receive the authentication code and encrypt the code prior to providing the encrypted authentication code to protocol/sequence controller **314**. NAT **62** may then provide an encrypted UUC **200** via a response signal to the UIA **16** (step **406**). That is, the encrypted UUC **200** may be provided by the NAT **62** to the UIA **16** via modulator/demodulator circuit, RF interface **306** and antenna **106, 108**. Optionally, the User also provides a PVC **202** via the UIA **16**, optionally by data-entering using a keypad **70** (step **406-A**), or by touch screen tapping, or by voice commands. The UIA **16** builds an encrypted packet with the UUC **200** (or optionally UUC **200**-PVC **202**) and converts the packet into a format compatible with the IS O/IEC 7813 standard for transmitting to the VP **12** associated with, or conjoined within, the RMN **14** via a Network **18** (step **408-A**). In one illustrative embodiment illustrated in FIG. **25** and FIG. **27**, the UUC **200** (or optionally UUC **200**-PVC **202**) packet may be forwarded in Track 1/Track 2 format from the UIA **16** conjoined with the Transaction Terminal **2**.

(259) Optionally, the PVC **202** may be provided to the UIA **16** conjoined with the Transaction Terminal **2** using a conventional merchant (e.g., POS) key pad **70**.

(260) VP **12** may then receive and decrypt the UUC **200** (or optionally UUC **200**-PVC **202**) packet for electronically comparing with registered UUCs **200** (or optionally the registered UUCs **200** and PVCs **202**) (step **410-A**). Once the VP **12** makes the electronic comparison using algorithmic methodologies known in the art, the VP **12** makes either a positive matching determination or a negative matching determination. If the matching determination is deemed to be failed and a negative matching determination is automatically output, wherein the User is not verified (step **418**) and User is notified of termination of the financial transaction (step **420**), which is deemed to be completed.

(261) Alternatively, if the VP **12** makes a positive matching determination (step **412-A**), outputting a VAC **206** (step **414-A**), wherein a Rule-Module **50** is invoked, comprising a UAR-Code **59**, and a UAR **15** is accessed (step **416-A**). Optionally, the VAC comprises a UAR-Code **59**. Optionally, the financial transaction is processed via a Registry Financial Account **65** being accessed; Alternatively, the financial transaction is processed via a plurality of Financial Accounts **65** of the User being accessed, wherein the Visible or Audible Signature(s) **81** are retrieved the RMN **14** and transmitted to the UIA **16**-Transaction Terminal **2** for presentation to, and selection by, the User (step **422**).

(262) Encryption/decryption component **318** may be further in communication with a secure platform **320** which stores the security keys necessary for decrypting the encrypted UUC **200** scanned from the NAT **62**. Upon appropriate request from protocol/sequence controller **314**, encryption/decryption component (e.g., circuitry **318**) may retrieve the appropriate security key, decrypt the UUC **200** and forward to protocol sequence controller **314** in any format readable to the Transaction Terminal **2**. Optionally, in one illustrative embodiment, the UUC **200** may be combined with the RMN-RC **61** and PVC **202** received from the keypad **70**, wherein the packet is encrypted and converted into a format compatible with the IS O/IEC 7813 standard for transmitting the RMN **14** via Network **18**. Further, wherein the UIA **16**-Transaction Terminal **2** receives a response from the RMN **14** or an Account Issuer platform **28** via Network **18** (e.g., transaction authorized or denied), the protocol/sequence controller **314** may provide the response for visibly and/or audibly communicating the response to User via Display **6**.

(263) UIA **16** may additionally include a USB interface **316**, in communication with the

protocol/sequence controller **314** and the Transaction Terminal **2**. In one embodiment, the USB interface may be a RS22 serial data interface. Alternatively, the UIA **16** may include a serial interface such as, for example, a RS232 interface in communication with the protocol/sequence controller **314** and the Transaction Terminal **2**. The USB connector **316** may be in communication with a personalization system (not shown) for initializing UIA **16** to certain application parameters. That is, the UIA **16** may be in communication with personalization system for populating internal platform **310** with a listing of security keys belonging to authorized NATs **62**, and for populating internal platform **320** with the security keys to decrypt UUCs **200** scanned from NATs **62**, enabling conversion of the UUC **200** into ISO/IEC 7813 format. In this way, UIA **16** may also be populated with a unique identifier (e.g., UIA-VC **204**) which may be used by NAT **62** to determine if UIA **16** may be authorized to receive a NAT **62** encrypted UUC **200**.

(264) The NAT **62** and the UIA **16** may then engage in mutual authentication. Where the mutual authentication may be unsuccessful, an error message may be provided to the User via Display **6** of the Transaction Terminal **2**, and the transaction may be aborted. Where the mutual authentication may be successful, the UIA **16** may optionally prompt the User via the Display **6** of the Transaction Terminal **2**, to provide the UIA **16** with a bid PVC **202** via the data-entry keypad **70** or touch screen.

(265) In an illustrative embodiment, the UIA **16** also has secure memory that can store and retrieve the unique secret encryption keys used to enable secure communications with the RMN **14** via the Transaction Terminal **2**. In this embodiment, this is battery backed-up RAM that is set up to be erased whenever the tamper-detect circuitry reports that tampering has been detected.

(266) To use encryption keys, a key management system must be employed to assure that both sender and receiver are using the same key. When using DES, an illustrative key management system is DUKPT, which is well known in the industry. DUKPT is designed to provide a different DES key for each transaction, without leaving behind the trace of the initial secret key. The implications of this are that even successful capture and dissection of a UIA **16** will not reveal messages that have previously been sent, a very important goal when the effective lifetime of the information transmitted is years. DUKPT is fully specified in ANSI X9.24. The DUKPT key table is stored in the secure memory.

(267) Each UIA **16**, in an illustrative embodiment, has a hardware verification code (UIA-VC) **204**, unique to each UIA **16**, and this UIA-VC **204** that is registered with the RMN **14** at the time of manufacture or distribution to an authorized Account Issuer. This makes the UIA **16** uniquely identifiable to the RMN **14** in all financial transactions from that device. This UIA-VC **204** is, in an illustrative embodiment, stored in write-once memory.

(268) UIA **16** physical security is assured by standard mechanisms known in the art. In an illustrative embodiment, these comprise tamper-detect circuitry, an enclosure that cannot be easily opened without visibly injuring the enclosure, erasable memory for critical secrets such as encryption keys, write-once memory for hardware verification, tight integration of all components, and “potting” of exposed circuitry.

(269) Information such as the amount of a transaction, the verification of a User, or other transaction-related information is displayed via the conjoined Transaction Terminal **2** using Display **6** with an integrated LCD screen. It is preferable that the Display **6** be connected securely to the other components in the UIA **16** to maintain security.

(270) Approval or cancellation of a financial transaction is done using the UIA keypad **70** or touch screen.

(271) Optionally, the UIA **16** also validates public key digital certificates. In one embodiment, public keys of a particular certifying authority are initially stored in the UIA **16** at the time of construction. This provides the mechanism to verify an Account Issuer's digital certificates that are signed by the certifying authority, or an Account Issuer's digital certificates that are signed by the certifying authority.

(272) Although an illustrative embodiment is described above, there are many different variations on specific **UIA 16** implementations. Fundamentally any device that is secure, can verify a User, an Account Issuer or an Account Issuer with a high degree of certainty, and can connect to the Master **RMN 14** via some form of communication line can serve as a **UIA 16**.

(273) In some embodiments, specifically the home use and public use instances, the **UIA-VC 204** is not used to verify either the **UIA 16** or the Account Issuer.

(274) In one embodiment, the **UIA 16-Terminal 2** has an updateable algorithm that selects certain alphanumeric data elements from the data stored on the **NAT 62**, including the **UUC 200**. The algorithm may also be synchronized with the **RMN 14** (including the **VP 12**), whereby a User has an additional unique identifier (**Adjusted UUC 399**) which is an algorithmic function their **UUC 200**, which may be single-use, multiple-use, or time-dependant based upon the algorithm's criteria, which is synchronized with the **RMN 14** (including the **VP 12**).

(275) For example, a **NAT 62** comprising data: **UUC #:** 1(2)34(5)6(7)8(9) J(o)e (V)(e)(r)(n)(e)(y) 6(9)3 P(a)dding Roa(d) S(an) Jos(e) Californ(i)(a) (954)32

(276) **Adjusted UUC 200** (optionally comprising additional data) transmitted to **RMN 14**=
(2579oVerny9adaneia954)

(277) In one embodiment, the **RMN 14** only stores the **Adjusted UUC 399** as used by the **VP 12** for verification of the User. If the **UIA 16-Terminal 2** is tampered with, or if there is the algorithm responsible for the **Adjusted UUC 399** is infiltrated, the **UIA 16-Terminal 2** transmits an alert to the **RMN 14**, which may then download an updated algorithm to the **UIA 16-Terminal 2**. As such, the **Adjusted UUC 399** data stored at the **RMN 14** may not be reverse-assembled into the original data resident on the **NAT 62**, or used by an unauthorized third-party.

(278) In one embodiment, such as where the **NAT 62** comprises a magnetic strip and interfaces with existing, in-market magnetic strip **UIA 16-Terminal 2** units, the **RMN 14** downloads algorithm updates via large numbers of distributed **UIA 16-Terminal 2** units, and enterprise servers (such as Systems' BankLink, POSway, and NT Connect **RMNs 14**). As such, the **NAT 62** data, including the **UUC 200**, is received and transmitted by the **UIA 16-Terminal 2** in a format compatible with existing payment **Network 18** protocols.

(279) The message protocol uses the **QTP** (Quick Transaction Protocol) over **UDP/IP** to provide the basic transport mechanism required as **QTP** combines minimum latency over Internet Protocol ("IP") **Networks 18** with light weight connections and a status mechanism designed for networks with large numbers of attached devices.

(280) The **RMNs 14** allow IP and non-IP **UIAs 16-Terminals 2** such as automatic teller machines (**ATMs**) and Point of Sale (**POS**) **UIAs 16-Terminals 2** to connect to processors and networks. In a typical non-TA installation, remote **ATMs** and **POS UIAs 16-Terminals 2** are connected directly to the central **RMNs 14** through leased line legacy networks. The **RMNs 14** then connect to the bank host as well as performing any protocol conversions, message concentration, and message translation. Alternatively, **UIAs 16-Terminals 2** communicate with a remote **RMN 14** over a leased line, dial, or wireless IP **Network 18**, thus eliminating the need for the legacy non-IP **Network 18**. The **UIA 16-Terminal 2** and **RMNs 14** communicate with each other via **QTP** over **UDP/IP**. This stack may be over any network interface (e.g. router based LAN, dial, wireless, etc.).

(281) The **QTP** link connecting a **UIA 16-Terminal 2** and a **RMN 14** consists of two primary types of **QTP** channels. The first **QTP** channel established is a supervisory **QTP** channel over which all management and control information between the **UIA 16-Terminal 2** and the **RMN 14** flows. Other **QTP** channels are established for transaction data flow (i.e. data **QTP** channels) and for the download of new configurations and **UIA 16-Terminal 2** software (i.e. the download **QTP** channel).

(282) The supervisory **QTP** channel should exist for there to be any data communication between a **UIA 16-Terminal 2** and the **RMN 14**. Messages over the supervisory channel may be of the following basic types: Information required to properly identify the **UIA 16-Terminal 2** to lower level **RMN 14** modules (e.g. serial number, **UIA 16-Terminal 2** device number, etc.) and to allow

access to low level modules where required (e.g. ping). Module specific statistical, status, and alarm information. Management control information (e.g. for re-initialisation of the **UIA 16-Terminal 2**, etc).

(283) Data QTP channels are used for the transfer of transaction data. Each data QTP channel represents a single virtual circuit as seen by the **UIA 16-Terminal 2** (e.g. a poll code or station address in the case of multi-drop operation, a line in the case of a single channel point-to-point protocol, a virtual circuit in the case of X.25, etc.).

(284) The download QTP channel is used for transferring new software downloads or configurations to the **UIA 16-Terminal 2**. It exists only when a download is in progress. A maximum of one download channel and one download can be active at any time. Downloads are done using the TFTP protocol as per RFC 1350 over the download QTP channel.

(285) Note that downloads may be treated like any other transaction data communications between the **UIA 16-Terminal 2** and the **RMN 14**. New software may be downloaded to a **UIA 16-Terminal 2** from a **RMN 14**, whereby downloaded file consists of a compressed software image for writing directly into the **UIA 16-Terminal 2's** flash memory. Each download consists of a complete software image, since it is not preferable to do partial updates of the software running on the **UIA 16-Terminal 2**.

(286) A supervisory QTP channel, optionally comprised within the **RMN 14**, is used for control of the **UIA 16-Terminal 2** and for transfer of informational, statistical, status, and alert information to the **RMN 14**. The **RMN 14** may requests the **UIA-VC 204**, enabling the **RMN 14** to verify the authenticity of the **UIA 16-Terminal 2**. As such, software updates and downloads from the **RMN 14** will not be sent to unauthorized, or trojan-horse, **UIAs 16-Terminals 2**.

(287) Registration

(288) Parties that wish to either originate or receive financial transactions should, in an illustrative embodiment, first register with the Rule-Module Nexus **14**, and its associated Verification Platform **12** and User Account Registry **15**. The verification and financial information registered with the **RMN 14** or **RMN-authorized Third-Party Platform 28** for a given party depends on the mode used to originate or receive settlement. A User, usually an individual person, should, in an illustrative embodiment, register at least one **UUC 200** (and optionally a **PVC 202**), as well as a Rule-Module **50** with Pattern Data **54** such as a plurality of proprietary Financial Accounts **65**, associated with at least one Execution Command **52** that can govern the accessing, deposit, display, deducting, and disbursing of Financial Data using at least one Financial Account **65**. The Financial Accounts **65** of a User are arranged within a User Account Registry (UAR) **15**, said **UAR 15** identified by a User Account Registry Code (UAR-Code) **59**. In an illustrative embodiment, the **UAR-Code 59** does not: identify a specific Financial Account **65** or specific Financial Account Number **65** of the User, nor; depend on a specific Financial Account **65** of the User being tagged as the "primary" account.

(289) For the User, registering verification data and Financial Accounts **65** can occur at home using a personal computer and/or at a merchant's point-of-sale location via a **UTA 62-Transaction Terminal 2** using a **Network 18** connection to the **RMN 14**, wherein Financial Account **65** data from magnetic stripe cards, bar-coded cards, paper checks, coupons, smart cards, and the like, are data-entered or electronically scanned. Registration may occur at a merchant's point-of-sale, over the Internet, or through a registration Transaction Terminal **2**. Data-entry of registration verification data and Financial Account **65**, along with associated data, can occur via: a keypad **70**; voice commands spoken into an audio receiver or microphone; swiping the magnetic stripe card; scanning a bar-code; reading a paper check with a magnetic ink character reader, and the like. The User's registration processes links any such data to the User's **UUC 200** (and optionally **PVC 202**) verification data, including the User's Rule-Modules **50**, within the **RMN 14**. The User is assigned and issued a **NAT 62**, stored with the **UUC 200** for that User.

(290) Further data which registered to the User may include: a driver's license number, a passport number, a debit account, a credit account, a checking account, a money-market account, a stored-

value account comprising pre-paid financial, and the like. Optionally, a stored value account with a participating Account Issuer may be pre-credited with funds, or financial, from the Account Issuer and for the use of which the User has pre-paid a premium to the Account Issuer.

(291) For verification data, the User registers by being assigned a UUC **200** which is stored into a NAT **62** issued to the User. Optionally, the User may also submit, or be assigned, a registration PVC **202**. The UIA **16** optionally confirms that the PVC **202** code is accurate, and in a preferable embodiment, scans the User's NAT **62** to determine that it and its stored UUC **200** are non-fraudulent. The UIA **16** then translates and compresses this UUC **200** (or optionally a UUC **200**-PVC **202**) encrypted packet into a format suitable for rapid electronic transmittal to the RMN **14**. In one embodiment, the User selects and enters a PVC **202** into the UIA keypad **70** or touch screen.

(292) In an illustrative embodiment, the User associates a plurality (or at least two) proprietary Financial Accounts **65** with the registration UUC **200** (and optionally a UUC **200**-PVC **202**) in the RMN **14** or RMN-authorized Third-Party Platform **28**, such as an associated Verification Platform **12**. In an illustrative embodiment, as described above, this is accomplished by automatically scanning a bar-code or a magnetic stripe through the data reader attached to the UTA **16**. In one embodiment, this bar-code or magnetic stripe comprises not only the User's Financial Account **65**, but also the verification of the Account Issuer or financial entity with which this account is associated.

(293) FIG. **11C** illustrates an embodiment wherein a User auto-populates an online UAR **15** with Financial Accounts **65** via a sample Internet registration sequence. Alternatively, a telephone registration process may be used, wherein instructions and data-entry may be audible or via the telephone keypad. Via a website, the User enters a UUC **200** (or UUC **200**-PVC **202**), optionally along with other identifying data such as a social security number (Step **510**). In addition, the User may engage in a sequence of challenge-response queries with the RMN **14** in order to more certainly ascertain the User's identity. With the User's consent, the RMN **14** pulls and displays a credit report comprising User's operative Financial Accounts **65** (Step **511**), whereby the User may indicate which of said Financial Accounts **65** are to be registered with the UAR **15** (Step **512**). The UAR **15** thereby automatically registers the selected Financial Accounts **65** via the RMN **14** (Step **512**). In this way, data-entry of "live" account data for each Financial Account **65** is avoided. In addition, the auto-population process may invoke Rule-Modules **50** registered via Account Issuers, manufacturers, and retailers which offer to the User certain rewards incentives, new Financial Accounts **65**, and other micro-merchandizing upsell offers to the User for their purchase or acceptance via the UAR **15** registration process. These Rule-Modules **50** may be invoked based on factors comprising any one of the following: a polling of User's credit report and their credit rating eligibility; an auction or arbitrage in which Account Issuers, manufacturers, and retailers bid via the RMN **14** to present their product/service advertisements or promotional offers via a Display **7** being used by the User during the UAR **15** registration process, or via subsequent email messages to the User with embedded invitations or offers. Similarly, a User may consent to "opt-in" to having another party auto-populate the RMN **14** with Rule-Module(s) **50** on their behalf, wherein the party comprises any one of the following: a merchant or payee, an account issuer, the RMN **14**, and the like. Such an "opt-in" may be accompanied by a rewards incentive to the User.

(294) FIG. **52** illustrates an embodiment wherein a User auto-populates an online UAR **15** with Financial Accounts **65** via a sample Internet registration sequence using a telephone registration process may be used, wherein instructions and data-entry may be audible or via the telephone keypad. Via a website, the User enters a UUC **200** (or UUC **200**-PVC **202**), optionally along with other identifying data such as a social security number (not shown). In addition, the User may engage in a sequence of challenge-response queries with the RMN **14** in order to more certainly ascertain the User's identity (not shown). With the User's consent, the RMN **14** pulls and displays a credit report comprising User's operative Financial Accounts **65** (Step **511**), whereby the User may indicate which of said Financial Accounts **65** are to be registered with the UAR **15** (Step **512**).

(295) In an illustrative embodiment, an attendant verifies that the User is the legally authorized signer on the Financial Account **65** by comparing personal, official photo identification such as a driver's license, passport, identification card, and like, to the name listed on the credit card, debit card, paper check and the like being used for registering the accounts.

(296) In an illustrative embodiment, for registering a rewards Financial Account **65**, if the User's NAT **62** is a cell phone with bar-code reading capabilities, the User's NAT **62** may scan the bar-code on a User's existing rewards card, to capture the account number of the User's rewards account with a certain merchant. Alternatively, the User may input the rewards account number manually via a UTA Keypad **70**, which may be conjoined with either a cell phone NAT **62**, or the User's personal computer.

(297) In another embodiment of the invention, registration comprises a message being provided to a potential User of the RMN **14**, said message may include, but not be limited to, any of the following: an electronic transmission (e.g., an email, an SMS text, an online advertisement, etc.); a printed mailer; an automated outbound voice message via telephone. Such a message may comprise: a response address for contacting the RMN **14** (physical mailing address; email address; phone number; etc.); a one-time, temporary Personal Verification Code **202**, preferably unique to the message, wherein no other message has the same PVC **202**; optionally an offer number or code. The potential User may then respond to the RMN **14** via any UTA **16** (personal computer, cell phone or land line), for example, wherein the potential User may be identified via any of the following: ANI (automatic number identification); the one-time, temporary PVC **202**, and/or; the offer code or number. The potential User may then be identified by the RMN **14** based on an reverse-lookup of the potential User's phone number provided via ANI, wherein the RMN **14** can perform an automatic match of the potential User's name, address, and phone number, to determine the identity of the potential User, and automatically research public databases to pull data about the potential User for identity confirmation. The RMN **14** may then present options to the User via an automated IVR (interactive voice response) system, prompting the potential User for responses via voice and/or dual-tone multi-frequency signaling keypad inputs. The RMN **14** may offer the potential User the option to accept an offer to register with the RMN **14**. For example, upon acceptance of the offer, the IVR system may prompt the potential User for responses to security questions for confirming the identity of the potential User, based on the database research ("What is your home address?"; "What was your previous home address?"; "When was your home purchased?"; "How much did you purchase your home for?"; "What is your email address?"; etc.), or the RMN **14** could ask the User to input their social security number, and then prompt the potential User for responses to security questions (e.g., "How much is your monthly mortgage?"; "How much is your monthly car payment?"; etc.). In these instances, the RMN **14** offers a plurality of answers to each questions, each with a corresponding number for selection (e.g., "When your home purchased? For the year 1991, press or say "1"; For the year 1993, press or say "2", etc.), wherein the potential User must input the correctly numbered answer to confirm their identity to the RMN **14**. Once the potential User's has correctly answered the questions, and their identity is thereby confirmed by the RMN **14**, the RMN **14** may prompt the potential User to consent to an automatic poll of their credit report, wherein a UAR **15** may be established and registered by the RMN **14** for the potential User and pre-populated with any or all of the potential User's active Financial Accounts **6**, based on the open Financial Account(s) **65** on the potential User's credit report(s). The User, now registered, may be prompted to select, for example: consenting to using default Rule-Module(s) **50** provided by the RMN **14** and/or a Third-Party Platform **28** (such as a merchant, an Account Issuer, etc.); consenting to having a NAT **62** mailed to them; consenting to keeping their previously assigned temporary PVC **202**, or selecting a new PVC **202** of their own choosing; designating a Primary PVC **200** or selecting a Secondary PVC **202**;

(298) The UTA **16** transmits the registration data to the RMN **14**. In an illustrative embodiment, the Master RMN **14** then inserts the UUC **200** (and optionally a UUC **200**-PVC **202**) packet into the

appropriate Verification Platform **12** and generates an User Verification Approval Code **206** that is unique to the User and is subsequently output by the Verification Platform **12** when issuing a positive matching determination from electronically comparing a User's bid UUC **200** (or optionally a UUC **200-PVC 202**) with registered UUCs **200** (or optionally with registered UUCs **200-PVCs 202**). From this point on, any time the User is verified by the Verification Platform **12** via submitted a UUC **200** (or optionally a UUC **200-PVC 202**) packet, the User Verification Approval Code **206** is forwarded to the Master Rule-Module Nexus **14** where it invokes at least one Rule-Module **50** for that User. In the Master Rule-Module Nexus **14** platform, a Rule-Module **50** is created that is identified by the User Verification Approval Code **206**. In one embodiment, the Verification Approval Code **206** identifies a User Account Registry **15**. In another embodiment, the Verification Approval Code **206** is identical the UAR-Code **65**, but does not: identify a specific Financial Account **65** or a specific Financial Account Number **65**.

(299) Before a new UUC **200** (or optionally UUC **200-PVC 202**) record is enabled to originate or invoke a Rule-Module **50**, the individual's submitted UUC **200** are checked against previously registered UUC **200** in the electronic Verification Platform **12** using the same UUC **200** comparison techniques as those used in the individual verification procedure. If a match is found for the newly submitted UUC **200** record, the UUC **200** record's status is set to "prior registration". If the prior registration check was executed as part of a registration request, the Gateway Platform **26** forwards a "registering individual with prior registration" warning to the Prior Fraud Platform **27**, indicating that the person has attempted to register with the RMN **14** or RMN-authorized Third-Party Platform **28** more than once.

(300) FIG. **31** illustrates an illustrative embodiment, the UUC **200** is encoded into a NAT **62**, which is mailed to either the User's home address or mailed to a pre-designated address for the User, such as a Federal Express® office. In another illustrative embodiment, the UUC **200** is encoded into a NAT **62** comprising a radio frequency identification chip, wherein said NAT **62** may be conjoined with a token of the User, said conjoining comprising any one of the following: being conjoined with a token of the User via an adhesive or snap-on attachment; being permanently conjoined with a token of the User via embedding into the token. Step **1**: RFID chips are now smaller than a grain of sand, costing less than 25 cents, some as little as 5 cents or 2½ cents. These chips provide the ability to transmit stored data for financial transactions, when powered by an interrogator. The lifespan of a passive RFID chip can exceed 20 years. Program chip with Nexus UUC (unique user code) and RC (routing code), using ISO 14443 and 13.56 MHz standards for standard RFID point-of sale readers. Step **2**: Embed the Nexus chip in a small, snap-on plastic end-cap. Step **3**: Send to customers to snap-on Proprietary Contactless Token. Proprietary Token now enabled for dual-mode: a) Proprietary payment function at proprietary merchant POS, and; b) RMN payment functions at standard POS retailers.

(301) FIG. **15A** illustrates another illustrative embodiment wherein the User receives a NAT **62** in real-time upon registration at a retail site. In one such example, the User registers verification data and Financial Accounts **65** with the RMN **14** (or Subset RMN **17**) via a UIA **16** at the point-of-sale (POS) Transaction Terminal **2** in a retail check-out aisle. As described above, the User's registration verification data and Financial Accounts may be input or scanned via a UIA **16** keypad **70** or the retailer's magnetic ink character reader at the POS Transaction Terminal **2**. Upon said registration data being transmitted to, and accepted for registration by, the RMN **14** (or Subset RMN **17**), thereby the RMN **14** (or Subset RMN **17**) transmits to either the POS Transaction Terminal **2** or the UIA **16**, a NAT-Dispensing Code **129** for the User. Said NAT **62**-dispensing code may optionally be only for one-time usage. The User may then use said NAT-Dispensing Code **129** at a NAT-Dispensing Kiosk **128** to which the RMN **14** (or Subset RMN **17**) has transmitted the User's UUC **200**. In an illustrative embodiment, the NAT-Dispensing Kiosk **128** may be co-located at, or proximally located to, the UIA **16** by which the User has registered Financial Accounts **65** and verification data. When the User inputs the NAT-Dispensing Code **129** into the NAT-Dispensing

Kiosk **128**, the User may be given the option to select a NAT **62** into which the User's UUC **200** may be data-stored by the NAT-Dispensing Kiosk **128**, said NAT **62** comprising any one of the following: an adhesively attachable NAT **62** for being conjoined with a token of the User; a temporary NAT **62**; a NAT **62** embedded into a token for the User, and; an embeddable NAT **62** which the User may have subsequently embedded into a token of the User.

(302) FIG. **15B** illustrates an illustrative embodiment of a registration process, wherein the User registers Pattern Data **54**, including a Primary PVC **200** (and optionally a Secondary PVC **202**) and a Financial Account **65** with the RMN **14** via a UIA **16** at the point-of-sale (POS) Transaction Terminal **2** in a retail check-out aisle. As described above, the User's registration verification data and Financial Accounts **65** may be input or scanned via a UIA **16** keypad **70** or the retailer's magnetic ink character reader at the POS Transaction Terminal **2**. Upon said registration data being transmitted to, and accepted for registration by, the RMN **14**, the RMN **14** transmits to either the POS Transaction Terminal **2** or the UIA **16**, an approval code verifying that the User has been successfully registered with the RMN **14**. The User may then use the Primary PVC **200** at a Self-Serve Kiosk (or Kiosk) **128** to which the RMN **14** has transmitted the User's Primary PVC **200**. In an illustrative embodiment, the Self-Serve Kiosk (or Kiosk) **128** may be co-located at, or proximally located to, the UIA **16** by which the User has registered Pattern Data **54**. When the User provides a bid Primary PVC **200** (or optionally, a bid Primary PVC **200**-Secondary PVC **202**) via a keypad **70** conjoined with the Self-Serve Kiosk (or Kiosk) **128**, the User may then continue registering Pattern Data **54**.

(303) In another illustrative embodiment of the invention, the User may register pattern data with the Rule-Module Nexus **14** (or Subset RMN **17**) via the internet or telephone, as described above, whereby the User may select a specific NAT-Dispensing Kiosk **128** in a particular location for receipt of the User's Unique User Code **200**, whereby the User may receive a NAT-Dispensing Code **129** from the Rule-Module Nexus **14** (or Subset RMN **17**) via email or telephone. The User may then go to said NAT-Dispensing Kiosk **128** for receipt of a NAT **62** into which the User's Unique User Code **200** has been data-stored.

(304) In another illustrative embodiment of the invention, the NAT-Dispensing Kiosk **128** may be used for dispensing a replacement NAT **62** for a User in the event that a User's NAT **62** or a User's UUC **200** has been compromised.

(305) In one embodiment, the Master RMN **14** validates the Account Transaction Data (or Transaction Data) **172** submitted during registration. This involves making certain that the Financial Account **65** being registered is a valid account and that the User is an authorized signer.

(306) In an illustrative embodiment wherein an Account Issuer or Payee Platform **28** receives electronic transfers of Financial Data and/or Account Transaction Data **172**, the Account Issuer or Payee, usually a corporate entity, must also register rule-modules with the RMN **14**, comprising a pattern data associated with an execution command.

(307) Pattern Data, in an illustrative embodiment, further comprises any one of the following: verification data unique to that Payee, such as a digital certificate: their Payee UTA-VC **204**; processing preferences for a Financial Account **65** of the User; a Master PAR (Payee Account Registry) **46**; a Subset PAR **47**; a PAR-Code **48**; rule-modules invoking preferences, tagging, Rankings **96**, and/or default Financial Account(s) of a payee; a payee legal name; a payee financial depository account; a payee Financial Account **65**; a payee financial depository account routing code, and; payee geographic positioning data; a private code; a unique payee code (UPC); the payee's legal name; a user interface apparatus hardware verification code; an employer identification number; Financial Account access authorization fields; an payee verification code; a transaction terminal identification code; an emergency code; a Financial Account **65**; an email address; a telephone number; a mailing address; authority of at least one employee of the payee; a digital certificate; a network credential; an Internet protocol address; a digital signature; an encryption key; electronic audible signature, and; an electronic visible signature.

(308) In an illustrative embodiment, execution command(s) further comprise invoking at least one of the following: accessing the rule-module nexus; accessing a user account registry; accessing a payee depository Financial Account **65**; presenting a Financial Account **65** of the payee; completing a financial transaction; authorizing settlement of the online transaction; presenting pattern data; presenting an execution command; presenting a rule-module; notifying an emergency authority upon rule-module nexus receiving an emergency code of the payee; accessing a third-party database; over-the-air provisionings; updating payee geographic positioning data, and; accessing a payee database.

(309) In an illustrative embodiment, accessing a UAR **15** comprises any one of the following: displaying a Financial Account **65**; querying a Financial Account **65**; editing a Rule-Module **50** associated with a Financial Account **65**; viewing Financial Data of a Financial Account **65**; a user manually selecting a Financial Account **65** for processing a financial transaction; a Rule-Module **50** from a Rule-Module Nexus **14** automatically selecting a Financial Account **65** for processing a financial transaction; Approval of a financial transaction; Settlement of a financial transaction; electronically querying data associated with a financial account; displaying an electronic audible signature of a Financial Account **65**; displaying an electronic visible signature of a Financial Account **65**; displaying an activity statement or account balance for a Financial Account **65**; aggregating invoices and billing statements for a Financial Account **65**; electronically querying data contained within a financial account; electronically invoking data associated with a financial account; electronically invoking data contained within a financial account; processing a financial transaction using a Financial Account **65**; transferring Financial Data from or to a Registry Financial Account **65** from or to another Registry Financial Account **65**; transferring Financial Data from or to a Registry Financial Account **65** from or to a Financial Account **65**, comprising a buy-sell; a gift; a scrip; intra-Financial Account **65** transfers between or among Financial Accounts **65** owned by same user, wherein there is no buyer-seller financial transaction or payment processing as may required with a standard purchase.

(310) Further, processing preferences may include: invoking a proprietary network; invoking a discount rate; invoking an interchange fee; invoking a settlement protocol; invoking a surcharge; invoking a processing partner, and; invoking a time period for settlement.

(311) Any Account Issuer may register also additional data that is unique to itself, comprising any one of the following: an alpha-numeric verification code, an Audible or Visible Signature **81**, a digital certificate, or a UIA-VC **204** to verify itself to the RMN **14**. Digital certificates are available from certifying authorities, and they provide the assurance that the entity with the certificate is the authentic owner of that verifier. These certificates comprise readable text and other information that describes the entity. This can include an Account Issuer the address, as well as the company name.

(312) This entity verification data is then linked to at least one User Financial Account **65** or an Account Issuer Account.

(313) In one embodiment, UTA-VC's **204** are unique numbers assigned to UIA **16** devices at the time of manufacture. A participating Account Issuer installing UIA **16** devices at the point of sale can register a User Interface Apparatus **16** with the RMN **14**. In an illustrative embodiment, this causes any transaction, either registration or purchase, flowing through those registered User Interface Apparatus **16** to automatically verify to the RMN **14** the participating Account Issuer which owns the UTA-VC **204**.

(314) Network Communications

(315) As used herein, "online" requires communications or communications means (or interconnecting communications means) external to a Nexus Access Token, comprising any one of the following: contact-based; contactless-based; wired, and; wireless. Contact-based communications comprise any one of the following: magnetic stripe reading; smart card reading; magnetic ink character reading; radio frequency transmission, and; the like. Contactless-based communications comprise any one of the following: radio frequency transmission; near-field

communication (i.e., Bluetooth); bar-code reading; infrared transmission, and; the like. Wired communications comprise any one of the following: X.25 network; land-line phone network; cable network; and the like. Wireless communications comprise any one of the following: local area network (“LAN”); wide area network (“WAN”); cellular network; WiFi network; global positioning network; satellite communications, and; the like.

(316) Communications via a Network **18** between the UTA **16** and a conjoined Transaction Terminal **2**, and the Verification Platform **12** associated with the RMN **14** occur via many different communication methods. Most depend on the particular communication networks already deployed by the organization or merchant that deploys the transmission authorization system.

Communication security over the Network **18** is provided by encryption using unique secret keys known only to that specific UIA **16** and the RMN **14**, and the DES encryption algorithm, in an illustrative embodiment, triple-encrypted. Triple encryption means successive encrypt/decrypt/encrypt operations using two distinct 56-bit DES keys. This provides significantly higher security than a single encryption operation with one 56-bit DES key. Alternately, a public/private key system may also be used to encrypt information that passes between UIA **16** and RMN **14**. Both DES and public key encryption is well known in the industry.

(317) In an embodiment the User Interface Apparatus **16** are connected via Ethernet **18** to a Local or Subset router **18**, which is itself connected to a network operations center (NOC) via frame relay lines. At least one Verification Platform **12** is located at the NOC. Messages are sent from UIA **16** to the Verification Platform **12** using TCP/IP over this network. In another embodiment, the User Interface Apparatus **16** are connected via a cellular digital packet data (CDPD) modem to a CDPD provider, who provides TCP/IP connectivity from the UIA **16** to an intranet **18** to which at least one Verification Platform **12** is attached.

(318) In yet another embodiment, a UIA **16** is connected via the Internet, as is at least one Verification Platform **12**. TCP/IP is used to transmit messages from UIA **16** to Verification Platform **12**. There are many different ways to connect UIA **16** to Verification Platform **12**, both tethered and wireless, that are well understood in the industry, including but not limited to: the Internet **18**; an intranet **18**; an extranet **18**; a Local or Subset area network (“LAN”) **18**; and a wide area network (“WAN”) **18**.

(319) Verification Platform

(320) The electronic Verification Platform (VP) **12** serves to verify the User in an electronic financial transaction. The Verification Platform **12** compares a User's bid UUC **200** scanned from the User's NAT **62** (or optionally from the User's bid UUC **200**-PVC **202** provided to the UTA **16**) with previously stored registration UUCs **200** (or with previously stored registration UUCs **200**-PVCs **202**) packets from registered Users, in order to verify the User. If a bid UUC **200** (or optionally a bid UUC **200**-PVC **202**) packet is successfully matched against a registered UUC **200** (or optionally a registered UUC **200**-PVC **202**) packet, and the Verification Platform makes User makes a positive matching determination, the User Verification Approval Code **206** which had been assigned to the User during initial registration is output and forwarded to the Master Rule-Module Nexus **14**. The User Verification Approval Code transmitted by the Verification Platform **12** is used by the Master Rule-Module Nexus **14** to locate the Rule-Modules **50** that are customized to that User, including the User Account Registry **15**

(321) As seen in FIG. **1**, a Firewall machine **40** connects the Verification Platform **12** to the Internet **18** or intranet **18**. Messages are sent to a Gateway Platform **26**, which is responsible for overseeing the steps required to process the financial transaction, including forwarding the financial transaction to the Verification Platform **12** and the Master Rule-Module Nexus **14**.

(322) In an illustrative embodiment, electronic messages transmitted between the UIA **16** and the Master RMN **14** are encrypted. For this, the financial transaction processor uses the Decryption Platform (DP) **22**, which utilizes the UIA-VC **204** of the UIA **16** to verify the encryption codes that is required to decrypt messages from the UIA **16**. Once decrypted, the verification of the User is

determined using Verification Platform **12**, which provides storage, retrieval and comparison of UUC **200** (or optionally a UUC **200-PVC 202**) packet.

(323) In another embodiment, the Verification Platform **12** provides periodic User re-verification queries. In this embodiment, in order for a User to extend an online session, the User is requested by the Verification Platform **12** to re-verify themselves using a User bid UUC **200** (or optionally a User bid UUC **200-PVC 202**).

(324) In another embodiment, an Account Issuer is also verified by the Verification Platform **12** using verification data comprising any one of the following: a digital certificate, an Internet protocol ("IP") address, a UUC **200**, a UIA-VC **204**, or any other code, text or number that uniquely identifies the entity. In this way, the Verification Platform **12** is enabled to provide the User with confirmation that the correct Account Issuer participated in the electronic financial transaction. Examples include confirming that the correct Account Issuer web site or remote Third-Party Platform **28** was accessed by the User, that the correct entity designee received the User's email or instant message, and the like.

(325) In another embodiment, the Verification Platform **12** platform is integrated with the Master Rule-Module Nexus **14** platform.

(326) In an illustrative embodiment, more than one Verification Platform **12** provides fault tolerance from either natural or man-made disasters. In this embodiment, each Verification Platform **12** uses a backup power generator, redundant hardware, mirrored platforms, and other standard fault tolerant equipment known in the industry.

(327) In one embodiment, the Verification Platform **12** compares bid verification comprising a Nexus Access Token Identifier (NAT-ID) and a UUC **200** of the user, with a registered NAT-ID and its associated UUC **200**, to determine if the UUC **200** has been stored on a NAT **62**, or is being used with a NAT, with which the UUC **200** is not associated. This may be used to prevent storing an authorized UUC **200** with an unauthorized NAT **62**. Alternatively, this may be used to prevent storing an unauthorized UUC **200** on an authorized NAT **62**.

(328) Verification of the User occurs using different methods, depending on the verification information that is provided by the UTA **16**. The Verification Platform **12** has subsystems for each type of information that is received by the Verification Platform **12**, and each subsystem is highly optimized to provide rapid verification as outlined below.

(329) In an illustrative embodiment, Verification Platform **12** comprises subsystems that can verify parties from the following information: UUC **200** data (or optionally, UUC **200-PVC 202** data); digital verification (digital certificates); UTA-VC **204**.

(330) In an illustrative embodiment, the Verification Platform (VP) **12** further comprises at least two Subset VP's **12**, all being are capable of verifying parties from their UUC **200** (or optionally a UUC **200-PVC 202**).

(331) In one embodiment, the records of parties identifiable from UUCs **200** (or optionally a UUC **200-PVC 202** combinations) are distributed equally across all Subset VP's **12**. In one embodiment, one Subset Verification Platform **13** is responsible for verifying Users with UUCs **200** (and optionally Users with PVCs **202**) numbered 1-10, another Subset Verification Platform **13** is responsible for verifying Users with UUCs **200** (or optionally PVCs **202**) numbered 11-20, and a third Subset Verification Platform **13** is responsible for verifying Users with UUCs **200** (or optionally PVCs **202**) numbered 21-30. For example, all messages from the UTA **16** comprising a UUC **200** (or optionally a PVC **202**) numbered 30 would be routed to Verification Platform **12** for verification of the User.

(332) In an illustrative embodiment, wherein a Verification Platform **12** receives a bid UUC **200** (or optionally a UUC **200-PVC 202**) packet from the Transaction Terminal **2** conjoined with a UTA **16**, for verification, a processor searches through its platform, retrieving all registered UUCs **200** that correspond to that particular bid UUC **200** (or correspond to that particular bid UUC **200-PVC 202**). Once all corresponding registered UUCs **200** are retrieved, the Verification Platform **12**

compares the bid UUC **200** (or optionally the bid UUC **200-PVC 202**) obtained from the electronic financial transaction to all retrieved registered UUCs **200** (or optionally registered UUCs **200-PVCs 202**). If a match occurs, the Verification Platform **12** makes a positive matching determination and outputs the User Verification Approval Code **206** to access the User Account Registry **15** and associated Rule-Modules **50** of the User. If no match is found, the Verification Platform transmits a “not identified” message back to Gateway Platform **26** and to the logging facility **42**.

(333) In one embodiment, there is a UUC **200** theft resolution step, wherein the User's UUC **200** is changed if the User's UUC **200** is determined to have been compromised or fraudulently duplicated.

(334) FIG. 5A is illustrative of an embodiment of the VP **12**. In this embodiment of the VP **12**, the Verification Platform **12** comprises an Identity Risk Factor Generation (IRFG) Platform **75**. The IRFG Platform **75** may optionally be configured to generate an identity risk factor associated with the User. The identity risk factor may optionally be associated with a level of risk of theft of the identity of the User by identity thieves. The graphical User interface may optionally be configured to provide access to and management of identity verification Rule-Module(s) **50** associated with the User.

(335) FIG. 5A also illustrates that a verification data may optionally comprises a driver's license of the User. According to an alternative illustrative embodiment of this embodiment, a verification data may optionally comprise a birth certificate of the User. The identity verification score assigned to each of a verification data may optionally be based upon a reliability of a verification data. The predetermined function may optionally comprise, for example, a summing function, a weighted summing function or the like. The VP **12** may optionally comprise a data storage device. Rule-Module **50** data of the User may optionally be stored in the data storage device. The total identity verification score may optionally be associated with Rule-Module **50** data of the User. The VP **12** may optionally comprise a Rule-Module **50** comprising a predetermined VAC **206** for the User. Alternatively, the VP **12** may comprise a Verification Access Code Generation (VACG) Platform **190**. The VACG Platform **190** may optionally be configured to generate a unique VAC **206** associated with the User for a single authorization to access a UAR **15**. Alternatively, the VACG **190** may optionally be configured to generate a unique VAC **206** associated with the User to authorize a Third-Party Platform **28** to access a RMN **12** or UAR **15** associated with the User. The VP **12** may optionally comprise a verification data transmission (VDT) Platform **191**. The VDT Platform **191** may optionally be configured to transmit at least the total identity verification score of the User externally to a Third-Party Platform **28** upon outputting or invoking a VAC **206**. The VDT Platform **191** may optionally be configured to transmit at least the total identity verification score of the User to the Third-Party Platform **28** upon further verification of a User VAC **206** of the User. The User VAC **206** may optionally comprise a social security number of the User.

(336) FIG. 5A also illustrates that the VP **12** may optionally comprise a Verification Log Platform **101**. The Verification Log Platform **101** may optionally be configured to record access attempts associated with the total identity verification score of the User by the Third-Party Platform **28**. The VP **12** may optionally comprise a Report Generation (RG) Platform **192**. The RG Platform **192** may optionally be configured to generate reports for displaying the record of access attempts associated with the total identity verification score of the User. Rule-Module **50** data of the User associated with the User may optionally be transmitted to the Third-Party Platform **28** upon outputting or invoking a VAC **206**. Rule-Module **50** data of the User associated with the User may optionally be transmitted to the Third-Party Platform **28** upon further verification of a VAC **206** of the User. The User VAC **206** may optionally comprise a social security number of the User.

(337) The VP **12** may optionally comprise any one of the following: being online, apart from the NAT **62**; being stored on a NAT **62**. The NAT **62** may optionally be configured to securely contain identity verification Rule-Module(s) **50** associated with the User. The NAT **62** may optionally

comprise, for example, a smart card. The identity verification Rule-Module(s) **50** associated with the User may optionally be encrypted on the NAT **62**. Uses of the identity verification Rule-Module(s) **50** securely contained on the NAT **62** may optionally be restricted by the User. Locations of where the identity verification Rule-Module(s) **50** is used may optionally be restricted by the User. Times of when the identity verification Rule-Module(s) **50** is used may optionally be restricted by the User. Types of transactions for which the identity verification Rule-Module(s) **50** is used may optionally be restricted by the User. Use of the identity verification Rule-Module(s) **50** for the transaction may optionally be prohibited when the identity verification Rule-Module(s) **50** is restricted by the User for the transaction. The identity verification Rule-Module(s) **50** for the transaction may optionally be used when the identity verification Rule-Module(s) **50** is not prohibited by the User for the transaction.

(338) FIG. 5A also illustrates that the VP **12** may optionally comprise approving access to a UAR **15** via a VAC **206** and a Rule-Module **50** gating or governing access to the UAR **15**. The Rule-Module **50** may optionally be configured to control parameters for accessing a UAR **15** order using the total identity verification score of the User. Additional Rule-Module **50** data of the User may optionally be accessed or invoked via the VAC **206**, to enable access to the UAR **15**.

(339) The VP **12** may communicate with a Display **6** to display communications to the User associated with the VP **12** or the RMN **14**.

(340) The VP **12** may optionally comprise the IRFG Platform **75**. The IRFG Platform **75** may optionally be configured to generate an identity risk factor associated with the User. The identity risk factor may optionally be associated with a level of risk of theft of the identity of the User by identity thieves. According to illustrative embodiments of this embodiment, the transaction may optionally comprise, for example, an application for credit, a purchase transaction or the like. The VP **12** may optionally comprise a graphical User interface. The graphical User interface may optionally be configured to provide access to and management of identity verification Rule-Module(s) **50** associated with the User.

(341) FIG. 5A also illustrates another embodiment of the VP **12**, wherein the Verification Platform **12** is configured to receive at least one verification data of the User. The VP **12** comprises an Identity Verification Score Assignment (IVSA) Platform **76** configured to assign an identity verification score to each of a verification data. The VP **12** comprises a total IVSA Platform **77** in communication with the IVSA Platform **76**. The VP **12** comprises an Identity Verification Score Generation (IVSG) Platform **77** configured to generate a total identity verification score of the User from the identity verification scores of each of a verification data and a predetermined function. The total identity verification score of the User is associated with a level of verification of the identity of the User. Alternatively, the total identity verification score of the User is compared to a minimum identity verification score associated with approving access to a UAR **15**. The VP **12** comprises an Identity Confidence Factor Generation (ICFG) Platform **78** in communication with the total IVSA Platform **77**. The ICVP **78** is configured to generate an identity confidence factor associated with the User in accordance with a validity of the identity of the User. The VP **12** comprises a Verification Platform **12** configured to compare the total identity verification score of the User to a minimum identity verification score associated with approving access to a UAR **15**. The VP **12** also comprises approving access to a UAR **15** via making a positive matching determination and outputting, or invoking, a VAC **206**, wherein said VAC **206** may comprise a UAR-Code **59**. The VP **12** is configured to make a positive matching determination when: the total identity verification score of the User is one of greater than and equal to the minimum identity verification score; the identity confidence factor of the User is greater than a predetermined identity threshold value. Additional verification data of the User may be received before making the matching determination when at least one of: the total identity verification score is less than the minimum identity verification score; the identity confidence factor of the User is less than the predetermined identity threshold value. The VP **12** is configured to generate a request for additional

verification data of the User before making a positive matching determination when the total identity verification score is less than the minimum identity verification score.

(342) Optionally, the VP **12** may be conjoined, co-located and/or integrated with the RMN **14** and/or the UAR **15**, in any combination thereof.

(343) Digital Identification Subsystem

(344) In an illustrative embodiment of the Verification Platform **12**, the Digital Identification subsystem comprises a plurality of processors, each of which is capable of verifying an entity, such as an Account Issuer, from their digital certificates. In this embodiment, digital certificates are used to perform digital verification of an entity. In an illustrative embodiment, these include corporate web site addresses and certifying authorities only. Where possible, computers provide digital certificates for verification of the Account Issuer, including a UIA-VC **204** for two-factor verification.

(345) Verifying that a particular digital certificate is valid requires a public key from the certifying authority that issued that particular digital certificate. This requires that the digital verification subsystem have a list of certifying authorities and the public keys used to validate the digital certificates they issue. This table must be secure, and the keys stored therein must be kept up to date. These processes and others relating to the actual process for validating digital certificates are well understood in the industry.

(346) UIA Hardware Verification (or Identification) Subsystem (UHV or UHI)

(347) In an illustrative embodiment of the Verification Platform **12**, UIA-VC's **204** are translated into entity verification by the UHI subsystem. This subsystem maintains a list of all User Interface Apparatus **16** manufactured. In an illustrative embodiment, when a particular User uses a UIA **16**, that User's geographic location is identified by their use of that particular UIA **16** during that electronic financial transaction session.

(348) In another embodiment, the UIA-VC **204** does not serve to verify either the User or an entity. This is the case in User Interface Apparatus **16** installed in public venues such as airport Transaction Terminals **2**, Automated Teller Machines in banks, or computers with User Interface Apparatus **16** for home use.

(349) User Verification Approval Code

(350) A User Verification Approval Code (VAC) **206** is an electronic message output by the Verification Platform **12** upon a positive matching determination of the User. In illustrative embodiments, the VAC **206** comprises any one of the following: a static code identifying the user account registry; a dynamic code comprising a unique code, such as a unique digital signature per transaction for storage and retrieval of data relating to processing of the financial transaction, and; a static code and a dynamic code. The VAC **206** informs the Master Rule-Module Nexus **14** that a User has been successfully identified, and instructs the Master Rule-Module Nexus **14** to invoke the Rule-Modules **50** for that particular User. In an illustrative embodiment, any time the User is verified by the Verification Platform **12** via submitted a UUC **200** (or optionally a UUC **200**-PVC **202**) packet, the User Verification Approval Code **206** is forwarded to the Master Rule-Module Nexus **14** where it identifies a UAR-Code **65**, invoking access to a User Account Registry **15** and invoking at least one Rule-Module **50** for that User. In the Master Rule-Module Nexus **14** platform, a Rule-Module **50** is created that is identified by the User Verification Approval Code **206**. In an illustrative embodiment, the Verification Approval Code **206** is identical to the UAR-Code **65**, but does not: identify a specific Financial Account **65**, identify a specific Financial Account Number **65**.

(351) Rule-Module Nexus

(352) Rule-Module Nexus (RMN) **14** serves to verify the Account Issuer and the User in a transaction, retrieve for verified parties a plurality of proprietary Financial Accounts **65**, and optionally Account Transaction Data (or Transaction Data or Ancillary Transaction Data) **172**, and perform the execution that will result in facilitating completing financial transactions, including

settlement of transactions. The Rule-Module Nexus **14** is comprised of an electronic Verification Platform **12**, a Master Rule-Module Nexus **14**, an internal Execution Platform **38**, a Firewall **40**, a Decryption Platform **22**, a Gateway Platform **26**, and a Logging Platform **42**.

(353) As seen in FIG. **2**, the Master RMN **14** is connected to a network, like the Internet **18** or intranet **18**, using a Firewall Machine (FW or FM) **40** that filters out all messages that are not from legitimate UIA **16** devices.

(354) In an illustrative embodiment, the messages are decrypted. For this, the transaction processor uses the Decryption Platform (DP) **22**, which utilizes the UIA-VC **204** of the UIA **16** to verify the encryption codes that is required to decrypt the message from the UIA **16**.

(355) Once decrypted, the verification of parties to the transaction is determined using the electronic Verification Platform **12**.

(356) Optionally, the RMN **14** may be conjoined, co-located and/or integrated with the VP **12** and/or the UAR **15**, in any combination thereof. Elements of the RMN **14** and/or the VP **12** may be conjoined, co-located and/or integrated with a NAT **62**, a UTA **16**, and/or a Terminal **2**.

(357) In an illustrative embodiment, once the User is identified by the Verification Platform **12**, the User Verification Approval Code **206** is forwarded to the Master Rule-Module Nexus **14**. The Master Rule-Module Nexus **14** instructs the Execution Platform **38** to take the necessary steps for executing the Execution Commands **52** that are associated with the Pattern Data **54** of the User registered with the Master Rule-Module Nexus **14**.

(358) Upon the VP **12** having made a positive matching determination regarding a User's authorization to accessing the RMN **12**, the User may view and edit Rule-Modules **50**.

(359) Rule-Modules

(360) The Master Rule-Module Nexus **14** is comprised of at least one Rule-Module **50** which comprises Pattern Data or an Execution Command which is “distinct” or “unique” to one registered User (hence, “User-Unique”). In an illustrative embodiment, at least one Rule-Module **50** is unique and exclusive to an individual User. In the event that a Rule-Module comprises pattern data and an execution command that is indexed to one or more registered Users, said Rule-Module is deemed “customized” to a User but not unique to that User (hence, “User-Customized”). As defined herein, User-customized does not necessarily mean that any Pattern Data **54** or the Execution Command **52** is unique to a User, but rather that they are indexed to or are assigned to a specific User. As such, the same Pattern Data **54** or Execution Command **52** may be assigned to several specific Users, and hence would not be unique to any one User.

(361) The Master Rule-Module Nexus **14** functions as a central storage facility for registering, indexing, updating, and invoking various Rule-Modules **50**, whereby the Rule-Modules **50** govern the deposit, the display, the deducting, and the dispensing of financial. In an illustrative embodiment, each of these Rule-Modules **50** is composed of at least two Pattern Data **54** which is associated with or electronically linked to at least one Execution Command. In an illustrative embodiment, said Pattern Data **54** minimally comprise any one of the following: a UUC **200** and a plurality of proprietary Financial Accounts **65**. Optionally, said Pattern Data may also comprise a PVC **202**.

(362) The Master Rule-Module Nexus **14** optionally stores User-customized Pattern Data **54** that is unassociated with any User-customized Execution Commands **52** and optionally stores User-customized Execution Commands **52** that are not associated with any User-customized Pattern Data **54**. Therefore, such unassociated Pattern Data **54** or Execution Commands **52** are optionally stored within the Master Rule-Module Nexus **14** until they are associated with a Pattern Data **54** or an Execution Command **52** together thereby forming an executable Rule-Module **50**.

(363) Once the Verification Platform **12** outputs a positive matching determination for a User, the User Verification Approval Code **206** is forwarded to the Master Rule-Module Nexus **14**. The Master Rule-Module Nexus **14** takes the User Verification Approval Code **206**, optionally along with the UTA-VC **204**, the UTA **16** location data and the Financial Transaction Request Message

(or Transaction Request or Transaction Request Message) **251**, and searches among the User's customized Rule-Modules **50** to invoke Pattern Data **54** and associated Execution Commands **52** relevant to the financial transaction being undertaken.

(364) Pattern Data (PD)

(365) As previously noted, Pattern Data **54** may be provided by the User, by the Master Rule-Module Nexus **14**, or by an authorized financial entity **28**, while the User provides at least one associated Execution Command **52**, to form a single Rule-Module **50**.

(366) Pattern Data **54** of a User is stored electronic data, which is customized to at least one User. For example, Pattern Data **54** may include any stored User-customized and User-unique electronic data, comprising any one of the following: a primary Personal Verification Code (PVC) **202**, which is optionally alpha-numeric; demographic information; an email address; a plurality of proprietary Financial Accounts **65**; a stored-value account comprising pre-paid or pre-earned financial; the User's date of birth; a secondary PVC **202**; a telephone number; Account Transaction Data **172**; a mailing address; purchasing patterns; a UUC **200**. The UUC **200** is unique to each User and is not shared between Users.

(367) Any such Pattern Data **54** may be provided to the Master Rule-Module Nexus **14** by: a User; a Master Rule-Module Nexus **14**; or an authorized third-party **28** such as an Account Issuer.

(368) Account Transaction Data **172** associated with a Financial Account **65** is any information pertaining to a User Account or an Account Issuer Account (respectively, User Account Data and Account Issuer Account Data). Such data includes any of the following: a number which uniquely locates or routes a transaction to a Financial Account **65**; a number which uniquely identifies the Financial Account **65**; User usage location; User usage frequency; User usage recency; User usage demographics, and; User usage volume of electronic financial transactions; a financial transaction processing preference of the Account Issuer associated with the Financial Account **65**.

(369) In an illustrative embodiment, within the Rule-Module Nexus **14**, a User Account Registry (UAR) **15**, which is identified by a User Account Registry Code (UAR-Code) **59**, contains a plurality of proprietary Financial Accounts **65** of a User, including associated Account Transaction Data (or Transaction Data) **172**, and associated Rule-Modules **50**. For example, in the event that the Financial Account **65** is a incentives account for rewards or scrip (e.g., a rewards incentives account), the value for each unit of Financial Data could be a dollar amount, a number of minutes of telephone calling time, points towards the purchase of a product or service, a percentage discount on current or future purchases, and the like. For rewards transactions, the Account Issuer then designates the number of financial to be disbursed to Users based upon the occurrence of predetermined criteria. This criteria may include a credit or debit of financial in the User's Financial Account **65** based on the User's purchasing patterns as a function of any of the following: time; demographics; frequency; recency, and; amount of expenditure.

(370) In another embodiment, within a User Account Registry **15**, the Master Rule-Module Nexus **14** stores and manages Financial Accounts **65** for participating Account Issuers, Users, and counter-party entities. Further, The Master Rule-Module Nexus **14** may comprise Execution Commands **52** to display the Financial Account **65** status, calculations, and adjustments, and the like for participating Account Issuers, counter-party entities, and Users.

(371) Execution Commands (ECs)

(372) An Execution Command **52** may be invoked by Pattern Data **54** with which it is associated. Execution Commands **52** stored within the Rule-Module Nexus **14** and executed by the Execution Platform **38**, may transmit electronic messages necessary for depositing, displaying, deducting, and/or disbursing Financial Data associated with Financial Accounts **65** of a User and, optionally, an Account Issuer. For example, the Execution Commands **52** may also include instructions or commands pertaining to the preserving the preferences of an Account Issuer for processing and/or completing of a financial transaction, comprising any one of the following: invoking criteria predetermined by the Account Issuer for declining the financial transaction; invoking criteria

predetermined by the Account Issuer for approving the financial transaction; invoking criteria predetermined by the Account Issuer for settlement of the financial transaction; invoking a proprietary network; invoking a discount rate; invoking an interchange fee; invoking a settlement protocol; invoking a surcharge; invoking a processing partner, and; invoking a time period for settlement. A processing partner is a Third-Party Platform **28**, in an illustrative embodiment, registered with the RMN **14**, comprising any one of the following: an Account Issuer (e.g., Wells Fargo Bank™); a merchant private label (e.g., Nordstrom's™); an acquirer (e.g., First Third Bank™); a credit association (Visa™); an intermediary (e.g., First Data Corporation™, GE Capital™); a debit processor (e.g., Interlink™), and; a credit company (e.g., American Express™). (373) Another illustrative embodiment of the invention includes a Financial Account **65** being used under predetermined circumstances comprising any one of the following: the number of units of Financial Data to be debited from a Financial Account **65** under which circumstances and the number of units of financial to be credited to an Account Issuer Account under which circumstances. Such Execution Commands **52** may include: a pre-calculated formula for surcharging a User's Financial Account **65** during a financial transaction, such that said surcharge is automatically disbursed to a financial counter-party (e.g., a non-profit charity or a checking account of a subordinated User) or deposited into another Financial Account **65** of the User (e.g., a savings account or brokerage account); a pre-designation that Financial Accounts **65** are to be displayed to the User such that the User can select which Financial Account **65** to invoke for the financial transaction; a pre-designation that Audible or Visible Signatures **81** are presented to the User on a UTA Display **6** or Transaction Terminal Display **6** such that the User may select which entity will be the counter-party of the financial transaction disbursement; a pre-designation that purchases from certain participating Account Issuers will automatically invoke a financial disbursement to at least one certain financial counter-party; a pre-designation that upon accumulation of certain types of financial, such as frequent-flyer miles or free phone minutes, such types of financial will be automatically disbursed to a predetermined financial counter-party; a pre-designation that upon accumulation of certain amounts of financial, there will be a disbursement to at least one predetermined financial counter-party; a pre-designation that upon one financial counter-party having received a certain quantity of financial transactions from the User, perhaps even within a certain timeframe, the User will be notified or further financial disbursement will automatically transfer to a different counter-party.

(374) In one embodiment of a scrip transaction, a Rule-Module **50** from the Master Rule-Module Nexus **14** comprises an Execution Command **52** which permits a merchant to contribute financial directly to a non-profit charity based upon a User's purchases. In such transactions, units of financial are electronically debited from the merchant account, and corresponding units of financial are electronically credited to the Financial Account **65** of the non-profit charity.

(375) The Execution Commands **52** in the Rule-Module Nexus's **14** may further provide several predetermined designations including any of the following: immediate cash discounts or premium charges to a User's Financial Account **65** during a commercial transaction; a deduction of financial units from a User's Financial Account **65**, and an immediate transaction thereof via electronic funds transfer (EFT) to an Account Issuer; a recurring debit based upon a predetermined interval of time, and; an accrual of financial which are credited towards a User's future purchase of a product or service.

(376) FIG. **5** is illustrative of an embodiment, wherein Rule-Modules **50** are registered to a User and a subordinated User, each having a plurality of Pattern Data **54** are associated with a plurality of Execution Commands **52**, including Global Execution Commands **55**.

(377) FIG. **6** shows another embodiment wherein only one Pattern Data **45** associated with one Execution Command **52**.

(378) FIG. **7A** is illustrative of an embodiment wherein a Pattern Data **54** is associated with a plurality of Execution Commands **52**, thereby forming a plurality of Rule-Modules **50**. In this

embodiment, the Pattern Data **54** comprises a UUC **200** associated with Execution Commands **52** comprising access to a plurality of proprietary Financial Accounts **65**. In one embodiment, accessing a proprietary Financial Account **65** comprises accessing a Rule-Module **50** associated with said proprietary Financial Account **65**.

(379) FIG. 7B shows another embodiment, wherein a plurality of Pattern Data **54** are associated with an Execution Command, again forming a plurality of Rule-Modules **50**.

(380) FIG. 7C shows another embodiment, wherein a Pattern Data **54** comprising a UUC **200** is associated with an Execution Command **52** comprising accessing a UAR **15**, said UAR **15** comprising a plurality of proprietary Financial Accounts **65**.

(381) Any User-customized Execution Command **52** may be provided to the Nexus **14** by a party comprising any one of the following: the User, the Nexus **14**, or an authorized third-party.

(382) Global Queries and Global Execution Commands

(383) In an illustrative embodiment of this invention, a Global Execution Command (GEC) **55** does not automatically compel or require all financial transactions of all Users to: be linked to any particular Account Issuer, and/or merchant, and/or product, and/or service, and/or; bypass or be diverted from the Account Issuer or Network **18** which might otherwise apply to a Financial Account **65** selected by any User during a financial transaction. As such, in an illustrative embodiment, there are no “default” GECs **55** governing all of the Financial Accounts **65** of all Users.

(384) An example would be as follows: upon the Verification Platform **12** making a positive matching determination from comparing the User's bid UUC **200** (or optionally a bid UUC **200-PVC 202**) packet with registered UUCs **200** (or optionally registered UUCs **200-PVCs 202**), the User's unique Verification Approval Code **206** is output to the Rule-Module Nexus **14**, matching Global Queries **53** and invoking User-customized Rule-Modules **50**. In an illustrative embodiment, the Verification Approval Code **206** matches the User Account Registry Code (UAR-Code) **59** which uniquely identifies the User Account Registry **15** of a User.

(385) In an illustrative embodiment, the submitted Verification Approval Codes **206** automatically matches to a set of Global Queries **53** in the Rule-Module Nexus **14**. For example, when a Verification Approval Code **206** is submitted, it may match automatically with Global Queries **53** such as the following: “What is the User's home zip code?”; “What that the User's most frequently used merchant?”; “What is the status of the User's Financial Account(s)?”. The answers to these Global Queries **53** are contained in the User-customized Pattern Data **54** which are statements that comprise data customized to the User. In this example, the Pattern Data **54** responses to the above Global Queries **53** are, respectively, as follows: “95401”; “Macy's”; “All payments are current”. In this embodiment, these Pattern Data **54** responses invoke Global Execution Commands (GEC) **55** which govern automatic response programs such as, respectively: “Notify the User to re-new subscription to Santa Rosa Symphony”; “Email the User an electronic coupon for discounted apparel and sports accessories at Macy's in Santa Rosa, CA”; “Mail the User an offer for reduced credit card interest rates because account is in good standing”. In this embodiment, therefore: Global Queries **53** and the Global Execution Commands **52** are actually non-specific, or non-customized, to any particular User; however, the Pattern Data **54** and Execution Commands **52** are unique to, or customized to, the specific User whose Verification Approval Code **206** has been submitted. This illustrative embodiment renders a platform architecture for the Rule-Module Nexus **14** that has: User-customized sub-modules with User-customized Pattern Data **54** and Execution Commands **52**; while the Global Queries **53** and the Global Execution Commands **55** sub-modules are not required to be customized to any one single User.

(386) Processing Financial Transactions Via a RMN and a UAR

(387) A financial transaction comprises a transfer of financial data from a Financial Account **65** to at least one other Financial Account **65**, and may comprise any one of the following: invoking a rule-module via a rule-module nexus; a buyer-seller exchange transaction wherein a user's financial

account balance is adjusted and a payee's financial account is correspondingly adjusted, wherein the user and the payee are different parties; transferring financial data from a financial account of the user to another financial account of the user, wherein there is no buyer-seller exchange; redeeming a pre-paid ticket transaction for venue admittance, wherein there is no buyer-seller exchange; redeeming a pre-paid membership benefit transaction for venue admittance, wherein there is no buyer-seller exchange; an electronic benefits transfer (EBT), wherein there is no buyer-seller exchange.

(388) In one embodiment, a buyer-seller exchange comprises the User buying a product or service from a seller, wherein the User's Financial Account **65** is debited and the seller's Financial Account **65** is correspondingly credited, with the seller being the payee. The seller may comprise an Account Issuer.

(389) FIG. **37** illustrates an overview of an embodiment of financial transactions via the RMN **14** and the UAR **15**, comprising any one of the following: Upon a User's consent, the Merchant Platform **28** sends a transaction authorization message to their Acquirer Processor **28** via existing telecommunications infrastructure Network **18** (Step **1**); The Acquirer Processor **28** determines that the authorization is a financial transaction bound for the RMN **14**, and forwards it via existing telecommunications infrastructure Network **18** to an Acquirer Processor Gateway Platform **26** or **28** (Step **2**); The Gateway Platform **26** or **28** routes the transaction to the RMN **14** which determines the Pattern Data **54** operative pursuant to the data of the transaction taking place, said transaction data comprising merchant name, location, time, products being purchase, transaction size, etc. (Step **3**); The RMN **14** then determines which Execution Commands **52** are operative to the transaction is taking place, comprising the appropriate live Financial Account(s) **65** to use given the Pattern Data **54** and the Execution Commands **52** registered to that User (Step **4**), and the Logging Platform (LP) **42** logs all elements of the transaction for future analysis (Step **4A**); The RMN **14** sends the information to the UAR **15** which reformats the original data into the appropriate live Financial Account **65** number (Step **5**); The UAR **15** either: delivers the message to the hosted Account Issuer Gateway **26** or **28** (Step **6A**), or if the Acquirer **28** leverages their own Account Issuer **28** Network **18** connectivity, back to the hosted Acquirer Gateway **26** or **28** (Step **6B**), which returns the reformatted message to the Acquirer Processor **28** (Step **6C**); The Live Financial Account **65** Authorization is forwarded to the appropriate Account Issuer Platform **28** by the hosted Issuer Gateway **26** or **28** (Step **7A**), or the proprietary Acquirer Processor's Gateway **28** (Step **7B**); The Account Issuer **28** returns the authorization message via the Network **18** channel that it received the request (Step **8**); If the transaction is returned to the hosted Issuer gateway then the information travels back through the Network **18** infrastructure in the same way it was originated (Steps **9A-9E**), If the transaction was routed by the Account Issuer **28** back via the Acquirer Processor **28**, then the Acquirer Processor **28** sends an information-only message to the RMN **14** as a record of the response from the Account Issuer **28** (Step **9F-9H**).

(390) In one embodiment, the Acquirer **28** (or Acquirer Processor **28** or Acquiring Bank **28**) is a financial institution that accepts credit and or debit card payments for products or services on behalf of a merchant, thereby processing the transaction information, coordinating and updating its accounts, and then relays the sales data directly to the Account Issuer **28**, which actually authorizes the sale in accordance with the User's Financial Account **65**. The authorization then is submitted back to the Acquirer **28**, which informs the merchant (or the payee) that the sale, or transaction, has been approved. In this embodiment, the payee may be the Acquirer via a Payee Depository Financial Account **65** may be designated by the Acquirer as a financial account of the Acquirer, from which the Acquirer subsequently disburses funds to the merchant.

(391) In another embodiment, a user may select an item or service for purchase while viewing a webcast, Internet advertisement, a television program (whether cable, satellite, or broadcast), or listening to a radio program (whether terrestrial or satellite radio). This selection may be in the form of a point-and-click action, wherein the user selects: a side-bar, banner, or pop-up

advertisement on a website; an item being advertised during a real-time television commercial, and/or; an item visibly or audibly presented during a real-time or previously broadcast television show. As such, the NAT **62** and/or UIA **16**-Transaction Terminal **2**, is used to tag the selected item(s), whereby a virtual shopping cart (VSC) **92** is compiled and stored via the NAT **62**, the UIA **16**, and/or remotely located RMN **14** with which the NAT **62** and/or the UIA **16** (herein NAT **62**-UIA **16**) is connected via Network **18**. The VSC **92** may also store data associated with the selected items: pricing, sizing, colors, product history, name of manufacturer, POS retail availability, etc. In one embodiment, the User is automatically verified by the VP **12** of the RMN **14** via a UUC **200** provided from a NAT **62**, wherein: a) for an internet embodiment, the UUC **200** may be a logon session cookie, a phone number, an IP address, etc. The RMN **14** thereby transmits SSL data directly to the NAT **62**-UIA **16** for display to the user via a pop-up window, without requiring API integration with a website of the payee and without requiring a form re-direct of the user away from the website, whereby a virtual shopping cart (VSC) **92** is displayed for the User. The User may also be prompted by the RMN **14** for input of a PVC **202**. The VSC **92** may compile all items selected by the User during the logon session, and enables the User at any time to proceed to purchasing the items using an online financial transaction processed via the RMN **14**. In another embodiment for the Internet, upon purchasing an item, the website's Third-Party Platform **28** blocks some or all subsequent advertising to the User during the logon session; b) for a television embodiment, the UUC **200** may be a set-top box code (e.g., a hardware code associated with the television hardware or software for that particular User), phone number or street address to which the television service is associated, etc. The User may also be prompted via pop-up window for input of a PVC **202**. Thereafter, upon the User selecting an item, a pop-up window may appear, providing a view of the User's VSC **92**, with related data on the item selected, without pausing the program being viewed. The VSC **92** may compile all items selected by the User during the viewing session, and enables the User at any time to proceed to purchasing the items using an online financial transaction processed via the RMN **14**. In another embodiment for the Internet, upon purchasing an item, the television service provider's Third-Party Platform **28** blocks some or all subsequent advertising to the User for a period of time. In all embodiments, the VSC **92** provides the User with the opportunity to receive more data on the item or related topics, such as an item selected may be a restaurant, whereby the User may seek for information on the history of the neighborhood, the type of food, make a reservation, etc.

(392) An overview of several example embodiments of financial transactions via the RMN **14** and the UAR **15**, comprises any one of the following:

Example 1: Auto-Account Selection at Physical-World POS, without Keypad **70**

(393) The User's NAT **62** interfaces with the POS UIA **16**-Terminal **2** via magnetic swipe or contactless enablement. Because the User's NAT **62** contains information such as the RMN-RC **61** that identifies the transaction packet as bound for the RMN **14**, the UIA **16**-Terminal **2** packages the available information regarding User, transaction, and merchant (such as the UIA-VC **204**) into a standardized message and forwards the Financial Transaction Authorization Request **251** to the RMN **14**. In many embodiments, the Financial Transaction Authorization Request **251** is a "thin-client" packet, comprising no live Financial Account **65** number or live Financial Account **65** financial data. The RMN **14** automatically determines by the Rule-Modules **50** registered to a User, the appropriate Financial Account **65** to select and invoke (e.g.: all grocery store transactions should be Visa® Debit, or all airline tickets should be American Express® Gold, etc.). The necessary information is translated by the RMN **14**, appended to the transaction packet, and passed along to Third-Party Platforms **28** further processing of the financial transaction. Upon receiving a Financial Transaction Authorization Response **252** from a Third-Party Platform **28**, the RMN **14** re-translates the transaction information into an "account number blind" (or a "blind", or "thin-client" Financial Account **65** number which is not a Live Financial Account **65** number) Financial Transaction Authorization Response **252**, which is forwarded to the Merchant **28**. To aid the User

and merchant in record keeping, a Visible Signature **81**, and the last four digits of the live Financial Account **65** number could be passed along to the UIA **16**-Terminal **2**, and printed on the receipt.

Example 2: Auto-Account Selection at Physical-World POS, with Keypad **70**

(394) The User's NAT **62** interfaces with the POS UIA **16**-Terminal **2** via magnetic swipe or contactless enablement. Additionally the User enters a PVC **202** via the Keypad **70** to secure the transaction. Because the User's NAT **62** contains information such as the RMN-RC **61** that identifies the transaction packet as bound for the RMN **14**, the UIA **16**-Terminal **2** packages the available information regarding User, transaction, and merchant (such as the UIA-VC **204**) into a standardized message and forwards the Financial Transaction Authorization Request **251** to the RMN **14**. RMN **14** determines: via the VP **12** if the User is authorized to access the RMN **14**, and; via the Rule-Modules **50** registered to a User, the appropriate Financial Account **65** to invoke. The necessary information is translated by the RMN **14**, appended to the transaction packet, and passed along to Third-Party Platforms **28** further processing of the financial transaction. Upon receiving an Financial Transaction Authorization Response **252** from a Third-Party Platform **28**, the RMN **14** re-translates the transaction information into an “account number blind” Financial Transaction Authorization Response **252**, which is forwarded to the merchant. To aid the User and merchant in record keeping, a Visible Signature **81**, and the last four digits of the live Financial Account **65** number could be passed along to the UIA **16**-Terminal **2**, and printed on the receipt.

Example 3: At Physical-World POS, with Keypad **70 and Manual-Account Selection**

(395) The User's NAT **62** interfaces with the POS UIA **16**-Terminal **2** via magnetic swipe or contactless enablement. Additionally the User enters a PVC **202** via the Keypad **70** to secure the transaction. In addition, the User enters an Audible/Visible Signature **81** via the Keypad **70** to identify a Financial Account **65** they are manually selecting for the RMN **14** to invoke. For example, the User could keypunch a two digit code via the Keypad **70**, comprising any one of the following: a “00” entry could mean, “Use the Default”; a “01” entry could mean Wells Fargo® Visa® debit; a “MC” could mean, “MasterCard®”, while all other options could be sequentially enrolled. The Audible/Visible Signature **81** is encrypted like the PVC **202**. Because the User's NAT **62** contains information such as the RMN-RC **61** that identifies the transaction packet as bound for the RMN **14**, the UIA **16**-Terminal **2** packages the available information regarding User, transaction, and merchant (such as the UTA-VC **204**) into a standardized message and forwards the Financial Transaction Authorization Request **251** to the RMN **14**. RMN **14** determines: via the VP **12** if the User is authorized to access the RMN **14**, and; via the Rule-Modules **50** registered to a User, the appropriate Financial Account **65** to invoke. The necessary information is translated by the RMN **14**, appended to the Financial Transaction Authorization Request **251**, and passed along to Third-Party Platforms **28** further processing of the financial transaction. Optionally, as determined by the RMN **14**, the PVC Association Platform **909** may also forward the PVC **202**, comprising any one of the following: forwarding the PIN to a debit processing Network **18**, and authorizing a PIN-enabled debit financial transaction; forwarding the PVC **202** to a debit financial transaction processing Network **18**, and authorizing a PIN-enabled debit financial transaction; forwarding the PVC **202** to a financial transaction processing Network **18** and authorizing a PIN-enabled financial transaction; forwarding the PVC **202** to a financial transaction processing Network **18** and authorizing a non-PIN enabled financial transaction. Upon receiving an Financial Transaction Authorization Response **252** from a Third-Party Platform **28**, the RMN **14** re-translates the transaction information into an “account number blind” Financial Transaction Authorization Response **252**, which is forwarded to the merchant. To aid the User and merchant in record keeping, a Visible Signature **81**, and the last four digits of the live Financial Account **65** number could be passed along to the UIA **16**-Terminal **2**, and printed on the receipt.

Example 4: At Physical-World POS, with Keypad **70 and Dynamic Display **7** for Manual-Selection**

(396) a. Return Message: The User's NAT **62** interfaces with the POS UIA **16**-Terminal **2** via magnetic swipe or contactless enablement. Additionally the User enters a PVC **202** via the Keypad

70 to secure the transaction. Because the User's NAT 62 contains information such as the RMN-RC 61 that identifies the transaction packet as bound for the RMN 14, the UIA 16-Terminal 2 packages the available information regarding User, transaction, and merchant (such as the UIA-VC 204), etc., into a standardized message and forwards the Financial Transaction Authorization Request 251 to the RMN 14. RMN 14 determines: via the VP 12 if the User is authorized to access the RMN 14, and; via the Rule-Modules 50 registered to a User, the appropriate Financial Account(s) 65 to invoke. The invoked Financial Accounts 65, optionally in ranked order of preference based upon the Rule-Modules 50 which the User, the merchant, and other authorized parties (e.g., Third-Parties 28, such as product manufacturers, rewards providers, etc.) have registered to the User, are presented via Audible/Visible Signatures 81 on a dynamic screen Display 7, from which the User selects their Financial Accounts 65. The necessary information is translated by the RMN 14, appended to the Financial Transaction Authorization Request 251, and passed along to Third-Party Platforms 28 for further processing of the financial transaction. Optionally, as determined by the RMN 12, the PVC Association Platform 909 may forward the PVC 202, comprising any one of the following: forwarding the PIN to a debit processing Network 18, and authorizing a PIN-enabled debit financial transaction; forwarding the PVC 202 to a debit financial transaction processing Network 18, and authorizing a PIN-enabled debit financial transaction; forwarding the PVC 202 to a financial transaction processing Network 18 and authorizing a PIN-enabled financial transaction; forwarding the PVC 202 to a financial transaction processing Network 18 and authorizing a non-PIN enabled financial transaction. Upon receiving an Financial Transaction Authorization Response 252 from a Third-Party Platform 28, the RMN 14 re-translates the transaction information into an "account number blind" Financial Transaction Authorization Response 252, which is forwarded to the merchant. To aid the User and merchant in record-keeping, a Visible Signature 81, and the last four digits of the live Financial Account 65 number could be passed along to the UIA 16-Terminal 2, and printed on the receipt. b. Merchant Host Matching: The User's NAT 62 interfaces with the POS UIA 16-Terminal 2 via magnetic swipe or contactless enablement. Additionally the User enters a PVC 202 via the Keypad 70 to secure the transaction. Because the User's NAT 62 contains information such as the RMN-RC 61 that identifies the transaction packet as bound for the RMN 14, the UIA 16-Terminal 2 packages the available information regarding User, transaction, and merchant (such as the UIA-VC 204) into a standardized message and forwards the Financial Transaction Authorization Request 251 to a Subset RMN 17 located at the Merchant Host Platform 28 for available payment options. Earlier the RMN 14 has distributed a file to the Merchant Host Platform's 28 Subset RMN 17, said file comprising registered UUCs 200, and associated RMs 50. In one embodiment, no Live Financial Account 65 numbers are distributed; only their associated Audible/Visible Signatures 81. The RMN 17 of the Merchant Host Platform 28 responds to the Terminal 2-UIA 16, presenting the operative Audible/Visible Signature(s) 81 via Display 7, determined pursuant to the transaction data, said transaction data comprising UPC 67, merchant name, location, time, products being purchase, transaction size, etc. The User makes their selection of Audible/Visible Signature(s) 81 from the dynamic screen Display 7, whereby the POS Terminal 2-UIA 16's packages an Financial Transaction Authorization Request 251, comprising User data (comprising UUC 200, or UUC 200-PVC 202), transaction data, and merchant data, and forwards the Financial Transaction Authorization Request 251 to the Master RMN 14. The RMN 14 determines: via the VP 12 if the User is authorized to access the RMN 14, and; via the Rule-Modules 50 registered to a User, the appropriate Financial Account(s) 65 to invoke. The necessary information is translated by the RMN 14, appended to the Financial Transaction Authorization Request 251, and passed along to Third-Party Platforms 28 for further processing of the financial transaction. Optionally, as determined by the RMN 12, the PVC Association Platform 909 may forward the PVC 202, comprising any one of the following: forwarding the PIN to a debit processing Network 18, and authorizing a PIN-enabled debit financial transaction; forwarding the PVC 202 to a debit financial transaction processing Network 18, and authorizing a PIN-enabled

debit financial transaction; forwarding the PVC **202** to a financial transaction processing Network **18** and authorizing a PIN-enabled financial transaction; forwarding the PVC **202** to a financial transaction processing Network **18** and authorizing a non-PIN enabled financial transaction. Upon receiving an Financial Transaction Authorization Response **252** from a Third-Party Platform **28**, the RMN **14** re-translates the transaction information into an “account number blind” Financial Transaction Authorization Response **252**, which is forwarded to the merchant. To aid the User and merchant in record-keeping, a Visible Signature **81**, and the last four digits of the live Financial Account **65** number could be passed along to the UIA **16**-Terminal **2**, and printed on the receipt.

Example 5: Handheld Device NAT **62**

(397) a. Handheld Device NAT **62** comprises no Audible/Visible Signature **81** Functionality: Same as Example 1, Example 2, Example 3, Example 4a, and Example 4b, depending upon POS Terminal **2**-UIA **16** and Merchant Platform **28** capabilities. The contactless chip, IR device or bar code functionality is read by the POS Terminal **2**-UIA **16** using an RFID interrogator or appropriate scanning functionality. b. Handheld Device NAT **62** comprises Audible/Visible Signature **81** Functionality: Same as Example 1 or Example 2 with additional capabilities. The User, before reaching the checkout line, selects from their PDA/cell phone NAT **62**, the stored Audible/Visible Signature(s) **81** they wish to use. No Live Financial Account **65** data need be stored on or access from the NAT **62**. This User selection is relayed to the RMN **14** via a wireless Network **18**, so that the RMN **14** is updated for this one-time selection and appends the Financial Account **65** upon receiving the subsequent Financial Transaction Authorization Request **251** to come from the POS Terminal **2**-UIA **16**. Depending upon the NAT **62** and its data-storage and communicating capabilities, the User may be prompted for Payee data, comprising any one of the following: a unique Payee code (UPC) **67**, a Payee's name, a Payee's depository account, and/or a Payee's bank routing code. In one embodiment, the NAT **62** may provide GPS coordinates, which are registered to a particular merchant location. Because the initiating Financial Transaction Authorization Request **251** is coming from a NAT **62**, it is possible that this transaction will be secured via any number of NATs **62** in the User's possession: using a fob NAT **62**, using a handheld cell/PDA NAT **62**, using a PIN-access code controlled cell/PDA NAT **62** access, and/or using the PVC **202** to append to the Financial Transaction Authorization Request **251**. After the User's NAT **62** is scanned via the POS Terminal **2**, the procedures for the Financial Transaction Authorization Request **251**, per the Example 1 through Example 4, apply. If the POS Terminal **2**-UIA **16** has a Keypad **70** or a touch-screen Display **7**, a PVC **202** can be used also. The RMN **14** is thus expecting the additional Financial Transaction Authorization Request **251**, comprising the transaction-related data from the POS Terminal **2**-UIA **16**, and thereby appends the Financial Account **65** to Financial Transaction Authorization Request **251**, as the User has requested via the NAT **62**. Thereafter, the procedures for processing the financial transaction follow per the Example 1 through Example 4. If the POS Terminal **2**-UIA **16** has a Keypad **70** or a touch-screen Display **7**, a PVC **202** can be appended also. c. Wallet Functionality, Dynamic Contactless Data: Same as Example 1 or Example 2 with additional capabilities. The User, before reaching the checkout line, selects from their PDA/cell phone NAT **62**, the stored Audible/Visible Signature(s) **81** they wish to use. No Live Financial Account **65** data need be stored on or access from the NAT **62**. This User selection updates the Subset RMN **17** residing on the NAT's **62** integrated chip and the Subset RMN **17** is updated for this one-time selection. When User's NAT's **62** data, comprising the UUC **200** and the User's selected Audible/Visible Signature **81**, is scanned via POS Terminal **2**-UIA **16**, the procedures per the Example 1 through Example 4 for processing the Financial Transaction Authorization Request **251**, are applicable. If the POS Terminal **2**-UIA **16** has a Keypad **70** or a touch-screen Display **7**, a PVC **202** can be appended also.

Example 6: Unattended POS Terminals (e.g.: Vending Machines)

(398) a. Online Connected: Same as Example 1 through Example 5c. Note that regardless of POS Terminal **2** capabilities, transactions are likely to be batched in order to reduce payment expense to

the vending machine operator. Therefore the RMN **14** is likely to have a Rule-Module **50** that limits the number of batch transactions that can be generated without an update to the Third-Party Merchant Platform **28**. This limitation is likely to be mirrored by the vending machine's own separate limitation on multiple transactions from a single NAT **62**. b. Intermittent Connection: Same as Example 1, Example 5a, Example 5b, and Example 5c. In this embodiment, the POS Terminal **2** is not capable of online PVC **202** validation via a remote VP **12**. However, using Example 5b enables the RMN **14** database to update its UAR **15** even before the unattended Terminal **2**-UTA **16** has presented the batched Financial Transaction Authorization Request **251**.

Example 7: Transaction Graphics

(399) Similar to Example. In this embodiment, the User wishes to purchase from something comprising a magazine, wall poster (e.g.: tickets to a concert) or the like. The handheld NAT **62** (e.g., cell phone, PDA, iPOD, etc.) either scans product/offering data from the magazine/wall poster/etc., or the User inputs said data derived from the magazine/wall poster/etc. via a Keypad **70** or touch screen Display **7** on the NAT **62**. The NAT **62** packages this data, appending the Audible/Visible Signature(s) **81** which the User's has selected from the stored options on the NAT **62**. The NAT **62** then forwards the Financial Transaction Authorization Request **251** to the RMN **14** for processing the transaction per Example 1 through Example 7 above. The RMN **14** responds to the NAT **62** with an electronic notification if the transaction is complete.

Example 8: E-Commerce

(400) FIG. **43** through **43B** illustrate embodiments of the invention, wherein the RMN **14** may hibernate the financial transaction. In this embodiment, to “hibernate” the financial transaction means that the RMN **14** may: a) store various data required for completing processing of the financial transaction, and; b) maintain Network **18** connection(s), either continuously or intermittently, with various parties to the financial transaction, such as with the user (e.g., payor) via the user's NAT **62**-UIA **16**, with the merchant (e.g., payee) via the merchant's Third-Party Platform **28**, and/or with the acquirer via the acquirer's Third-Party Platform. As such, the RMN **14** remains as a trusted intermediary, providing the necessary continuity for data storage, computer operations, Network **18** connections, and/or party verification, such that the financial transaction is presented to an acquirer Third-Party Platform **28**, or other payment processing Third-Party Platform **28**, as if it were executed in one single or “atomic” step, even though multiple intervening steps or computer operations may have occurred, even if they actually occurred non-contiguously, such as if there were gaps in time between: steps in the transaction; computer operations in the transaction, and/or; Network **18** connections with the RMN **14**. As such, the RMN **14** provides the necessary continuity so that a series of computer operations or logon sessions, perhaps occurring over different periods of time, appear to a Third-Party Platform **28** to have been executed as one large, atomic operation.

(401) In another embodiment of the invention, a logon (or login) session is the period of time elapsed between a user logging in and logging out of an online computer system, wherein the communications Network comprises Internet connectivity, and the user's NAT **62** and a UIA **16** is used. As such, Logon begins when the user visits an Internet website “logon page”, which is a website with which the user has pre-registered (e.g., on www.facebook.com, www.twitter.com, www.google.com, www.nexuspayments.com, etc.), and therein logs on (or logs in) by entering access codes via the Keypad **70** or via the Display **6** (e.g., visual widgets of alpha-numeric-symbolic character for selecting via a computer mouse or via a touch screen, etc.), said access codes comprising at least one of the following: (i) a pre-registered name and/or password; (ii) at least one pre-registered PVC **202**. In addition, or alternatively, the user may logon via providing their UUC **200**, optionally along with at least one PVC **202**. Once the user is thereby verified by the website's server (e.g., Third-Party Platform **28** or RMN **14**), the server transmits a “cookie”, which is a text string which may be in binary code (e.g., ID=0100111001), thereby stored on Memory **320** or **310** of the user's NAT **62**-UIA **16**. As such, the browser then returns the cookie to

the server the next time either that same website is viewed, or the next time the user views a website which has an agreement to share or acknowledge the cookie which the user's NAT **62-UIA 16** had received from the "login page". If the cookie was provided by the RMN **14** (e.g., www.nexuspayments.com), the cookie may function as the user's UUC **200** for Internet access, until such time as the cookie is deleted from the user's NAT **62-UIA 16**, or until such time as the RMN **14** has terminated the cookie's validity (e.g., the cookie passes an expiration date pre-set by RMN **14**). If the cookie was provided by a Third-Party Platform **28** with which the RMN **14** has agreed to share data via a secure connection, then the Third-Party Platform **28** (e.g., www.facebook.com), may transmit confirmation to the RMN **14** that the user has been verified. This acknowledgement may be in the form of a UUC **200** of the user, which the RMN **14** would recognize. Here again, this cookie the cookie may function as the user's UUC **200** for Internet access, until such time as the cookie is deleted from the user's NAT **62-UIA 16**, or until such time as the Third-Party Platform **28** has terminated the cookie's validity (e.g., the cookie passes an expiration date pre-set by Third-Party Platform **28**). Whether provided by the RMN **14** or the Third-Party Platform **28**, once the cookie has been deleted or invalidated, the user is deemed to have logged off, and as such the logon session is terminated. Alternatively, the invention may use cookieless forms authentication, wherein the authentication ticket being passed is in a query string, pursuant to ASP.NET 2.0.

(402) In one embodiment of the invention, a "website widget" (or "widget") means a small piece of software code installed on a website, and visible in the form of an icon image (e.g., a click-on button labeled "I'll Buy"). Once selected by the user, the widget provides an interactive experience for the user via launching a pop-up window which transmits secure socket layer data via HTTPS or S-HTTP to the user's browser, and provides a data-entry box for the user to input a PVC **202**. Additionally, the secure socket layer data may comprise presenting the user's Private Code **79** via Display **6**, which confirms that the user has a secure, authentic, real-time connection to the RMN **14** via communications Network **18**.

(403) In one embodiment of the invention, the secure socket layer data ("SSL") uses two keys to encrypt data (e.g., DUKPT), a public key and a private key, wherein the uniform resources locator ("URL") of the SSL connection starts with Hypertext Transfer Protocol Secure ("HTTPS"), over which any amount of data can be sent securely. In one embodiment of the invention, SSL creates a secure connection between the user's NAT **62-UTA 16** and the RMN **14**, wherein secure HTTP ("S-HTTP") is designed to transmit individual messages securely. As such, secure socket layer data comprise at least one of the following: HTTPS data, and; S-HTTP data.

(404) In one embodiment of the invention, there is no requirement for Third-Party Platform **62** website integration via an Application Programming Interface ("API"). API integration occurs wherein there is a requirement for a defined set of Hypertext Transfer Protocol ("HTTP") request messages, along with a definition of the structure of response messages, which could be an Extensible Markup Language (XML) or JavaScript Object Notation (JSON) format.

(405) In one embodiment of the invention, there is no requirement for: Simple Object Access Protocol (SOAP) based services, or; Representational State Transfer (REST) style communications: the combination of multiple services into new applications known as mashups.

(406) In another embodiment of the invention, there is no requirement for a form re-direct of the user. "Form re-direct" is defined as a sequence of events, wherein: a) if the user requests a page that requires authenticated access and that user has not previously logged on to the site, then the user would have to be redirected to a configured logon page, often in a separate website; b) such a logon page prompts the user to supply credentials, typically a user name, a password, an address, a credit card number, etc.; c) These credentials would then have to be passed to a server and validated against a user store, such as a SQL Server database, and; d) After the user's credentials are authenticated, the user is redirected to the originally requested page. Forms authentication processing is handled by the FormsAuthenticationModule class, which is an HTTP module that

participates in the regular ASP.NET page-processing cycle via ASP.NET 2.0.

(407) In another embodiment of the invention, the secure logon session comprises an active communications connection via a Network **18** and the RMN **14** and optionally a user's NAT **62**-UIA **16**, which is associated with cryptographic keys and other security technologies, to protect the integrity of the communications and the data being transmitted, along with preventing unauthorized access to said communications and said data.

(408) E-Commerce with Widget

(409) FIG. **44A** through FIG. **44M** illustrate various embodiments of the invention using a website widget, as described in hereinbelow.

(410) In one embodiment of the invention, the website of a merchant's Third-Party Platform **28** comprises a widget which, when selected by the user, opens a secure Network **18** connection to the RMN **14**. Said widget may sometimes be referred to herein as a "Nexus 'I'll Buy' button or icon". The Nexus "I'll Buy" button takes advantage of a browser's capability to create a backchannel communication wherein the RMN **14** has an independent connection directly between the user's NAT **62**-UIA **16** and the RMN **14** (this is a triangle, point-to-point connection, between the RMN **14** and the user's NAT **62**-UIA **16**, and also between RMN **14** and the merchant's Third-Party Platform **28** e-commerce website, as opposed to the standard brokered, pass-through connection the RMN **14** requiring API integration with the merchant's e-commerce website Third-Party Platform **28**. As such, SSL data (or HTTPS data or S-HTTP data) is sent directly from the RMN **14** to the user's NAT **62**-UIA **16** via a pop-up window. As such, the RMN **14**: a) Bypasses the standard e-commerce website check-out page and its required form-fillings; b) Avoids the need for API integration with the merchant's e-commerce website Third-Party Platform **28**; c) Avoids a form re-direct, so the user now does not have to leave the e-commerce website Third-Party Platform **28** to complete the transaction. As a result, this embodiment eliminates: a) significant RMN **14** software development which would otherwise be required for API integration; b) significant Third-party Platform **28** software development from the merchant's e-commerce site which would otherwise be required with a form re-direct; c) transaction delays wherein the consumer would otherwise have to leave the merchant's e-commerce website to complete the transaction. In one embodiment, the RMN **14** expedites transaction processing by invoking at least one rule-module **50** that can provide automatic shipping/delivery preferences, so this information does not need to be generated or entered by the user with each transaction at a new e-commerce website. As such, using a single SSL data transmission from the RMN **14**,

(411) In another embodiment the user's logon session commences via the website of a Third-Party Platform **28**, wherein the user provide verification credentials via a UIA **16** to the Third-Party Platform **28**. In this embodiment, the Third-Party Platform **28** has a trusted relationship and a Network **18** connection with the RMN **14**, wherein the Third-Party Platform **28** transmits a UUC **200** to the RMN **14** on behalf of the user. This UUC **200** could be a temporary, time- or use-limited UUC **200**, or it could be a permanent UUC **200** associated with the user. As such, the user's UIA **16** receives and stores a cookie from the Third-Party Platform **28** or the RMN **14**, said cookie comprising either the UUC **200** or another UUC **200** also associated with the user. As such, the user's UIA **16** functions as the user's NAT **62**, wherein they are essentially conjoined. When the user then browses a Third-Party Platform **28** e-commerce website and selects a Nexus "I'll Buy" widget, the RMN **14** commences an SSL data exchange, which may comprise presenting the user's Private Code **79** to the user via the Display **6** on the UIA **16**-NAT **62**, so that the user is prompted to input a PVC **202**. Thereafter, the user may browse any Third-Party Platform **28** e-commerce website and select the Nexus "I'll Buy" widget to commence a financial transaction, and with minimal or not further data input by the user, especially if, for example, the user has pre-registered rule-modules **50** which automatically invoke: a preferred or default Financial Account **65**; a preferred or default mailing/shipping address; a preferred or default shipping service (e.g., FedEx 2-day air). Alternatively, the user may only have to select an Account Signature **81** presented via

Display 6.

(412) In another embodiment, the user commences a logon session via the RMN **14** website, wherein if the user's NAT **62**-UIA **16** comprises a UUC **200** cookie, the RMN **14** may present the user with the user's Private Code **79** and prompt the user for input of a PVC **202**.

(413) In another embodiment, the user browses the Internet using a NAT **62**-UIA **16** and commences a logon session via selecting the Nexus "I'll Buy" widget on a Third-Party Platform **28** e-commerce website. If the RMN **14** detects a UUC **200** (e.g., in the form of a cookie), the RMN **14** may present the user's Private Code **79**, and prompt the user to input their PVC **202**.

(414) In another embodiment, the user browses the Internet using a UIA **16** and commences a logon session via selecting the Nexus "I'll Buy" widget on a Third-Party Platform **28** e-commerce website. If the RMN **14** does not detect a UUC **200** (e.g., in the form of a cookie), the RMN **14** may prompt the user to input their PVC **202** (Primary PVC and/or Secondary PVC). The RMN **14** may thereby present the user's Private Code **79** via the Display **6** of the UIA **16**, and optionally transmit a UUC **200** cookie for receipt and storage by the user's UIA **16**.

Additional e-Commerce Embodiments

(415) a. Direct to Website: If the Terminal 2-UIA **16** has scanning capabilities, a NAT **62** may be scanned for the UUC **200** (or, optionally the UUC **200**-PVC **2020**) at the website's checkout web page, which invokes a hyperlink to the RMN **14**. Alternatively, at the website's checkout web page, via Display **7** on the Terminal 2-UIA **16** (comprising computer screen, television screen, cell phone screen, etc.), the User selects an icon or text which hyperlinks to the RMN **14**. In one embodiment, when prompted, whether being scanned from the NAT **62** or entered by the User via a Keyboard **70** or touch screen Display **7**, the entered UUC **200** is one reserved for virtual POS, or e-commerce internet, transactions. A Private Code **79** may be returned by the RMN **14** for presentation to the User via the Display **7** to validate to the User that the authentic RMN **14** has been contacted. In the event that Audible/Visible Signature(s) **81** are stored in a NAT **62** itself, or in a NAT **62** conjoined with the Terminal 2-UTA **16** (such as a home computer), the User may select an Audible/Visible Signature(s) **81** either via entry on Keypad **70**, touch screen Display **70**, or audio input.

Alternatively, the RMN **14** may return Audible/Visible Signature(s) **81** for presentation to the User, for the User's selection. In another embodiment, upon activating the hyperlink to the RMN **14**, the Rule-Module(s) **50** invoked pertaining to the data (comprising UPC **67**, merchant name, location, time, products being purchase, transaction size, etc.) in the Financial Transaction Authorization Request **251** arriving from the web site, may include invoking a default or preferred

Audible/Visible Account Signature **81** or Financial Account **65**, whereby processing of the financial transaction proceeds as in Example 1 through Example 7. Alternatively, the Financial Transaction Authorization Request **251** arriving from the website, may invoke a menu of a plurality of Visible Signatures **81** from the RMN **14** for presentation to the User via the Display **7**, from which the User can select to designate the Financial Account **65** which the RMN **14** should invoke upon receiving the Financial Transaction Authorization Request **251** with transaction data. Thereafter, processing of the financial transaction proceeds as in Example 1 through Example 7.

b. Via Home Banking: The User's home banking website validates the User's identity before shopping begins. At the time of Virtual POS shopping, this User's identity validation may be forwarded to the VP **12** of the RMN **14**, comprising the UUC **200**, if the home banking website is appropriately validated to the VP **12** of the RMN **14**. Thus the home banking website may provide a portal through which the User shops on an e-commerce website, and/or a hyperlink to communicate directly with the RMN **14**.

The e-commerce site might prompt the User for a one-time only, or short duration, PVC **202** provided to the User by RMN **14** via the home banking site, optionally along with a Private Code **79** to validate the RMN **14** to the User. Alternatively, the User's may be prompted to re-validate his identify by the RMN **14** in the event the User signals an intention to shop online, an alteration to their default or preferred Financial Account **65** as might otherwise apply to the transaction pursuant to Rule-Modules **50**, or a time limit expiration of the original banking site's User identity

verification. As in Example 8a, subsequently the RMN **14** may invoking a default Financial Account **65**, or the presentation of Visible Signatures **81** via the Display **7** for selection of a Financial Account **65**. The merchant's website shopping cart could then proceed with processing the financial transaction via the RMN **14** per Example 1 through Example 8a. c. Via handheld device: The same as Example 8b, except that the User communicates with, validates their identify through, and receives any short duration PVCs **202**, directly from the RMN **14**. d. Via cross-browser extension: FIG. **51** is illustrative of an embodiment of the invention wherein while a user is visiting a Payee/Merchant Website or Webpage **530**, the web Cross-Browser Extension GUI Widget **201** is clicked on to invoke (Step **500**) the Pop-Up Window **516**, without a form re-direct away from the Payee/Merchant Website or Webpage **530** and without requiring API integration with the Payee/Merchant Website or Webpage **530**, for displaying the user's payment-related data and for selecting a Financial Account **65** via Financial Account Signature **81**, with a preferred Financial Account **65** being indicated/selected (Step **550**). Also shown in this embodiment and included in the user's displayed payment-related data is a User-Customized Interactive Advertisement or electronic Coupon **520**, in this instance appearing in the Pop-Up Window **516**. Also shown in this embodiment is the Intelligent Search and Extrapolating Agent **580** which can be invoked (Step **560**) to conduct online data retrievals reflecting User-customized priorities.

(416) FIG. **51A** is illustrative of an embodiment of the invention wherein while a user is visiting a Payee/Merchant Website or Webpage **530**, the web Cross-Browser Extension GUI Widget **201** is clicked on to invoke (Step **500**) the Pop-Up Window **516**, without a form re-direct away from the Payee/Merchant Website or Webpage **530** and without requiring API integration with the Payee/Merchant Website or Webpage **530**, for displaying the user's payment-related data and for selecting a Financial Account **65** via Financial Account Signature **81**, with a preferred Financial Account **65** being indicated/selected (Step **550**). Also shown in this embodiment and included in the user's displayed payment-related data is a User-Customized Interactive Advertisement or electronic Coupon **520**, in this instance appearing in the Pop-Up Window **516**. Also shown in this embodiment is the Intelligent Search and Extrapolating Agent **580** which can be invoked (Step **560**) to conduct online data retrievals reflecting User-customized priorities. Also shown in this embodiment of the invention is, upon the user selecting a Financial Account **65** from within the Pop-Up Window and upon an On-Line Checkout Form **610** from the website or web application of the payee being presented on the display, the web cross-browser extension invokes an auto-complete function (Step **555**) to automatically populate fields of the payee's On-Line Checkout Form **610** with the user's payment-related data.

(417) FIG. **51B** is illustrative of an embodiment of the invention wherein the intelligent software agent is operable to track the user's Internet browsing (optionally via the cross-browser extension) and is operable to automatically conduct on-line data retrievals (Step **560**) reflecting user-customized Web browsing priorities and/or preferences for Competitive Product and/or Service Prices **590** across a plurality of on-line merchants, reward incentives providers, banks, internet payment providers, barter providers, insurance companies, credit account organizations, and/or brokerage firms. Also shown in this embodiment is the Payee/Merchant Webpage **530**, the web Cross-Browser Extension GUI Widget **201** which invokes (Step **500**) the Pop-Up Window **516**, and the displayed Competitive Product and/or Service Prices **590**.

(418) FIG. **51C** is illustrative of an embodiment of the invention wherein while a user is visiting a Payee/Merchant Website or Webpage **530**, the web Cross-Browser Extension GUI Widget **201** is clicked on to invoke (Step **500**) the Pop-Up Window **516**, without a form re-direct away from the Payee/Merchant Website or Webpage **530** and without requiring API integration with the Payee/Merchant Website or Webpage **530**, for displaying the user's payment-related data and for selecting a Financial Account **65** via Financial Account Signature **81**. Also shown in this embodiment and included in the user's displayed payment-related data is a User-Customized Interactive Advertisement or electronic Coupon **520**, in this instance appearing in the Pop-Up

Window **516**. Also shown in this embodiment is the Intelligent Search and Extrapolating Agent **580** which can be invoked (Step **560**) to conduct online data retrievals reflecting User-customized priorities. This illustrative embodiment of the invention also depicts the intelligent tracking and extrapolating software agent is operable to track the user's Internet browsing (optionally via the cross-browser extension) and is operable to automatically conduct on-line data retrievals reflecting user-customized Web browsing priorities and/or preferences to extrapolate (Step **560**) from the User's existing preferences and online activity patterns to automatically and periodically recommend to the User new data, such as displaying Extrapolated Suggestions for New Products and/or Services **595** available from any number of Payee/Merchant Websites or Webpages **530** or Internet Account Issuers. Examples include retrieval of new types of music, books, and investment opportunities that reflect the User's preferences.

Example 9: Via Landline Phone (VRU, Telephone Order)

(419) Same as Example 8b or Example 8c with the exception that the User's UUC **200** and PVC **202** may be entered via options comprising the telephone's Keypad **70**, audio receiver, touch screen Display **7**.

(420) FIG. **17** illustrates an illustrative financial transaction, characterized by verifying the User providing a bid UUC **200** from a scanned NAT **62** (and optionally a bid PVC **202** entered via a keypad **70**) via the UIA **16** of an Account Issuer. The UIA **16**-Transaction Terminal **2** transmits the UUC **200** (or optionally a UUC **200**-PVC **202**) packet to the Master RMN **14** for verification, along with the UIA-VC **204**. The User is thus verified (or identified) through the bid UUC **200** (or optionally a UUC **200**-PVC **202**) packet submitted to the Verification Platform **12**, while the Account Issuer is verified (or identified) through the UIA-VC **204** of the UIA **16**.

(421) The VP **12** outputs a VAC **206** upon verifying the User, wherein a UAR **15** is identified and accessed for presenting a plurality of Financial Accounts **65** of the User via a Display **6** on a UIA **16**-Transaction Terminal **2**.

(422) The User selects a presented Financial Account **65** via the UIA **16**-Transaction Terminal **2**, wherein Transaction Data **172** is entered and appended via the Transaction Terminal **2** either using an electronic cash register or manually. The User then either authorizes or cancels the transaction using the keypad **70** of the UIA **16**-Transaction Terminal **2**. Once the financial transaction is authorized, the UIA **16**-Transaction Terminal **2** transmits to the Account Issuer platforms **28** via the RMN **14**. The Master RMN **14** preserves a forwards the transaction to the financial responsible party(ies) for completing the transaction, optionally including execution and settlement, said party(ies) comprising any one of the following: the Master RMN **14**, a participating Account Issuer platform **28**, and the like.

(423) Execution of the transaction may result in a declined transaction due to lack of financial or other problem condition reported by the Account Issuer. If the transaction is declined, the Master RMN **14** or the Account Issuer platform **28** may transmit the decline notification back to the UIA **16**-Transaction Terminal **2**, canceling the transaction.

(424) As further described below, FIG. **4B** through FIG. **4D** show, respectively: an embodiment of a Financial Transaction Authorization Request **251** packet (or message); an embodiment of a financial transaction response packet (or message); an embodiment of the construction of a financial transaction response packet (or message).

(425) FIG. **4B** shows an embodiment of a Transaction Request Message (or Transaction Request or Transaction Request Message) **251**: A Transaction Request Message **251** originates from the UIA **16**-Transaction Terminal **2** and comprises information and data that is required for the processing of an electronic transaction. In one embodiment these data comprise any one of the following: a bid UUC **200** (or optionally a UUC **200**-PVC **202**) packet, a RMN-RC **61**, and a UIA-VC **204**; in an alternative embodiment, the Transaction Request Message **251** includes additional Transaction Data **172**, comprising any one of the following: amount of purchase, the User's selected Financial Account **65**, and other instructions.

(426) FIG. 4C shows an embodiment of a Financial Transaction Response Message (or Transaction Response or Transaction Response Message) **252**: A Transaction Response Message **252** originates from the RMN or an Account Issuer platform **28** and comprises information and data that confirms the success or failure of a requested electronic transaction, in an illustrative embodiment, also including the Visible or Audible Signature **81** of the Account Issuer.

(427) FIG. 4D shows an embodiment of construction of a Transaction Response Message **252**, wherein the User (or Buyer) is verified (or identified). Optionally, this may include an Account Issuer entity code if the Account Issuer audible or visible signature is already stored on Secure Memory **320** or **310** of the UIA **16**, or if a new Account Issuer audible/visible signature is to be stored on Secure Memory **320** or **310** of the UIA **16**, and played back or displayed to the User. In one embodiment, a registered Audible Signature **81** of an Account Issuer is a series of notes, such as a MIDI sequence, or a sample audio waveform, such as a 44.1 kHz 16-bit stereo audio stream. Each Audible/Visible Signature **81** is associated with an Account Issuer through an Audible/Visible Signature **81** code. The RMN **14** may thereby present Audible/Visible Signatures **81** to the User via the UIA **16**-Transaction Terminal **2** to identify an Account Issuer party or a Financial Account **65** to the User.

(428) FIG. 4E shows a flow chart of the operation of the User Interface Apparatus and the Transaction Terminal (or Terminal) for generating a Financial Transaction Authorization Request **251** packet. FIG. 4F shows a flow chart depicting the data encryption and sealing process at the User Interface Apparatus **16**.

(429) FIG. 4G shows a flow chart depicting the data decryption and counter party identification process at the Rule-Module Nexus **14**. FIG. 4H shows a flow chart depicting the data encryption and sealing process at the Rule-Module Nexus **14**. FIG. 4I shows a representational diagram of the Transaction Request Packet (or Request Message) **251**, and the mandatory and optional data it contains. FIG. 4J shows a representational diagram of the Transaction Response Packet (or Response Message) **252**, and the mandatory and optional data it contains.

(430) FIG. **13** shows an embodiment of a rewards or coupon transaction executed via the Rule-Module Nexus **14**: a transaction is sent from the UIA **16**-Transaction Terminal **2** to the RMN **14**. Upon a User's UUC **200** (or optionally a UUC **200**-PVC **202**) packet being verified with a positive matching determination from the VP **12**, the VAC **206** is output for accessing the UAR **15** and retrieving the User's rewards Financial Account Number **65** (e.g., Safeway™ Club number), optionally including associated purchasing Financial Account Number **65** (e.g., Visa™ account number) and optionally including related Transaction Data **172** (e.g., Visible Signature **81**; list of items recently purchased). In one embodiment, the Visible Signature(s) **81** for the rewards Financial Account **65**, and optionally a credit Financial Account **65** for payment, are transmitted to the UIA **16**-Transaction Terminal **2** for display to the User. Upon the User's selection of the rewards Financial Account **65**, and optionally the credit Financial Account **65** for payment in the event of a purchase, the online transaction is sent to the Account Issuer platforms **28** for further processing. Specifically, portions of the Financial Transaction Request packet (or message) **251** are forwarded to the (Authorized Issuer)-Rewards Processor platform **28**, and in the event of a purchase, to the payment account's Account Issuer platform **28** (not shown). The Authorized Issuer-Rewards Processor platform **28** generates predetermined rewards, coupons or incentives; in addition, optionally, the data for the Financial Account **65** for payment is sent to the Authorized Issuer-Acquirer platform **28** for authorization and settlement (not shown).

(431) Processing rewards or loyalty financial transactions via the RMN **14** enables avoiding the restrictions of current loyalty programs which work either: by issuing a new, unique number to a customer, or; by registering an existing number such as a live Financial Account **65** number. Under such current loyalty programs, a lost/replaced card, or even a consumer pulling out a different card each time they checkout creates havoc for the loyalty program.

(432) By contrast, the RMN **14** and the UAR **15** enable access to a plurality of Registry Financial

Accounts **65**. In one embodiment, if the merchant used the UUC **200**, or a portion thereof, as the loyalty account number, while at the same time the consumer used a Registry Financial Account **65** accessed via using the NAT **62**, the merchant could always be able to tracking the consumer purchase behavior.

(433) In one embodiment, if the loyalty program is issuing its own unique account numbers, the program could simply register a consumer via their UUC **200**, or a portion thereof, in order to integrate with the RMN **14** and the UAR **15**. Then the merchant could use this single loyalty program account number on their system for that consumer, and have the RMN **14** aggregate multiple family members transactions to that number, regardless of the payment method or Registry Financial Accounts **65** being used by any of the consumer's family members.

(434) In another embodiment, when the consumer used any payment token that may or may not be a NAT **62** registered to RMN **14**, a query could be made to the RMN **14** to determine if that payment token could be matched back to a particular UUC **200**.

(435) FIG. **14** shows an embodiment of an electronic check transaction executed via the Rule-Module Nexus **14**: the merchant UIA **16**-Transaction Terminal **2** at point of sale sends a UUC **200** (or optionally a UUC **200**-PVC **202**) packet to the RMN **14** for a User's verification; a positive matching verification enables the UAR **15** to retrieve the e-Check Financial Account **65** for transmitting the e-Check Visible Signature **81** to the UIA **16**-Transaction Terminal **2**. Upon the User's selection of the e-Check Financial Account **65**, the transaction may be passed along to a Authorized Issuer-Check Verification Processor platform **28** where certain high-risk e-Check transactions are vetted by in order to weed out previously tagged bad check writers, and to avoid a decline at the point of sale; the e-Check transaction may then be batched and sent to the Authorized Issuer-ACH Network platform **28** for processing; the ACH Network platform **28** then doles out ACH credits and debits to the appropriate Originating Depository Account Issuer platforms (ODAAuthorized Issuer) **28** and Receiving Depository Account Issuer platforms (RDAAuthorized Issuer) **28**.

(436) FIG. **15** and FIG. **16** show embodiments of credit and debit transactions executed via the Rule-Module Nexus **14**: a User's Unique User Code (UUC) **200** is scanned from a Nexus Access Token (NAT) **62** by the User Interface Apparatus (UIA) **16**; optionally, the User enters a Personal Verification Code (PVC) **202** into the UIA **16**, wherein the UUC **200** (or optionally a UUC **200**-PVC **202**) are encrypted; the UUC **200** (or optionally a UUC **200**-PVC **202**) packet is transmitted to the Rule-Module Nexus (RMN) **14** for User verification and, upon verifying the User, Financial Account **65** access via the User's UAR **15**. The RMN **14** invokes Rule-Modules **50** of the User, transmits Visible Signatures **81** of a plurality of proprietary Financial Accounts **65** to the UIA **16**, optionally with Account Transaction Data **172** along with the User's name and Private Code **79** for presentation to the User via Display **6**. In this embodiment, the User views the Private Code **79** and Financial Accounts **65** on the Display, and selects a Financial Account **65** either by entering an alpha-numeric code, or touching a Visible Signature **81** of a Financial Account **65**. The UIA **16**-Transaction Terminal **2** constructs a Financial Transaction Request Message **251** either automatically if the UIA **16** is integrated with the Transaction Terminal-Electronic Cash Register (ECR) **2**, or; manually by a cashier if the UIA **16** is not integrated directly to the Transaction Terminal-ECR **2**, which is then transmitted via the Network **28** to an Account Issuer (Authorized Issuer)-Merchant platform **28**. The UIA **6**-Transaction Terminal **2** at Authorized Issuer-Merchant either transmits the encrypted financial transaction to a modem bank at the Authorized Issuer-Processor platform **28** which handles Authorized Issuer-Merchant's credit transactions on behalf of Authorized Issuer-Merchant's Acquiring Bank platform **28**, or, transmits the encrypted financial transaction to the RMN **14**, which then forwards it to the Authorized Issuer-Processor platform **28**; the selected Financial Account Data (or Account Transaction Data) **172** tells the Authorized Issuer-Processor platform **28** to route the Transaction Data (e.g., issuing bank code and transaction amount) **172** to the appropriate online Network **18**, with a query as to whether the account is valid

and the transaction can be authorized; the account data tells the Authorized Issuer-Processor platform **28** to route the query to the Authorized Issuer-Issuing Bank platform **28**, which confirms that the account is valid and that the credit balance is sufficient to cover the amount of the transaction, wherein the Authorized Issuer-Issuing Bank **28** platform responds affirmatively to the Authorized Issuer-Processor platform **28**; the Authorized Issuer-Processor **28** platform hands off the approval to the Authorized Issuer-Acquirer Bank platform **28**, which routes the approval to Authorized Issuer-Merchant platform **28** and marks Authorized Issuer-Merchant's account for settlement.

(437) FIG. **15** also shows an embodiment of User verification further comprising a User facial photograph display, wherein upon the VP **12** outputting a positive matching determination that the User is authorized to access the rule-module database, the RMN **14** transmits the User's registered facial photograph **7** for the UIA **16** or the Transaction Terminal **2** to present via a Display **7** for a third-party present, such as a cashier, in real-time during the financial transaction, to visually compare the User's actual face with the User's displayed facial photograph **7**, for affirming or denying that the authentic User is providing the UUC **200** and/or PVC **202**. Illustrative embodiments of this transaction include purchasing alcohol, tobacco, and/or medications, where the verifying the identify of the User may be important.

(438) FIG. **16A** to FIG. **16D** show a flow charts of embodiments for accessing a User Account Registry via a Verification Platform.

(439) FIG. **16B** is illustrative of one embodiment of the Verification Platform **12**, comprising any one of the following: verification via one factor, and; verification via a plurality of factors. Said factors comprise verification data, such as a UUC **200**, a PVC **202**, a photograph display, and the like. In one embodiment, verifying access authorization is electronic and automatic via Verification Platform **12**. In another embodiment, verifying access authorization is electronic and manual, via the RMN **14** transmitting the User's registered facial photograph **7** for the UTA **16** or the Transaction Terminal **2** to present via a Display **7** for a third-party present, such as a cashier, in real-time during the financial transaction, to visually compare the User's actual face with the User's displayed facial photograph **7**, and to affirming or denying, via manual data-entry to the Terminal **2**, that the authentic User is providing the UUC **200** and/or PVC **202**.

(440) In another illustrative embodiment, the invention comprises a method for providing approval of an age-restricted or identity-restricted transaction conducted by a User, comprising any one of the following: receiving, at an unattended age verification station (UIA **16** or UIA **16**-Terminal **2**), at least one UUC **200** (or UUC **200**-PVC **202**) proffered by the User via a UIA **16** (or UIA **16**-Terminal **2**); sending the at least one UUC **200** (or UUC **200**-PVC **202**) to at least one RMN **14**, wherein the at least one database has at least one UUC **200** (or UUC **200**-PVC **202**) record stored therein, wherein the at least one UUC **200** (or UUC **200**-PVC **202**) record contains a reference of the User's age; comparing, at the at least one database, the at least one UUC **200** (or UUC **200**-PVC **202**) to the at least one UUC **200** (or UUC **200**-PVC **202**) record stored in the at least one database; making a first determination, at the at least one database, whether the at least one UUC **200** (or UUC **200**-PVC **202**) matches the at least one UUC **200** (or UUC **200**-PVC **202**) record stored in the at least one database; in the event the at least one UUC **200** (or UUC **200**-PVC **202**) matches the at least one UUC **200** (or UUC **200**-PVC **202**) record stored in the at least one database, making a second determination whether the User's age information in the reference meets at least one system parameter; in the event that the age mapped to the UUC **200** (or UUC **200**-PVC **202**) record meets the at least one system parameter, approving the age-restricted or identity-restricted transaction at the at least one database; and receiving, at the unattended age verification station (UIA **16** or UIA **16**-Terminal **2**), confirmation of the age-restricted or identity-restricted transaction approval.

(441) In one embodiment, the at least one system parameter is defined by the age verification station (UIA **16** or UIA **16**-Terminal **2**).

(442) In one embodiment, the steps of sending and receiving comprise sending and receiving encrypted data.

(443) In one embodiment, sending the at least one UUC **200** (or UUC **200-PVC 202**) to the at least one RMN **14** further comprising sending at least one of a location for the unattended age verification station (UIA **16** or UIA **16-Terminal 2**), a time of age-restricted or identity-restricted transaction, a transaction type, a reference code, a detailed access record, and blood alcohol content of the User.

(444) In one embodiment, a system user's information is stored in the at least one RMN **14**.

(445) In one embodiment, transaction information pertaining to the age-restricted or identity-restricted transaction is compared to the system user information stored in the at least one RMN **14**.

(446) In one embodiment, the unattended age verification station (UIA **16** or UIA **16-Terminal 2**) is selected from the group consisting of a vending machine, a kiosk, a PC, a wireless device, a computing device, and a storage device.

(447) In one embodiment, the unattended age verification station (UIA **16** or UIA **16-Terminal 2**) is an unattended point of sale device.

(448) In one embodiment, the age-restricted or identity-restricted transaction comprises a financial transaction.

(449) In one embodiment, the financial transaction settlement is conducted via a financial transaction system, wherein the financial transaction system comprises the at least one UUC **200** (or UUC **200-PVC 202**) being used to authenticate the financial transaction settlement.

(450) In one embodiment, the second determination comprises a reliability determination and includes assessing whether or not the at least one UUC **200** (or UUC **200-PVC 202**) is being used fraudulently.

(451) In one embodiment, approval of an age-restricted or identity-restricted transaction conducted by a User, comprises any one of the following: receiving, at an unattended age verification station (UIA **16** or UIA **16-Terminal 2**), at least one UUC **200** (or UUC **200-PVC 202**) proffered by the User via a UIA **16** (or UIA **16-Terminal 2**); receiving, at the unattended age verification station (UIA **16** or UIA **16-Terminal 2**), at least one UUC **200** (or UUC **200-PVC 202**) proffered by the User; sending the UUC **200** (or UUC **200-PVC 202**) proffered by the User from the unattended age verification station (UIA **16** or UIA **16-Terminal 2**) to at least one RMN **14**; wherein the at least one database has at least one UUC **200** (or UUC **200-PVC 202**) record stored therein, wherein the at least one UUC **200** (or UUC **200-PVC 202**) record contains a reference of the User's age, and wherein the at least one database has at least one UUC **200** (OR UUC **200-PVC 202**) record; comparing, at the at least one database, the at least one UUC **200** (OR UUC **200-PVC 202**) to the at least one UUC **200** (OR UUC **200-PVC 202**) record; making, at the at least one database, a first determination whether the at least one UUC **200** (OR UUC **200-PVC 202**) matches at least UUC **200** (OR UUC **200-PVC 202**) record; in the event the at least one UUC **200** (OR UUC **200-PVC 202**) matches the at least one UUC **200** (OR UUC **200-PVC 202**) record, making a second determination whether the reference meets at least one system parameter; in the event the reference meets the at least one system parameter, pre-approving, at the at least one database, the at least one UUC **200** (or UUC **200-PVC 202**) record as a potential matching UUC **200** (or UUC **200-PVC 202**) record; receiving, at the unattended age verification station (UIA **16** or UIA **16-Terminal 2**), the potential matching UUC **200** (or UUC **200-PVC 202**) record from the at least one database; comparing, at the unattended age verification station (UIA **16** or UIA **16-Terminal 2**), the at least one UUC **200** (or UUC **200-PVC 202**) with the potential matching UUC **200** (or UUC **200-PVC 202**) record; making, at the unattended age verification station (UIA **16** or UIA **16-Terminal 2**), a third determination whether the at least one UUC **200** (or UUC **200-PVC 202**) matches the potential matching UUC **200** (or UUC **200-PVC 202**) record; in the event that the at least one UUC **200** (or UUC **200-PVC 202**) matches the potential matching UUC **200** (or UUC **200-PVC 202**) record; making, at the unattended age verification station (UIA **16** or UIA **16-Terminal 2**), a fourth

determination whether the matching UUC **200** (or UUC **200-PVC 202**) record was pre-approved at the at least one database; and in the event the matching UUC **200** (or UUC **200-PVC 202**) record was pre-approved at the at least one database, approving, at the unattended age verification station (UIA **16** or UIA **16-Terminal 2**), the age-restricted or identity-restricted transaction.

(452) In one embodiment, the age verification station (UIA **16** or UIA **16-Terminal 2**) sends to the at least one RMN **14** confirmation of completion of the age-restricted or identity-restricted transaction.

(453) In one embodiment, a system user information is stored in the at least one RMN **14**.

(454) In one embodiment, a transaction information pertaining to the age-restricted or identity-restricted transaction is compared to the system user information stored in the at least one RMN **14**.

(455) In one embodiment, providing approval of an age-restricted or identity-restricted transaction conducted by a User, comprises any one of the following: receiving, at an unattended age verification station (UIA **16** or UIA **16-Terminal 2**), at least one UUC **200** (or UUC **200-PVC 202**) proffered by the User via a UIA **16** (or UIA **16-Terminal 2**); receiving, at the unattended age verification station (UIA **16** or UIA **16-Terminal 2**), at least one age-restricted or identity-restricted action proposal proffered by the User; associating at least one proposal code with the at least one age-restricted or identity-restricted action proposal; sending at least one UUC **200** (or UUC **200-PVC 202**) proffered by the User from the unattended age verification station (UIA **16** or UIA **16-Terminal 2**) to at least one RMN **14**; wherein the at least one database has at least one UUC **200** (or UUC **200-PVC 202**) record stored therein, wherein the at least one UUC **200** (or UUC **200-PVC 202**) record contains a reference of the User's age, and wherein the at least one database has at least one UUC **200** (OR UUC **200-PVC 202**) record; sending the at least one proposal code affiliated with the age-restricted or identity-restricted action proposal proffered by the User from the unattended age verification station (UIA **16** or UIA **16-Terminal 2**) to the database; comparing, at the at least one database, the at least one UUC **200** (or UUC **200-PVC 202**) to the at least one UUC **200** (or UUC **200-PVC 202**) record; making a first determination, at the at least one database, whether the UUC **200** (or UUC **200-PVC 202**) matches the at least one UUC **200** (or UUC **200-PVC 202**) record; in the event the at least one UUC **200** (or UUC **200-PVC 202**) matches the at least one UUC **200** (or UUC **200-PVC 202**) record, making a second determination whether the reference meets a parameter, wherein the parameter is indicated by the at least one proposal code; in the event the reference meets the parameter, approving the age-restricted or identity-restricted transaction at the at least one database; and receiving, at the unattended age verification station (UIA **16** or UIA **16-Terminal 2**), confirmation of the age-restricted or identity-restricted transaction approval.

(456) In one embodiment, enrolling a system user in a transaction system comprises any one of the following: providing at least one unattended age verification station (UIA **16** or UIA **16-Terminal 2**), wherein the least one unattended age verification station (UIA **16** or UIA **16-Terminal 2**) is configured to access at least one RMN **14**, wherein the at least one database has at least one UUC **200** (or UUC **200-PVC 202**) record stored therein, wherein the at least one UUC **200** (or UUC **200-PVC 202**) record is associated with the system user and contains a reference of the User's age; prompting the system user to proffer an enrollment data comprising age information, personal information, and at least one UUC **200** (or UUC **200-PVC 202**)s; transmitting to the at least one RMN **14** the enrollment data; and storing in the at least one RMN **14** the enrollment data.

(457) In one embodiment, the reference is a record of at least one of a government issued identification data, a driver's license data, a passport data, a birth certificate data, and a credit data.

(458) In one embodiment, the enrollment data comprises at least one of a driver's license data, a social security number, an address, and a phone number.

(459) In one embodiment, the enrollment data is acquired via a process selected from the group consisting of swiping, scanning and hand-keying.

(460) In one embodiment, the system user additionally proffers a User UUC **200** (OR UUC **200-**

PVC 202).

(461) In one embodiment, the User UUC **200** (OR UUC **200-PVC 202**) comprises at least one of a unique number and a reasonably unique number.

(462) In one embodiment, the enrollment data comprises a digital image of the at least one UUC **200** (or UUC **200-PVC 202**).

(463) In one embodiment, the enrollment data comprises a record maintenance UUC **200** (OR UUC **200-PVC 202**).

(464) FIG. **15C** illustrates an illustrative embodiment using verification data of a 1° PVC (and optionally a 2° PVC), wherein the settlement process is shown in more detail. The User **10** will select a UTA **16** or Transaction Terminal **2** in order to make a Financial Transaction Authorization Request **251**, which may include a Money Transfer Kiosk (MT-Kiosk), as may be found in a Western Union retail site. The User verification data, RMN-RC **61** and the Financial Transaction Authorization Request **251** may be encrypted in a Transaction Packet **11a** which is transmitted to an EFT (electronic funds transfer) Network **29**. A Merchant Acquirer Platform **38a** provides identifying and transaction information in an electronic Merchant Packet **11b** which is also transmitted, as appropriate to the EFT Network **29**. The combined Transaction Packet **11a** and Merchant Packet **11b** are then transmitted via the RMN **14** for processing by a Stored Value Processing System (SVPS or SVP Systems) **50a**, which may optionally be within, or external to, the RMN **14**. The SVPS **50a** initiates an Authorization Process **53a** and a Settlement Process **54a** using the User Account Registry (UAR) **15**. The Account Issuer(s) Platforms **28** then transmit the funds via electronic transfer via the Federal Reserve ACH System **62** via Third-Party ACH Platform **62**, wire transfer via Wire Server (or Platform) **51** or via a proprietary ATM/POS System (e.g., Visa®, MasterCard®, Pulse®) **64** depending on the identity provided by the Account Issuer. The funds are then credited to the Merchant Bank(s) **66, 68**.

(465) FIG. **15D** illustrates an illustrative embodiment of the sub-routing number and consolidation account data routine. The Issuer Bank **58, 60a** may obtain a sub-routing number from the Federal Reserve Bank **62** and create a stored value consolidation Financial Account **65**. The SVPS **50a** may process transactions which are routed to the sub routing number. Internally the SVPS **50a** may maintain a stored value consolidation Financial Account **65** for each Issuer Bank **58, 60a**. Both accounts, consolidation accounts at the Issuer Bank **58, 60a** and the consolidation account maintained by the SVPS **50a**, may tally as stored value processing updates the internal consolidation account in real time upon a transaction processing, sending batch updates to the Issuer Bank **58, 60a**. At the end of a processing day, the SVPS **50a** may send an “on us” type of ACH transaction to the Issuer Bank **58, 60a**. Specifically, the SVPS **50a** may maintain account and transaction information, whereas the stored value consolidation Financial Account **65** may represent the total balance of all stored value processing Financial Accounts **65** with limited net daily charge transactions. Optionally, no individual account transactions or balances are reflected in the SVPS **50a**.

(466) FIG. **16** also shows an embodiment of the Subset RMN Platforms **17** co-located with an Authorized Issuer-Processor **28**, including private label “house accounts”, Visa™ MasterCard™, Discover™, Interlink™ and American Express™. Upon the RMN's **14** verification of the User's bid UUC **200** (or optionally the User's bid UUC **200-PVC 202**) data, the credit/debit card and merchant point-of-sale financial transactions are processed through both the RMN **14** authorized partner platforms **28** and third-party, non-partner Authorized Issuer-Acquirers platforms **28**. The Master RMN Platforms **14** may also update the Subset RMN Platforms **17**.

(467) In another illustrative embodiment of the invention, upon the Verification Platform's **12** successful verification of the User from their bid UUC **200** (or optionally from their bid UUC **200-PVC 202**), other embodiments of Execution Commands **52** governing electronic transmission access comprise permitting the User: to access their health insurance account and validate their prepaid benefits to a health-care provider prior to being admitted to a hospital; to access their

prepaid entertainment account and validate to admittance personnel their eligibility to attend an entertainment event, such as a live music concert on a predetermined day, at a predetermined time and to sit in a predetermined seat; to access their prepaid video club rental account and validate to an Account Issuer their eligibility to rent videos under their pre-paid membership account; to access their dining club privileges at a restaurant using their prepaid membership account for recurring charges.

(468) FIG. 17 shows a flow chart of registration and transaction processing with the Rule-Module Nexus.

(469) FIG. 18 shows a flow chart of registration and transaction processing with the Rule-Module Nexus 14, User Account Registry 15, and Third-Party Platforms 28.

(470) In another embodiment of the invention, the Execution of a Rule-Module (RM) 50 and/or an Execution Command (EC) 52 by the Execution Platform (Execution Platforms) 38 may result in a declined financial transaction due to unidentifiable data comprising any one of the following: bid UUC 200; bid UUC 200-bid PVC 202; a lack of an identifiable Account Issuer platform 28; a closed or inoperative participating Account Issuer platform 28; a closed Financial Account 65; a prior fraud record, and/or; some other immediately detectable problem condition. If the transmission is declined, the Master Rule-Module Nexus 14 or the Verification Platform 12 transmits the decline notification back to the UIA 16.

(471) FIG. 23 through 28 show the process flow of a financial transaction via the Rule-Module Nexus 14, including: the User presents a Nexus Access Token (NAT) 62 to the User Interface Apparatus (UIA) 16 for scanning of a bid UUC 200. Optionally, the User also inputs a bid Personal Verification Code (PVC) 202 into a User Interface Apparatus (UIA) 16. The UIA 16, optionally integrated with a POS-Transaction Terminal 2, builds an encrypted packet for a Transaction Request 251, wherein the UUC 200 (or optionally a UUC 200-PVC 202), a UIA Verification Code (UIA-VC) 204 and a reply key are encrypted. Additional Account Transaction Data 172 is appended to the packet from the POS-Transaction Terminal 2, wherein the Merchant Subset Platform (MSP) forwards the encrypted packet to the Rule-Module Nexus 14. The Rule-Module Nexus 14 platforms check the sequence number of the UIA-VC 204, decrypt the UUC 200 (or optionally a UUC 200-PVC 202), using a Message Authentication Code (MAC) for checking the integrity of the encrypted message. The Verification Platform (VP) 12 then compares the bid UUC 200 (or optionally the bid UUC 200-PVC 202) with registered UUCs 200 (or optionally with registered UUCs 200-PVCs 202), to output a matching determination. If there is a positive matching determination, VP 12 issues a Verification Approval Code (VAC) 206, and the RMN 14 invokes at least one Rule-Module 50 of the User, including Pattern-Data 54 and Execution Command 52, to access Financial Account(s) of the User.

(472) In this illustrative embodiment, a transaction is initiated from the UIA 16 when the User's UUC 200 is scanned from his NAT 62 (and optionally, the User also enters his PVC 202) and, along with optionally appending the financial Transaction Data 172 from the POS-Transaction Terminal 2, whereby a Financial Transaction Authorization Request 251 (TRQ1) is transmitted to RMN 14; The RMN 14 transmits a reply packet (TRP1) including User account information is sent back to UIA 16 and POS-Transaction Terminal 2; Once the User has chosen the Financial Account 65, a second Financial Transaction Authorization Request 251 (TRQ2) with transaction amount and selected account is sent to the RMN 14; The RMN 14 will process the transaction with Account Issuer processor, and reply back (TRP2) the results to UIA 16 and POS-Transaction Terminal 2. POS-Transaction Terminal 2 will submit the results to merchant main server (TRQ3).

(473) FIG. 18 through FIG. 18I are illustrative of using Rule-Modules 50 to improve, or optimize, the benefits of financial transactions via the NAT 62 and/or the RMN 14.

(474) In FIG. 18A through FIG. 18E, Rule-Modules 50 within a UAR 15 or a Subset UAR 19 further comprising Ranking 96 the plurality of proprietary Financial Accounts 65, wherein predetermined criteria is used for tagging and Ranking 96 the plurality of proprietary Financial

Accounts **65** in a certain order. In an illustrative embodiment, the predetermined criteria for the Ranking **96** further comprises any one of the following: improving a transaction benefit for a payee (e.g., merchant), an Account Issuer, and/or the user. Improving a transaction benefit further comprises any one of the following: increasing efficiency; increasing speed; increasing profit; increasing security; decreasing cost; increasing reward incentives, and; invoking a surcharge predetermined by the user.

(475) In an illustrative embodiment, the Ranking **96** further comprises presenting to the user via a UTA **16** a display comprising any one of the following: a default Financial Account **65**; a plurality of Financial Accounts **65**, with visibly distinct indicators for the respective Rankings **96** of the Financial Accounts, and; a plurality of Financial Accounts **65** displayed in sequence as a function of their respective Rankings **96**.

(476) FIG. **18C** is illustrative of an embodiment wherein the Merchant, working with the Acquirer, registers their preferences too, telling the Rule-Module Nexus **14** what devices their POS environment can accept and also telling the Rule-Module Nexus **14** what payment brands they have contracts for acceptance. With that list, the Merchant can identify their business preferences including such things as payment size, security, etc. The Merchant now has far greater control, and opportunities for competition among its payment providers.

(477) FIG. **18D** is illustrative of an embodiment wherein the consumer tells the Rule-Module Nexus **14** their preferences. The Issuer or consumer, or both, registers the available payment tools with the Rule-Module Nexus **14**, wherein they tell how, when, and where they want those devices to operate. They also register a contactless device(s) that they may have received from their Issuer or they may have purchased by themselves.

(478) FIG. **18H** illustrates an illustrative embodiment of the RMN **14** (or Subset RMN **17**) with a Payee Account Registry (PAR) **46** (or Subset PAR **47**), and the Rule-Modules **50** for Ranking **96** the preferences of the Payee for processing a financial transaction, and/or improving the benefits of a financial transaction.

(479) In the event that a Rule-Module **50** is self-conflicting, or is in conflict with another Rule-Module **50**, the RMN **14**, in an illustrative embodiment, invokes a weighting software formula which sorts the conflicting Rule-Modules **50**, weights the conflicting Rule-Modules **50** pursuant to certain criteria, and reconciles any conflicts by determining which Rule-Module(s) **50** will be given greater weight for the processing of the financial transaction.

(480) In embodiments illustrated in FIGS. **32A**, **32B-1**, **32B-2**, **32C-1** and **32C-2**, transactions are transmitted to centralized National/International (or Global) Gateway Platform **28** and co-located Master RMN **14** (and/or Master UAR **15**) in one of two ways: either using an Acquirer Platform **28** or directly from a Merchant Platform **28**. The latter scenario is the most common but does not represent the greatest single point of merchant volume.

(481) FIG. **32A** illustrates an overview of an embodiment of a centralized National/International Gateway Platform **28** and co-located Master RMN **14** system architecture.

(482) FIG. **32B-1** illustrates an embodiment of an Financial Transaction Authorization Request **251** being processed via the Acquirer Platform **28** within a centralized National/International Gateway Platform **28** and co-located Master RMN **14** system architecture, and the following may occur:

(483) The merchant POS system, comprising a UIA **16** and Terminal **2**, creates an Financial Transaction Authorization Request **251** (Step **601**) based upon a standard message format and presents it to their Acquirer Platform **28**. The message format will either be similar to a private label transaction or an existing payment Third-Party Network **18** (e.g. MasterCard®) that has allowed the Master RMN **14** (and/or Master UAR **15**) to use a section of their data string, or routing code, series to access their National/International Gateway **28**.

(484) In one embodiment, a NAT **62** having a 14-digit UUC **200** with a Bank Identification Number (“BIN”) beginning with “79” that can be accepted just like any MasterCard® credit transaction by downloading a UIA **16**-Terminal **2** software upgrade with instructions to format a

transaction with a UUC **200** beginning with “79” as if the transaction had an data string starting with “5”, which is the pre-set routing code to MasterCard®. Acquirer Platforms **28** also arrange routing to MasterCard® so that a “79” is routed as if it were a standard MasterCard® “5”.

(485) The Acquirer Platform **28** evaluates the message (Step **602**) and: If it is not a transaction for processing via the Master Master RMN **14** (and/or Master UAR **15**), the message is routed directly to the Payment Networks **18** and/or the Account Issuer Platforms **28** (Step **602A**), or; If the transaction is designated to be further processed via a Master RMN **14** (and/or Master UAR **15**), the message is routed to the National/International Gateway Platform **28** (Step **602B**) and then to the Master RMN **14** (and/or Master UAR **15**) (Step **603B**), which may be co-located with the National/International Gateway Platform **28**.

(486) The Master RMN **14** (and/or Master UAR **15**) further processes and translates the message (Step **604**) and transmits the transaction: Back to National/International Gateway Platform **28** with live Financial Account **65** data from the queried UAR **15** (Step **604B**), or; Back to the Acquirer Platform **28** with live Financial Account **65** data from the queried UAR **15** (Step **604C**), so that the Acquirer Platform **28** can use its own gateway infrastructure to route the transaction onto the Payment Networks **18** and/or the Account Issuer Platforms **28** (Step **605C**). This may be an illustrative embodiment for an Acquirer Platform **28** that is attempting to maximize the transaction volume being processed on its gateway infrastructure. Since processing is a scale business with a high proportion of costs fixed, the Acquirer Platform **28** with extensive existing infrastructure is always trying to capture additional transaction volume, or; transmits the financial transaction onto the appropriate Payment Networks **18** and/or the Account Issuer Platforms **28** (Step **605B**).

(487) FIG. **32B-2** illustrates an embodiment of an authorization response being processed via the Acquirer Platform **28** within a centralized National/International Gateway Platform **28** and co-located Master RMN **14** system architecture, and the following may occur:

(488) If the Financial Transaction Authorization Request **251** of the financial transaction did not come from the National/International Gateway Platform **28** using the Master RMN **14** (and/or Master UAR **15**), the Payment Networks **18** and/or the Account Issuer Platforms **28** transmit the transaction authorization response, comprising live Registry Financial Account **65** data and Master RMN **14** (and/or Master UAR **15**) data, optionally comprising the UUC **200**, to the Acquirer Platform **28** (Step **701B**). The Acquirer Platform **28** strips the live Registry Financial Account **65** data from the return authorization message and provides the set tracking data in the message (Step **702B**) before passing it to the merchant. The Acquirer Platform **28** may also send a copy of the authorization response provided by the Payment Networks **18** and/or the Account Issuer Platforms **28** to the Master RMN **14** (and/or Master UAR **15**) (Step **703B**) for record keeping and risk management.

(489) If the Financial Transaction Authorization Request **251** did indeed come from the National/International Gateway Platform **28** using the Master RMN **14** (and/or Master UAR **15**), the authorization response is transmitted to the National/International Gateway Platform **28** (Step **701A**), which transmits the transaction authorization response, comprising live Registry Financial Account **65** data and the Master RMN **14** (and/or Master UAR **15**) data, optionally comprising the UUC **200**, to the Acquirer Platform **28** (Step **702A**). The Acquirer Platform **28** may then simply strip the live Registry Financial Account **65** number from the return authorization message and provide the set tracking data in the message (Step **705A**) before passing it to the merchant (Step **703A**). No copy of the authorization is necessary since the Master RMN **14** (and/or Master UAR **15**) would capture that information as part of being in the return path (Step **704A**).

(490) In either embodiment above, the Acquirer Platform may build its settlement file (Step **705B**) by comparing the clearing messages provided by the merchant (carrying: the UUC **200**; a financial transaction tracking number; the last four digits of the live Registry Financial Account **65** number used; and optionally other tracking tools) with the Acquirer Platform's **28** own internal records, which include the live Registry Financial Account **65** number.

(491) FIG. 32C-1 illustrates an embodiment wherein the Merchant Platform **28** is handling its own gateway activities, then the following may occur for the Financial Transaction Authorization Request **251**:

(492) The merchant POS system, comprising a UTA **16** and Terminal **2**, creates an Financial Transaction Authorization Request **251** (Step **801**) based upon a standard message format and presents it to its own Merchant Gateway Platform **28**, wherein a determination is made as to whether the transaction is to be processed via a Master RMN **14** (and/or Master UAR **15**) (Step **802**). In one embodiment, if the transaction is not to be processed via a Master RMN **14** (and/or Master UAR **15**), the message may be routed directly to the appropriate Payment Networks **18** and/or the Account Issuer Platforms **28** (Step **802A**). In another embodiment, the Financial Transaction Authorization Request **251** may be routed directly to the National/International Gateway Platform **28** (Step **802B**), whereby the National/International Gateway Platform **28** determines (Step **803**) whether or not the transaction is designated for processing via the Master RMN **14** (and/or Master UAR **15**). If the transaction is not to be processed via the Master RMN **14** (and/or Master UAR **15**), the message is transmitted to the appropriate Payment Networks **18** and/or the Account Issuer Platforms **28** (Step **803C**).

(493) Alternatively, if the transaction is indeed designated for processing via the Master RMN **14** (and/or Master UAR **15**), the Financial Transaction Authorization Request **251**, comprising a UUC **200**, is transmitted from the National/International Gateway Platform **28** to the Master RMN **14** (and/or Master UAR **15**) (Step **803B**), which may optionally be co-located therein.

(494) The message is then further processed and translated via the Master RMN **14** (and/or Master UAR **15**) (Step **804**), whereby, post-translation, it may then be: returned to the National/International Gateway Platform **28** (Step **804B**) for transmittal to the Payment Networks **18** and/or the Account Issuer Platforms **28** (Step **805B**), or; transmitted directly to the Acquirer Platform **28** (Step **804D**) and then transmitted to the Payment Networks **18** and/or the Account Issuer Platforms **28** (Step **805D**).

(495) FIG. 32C-2 illustrates an embodiment wherein the Merchant Platform **28** is handling its own gateway activities, then the following may occur for the authorization response message:

(496) If the Financial Transaction Authorization Request **251** was sent to the Payment Networks **18** and/or the Account Issuer Platforms **28** via the National/International Gateway Platform **28**, then the return authorization message is returned to the National/International Gateway Platform **28** containing the live Registry Financial Account **65** data along with Master RMN **14** (and/or Master UAR **15**) data, optionally comprising the UUC **200** (Step **901A**). The National/International Gateway Platform **28** strips the return authorization message of the live Registry Financial Account **65** number and inserts the necessary tracking data in the message (Step **903**) before transmitting it to back the merchant (Step **903A**). Separately a record message is sent to the Master RMN **14** (and/or Master UAR **15**) (Step **902B**) for record keeping and risk management, and a message is transmitted to the Acquirer Platform **28** (Step **904A**) so that clearing files can be matched and settlement calculated.

(497) In another embodiment, if the Financial Transaction Authorization Request **251** was not sent to the Payment Networks **18** and/or the Account Issuer Platforms **28** via the National/International Gateway Platform **28**, then the return authorization message is returned to the Acquirer Platform **28** containing the live Registry Financial Account **65** data along with Master RMN **14** (and/or Master UAR **15**) data, optionally comprising the UUC **200** (Step **901B**). The Acquirer Platform **28** strips the return authorization message of the live Registry Financial Account **65** number and inserts the necessary tracking data in the message (Step **902**) before transmitting it to back the merchant (Step **902B**). Separately a record message is sent to the Master RMN **14** (and/or Master UAR **15**) (Step **903B**) for record keeping and risk management.

(498) The Master RMN **14** (and/or Master UAR **15**) may be co-located with a National/International Gateway Platform **28** such as MasterCard®, Visa®, American Express®,

Discover® Financial Services, FDC® (First Data Corporation), TSYS®, or Certegy®. In this embodiment, This centralized system architecture enables single-site data maintenance and transaction security for the RMN **14** and UAR **15**, along with rapid access to a large community of Acquirer Platforms **14**, the largest of which are already connected to the major National/International Gateway Platforms **28**. This embodiment also offers the opportunity for the RMN **14** to return authorization messages to the Acquirer Platform **28** for it to gateway to the Payment Networks **18** and/or the Account Issuer Platforms **28**.

(499) In an embodiment in a physical POS merchant environment, wherein the Terminal 2-UTA **16** has bar-code scanning capabilities, the NAT **62** may comprise a card comprising a barcode presentation of the UUC **200**, or a mobile with a dynamic screen Display **7** that presents a barcode image of the UUC **200**. In addition, the User may input a PVC **202** into the mobile NAT **62**, which then communicates directly with the RMN **14** to verify the User and transmit a one-time UUC **200** or the User's standard UUC **200** to the mobile NAT **62**, for presentation as a barcode on the NAT **62** display for scanning by the Terminal 2-UIA **16**.

(500) In another embodiment, the User enters via audio receiver or Keypad **70**, a Primary PVC **200** (1° PVC **202**) and Secondary PVC **202** (2° PVC **202**) via the mobile NAT **62**, wherein a request is sent to the RMN **14** which returns a single-use UUC **200** for presentation as a barcode via the dynamic Display **7** of the NAT **62**, which is scanned by the physical POS Terminal 2-UIA **16**.

(501) Gifting Transactions Via the Rule-Module Nexus and the User Account Registry

(502) FIG. **39** is illustrative of another embodiment of the invention, wherein a first user gifts a smart-product to a second user via the UAR **15**, and via the RMN **14** as a trusted third-party. A first user ("User A") sends a request message transmitted via the RMN **14** (Step **1**) to a second user ("User B") (Step **2**), requesting permission of User B to gift a smart-product to User B. User B then sends a response message transmitted via the RMN **14** (Step **3**), either accepting or declining User A's gifting request. In the event of a decline, the process is completed and terminates, with notification of the decline being sent to User A (Step **3a**). In the event of an acceptance, the RMN **14** invokes a Rule-Module **50** to process pan-portfolio analytics via the UAR **15** of User B (Step **4**). At this point, for further details on processing pan-portfolio analytics, refer to FIGS. **38A**, **38B**, and **38C**. Upon detecting a pattern from the financial transactions and financial data surveyed via the UAR **15** of User B, the detected pattern may optionally (depending on whether it is required by the Rule-Modules **50** of User B) be transmitted to User B (Step **5a**) for User B's approval or declination to permit the RMN **14** to transmit the detected pattern to User A. User B's acceptance or declination is then transmitted to the RMN **14** (Step **5b**). In the event of a decline, for example, User B's Rule-Modules **50** may permit a portion of the pattern to be transmitted by the RMN **14** to User A. Said portion may reflect User B's Rule-Modules **50** instructions that certain personal or private patterns be deleted from the pattern being forwarded by the RMN **14** to User A. For example, User B may elect to have deleted any of the following: medication purchases, controversial book title purchases, expensive purchases over a certain dollar amount, etc.

Alternatively, such restrictions may be automatically invoked via User B's Rule-Modules **50**, such that the detected pattern transmitted to User B for approval in Step **5a** would thereby already incorporate such deletions. For example, in the event that User B does not require manual review and approval of the detected pattern (e.g., because User B's Rule-Modules **50** automatically screen for certain patterns per above), or in the event that User B approves the transmittal of the detected pattern to User A (via Step **5b**), at this point User A may be notified by the RMN **14** of the pattern of User B (Step **5c**), whereby User A can then purchase a smart-product for User B. User A then purchases a smart-product from Merchant Z **28** (Step **6a**), whereby processing the financial transaction occurs (Step **6b**). Refer to FIG. **22** for processing the financial transaction via the RMN **14**. The gifted smart-product is then provided to User B (Step **7**). The smart-product may be Gift-Branded **1001** in the name of User A via any visible or audible means: a stamp, a card insert, an adhesive note, a recorded message, etc. As such, User B thereby has a durable record attributing the

gifted smart-product to User A.

(503) FIG. 40A is illustrative of another exemplary embodiment of the invention, wherein Financial Data 172 or a Financial Account 65 is gifted to User B by User A via the UAR 15 and the RMN 14 as a trusted third-party. User A transmits a request via the RMN 14 (Step 1) to User B (Step 2), requesting permission to gift a Financial Data 172 or a Financial Account 65 to User B. User B transmits a reply via the RMN 14 (Step 3) to User A (Step 4) accepting or declining the request. In the event that User A's gifting request has been accepted by User B, User B's Rule-Modules 50 may require that User A's gifted Financial Data 172 be credited to either an existing or new Financial Account 65 within User B's UAR 15. For example, User B may want the gifted Financial Data 172 to accrue additional rewards benefits within an existing Financial Account 65 of User B (e.g., frequent flyer miles, free or discounted stays at hotels, etc.). User A then transmits a message to the RMN 14 (Step 5), authorizing the RMN 14 (Step 6a) to instruct Issuer A 28 (Step 6b) to debit a Financial Account 65 in User A's UAR 15, and either credit the corresponding Financial Data 172 to an Issuer B (Step 7) existing or new Financial Account 65 of Issuer B (Step 8a) within User B's UAR 15. The RMN 14 is then informed of the transfer (Step 8b), and then transmits notification to User B (Step 9a) and User A (Step 9b). User B then completes an online financial transaction using the gifted Financial Data 172 via Merchant Z (Step 10a). Refer to FIG. 22 for processing the financial transaction (Step 10b) via the RMN 14 and UAR 15. The RMN 14 then notifies User A of User B's usage of the gifted Financial Data 172 (Step 11).

(504) FIG. 40B is illustrative of details of Step 8a from FIG. 40A, including a new Gift-Branded 1001 Financial Account 65, and new Gift-Branded 1001 Financial Data 172 within an existing Financial Account 65. The Gift-Branding 1001 provides User B with a visible or audible notification that the gift is attributable to User A.

(505) FIG. 41 is illustrative of an embodiment of the invention wherein Gifted Financial Data 172 is visibly displayed on a UTA 16 Display 6 via Visible Signatures 81, with Gift-Branding 1001 for User A through User "N+1" on both new and existing Financial Accounts 65. As such, User B is notified at a time of usage, the attribution of each Gift Financial Data 172 to each gifter, such that User B may select which Gifted Financial Data 172 he may want to use in any particular financial transaction. In the event that a plurality of new gifted Financial Accounts 65 are created for User B, presentation of the Financial Accounts 65 may be ordered in a manner comprising any of the following: alphabetical order by name of the Gift-Branding 1001; dollar amount of the gifted Financial Data 172; date of creation of the Financial Account 65 (e.g., oldest or newest first); closest social networking proximity to User B (e.g., closest friend or newest acquaintance first); etc.

(506) FIG. 42 is illustrative of another exemplary embodiment of the invention, wherein Financial Data 172 or a Financial Account 65 is gifted to User B by User A via the UAR 15 and the RMN 14 as a trusted third-party. User A transmits a request comprising gifting instructions (e.g., dollar amount, specific merchant, specific geography, specific product category, etc.) via the RMN 14 (Step 1) to User B (Step 2), requesting permission to gift a Financial Data 172 or a Financial Account 65 to User B. User B transmits a reply via the RMN 14 (Step 3) accepting the request. In the event that User A's gifting request has been accepted by User B, User B's Rule-Modules 50 may require that User A's gifted Financial Data 172 be credited to either an existing or new Financial Account 65 within User B's UAR 15. For example, User B may want the gifted Financial Data 172 to accrue additional rewards benefits within an existing Financial Account 65 of User B (e.g., frequent flyer miles, free or discounted stays at hotels, etc.). The RMN 14 (Step 4a) then transmits via the UAR 15 (Step 4a) instructions to Issuer A 28 (Step 4b) for debiting a Financial Account 65 in User A's UAR 15, and either credit the corresponding Financial Data 172 to an Issuer B (Step 5) existing or new Financial Account 65 of Issuer B (Step 6a) within User B's UAR 15. The RMN 14 is then informed of the transfer (Step 6b), and then transmits notification to User A (Step 7a) and User B (Step 7b). User B then completes an online financial transaction using the gifted Financial

Data **172** via MerchantZ (Step **8a**). Refer to FIG. **22** for processing the financial transaction (Step **8b**) via the RMN **14** and UAR **15**. The RMN **14** then notifies User A of User B's usage of the gifted Financial Data **172** (Step **9**).

(507) In another illustrative embodiment, User B invites User A via the RMN **14** to join User B's gifting network, whereby if User A accepts, the RMN **14** is authorized by User B to transmit at least one pattern detected via pan-portfolio analytics from a survey by the RMN **14** of financial transactions of User B's UAR **15**. User A can then gift a smart-product to User B at any time. In addition, User B may invite User A via the RMN **14** to gift a branded financial account or to gift branded financial data to User B at any time. In this embodiment, as long as User B's invitation remains open, User A may gift to User B at any time without making a specific request of User B.

(508) Processing Financial Transactions Via Physical POS and Via Virtual POS

(509) In one embodiment, the RMN **14** distinguishes between financial transactions at the physical, "brick and mortar" or "real-world" point-of-sale merchant ("Physical POS"), and financial transactions for the remotely located, "virtual-world" point-of-sale merchant via internet, mail order, telephone order ("Virtual POS"). In this embodiment, the RMN **14** enables for a User: a distinct UUC for Physical POS (Physical POS UUC **205**), and; a distinct UUC for Virtual POS (Virtual POS UUC **203**). This embodiment streamlines logistics, enabling one distribution channel to be used for distributing the Virtual POS UUC **203** and the Physical POS UUC **205**, since they could be both associated with the same User and that User's NAT **62**. In one embodiment, Virtual POS and/or Physical POS financial transactions may be distinguished via the RMN **14** and/or the VP **12** comparing a bid UTA-VC **204** and/or a bid Payee Verification Code ("Payee-VC **209**") with data stored in a Virtual POS Registry **210** and/or a Physical POS Registry **211**, at least one said registry comprising any one of the following: a registered UTA-VC **204**; a registered Payee-VC **209**; a registered Virtual POS UUC **203**; a registered Physical POS UUC **205**.

(510) In one embodiment in a Physical POS, a printed receipt for a financial transaction processed via the RMN **14** comprises any one of the following: the NAT **62** owner's name as provided by the RMN **14**; a portion of the UUC **200**; a portion of the live Financial Account **65** number which was used for the transaction; the expiration date of the live Financial Account **65** number; the nomenclature for the live Financial Account **65** (e.g.: Chase® Visa®, American Express® Optima® Business Account, etc.), and; the financial transaction identification number.

(511) In one embodiment, a financial transaction via the RMN **14** may default whenever possible to requiring a User to enter a PVC **202** in conjunction with the UUC **200**. Similarly, the RMN **14** may default at registration to limit the number or value of financial transactions that can be completed by a User without a PVC **202**. When the threshold is hit, the User may either answer an e-mail or an automated phone validation. In one embodiment, the NAT **62** token may be registered by default to domestic United States use only, but could be changed by country code addition.

(512) In one embodiment for Virtual POS, a User may enable their use of the RMN **14** and UAR **15** for financial transactions by: (i) shopping via a RMN **14** affiliated registration/aggregation web portal of an Account Issuer Platform **28** using their proprietary identity control mechanism; (ii) notifying the RMN **14** via a website, or; (iii) notifying the RMN **14** via telephone with live or automated attendant. In the case of (ii) and (iii) the User may provide verification data comprising any one of the following: part or all of a Virtual POS UUC **203** from the back of the NAT **62**; the "risk key" (Risk Key **207**); their PVC **202**, and; total dollar amount they want to be able to spend in the next chosen time period (e.g., number of hours or until a certain date).

(513) The Risk Key **207** may comprise any one of the following: a unique DCVC3 (Dynamic card validation code); a unique Static CVC (card validation code); a unique DCVV (Dynamic card validation value), and/or; a unique Static CVV (card validation value). In the embodiment where the Risk Code **207** is a CVV, this is a short data string which User of the NAT **62** provides during a Virtual POS financial transaction number to verify that the NAT **62** is their possession. The Risk Key **207** provides a cryptographic check of the Virtual POS UUC **203** associated with that NAT **62**,

and may not necessarily be part of the Virtual POS UUC **203** itself.

(514) FIG. 3E is illustrative of embodiments of a NAT **62** with a Virtual POS UUC **203** and a Physical POS UUC **205**, further comprising any one of the following: Risk Key **207**, a USB port; magnetic stripe; bar code; magnetic ink characters; and embedded RFID chip.

(515) FIG. 3F is illustrative of an embodiment of the a NAT **62** comprising a plurality of magnetic stripes (or strips), wherein each magnetic stripe comprises a different UUC **200** of the same user, whereby the user may select one UUC **200** to be scanned for purposes of processing financial transactions via the RMN **14** to invoke rule-modules **50** for business versus personal accounting purposes, or to designate one UUC **200** for non-PVC transactions (e.g., in a restaurant wherein no UTA **16** with a Keypad **70** for user input of a PVC **202**), wherein certain rule-modules **50** are invoked with spending limits to preclude fraudulent charges by unauthorized users.

(516) In another embodiment comprising shopping via Virtual POS, a User may be allowed to make a restricted number of financial transactions worth up to prescribed dollar value without the User invoking a PVC **200**. This allows for “an impulse purchase”. However, no additional transactions will be allowed until the User's invokes their PVC **200**. An automated outreach from the RMN **14** may contact the User via a pre-registered User phone number or User e-mail address to require validation comprising Pattern Data **54** such as a PVC **202**.

(517) In another embodiment, for Virtual POS, the following may occur: At checkout via an internet web site, telephone order, or mail order, the User may be asked to provide the Virtual POS UUC **203** in full as located on the NAT **62**, including the Risk Key **207**. In this fashion the merchant's check out system has to do nothing different for processing a financial transaction via the RMN **14** than it does for every other standard payment device. A merchant may increase their security, if they desired, by: asking for a portion of the Physical POS UUC **205**, or PVC **202**. This would, in essence provide one more level of validation that the NAT **62** is in the hands of the authorized User.

(518) In another embodiment, the RMN **14** may invert the current internet security approach of some major payment brands which currently invoke monetary penalties on merchants that fail to prove that they have secure systems. Instead, the RMN **14** may decide to only allow merchants with secure systems to even ask for a portion of the PVC **202**.

(519) In another embodiment, the merchant Third-Party Platform **28** or the RMN **14** may query for: a Secondary PVC **202** which is designated for Virtual POS shopping only, and/or; an alternative data string designated for Virtual POS use only. Since the Virtual POS UUC **203** is never printed on a Physical POS receipt, the only way it could be collected is if the NAT **62** were stolen or duplicated, whereby any subsequent unauthorized financial transactions could be identified via several means, including by the User notifying the RMN **14**, by a software security neural network of the RMN **14** or an Account Issuer Platform **28**. In this embodiment, with only the Virtual POS UUC **203** being available for internet, telephone order, and mail order financial transactions, a break-in of a Virtual POS merchant or Account Issuer Platform **28** provides only data strings which the RMN **14** restricts for use in the Virtual POS environment and which cannot be used for Physical POS financial transactions via the RMN **14**.

(520) Processing Mobile Financial Transactions Via a RMN and a UAR

(521) As used herein, electronic “Positioning System” may refer to and include electronic positioning communications and systems, electronic navigating communications and systems, electronic tracking communications and systems, electronic mapping communications and systems, and electronic guiding communications and systems. Such systems and their related software and hardware components, known in the art, comprise any one of the following: Global Positioning System (GPS); Standard Positioning Service (SPS); Differential GPS (DGPS); basic local area DGPS (LADGPS), wide area DGPS (WADGPS) such as WAAS, ARGOS satellite system, Cell phone Networks **18** systems such as Cell-ID, AOA, TDOA, TOA, E-OTD, A-FLT and OTDOA, E-CID, U-TDOA, A-GPS; Matrix, Enhanced-GPS (E-GPS); Polaris Wireless Location Signatures

(PWLS); TV-GP; Precision Location Technology (PLT), based on time of arrival (TOA); Ekahau Positioning Engine (EPE); AeroScout; PanGo Locator; AXCESS tracking; InnerWireless RF; Ultra Band Width (UWB)-based tracking; Laser Positioning System; Microwave-based tracking systems; Radar/Sonar positioning; Universal Time Coordinated (UTC), and; Acoustic Positioning Systems (APS).

(522) As an overview of several embodiments for mobile financial transactions via the RMN **14**, the Financial Account(s) reside remote from the mobile NAT **62** in an online UAR **15**, accessible to the NAT **62** via Network **18** connectivity, while the NAT **62** may store a registry of Audible or Visible Signatures **81**.

(523) In one embodiment, the User may enter their Primary PVC **200** and Secondary PVC **202** into a “public” cell phone/PDA or other mobile device with Network **18** connectivity, such as when renting a mobile device or borrowing a mobile device of a friend. The Primary PVC **200**-Secondary PVC **202** may be transmitted to the RMN **14** using SMS or other message transmitted via Network **18**. When the VP **12** verifies the User's authority to access the RMN **14**, the RMN **14** may then transmit a single-use, or limited-use, Secondary UUC **200** to the “public” mobile device via SMS, whereby the RMN **14** temporarily enables the “public” mobile device to function, in real-time, as the User's own private NAT **62** by presenting a barcode of the Secondary UUC **200** via the mobile device's Display **7**, for scanning by a Terminal 2-UIA **16**.

(524) In an alternative embodiment, for additional security, a request for a Secondary UUC **200** or a Secondary PVC **202** may also be generated from a kernel software program running on the mobile device NAT **62**, whereby the mobile device NAT **62** generates a Financial Transaction Authorization Request **251** via SMS message transmitted over a Network **18** to the RMN **14**. The RMN **14** generates a single-use, or time-limited use, Secondary UUC **200** or Secondary PVC **202** that is time-synchronized between the NAT **62** and the RMN **62**, and is transmitted to the NAT **62** over a Network **18** via Financial Transaction Authorization Response **252**. In one instance, the User's NAT **62** is scanned by the Terminal 2-UIA **16** for a single-use Secondary UUC **200** which was just SMS-issued by the RMN **14**, and presented via barcode image on the NAT's **62** Display **7**. In an alternative instance, the User's NAT **62** displays a Secondary PVC **202** which was just SMS-issued by the RMN **14**, and which the User then enters via the POS Terminal 2-UIA **16** for that transaction only.

(525) In another embodiment, where neither the User's NAT **62** nor the POS Terminal 2-UIA **16** enables storage of, or direct Keypad **70** entry of, a the User's Audible or Visible (Audible/Visible) Signature **81**, the mobile device NAT **62** may transmit a Financial Transaction Authorization Request **251** to the RMN **14**, comprising any one of the following: the PVC **202** inputted using audio commands or Keypad **70** entry on the NAT **62**, and; the UUC **200** stored via the NAT's **62** Subscriber Identification Module (SIM Card), whereby the RMN **14** returns a Financial Transaction Authorization Response **252**, comprising an Audible or Visible (Audible/Visible) Signature **81** in a barcode image presented via the NAT's **62** dynamic Display **7**, for scanning by the POS Terminal 2-UIA **16**. The Visible Signature **81** is appended to the Terminal 2-UIA **16** transaction data, and the packet is forwarded as a Financial Transaction Authorization Request **251** and the transaction processing proceeds from there as described elsewhere herein in other embodiments.

(526) FIG. **29A** through FIG. **29H** are illustrative of illustrative embodiments of financial transactions using mobile payment systems via the RMN **14**.

(527) FIG. **29A** is an example of steps in a mobile payment system using the RMN **14**. A charging session is defined by the time interval starting from when a payment contract is agreed by all parties until the payment agreement is no longer valid. Each charging session can have only one mobile user, one content provider or merchant/payee. There may be numerous payment transactions in a charging session, depending on the transaction type used. A charging session is represented by a unique, persistent session identification data type object, which is related to every

operation performed during the life time of a charging session.

(528) The RMN **14** provides an interface for the creation of a charging session contract. The RMN **14** extracts relevant information from the user, the NAT **62** and the UAR **15** to construct the payment agreement between the parties involved, including the settlement type(s) for the payment transaction. The contract is defined by the charging WSDL interface as a complex data type, which may be extended to fit any kind of service and payment scenario. Once a charging session is no longer required, a charge and release operation is performed to charge any remaining amounts and followed by releasing the charging session. All session details are now obsolete and the session id will be invalid, therefore this information will be removed from the payment service persistent storage.

(529) In FIG. **29B**, a mobile payment system is illustrated using a real GSM mobile Network **18** for communication. The overall system consists of three distributed systems: the mobile NAT **62** (or NAT **62**-UTA **16**) operated by the mobile User, the Payee or merchant server and, the RMN **14**. Operator billing represents the payment mechanism used by mobile Network **18** operators to acquire payment from their customers for services provided. A Payee using this payment method with the Web Services framework would be requesting that the RMN **14**, or the Network **18** operator such as a cell phone service provider, will bill the mobile NAT **62** User according when the service provided in being used. For example, the RMN **14** may provide aggregated billing services to the User, spanning across services and Networks **18** used by the NAT **62**. The RMN **14** will then forward that payment on to the Payee after deducting a payment service charge. The amount to be billed may be deducted from a mobile NAT **62** user's pre-paid account or added to their next post-paid billing aggregation invoice from the RMN **14**. Alternatively, if the Network **18** operator, such a cell phone company, provides the invoice, costs may be added to the User's phone bill. The RMN **14** in this case may be conjoined with the network operator or may be independent with access to the Network **18** operator's third-party Platform(s) **28** and payment infrastructure. This illustrative embodiment enables NAT **62** Users to use SMS text messaging to anonymously pay for access to digital entertainment and content. Reverse SMS billing via the RMN **14** means that the owner of the recipient NAT **62** phone rather than the message sender is charged for the cost of the SMS message received. This provides an alternative payment option not connected to mobile Network **18** operators' infrastructure, or Account Issuers.

(530) FIG. **29C** is illustrative of an alternative mobile payment using the implemented Web Services framework with the additional functionality on the mobile NAT **62** due to the introduction of a J2ME application. This expands the communication capabilities, by using TCP sockets, between the mobile NAT **62** and the Payee Platform **28** via the RMN **14**, allowing the integration of more complex services into this payment domain, such as video streaming services.

(531) In another embodiment, the RMN **14** provides aggregated billing services, avoiding the need to send a User an SMS each time a Payee bills a user. Instead, Users are billed when the UUC **200** from a NAT **62** and, optionally, their PVC **202**, are verified by the RMN **14**. The RMN **14** provides this verification to the Payee(s) pre-registered with the User's UAR **15**, whether via a Payee's Internet **18** website or WAP site, or at a Payee physical point-of-sale (POS). Optionally the RMN **14** batches the User's billings for invoicing on a periodic basis.

(532) In a POS merchant environment, the User's NAT **62** may be scanned by the UTA **16**-Terminal **2**, which transmits the UUC **200** (or UUC **200**-PVC **202**) packet, a description of the goods and the payment amount to the RMN **14**. The RMN **14** verifies the User via the VP **12**, invoking a VAC **206** and accessing a plurality of the User's proprietary Registry Financial Accounts **65** via the UAR **15**, PAR **46**, CCR **30**, and/or NRR **31**. Upon invoking the Rule-Modules **50**, including any Ranking **96** of Financial Accounts **65** to improve a transaction benefit, the RMN **14** may present the User with a plurality of proprietary Financial Accounts **65** from which to select for transaction approval and processing. Alternatively, the RMN **14** may invoke a pre-registered default Financial Account **65**, transmitting an Unstructured Supplementary Services Data (USSD) communication to the

Account Issuer of the default Financial Account **65**, and transmits to the User an "invoice" and amount. The User then may authorize the transaction by data key-entry and/or voice-command(s). The User may then be billed, optionally in an aggregated batched invoice, via a pre-paid account via the RMN **14** or via a Third-Party Platform **28**, or; optionally via post-pay against a Financial Account **65** with an Account Issuer. In an illustrative embodiment this embodiment supports Web Services architecture and the open standards; XML, SOAP, WSDL and UDDI.

(533) FIG. **29D** through FIG. **29F** are illustrative of embodiments wherein the RMN **14** manages the flow of identity information between or among the Third-Party Platforms **28**, the NAT **62**, and/or the UTA **16**-Terminal **2**, for the purpose of processing a financial transaction using a NAT **62**, a UUC **200** (or UUC **200**-PVC **202**) packet for the RMN **14** verifies the User via the VP **12**, thereby invoking a VAC **206** and accessing a plurality of the User's proprietary Financial Accounts **65** via the UAR **15**, PAR **46**, CCR **30**, and/or NRR **31**. In SPF, the RMN **14** is assists in creating and maintaining system management models. The RMN **14** makes system management capabilities available to SPF through a publishing/discovery mechanism making transparent the changes in the system management interface, which may come as a result of components reconfiguration and upgrades (static changes) and/or service mobility (dynamic changes). The functionality of the RMN **14** may optionally consist of three tasks: interfacing with Third-Party Platforms **28** information sources, making queries against those information sources, and preparing received data for use in SPF. With its interfacing capabilities, the RMN **14** promotes SPF into a generic (system independent) plugin component. Because the RMN **14** gives essential plug-and-play capabilities to SPF, it is preferable that it supports interactions with various sources of information, including: UDDI Registries, for the discovery of message-based management operations (management services), wherein the management interface will, in an illustrative embodiment, be published in a the RMN **14** private UDDI registry that caters to SPF as the client; ODP Traders, for the discovery of RPC-based management operations; Directory services, which offer generic information storage and discovery facilities for use in implementing UDDI Registries and other mechanisms of resources registration, tracking and discovery.

(534) FIG. **29G** through **29G-7** are illustrative of embodiments where the RMN **14** is an intermediary acting, receiving and transmitting requests on behalf of Third-Party Platforms **28** for the purpose of processing a financial transaction using a NAT **62**, a UUC **200** (or UUC **200**-PVC **202**) packet for the RMN **14** verifies the User via the VP **12**, thereby invoking a VAC **206** and accessing a plurality of the User's proprietary Financial Accounts **65** via the UAR **15**, PAR **46**, CCR **30**, and/or NRR **31**. The RMN **14** may optionally provide: translation of the data transmissions where needed; protocol enhancements; transcoding; and/or other optimization or transformation functions.

(535) FIG. **29G-8** is illustrative of an embodiment wherein the RMN **14** a counter part component to an MMS Client Platform **28** in the Network **18**. The RMN **14** is responsible for providing an access to other messaging systems **28**. For example, an MMS client **28** or Payee submits the message to the RMN **14** which resolves the MM target address(es) and then routes forward the MM to each target Third-Party Platform **28** for the purpose of processing a financial transaction using a NAT **62**, a UUC **200** (or UUC **200**-PVC **202**) packet for the RMN **14** verifies the User via the VP **12**, thereby invoking a VAC **206** and accessing a plurality of the User's proprietary Financial Accounts **65** via the UAR **15**, PAR **46**, CCR **30**, and/or NRR **31**.

(536) FIG. **29H** is illustrative of an embodiment wherein an authentication process within a single sign-on environment works via the RMN **14** for the purpose of processing a financial transaction using a NAT **62**, a UUC **200** (or UUC **200**-PVC **202**) packet for the RMN **14** verifies the User via the VP **12**, thereby invoking a VAC **206** and accessing a plurality of the User's proprietary Financial Accounts **65** via the UAR **15**, PAR **46**, CCR **30**, and/or NRR **31**.

(537) FIG. **29H** is also illustrative of illustrative steps in this embodiment, including: 1) The end user logs in to the RMN **14** server by entering the URL in a web browser; 2) the RMN **14** software

authenticates the end user and returns a session token; 3) The session token is what enables single sign-on to work. This token is provided as an applet parameter and is used throughout the authentication process. The users may not be asked for their credentials again as long as the session token is present; 3A) the RMN **14** invokes Rule-Modules **50** of the User, including accessing a plurality of the Financial Accounts **65** of the User, whereby the RMN **15** presents to the User via a NAT **62**-UTA **16**, a default Financial Account **65** or a plurality of proprietary Financial Accounts **65** for selection to process the financial transaction; 4) the User accesses an Instant Messenger applet URL from a browser and chooses a method to invoke the client; 5) User selects and/or approves a Financial Account **65** for processing the financial transaction; 5A) the browser invokes Java Web Start or the Java plugin as appropriate. Java Web Start or the Java plugin downloads the necessary Instant Messenger resource files and starts Instant Messenger; 6) Instant Messenger requests authentication to the Instant Messaging server using the session token; 7) The Instant Messaging server asks Access Manager to validate the session token. If the session is valid, Instant Messenger displays the end user's contact list and the end user can then use Instant Messenger services: chat, alerts, polls, etc; 8) The Instant Messaging server, in an illustrative embodiment, queries LDAP directly to get or set end-user information, such as contact lists or subscriptions.

(538) FIG. **30** to FIG. **30D** illustrate illustrative embodiments of the invention used in conjunction with automatic detection and/or automatic transmission of Payee data to a NAT **62** (or NAT **62**-UIA **16**), whether at a physical point-of-sale merchant or in a residential context. This process enables a User to pay for goods and/or services without the requirement to use the NAT **62** to search for, and/or manually enter, Payee data such as a unique Payee code (UPC) **67**, a Payee's name, a Payee's depository account, and/or a Payee's bank routing code.

(539) FIG. **30** is an illustrative illustration of steps involved in a financial transaction processed using Over-the-Air (OTA) Provisioning and Protocols via the RMN **14** or Subset RMN **17**, used in conjunction with automatic detection and/or automatic transmission of Payee data to a NAT **62** (or NAT **62**-UIA **16**). When a Payee registers Payee data with the RMN **14** or Subset RMN **17**, the Payee, in an illustrative embodiment, associates their registration Rule-Module(s) **50**, including Pattern Data **52** and Execution Commands **54**, within a Payee Account Registry (PAR) **46**. OTA Provisioning and Protocols hardware and software components are known in the art. In an illustrative embodiment, a SIM ("Subscriber Identification Module") is incorporated into a NAT **62** and the communication between the SIM card and the RMN **14** can be done by Short Message System (SMS) exchange and in this case named the SMS channel. In an illustrative embodiment, the NAT **62** is phase 2+ in the GSM standard and is SIM Tool Kit compliant.

(540) In one embodiment, illustrated in FIG. **30A** the Payee data may be transmitted to, or detected by, a NAT **62** from a Payee Terminal **2** or a Payee UIA **16** via any Network **18**, including internet, intranet, a satellite network, a radio network, and near field communications (NFC), such as a local WiFi field. In one such embodiment, Over-the-Air (OTA) provisioning is used to provide Payee data, pre-registered with the RMN **14** or Subset RMN **17**, directly from a Payee UIA **16** or Payee Terminal **2** to a NAT **62**. In one embodiment, the User manually initiates the NAT **62** electronic detection, and optionally downloading and/or storing, of Payee data directly from a Payee UIA **16** or Payee Terminal **2** to the NAT **62**. Alternatively, in another embodiment, the User may elect to have the NAT **62** set in an "always on" electronic detection mode, whereby the NAT **62** automatically detects, and optionally downloads and/or stores, Payee data directly from a Payee UIA **16** or Payee Terminal **2** to a NAT **62**. to the RMN **14** or Subset RMN **17**. This OTA Provisioning via the RMN **14** or Subset RMN **17** may electronically inform or populate the NAT **62** with Payee data to enable the User to conduct an online electronic financial transaction with the Payee in payment for goods and/or services without the User having to manually data-enter Payee data required for such a transaction. For example, such required Payee data may include a UPC **67**, a Payee depository account (which may also be an acquirer, or acquiring bank, designated Payee depository account), a Payee bank routing code, a Payee's roster of goods and/or services and

associated pricing or costs. OTA is known in the art for the transmission and reception of application-related information in a wireless communications system, and is commonly used in conjunction with the Short Messaging Service (SMS), which allows the transfer of small text files even while using a mobile phone for more conventional purposes. In addition to short messages and small graphics, such files can contain instructions for subscription activation, banking transactions, ring tones, and Wireless Access Protocol (WAP) settings. OTA messages are, in an illustrative embodiment, encrypted to ensure User privacy and data security. Such OTA provisioning, in an illustrative embodiment, via the RMN **14** and using pre-registered Rule-Modules **50**, enables a NAT **62** (or NAT **62**-UIA **16**) to select, download and install contents and applications on a NAT **62** mobile device. The software bundles used are, in an illustrative embodiment, compatible with the J2EE Client Provisioning Specification (JSR **124**), for ease of uploading those bundles to the provisioning server, in an illustrative embodiment, located at the RMN **14** or Subset RMN **17**. These bundles may be delivered to NAT **62** mobile phones through WAP-Push, a technology known in the art. An Application ID can uniquely identify a particular application in the NAT **62**. The NAT **62** can inform the RMN **14** of application IDs in a session. In one embodiment, the User manually initiates the NAT **62** transmittal of positioning data, including real-time NAT **62** positioning coordinates, via the RMN **14** or Subset RMN **17**. In one embodiment, the User may press a key on a NAT **62** cell phone or PDA having a pre-set speed-dial number or a pre-set approval command. Alternatively, in another embodiment, the User may elect to have the NAT **62** set in an "always on" OTA provisioning data mode, whereby the NAT **62** automatically communicates with a Payee via the RMN **14** or Subset RMN **17**, including a Payee UIA **16**-Terminal **2**. The NAT **62** may optionally be populated with, download and/or store any portion of said Payee data, in an illustrative embodiment, depending on Payee Rule-Modules **50** pre-registered within a PAR **46**.

(541) FIG. **30B** is an illustrative illustration of steps involved in a financial transaction processed using a Positioning System via the RMN **14** or Subset RMN **17**, used in conjunction with automatic detection and/or automatic transmission of Payee data to a NAT **62** (or NAT **62**-UTA **16**). When a Payee registers Payee data with the RMN **14** or Subset RMN **17**, the Payee, in an illustrative embodiment, associates their registration Rule-Module(s) **50**, including Pattern Data **52** and Execution Commands **54**, within a Payee Account Registry (PAR) **46**.

(542) In one embodiment, illustrated in FIG. **30C** and FIG. **30D**, said Payee data is, in an illustrative embodiment, associated with Payee-specific and Payee-unique geographic positioning data (or positioning data), such that a point-of-sale (POS) physical locality of the Payee is identified for triangulation by an electronic Positioning System, such as a Global Positioning System. As such, when a NAT **62**, containing a Positioning System receiver, is located in an appropriate proximity of the Payee's POS location, such as a store or restaurant, the NAT **62** Positioning System receiver may electronically inform or be detected by an electronic Positioning System via the RMN **14** or Subset RMN **17**. This detection may enable the RMN **14** or Subset RMN **17**, using triangulation or other similar positioning communications, systems and/or methods, to electronically associate the location coordinates of the NAT **62** with corresponding location coordinates of a Payee, whereby the RMN **14** or Subset RMN **17** may invoke Payee Rule-Module(s) **50** to electronically inform or populate the NAT **62** with Payee data to enable the User to conduct an online electronic financial transaction with the Payee in payment for goods and/or services without the User having to manually data-enter Payee data required for such a transaction. For example, such required Payee data may include a UPC **67**, a Payee depository account, a Payee bank routing code, a Payee's roster of goods and/or services and associated pricing or costs. In one embodiment, the User manually initiates the NAT **62** transmittal of positioning data, including real-time NAT **62** positioning coordinates, via the RMN **14** or Subset RMN **17**. In one embodiment, the User may press a key on a NAT **62** cell phone or PDA having a pre-set speed-dial number or a pre-set approval command. Alternatively, in another embodiment, the User may elect to have the NAT

62 set in an “always on” positioning data mode, whereby the NAT **62** positioning coordinates are automatically transmitted via the RMN **14** or Subset RMN **17**. Once the NAT **62** positioning data is received by the Positioning System via the RMN **14** or Subset RMN **17**, the Payee positioning data pre-registered with the RMN **14** or Subset RMN **17**, may be associated with the real-time NAT **62** positioning data, whereby Payee data may be transmitted or detected by the NAT **62**. The NAT **62** may optionally download and/or store any portion of said Payee data, in an illustrative embodiment, depending on Payee Rule-Modules **50** pre-registered within a PAR **46**.

(543) In one embodiment, illustrated in FIG. **30D** and in FIG. **48**, the NAT **62** further comprises an RFID interrogator for scanning a product's embedded, or adhered, RFID chip for Product-Related Data (PRD) **91** comprising any one of the following: product name, brand, and manufacturer; unique package code; unique product inventory/tracking code, and; product price. In an alternative embodiment, the NAT **62** may further comprise a bar-code reader for scanning a product's Universal Product Code bar-code label. Each product scanned by the NAT **62** adds the product PRD **91** to the User's “virtual shopping cart” (VSC) **92**, which is aggregated product purchasing data stored either on the NAT **62** or on a RMN **14** or subset RMN **17**. The VSC **92** enables a real-time cumulative total of products selected by the User for purchase; products added to the VSC **92** may also be deleted by the User, using a Display **7**, a keypad **70**, or audio commands via the NAT **62** or UTA **16**-Transaction Terminal **2**. Upon a User concluding product selection(s), the VSC **92** tabulates the total purchase price and product itemization for presentation to the User via the NAT **62**. The User may then approve the purchase via the NAT **62** for purposes of processing the financial transaction, with the Payee data being invoked by the NAT **62** via an electronic Positioning System, as described in other embodiments above. As such, in one embodiment within a point-of-sale retailer, the User may thereby conduct real-time product selections and purchases while in a store aisle, avoiding the need to wait for, or pass through, any standard checkout lane, whether self-serve or cashier-attended. In addition, product purchasing data in the VSC **92** may be communicated by the RMN **14** to a Third-Party Platform **28** of a retailer or manufacturer, for purposes of product inventory, tracking, and logistics.

(544) FIG. **47** illustrates an embodiment, wherein the RMN **14** may transmit the product purchasing data to a remotely located merchant Terminal **2**, and the merchant's Third-Party Platform **28** is thereby informed of the completed purchase, and assembles the purchased product(s) for pick-up by the user. For example, this may occur remotely from a retail site, such as a restaurant or supermarket, wherein the user selected items for purchase from a retail site designated via physical address, phone number, or Positioning System coordinates, thereby completes the financial transaction off-site via the user's NAT **62**-UIA **16**, and then drives to the restaurant or supermarket to pick-up their pre-paid order.

(545) In one embodiment, the User, remotely located from a POS retail site, initiates a logon session with the RMN **14** via a UUC **200** provided from a NAT **62**-UIA **16** connected to a Positioning System via Network **18**, wherein the User selects a restaurant based on criteria (type of food, price of meal, brand name, etc.), for which the RMN **14** then detects the most proximal restaurant to the User pursuant to said criteria, along with Positioning System coordinates, and related data for processing an online financial transaction via the RMN **14**, including: Payee Depository Account, a roster of Payee-related products and services along with pricing. Said roster (e.g., the restaurant's menu) may then be presented to the User via NAT **62**-UIA **16**, whereby the User may select item(s) to order for subsequent pick up by the User or delivery to the User.

(546) FIG. **48** illustrates an embodiment, wherein the User is on-site at a merchant store, and wherein upon completion of the financial transaction via the user's NAT **62**-UIA **16**, the RMN **14** transmits a confirmation code to the user's NAT **62**-UIA **16**. Said confirmation code, which may be presented via Display **7** as visibly readable by a person (e.g., an alpha-numeric-symbolic characters), and/or as computer-readable (e.g., a scannable bar-code), for confirmation to the merchant's clerk or an automated self-checkout UIA **16**-Terminal **2** that the products have been paid

for and that the financial transaction has been successfully completed. In the example of an automated self-checkout UIA **16-Terminal 2**, the products may be weighed on a scale, wherein the confirmation code comprises data on the combined weight of the products purchased, based on their universal product codes and their associated, respective individual weights.

(547) Mobile and Web-Based Financial Transactions Using the Rule-Module Nexus and an Online Account Registry

(548) FIG. **44C** is illustrative of an embodiment wherein the “BuyIt!” widget is displayed on a web page of the hosting site by providing a piece of HTML and/or javascript that calls out to the RMN server and renders a widget on the hosting site's page adjacent to the item to be purchased. The item's details are encoded into the javascript that calls the RMN server. The user must have authenticated to the hosting site in order for the widget to appear on the site. (This is required because the user's Nexus ID is stored in the user's profile on the hosting site, and must be passed back to the RMN server during the HTML/javascript call). The encoded information includes: The ID of the site making the call; The product to be purchased; The total amount of the transaction; The Nexus ID of the user. The ID of the hosting site making the call is validated against the IP address and DNS name of the site on file with the RMN servers before returning the widget code in the HTTP response stream.

(549) FIG. **44D** is illustrative of an embodiment wherein, if the hosting site is determined to be valid by review of the ID, IP address, and DNS name of the server submitting the request to the RMN, then the RMN returns a token that the widget uses to pop open a window with a direct connection to the RMN server. Because the window is being opened in direct response to a click by the user, popup-blockers will not trap this window. The window that is popped up by the widget contains the Login Image for the user ID that was passed into the RMN by the HTML/javascript provided by the hosting site. The user is presented with a personalized login image that they selected when they registered with Nexus. This login image is intended to help the user be certain that the authentication request they are seeing is genuinely from the Nexus system, and not a “phishing” scam from identity thieves. If the user recognizes the Login Image, then they will enter their PIN to authenticate directly with the RMN system.

(550) FIG. **44E** is illustrative of an embodiment wherein, once the user has authenticated with the RMN server, the RMN server interrogates the Nexus wallet and rules for the user and generates a set of Tender Suggestions. Per the Payee (i.e., the hosting site) and Payor (i.e., the user) preferences in the RMN, the RMN determines which types of financial accounts (Tenders) are registered by the Payor and accepted by the Payee (e.g., Visa, M/C, ACH, etc.), and which rewards apply to the transaction (e.g., discounts offered by Payee, or reward points accumulated by Payor), and then presents to Payor their Nexus wallet with the accounts and associated benefits that can be used for this transaction. In this example, the user selects the MasterCard listed as the best option by the RMN, based on the Payor's established preferences, and the Payee's accepted tender types.

(551) FIG. **44F** is illustrative of an embodiment wherein, once tender account has been selected and submitted to the RMN, the RMN formats a payment transaction for that tender type into a format recognized by the Merchant Acquirer used by the hosting site, and submits the transaction to the hosting site. The response from the Merchant Acquirer is then reported back to the user. A record of this transaction is stored in the RMN transaction log, and a notification is sent to both the Payor and Payee detailing the transaction ID, approval code, amount, and tender type of the purchase.

(552) FIG. **44G** is illustrative of an embodiment wherein, once an item has been purchased via the widget, subsequent invocations of the widget (submissions of the HTML/javascript by the same vendor with the same user ID and product ID) will result in a changed presentation of the widget to signify that the user has already bought this item from this merchant. However, clicking the widget delivers the same behavior, permitting the user to make additional purchases if so desired.

(553) FIG. **45** is illustrative of an embodiment wherein: (Step 1) Customer C agrees to purchase of \$X from Merchant M; (Step 2) M submits payment request for \$X to RMN **14**, indicating

Merchant ID, Customer ID, Transaction Amount, and Ancillary information. Customer ID may be any valid, active NAT's **62 UUC 200**, including cell phone number, email address, etc.; (Step 3) RMN **14** logs payment request and assign a transaction identifier T and timestamp to the request; (Step 4) RMN **14** looks up the registered user U based on the customer identifier (UUC **200**) in T; (Step 5) RMN **14** sends authentication request to C, and U receives request via cell phone page, web browser, email, etc.; (Step 6) U connects to the RMN **14** via cell phone or web browser and sends message to RMN **14** approving the transaction and submits their PIN (PVC **202**) for authentication; (Step 7) RMN **14** interrogates UAR **15** and Rule-Modules **50** for U and generates a Tender Suggestion; (Step 8) Tender suggestion is submitted to Merchant Acquirer A in standard ISO-formatted message envelope; (Step 9) Acquirer A processes transactions and returns an authorization to the RMN **14**; (Step 10) RMN sends confirmation message to M; (Step 11) RMN **14** sends confirmation message to C.

(554) FIG. **45** through **46Q** illustrate embodiments wherein the User using their NAT **62-UIA 16**, communicates via Network **18** with the RMN **14** and a counter-party User for processing a financial transaction.

(555) FIG. **46A** illustrates an embodiment wherein upon the User (whether Payor or Payee) launching a RMN **14** mobile e-commerce software application which was downloadable to a NAT **62-UIA 16**, the User (whether Payor or Payee) is prompted to provide verification data by entering a PVC **202**. This controls User (whether Payor or Payee) access to confidential information which this downloadable application may view and present to the User (whether Payor or Payee) via Display **6**. Preferably, the RMN **14** recognizes the Thin-Client UUC **200** provided from the NAT **62**, whereby the User (whether Payor or Payee) is also presented with a Private Code **79** (here an image of a white balloon in a green field) which the User (whether Payor or Payee) previously selected when they registered with RMN **14**. This Private Code **79** validates to the User (whether Payor or Payee) that the prompt for entering verification data is genuinely from the RMN **14**, and not a "phishing" scam from an unauthorized Third-Party Platform **28**.

(556) FIG. **46B** illustrates Step **1** of an embodiment, wherein if the User (whether Payor or Payee) recognizes the Private Code **79**, then the User (whether Payor or Payee) can be satisfied that this is a legitimate RMN **14** authentication request. The User (whether Payor or Payee) then enters a PVC **202** using the Keypad **70** presented on Display **6**.

(557) FIG. **46C** illustrates Step **2** of the embodiment begun in FIG. **46B**, wherein if this is the first time the User (whether Payor or Payee) has launched the downloadable application from the RMN **14**, the User (whether Payor or Payee) is prompted to configure the downloadable application with verification data which the User (whether Payor or Payee) has pre-registered with the RMN **14**. The information on this screen is saved, and is protected by the User's PVC **202** (whether Payor or Payee) so that it is not visible or changeable by unauthorized users. The verification data collected and stored in this screen are outlined as follows: A) The User's Thin-Client UUC **200** (whether Payor or Payee) must be provided to the RMN **14** for verification by the VP **12** (using the corresponding Nexus PVC **202**) in order to be saved in this application. The Thin-Client UUC **200** may be any valid Pattern Data **54** which the User (whether Payor or Payee) has in their pre-registered Rule-Module(s) **50**, including mobile phone number, email address, or SIM card number, etc.; B) The User (whether Payor or Payee) may choose to store their PVC **202** in the application to streamline their Transaction Request(s) **251**, or they may choose not to. If the User (whether Payor or Payee) chooses to store their PVC **202** in the application, then the PVC **202** will automatically be submitted to the VP **12** with any Transaction Request **251** generated by the application. If the User (whether Payor or Payee) does not, then the PVC **202** is preferably entered each time a Transaction Request **251** is generated. Preferably, the User (whether Payor or Payee) should enter the PVC **202** each time the application launches in order to prevent unauthorized access to the sensitive data in this application; C) The cell phone number of the NAT **62-UTA 16** may be used as a Thin-Client UUC **200**; D) The Positioning System coordinates may also be transmitted (if

available) to the VP 12 as an extra fraud-prevention measure. As such, the VP 12 can determine if there is a pattern of fraud, as may be the case when more than one Transaction Request 251 is submitted within a proximal time period using the same Thin-Client UUC 200 from different Global Positioning coordinates which may be substantially far apart from each other geographically; E) Default comment verbiage may be entered to streamline Transaction Request 251 data entry.

(558) FIG. 46D illustrates Step 3 of the embodiment begun in FIG. 46B, wherein provided that the User's (whether Payor or Payee) verification data has been successfully validated by the VP 12 and that the User (whether Payor or Payee) has successfully configured the downloadably software application on the NAT 62-UIA 16, then the home page of the application is displayed from the RMN 14, wherein the User (whether Payor or Payee) may select "Request Payment."

(559) FIG. 46E illustrates Step 4 of the embodiment begun in FIG. 46B, wherein the "Request to Receive Payment" screen is presented on Display 6 of NAT 62-UIA 16. In the current illustration depicted herein of the "Request to Receive Payment" screen, the Payee is User and is the party who is initiating the Transaction Request 251. The Payor is the party who will receive this Transaction Request 251. This screen is used to enter a request for payment to be processed via the RMN 14. In this embodiment, a request for payment preferably contains the following mandatory information: a) The identity of the User/Payor; b) The identity of the User/Payee; c) The dollar amount of the transaction. The User/Payee's identity is already known to this application, as was previously established when the User/Payee configures the application for use on the Settings screen, per FIG. 46A through FIG. 46D hereinabove. The User/Payor identity and the dollar amount therefore is preferably be entered on this screen.

(560) FIG. 46F illustrates Step 5 of the embodiment begun in FIG. 46B, wherein the User/Payee enters the dollar amount of the payment Transaction Request 251 (in this example, \$177.43).

(561) FIG. 46G illustrates Step 6 of the embodiment begun in FIG. 46B, wherein the User/Payee enters a Thin-Client UUC 200 associated with the Payor. A Payor's Thin-Client UUC 200 may for example, be any of the following, as long as the Payor has pre-registered said Thin-Client UUC(s) 202 with the RMN 14: a) An email address controlled by the Payor; b) A mobile phone number controlled by the Payor; c) A Thin-Client UUC 200 associated with a NAT 62 of the Payor. In this example, the User/Payee has entered the mobile phone number of the Payor (817-555-1964).

(562) FIG. 46H illustrates Step 7 of the embodiment begun in FIG. 46B, wherein the User/Payee enters a comment to remind the User/Payor of which transaction this is. In this example, the transaction is for the sale of a Deluxe Blue Product, so the Payee enters that as a comment, along with any other disambiguating information such as an order summary, or anything else the User/Payee and User/Payor have agreed upon.

(563) FIG. 46I illustrates Step 8 of the embodiment begun in FIG. 46B, wherein the User/Payee then reviews the Transaction Request 251 and, if satisfied, clicks the "Send" button (A). If the User/Payee has not opted to store their PVC 202 in the application (as shown in the Settings screen), then the User/Payee is first prompted to enter the PVC 202. The Transaction Request 251, along with the Thin-Client UUC 200 and PVC 202 of the User/Payee, is then transmitted to the VP 12 of the RMN 14 for verification. If the submitting User/Payee's Thin-Client UUC 200 and PVC 202 are successfully verified, then the Transaction Request 251 is logged into the RMN 14, where a transaction ID and timestamp are assigned. The transaction ID is returned to the User's application on the user's NAT 62-UTA 16 as a receipt of the Transaction Request 251, and is stored in the RMN 14 for this User/Payee, where it is visible in the History screen of the User/Payee's downloadable application.

(564) FIG. 46J illustrates Step 9 of the embodiment begun in FIG. 46B, wherein upon receiving the Transaction Request 251, the RMN 14 looks up the Thin-Client UUC 200 of the intended recipient of the Transaction Request 251 (in this case, the Payor). If the Thin-Client UUC 200 for the Payor passed in with the Transaction Request 251 corresponds to a valid, active User registered with the

RMN **14**, then the RMN **14** generates a notification to the intended recipient (in this case, the Payor). Notifications can be received in a variety of modes, as defined by the User/Payor's preferences pursuant to their pre-registered Rule-Modules **50** within the RMN **14**. One the methods of notification is through real-time (i.e., "push") alerts on the User/Payor's mobile phone that pop-up regardless of what the User/Payor is doing on the phone. In this example, the User/Payor is receiving such a notice on their mobile phone NAT **62-UIA 16** while they are using a facebook-type of application. The pop-up alerts the User/Payor that they have received a payment Transaction Request **251** from the RMN **14** (herein shown as "Nexus Payments", and provides the User/Payor the opportunity to respond immediately. If the User/Payor clicks "Close," then the alert disappears. If the User/Payor click "Respond," then the e-commerce downloadable software application of the RMN **14** is launched on that User/Payor's NAT **62-UIA 16**. In this example, the User/Payor clicks "Respond" (A). In this illustration, the User/Payor has previously been verified by the VP **12** of the RMN **14**, per FIG. **46A** through FIG. **46D** hereinabove.

(565) FIG. **46K** illustrates Steps **9a** and **9b** of the embodiment begun in FIG. **46B**, wherein upon the User/Payor just previously clicked "Respond", it is preferable the User/Payor enter their PVC **202** once User/Payor views their Private Code **79**, herein depicted as a security lock.

(566) FIG. **46L** illustrates Step **10** of the embodiment begun in FIG. **46B**, wherein the User/Payor's UIA **62-UIA 16** loads the e-commerce downloadable software application from the RMN **14**, and the Transaction Request **251** (or Payment Request) Summary screen is displayed. The User/Payor may reject the transaction by clicking the "Reject" button, or approve it by clicking the "Approve" button. If the User/Payor clicks the "Reject" button, the transaction is cancelled, and the User/Payee is notified via the RMN **14** that the User/Payor has rejected the transaction. Once the User/Payor clicks the "Approve" button, then the transaction is approved.

(567) FIG. **46M** illustrates Step **11** of the embodiment begun in FIG. **46B**, wherein the User/Payor's Rules Module(s) **50** for in the RMN **14** invoke Financial Account(s) **65** from within the User/Payor's UAR **15** to present Account Signature(s) **81** for review and selection by the User/Payor. Per the User/Payee and User/Payor Rule-Modules **50** in the RMN **14**, the RMN **14** determines which types of financial accounts (Tenders) are registered by the User/Payor and accepted by the User/Payee (e.g., Visa, M/C, ACH, etc.), and which rewards apply to the transaction (e.g., discounts offered by User/Payee, or reward points accumulated by User/Payor), and then presents to Payor Account Signatures **81** and associated benefits that can be used for this transaction. In this example, the user selects the Account Signature **81** listed as the best option by the RMN **14**, based on the User/Payor's established preferences, and the User/Payee's accepted tender types.

(568) FIG. **46N** illustrates Step **12** of the embodiment begun in FIG. **46B**, wherein the User/Payor's Account Signature **81** tender selection is submitted to the RMN **14**, and confirmed. The RMN **14** then formats a Transaction Request **251** in the ISO-standard message envelope for financial transactions for submittal to the Merchant Acquirer's Third-Party Platform **28** associated with User/Payee's selected Financial Account **65**.

(569) FIG. **46O** illustrates Step **13** of the embodiment begun in FIG. **46B**, wherein upon the financial transaction being processed by the Merchant Acquirer's Third-Party Platform **28**, a message is returned to the RMN **14** indicating the success or failure of the financial transaction. The RMN **14** then preferably sends an electronic message to both the User/Payor and User/Payee notifying each of them of the result of the submitted payment. Messages can be received in a variety of modes, as defined by User/Payor and User/Payee Rule-Modules **50**. One the methods of messaging is through real-time (i.e., "push") alerts on the User's (whether Payor or Payee) mobile phone NAT **62-UIA 16** that pop up regardless of what the User (whether Payor or Payee) is doing on the phone. In this example, a User (whether Payor or Payee) is receiving the message on their mobile phone NAT **62-UIA 16** while aa facebook-type software application is being used. The pop-up alerts the user that the payment transaction just authorized has been completed successfully. If

the User (whether Payor or Payee) clicks “Close,” then the alert disappears. If the User (whether Payor or Payee) clicks “View,” then the RMN **14** e-commerce downloadable software application is launched and the history of this transaction is displayed. In this example, the User (whether Payor or Payee) clicks “Close” (A), since the message is self-evident.

(570) FIG. **45** and FIG. **46** are illustrative of embodiments of the invention comprising: 1.

User/Payor comes to an agreement with User/Payee for User/Payor to pay Payee \$X for goods and/or services rendered. 2. In order to initiate this payment transaction, a Payment Request (or Transaction Request **251**) must be generated in the RMN **14**. a. The Transaction Request **251** contains the following required elements (other optional elements may be included in the request, and they will be passed on solely for informational purposes): i. User/Payor Verification Data: The User/Payor may be logged onto the RMN **14** website, or has activated an option on their NAT **62**-UIA **16** cell phone (e.g., a bookmark or “speed-dial” for RMN **14** connection via Network **18**) which connects to RMN **14** so that the payment can be initiated; User/Payor is verified by Thin-Client UUC **200** provided from NAT **62**; RMN **14** thereby authenticates itself to the User/Payor via displaying the User/Payor's Private Code **79**; User/Payor is then prompted to enter their PVC **202** via the NAT **62**-UIA **16**. ii. User/Payee ID Verification Data: The User/Payee is preferably verified by Thin-Client UUC **200** as input by the User/Payor (e.g., email address, phone number, URL, etc.), whereby this UUC **200** has been pre-registered by the Payee with RMN **14**. iii. Transaction Amount: The dollar amount to be paid by User/Payor to User/Payee. b. Two examples for initiating the payment transaction are illustrated herein: User/Payor-Initiated and User/Payee-Initiated. i. Payor-Initiated: In this case, User/Payor is the Request Initiator and submits the Transaction Request **251** (i.e., “I want to pay you \$10”). In order to be permitted to submit the Transaction Request **251** to the RMN system, the User/Payor must be verified by the RMN **14** (via UUC **200** and optionally the PVC **202** entry), thus establishing the validity of the User/Payor identity listed in the Transaction Request **251**. [To authenticate to the RMN **14**, the Payor is may be logged onto RMN **14**, or have activated an option on their mobile phone (e.g., a mobile browser bookmark or custom Nexus e-commerce application) which connects to RMN. Payor is uniquely identified by a valid, authenticated Nexus token. Nexus thereby authenticates itself to the Payor via displaying the Payor's unique private security code (e.g., “Pizza Slice”, as in Nexus demo); Payor is then prompted to enter their PVC **202**.] ii. Payee-Initiated: In this case, the User/Payee is the Request Initiator and submits the Transaction Request **251** (i.e., “I want you to pay me \$10”). In order to be permitted to submit the Transaction Request **251** to the RMN **14**, the User/Payee must be verified by the RMN **14** (via UUC **200** and optionally the PVC **202** entry), thus establishing the validity of the User/Payee identity listed in the Transaction Request **251**. [The same authentication protocol is followed for the User/Payee as for the User/Payor, as defined above.] 3. The Transaction Request **251** is recorded in the RMN **14**, and a unique transaction identifier and timestamp are assigned to the Transaction Request **251**. 4. The RMN **14** attempts to look up the Request Recipient (User/Payee). In one embodiment, the User/Payee has not yet provided their pre-registered UUC **200** (and optionally PVC **202**) to the VP **12** for verification, and therefore would be a non-validated party in the Transaction Request **251** (i.e., the one who did not submit the Transaction Request **251**), based on the UUC **200** provided by the User/Payor in the Transaction Request **251**: a. If the Transaction Request **251** was submitted by the User/Payor, then the RMN **14** looks up the User/Payee as the Request Recipient. b. If the Transaction Request **251** was submitted by the User/Payee, then the RMN **14** looks up the User/Payor as the Request Recipient. 5. If a valid, active UUC **200** (e.g., URL, email address, phone number, etc.) is verified by the VP **12** for the Request Recipient (User/Payee), then the RMN **14** transmits an electronic message to the Request Initiator (User/Payor), containing the clear name associated with the User/Payee's UUC **200** which has been pre-registered with RMN **14**, thereby authenticating the User/Payee to the User/Payor, and informing the User/Payor that a Transaction Request **251** has been submitted to which they are a party. [The electronic message contains the data from the Transaction Request **251** with the clear

name of the Request Initiator (User/Payor), the \$\$ amount of the transaction, and a link to the RMN system.]. Note that the “clear name” is the legal personal or business name of the User/Payee or User/Payor. 6. The RMN **14** also sends an electronic message to the Request Recipient (User/Payee) via any NAT **62**-UTA **16** connected to the RMN **14** using Network **18** (e.g., Cell Phone, pager, personal computer, etc.), as defined by the Request Recipient (User/Payee)'s Rule-Modules **50**, wherein the message contains Transaction Request **251** with the clear name of the User/Payor, the \$\$ amount of the transaction, and a link for login to the RMN **14**. The RMN **14**, using the User/Payee's UUC **200**, then preferably transmits to the Request Recipient (User/Payee) their Private Code **79** via NAT **62**-UTA **16**, confirming that User/Payee is connected to the authentic RMN **14**; RMN **14** then prompts the Request Recipient (User/Payee) to accept the Transaction Request **251** via input of their User/Payee PVC **202**. 7. Per the User/Payee and User/Payor preferences in the RMN **14**, the RMN **14** determines which types of Financial Accounts **65** are accepted by the User/Payee (e.g., Visa, M/C, ACH, etc.), and which rewards apply to the transaction (e.g., discounts offered by User/Payee, or reward points accumulated by User/Payor), and then presents to User/Payor their Account Signatures **81** with the accounts and associated benefits that can be used for this transaction. The Payor then selects the final account and accepts the transaction terms (e.g., \$\$ amount, designated Payee, financial account to use). a. If this is a genuine P2P (person to person) transaction, without a standard merchant and a acquirer Third-Party Platform **28**, wherein the RMN **14** must confirm both the User/Payor and the User/Payee are members of the same closed loop transfer agent (“Transfer Agent”) (e.g., Checkfree, etc.) Third-Party Platform **28**, with which they would have had to pre-register with the RMN **14**. The RMN **14** then sends an electronic message to the closed loop Transfer Agent that two of the members of the Transfer Agent are initiating a P2P transaction via the RMN **14**. 8. The RMN **14** then converts the Tender Suggestion (e.g., Account Signature **81** along with Request Message or Transaction Request **251**) from the RMN **14** proprietary electronic message format into a Payment Transaction in an ISO-formatted message packet and transmits it to a Merchant Acquirer Third-Party Platform **28**. Here are two illustrative embodiments for submission of the payment transaction to the Acquirer: a. User/Payee Is Merchant: The User/Payee is actively registered with a Merchant Acquirer Third-Party Platform **28**, which connects to the RMN **14** via Network **18**. In this case, the Payment Transaction is transmitted by the RMN **14** to the Merchant Acquirer Third-Party Platform **18** associated with the User/Payee's active registration therewith. b. User/Payee Is Not A Merchant: The Payee does not have a relationship with a Merchant Acquirer Third-Party Platform **28**. As such, the Payment Transaction is submitted to a default Merchant Acquirer Third-Party Platform **28** with which all said Users thereby automatically establish an active registration upon said Users registering Rule-Module(s) **50** with the RMN **14**. 10. The Merchant Acquirer Third-Party Platform **28** processes the Payment Transaction and returns an authorization confirmation to the RMN **14**. 11. The RMN **14** transmits a Payment Confirmation electronic message to the User/Payee. 12. The RMN transmits a Payment Confirmation electronic message to the User/Payor.

Processing Financial Transactions Via Arbitrage or Auction Rule-Module(s)

(571) In one illustrative embodiment, processing a User's financial transaction comprises invoking a Rule-Module **50** governing an automated arbitrage, or auction, software program. In one example of this embodiment, a first Account Issuer competes against at least a second Account Issuer to offer the User, or for the User to select, a particular Financial Account **65** for settlement of the financial transaction, either concurrent with a financial transaction or on a post-dated basis.

(572) Another embodiment of Rule-Module **50** may invoke a rewards or loyalty program comprising any one of the following: providing for the User, or offering the User, split-tender processing with the financial transaction via a rewards or loyalty Financial Account **65**; providing for the User, or offering the User, a concurrent or post-dated coupon, printed or electronic, with the financial transaction. Said rewards or loyalty program may comprise input from a product manufacturer. In one embodiment, a printed receipt associated with a User's financial transaction

comprises any one of the following: a coupon for a manufacturer's product(s); a coupon for a service provider's offering(s); a tabulation of discounts provided for the instant financial transaction. In another embodiment, an electronic coupon associated with a User's financial transaction comprises any one of the following: an email for a manufacturer's product(s); an email for a service provider's offering(s); an email tabulation of discounts provided for the instant financial transaction.

(573) In another embodiment, the UTA-VC **204** is appended to the transaction data of the Financial Transaction Request Packet **251** and a Rule-Module **50** of the RMN **14** thereby invokes a real-time rewards discount during a financial transaction. The RMN **14** may invoke a rewards Financial Account **65** with which processing of a split-tender financial transaction proceeds, wherein via the RMN **14**, the User's purchase is partially split between, for example, a Wells Fargo® Visa® Debit Financial Account **65**, and a Safeway® Club Discount rewards Financial Account **65**. Alternatively, the RMN **14** may invoke a real-time coupon which is transmitted to the User's mobile NAT **62** via a barcode image for presentation on the NAT's **62** Display **7** and scanning via the POS Terminal **2-UTA 16**, with which processing of a split-tender financial transaction proceeds, wherein via the RMN **14**, the User's purchase is partially split between, for example, a BofA® MasterCard® Debit Financial Account **65**, and a Kellogg's® cereal coupon deduction from the retail price.

(574) In one example, an arbitrage or auction Rule-Module **50** may automatically invoke a Financial Account **65**, whether for payment, rewards, or coupons, as the default option which the User can select for processing the financial transaction, based upon a competitive bidding process (arbitrage or auction) amongst Account Issuers, product manufacturers, and other parties via the RMN **14**. The auction may be taking place in real-time during the financial transaction, or at any previous time wherein the results are invoked by Rule-Modules **50** when a financial transaction is being processed.

(575) In another embodiment, an arbitrage or auction Rule-Module **50** may automatically invoke a plurality of Financial Accounts **65** as default options from which the User can select for processing the financial transaction. In this embodiment, the Financial Account(s) **65** may be displayed to the User in a Ranking **96** manner, comprising any one of the following: sequential display, a Financial Account **65** at a time, until the User makes a selection; in parallel, a plurality of Financial Accounts **65** at a time or concurrently, optionally including visible indicators. Said visible indicators may comprise any one of the following: the respective Rankings **96** of the Financial Accounts **65**; the offered incentives for selecting the respective Financial Accounts **65**.

(576) In one embodiment, the User may elect to decline any offered Financial Account(s) **65**, in which case a Rule-Module **50** invokes at least one additional Financial Account **65** to offer the User for processing the financial transaction.

(577) In one embodiment, an arbitrage or auction Rule-Module **50** computes the automatic selection of a Financial Account **65** to offer the User based up predetermined criteria comprising any one of the following: improving a transaction benefit for an account issuer; improving a transaction benefit for the user, and; improving a transaction benefit for a payee.

(578) In one embodiment, an arbitrage (or auction) Rule-Module **50** comprises a computational program where an Account Issuer can bid per financial transaction, or per purchased product or service, to buy space on a retailer or a manufacturer's printed or electronic receipts which results from a financial transaction. In another embodiment, an Account Issuer and/or a manufacturer split revenues with a retailer where a User is conducting a financial transaction. The exact amount of the split is may reflect the several weighted factors, of traffic, purchase-through-rates and other aspects.

(579) An Account Issuer signs up via the Rule-Module Nexus **14** and, once approved, is able to select the format of coupons, the retailer(s), the type of retailer(s), the User(s), and/or the type of User(s) for which an Account Issuer wants to offer competitive Financial Account(s) **65**, competitive transaction processing and/or competitive coupons/rewards.

(580) In an alternative embodiment whether via the Internet or at the point of sale (POS), an Account Issuer can bid on keywords which a User may input or click on when Internet browsing. When a User clicks or inputs a keyword on which an Account Issuer has the highest bid, the Account Issuer can issue an electronic or printed coupon, make an adjustment to Financial Data in a User's rewards/loyalty Financial Account **65**, and/or influence the Ranking **96** of a Financial Account **65** which may be offered to the User for selection in processing the financial transaction. At a point when the User is shopping either online or at a point-of-sale, the Account Issuer may be paid by a retailer or manufacturer via the Rule-Module Nexus **14** for having influenced the User's selection of a product, retailer or Financial Account **65**.

(581) In another embodiment, an Account Issuer such as a retailer or manufacturer can bid on keywords relating to Ancillary Transaction Data **172** of a User's financial transaction over the Internet or at POS. Prior to, or concurrent with, a User making a purchase, a keyword associated with that purchase may be the subject of such bids, said keyword comprising any one of the following: the name or type of product; the geographic location of a purchase; the name or type of retailer; the dollar amount of the purchase; the manufacturer of the product; and the like. When a User purchases a product on which an Account Issuer has the highest bid, the Account Issuer can issue an electronic or printed coupon, make an adjustment to Financial Data in a User's rewards/loyalty Financial Account **65**, or influence the Ranking **96** of a Financial Account **65** which may be offered to the User for selection in processing the financial transaction. At a point when the User is shopping either online or at a point-of-sale, the Account Issuer may be paid by a retailer or manufacturer via the Rule-Module Nexus **14** for having influenced the User's selection of a product, retailer, or Financial Account **65**.

(582) In another embodiment, an Account Issuer can bid on keywords relating to Ancillary Transaction Data **172** of a User's financial transaction over the Internet or at POS. Prior to, or concurrent with, a User making a purchase, a keyword associated with that purchase may be the subject of such bids, said keyword comprising any one of the following: the name or type of product; the geographic location of a purchase; the name or type of retailer; the dollar amount of the purchase; the manufacturer of the product; and the like. When a User purchases a product on which an Account Issuer has the highest bid, the Account Issuer can issue an non-coupon advertisement, on a printed receipt, or in an electronic receipt, associated with the financial transaction. At a point when the User is shopping either online or at a point-of-sale, the Account Issuer may be paid by a retailer or manufacturer via the Rule-Module Nexus **14** for having influenced the User's selection of a product, retailer, or Financial Account **65**.

(583) In an illustrative embodiment, via the Rule-Module Nexus **14**, an Account Issuer can either bid low on the same term, looking for keywords with a large bid gap (more on that later) or an Account Issuer can bid on very specific keywords with little competition. The Rule-Module Nexus **14** pricing computations may automatically adjust the cost of a keyword. If the Rule-Module Nexus **14** data shows that a keyword is less likely to turn into actionable business results (Purchase-Through-Rates)—such as a related online or POS sale or financial transaction, registrations, phone calls, or newsletter signups—the Rule-Module Nexus **14** may reduce the price an Account Issuer pays for that keyword.

(584) The Rule-Module Nexus **14** may provide statistics to an Account Issuer, reflecting basic User selection(s) and revenue information as well as monthly totals. The Rule-Module Nexus **14** may also administer bidding on keywords via a software engine comprising a formula or algorithm such as the example below:

$$(585) s = \sqrt{\frac{1}{N} \cdot \text{Math.} \sum_{i=1}^N (x_i - \bar{x})^2}$$

(586) In another illustrative embodiment, NAT **62** transmits a data packet to the RMN **14** which invokes a Rule-Module **50** for processing a financial transaction, said data packet comprising any one of the following: the User's geographic location based on GPS coordinates, the UTA-VC **204**

merchant's address, descriptions of products being purchased, and the like. As such, a rewards Rule-Module **50** may be invoked to offer the User a coupon or advertisement that is customized for the User's real-time financial transaction, comprising any one of the following: User location, time of day/night, products being purchased, and the like.

(587) User Account Registry (UAR) for Financial Accounts

(588) FIG. **11A** illustrates two proprietary Financial Accounts **65** within a UAR **15** registered to a User of the RMN **14**. The Rule-Modules **50** comprise which govern the Financial Account **65**; the Account Transaction Data **172** optionally stores information relating to a particular transaction. As is known in the art, the processing mechanisms and data structure methods can be structured in a variety of ways.

(589) In an illustrative embodiment, the UAR **15** comprising "live" account data is "online", whereby it is remotely located from a NAT **62**, a UTA **16**, and a Terminal **2**. As such, whether during a financial transaction originating by a User via Point-of-Sale (POS), Internet, Mail Order, or Telephone Order, the UAR **15** is only accessible by the User via Network **18** connectivity.

(590) FIG. **11B** illustrates another embodiment, wherein an electronic Audible/Visible Signature **81** is matched with a Financial Account **65** of a UAR **15**. In one embodiment, the Audible/Visible Signature **81** is "thin-client", comprising no "live" financial data and no "live" Financial Account **65** number, and may be stored with a NAT **62** to facilitate a User's selection of a Financial Account **65** for processing an online financial transaction. Note that the Audible/Visible Signature **81** may comprise any one of the following: a visible display of an electronic Visible Signature **81**; a visible text message, and; an audible signal of an electronic Audible Signature **81**.

(591) In one illustrative embodiment of the invention, the UAR **15** within the RMN **14** comprises two Financial Accounts **65** comprising Rule-Modules **50** which govern, for example, a per purchase spending limit, a time of day use, geographic location, type or nature of financial transactions, a day of week use, certain merchant use and/or the like, wherein an additional verification may be required when using the NAT **62** outside of the restriction. The restrictions may be personally assigned by the User, the Account Issuer or the RMN **14**. For example, in one illustrative embodiment, the account may be established such that purchases above \$X (i.e., the spending limit) must be verified by the User using a special secondary Personal Verification Code (PVC) **202** which may be recognized by the RMN **14** as being unique to the User and the correlative financial transaction wherein the Financial Account **65** is selected. The restrictions may also be restricted by other User parameters, such as, geographic region (NAT **62** may only be used in a certain geographic region), product type (NAT **62** may only be used to purchase a certain product type), or Internet use only (NAT **62**) transactions.

(592) In yet another embodiment, settlement of a financial transaction occurs on a predetermined time schedule, comprising any one of the following: net-30 settlement terms; recurring monthly charges; and the like. In one embodiment, the financial amounts from an online transaction are deposited into an escrow account for an Internet Account Issuer or a User, instead of being directly credited to or debited from a User's Financial Account **65**. This may be the case where a purchase of a product is made over the Internet **18** using an online auction provider such as eBay™, wherein funds may only be released upon delivery or receipt of the purchased product.

(593) In FIG. **18**, another illustrative embodiment is shown in which there is a UAR **15** comprising a plurality (e.g., at least two) proprietary Financial Accounts **65** of a User and/or at least one Account Issuer Account in the Rule-Module Nexus **14**.

(594) In FIG. **18I**, another illustrative embodiment is shown including a RMN **14** Platform and Service Stack. This embodiment also is illustrative of exception item support or lifecycle management. Here, putting the RMN **14** in front of the Account Issuer or acquirer of record, platform **28**, ensures that the Acquirer retains the ability to adjudicate exception items. In addition, because credit transactions could now be "pre-authorization secured with PVC **202**" via the RMN **14**, the volume of chargebacks could drop. In an illustrative embodiment, Users may "seal" their

Financial Transaction Authorization Requests **251** with a PVC **202** or other security tool managed via the RMN **14**, exception items should be minimized when compared to current market behavior. In the event that exceptions and charge-backs occur, the tracking of these items and data support may occur via the RMN **14** to more accurately account for transactions and provide an audit trail, particularly with respect to Point of Sale transactions such as rewards or split-tender processing previously not supported (e.g. United Airlines® Mileage Plus Points). In the event of split-tender processing, settlement of the financial transaction is divided according to Rule-Modules **50** in the RMN **14**, between or among pre-registered Financial Accounts **65** of the User. In an illustrative embodiment, the RMN **14** data flow supports existing infrastructure (e.g. Visa® Reengineering Disputes Infrastructure) to ensure that handling of exception items is a value-added service Acquirers may provide their merchant customers.

(595) In one embodiment, there may be at least one electronic Master User Account Registry **15** platform comprising all of the Financial Accounts **65** in the Rule-Module Nexus and there may be at least one electronic Local or Subset User Account Registry **19** platform comprising a sub-set of the Financial Accounts **65** in the Rule-Module Nexus. In another embodiment, a UAR **15** comprising Financial Accounts **65** of the User reside within Third-Party Platforms **28**, external to the Master RMN **14**.

(596) In one embodiment of a Master UAR **15**, a RMN **14** provides “pan-payment portfolio” rewards and anti-fraud programs.

(597) FIG. **13A** illustrates one such embodiment, wherein the RMN **14** and Master UAR **15** may convert, exchange, translate and/or aggregate financial data via rewards or loyalty programs, optionally with communications with Third-Party Platforms **28**. In one example, the Master UAR **15** and the RMN **14** operate in conjunction with a software neural network and may electronically aggregate and project purchasing patterns within a Registry Financial Account **65**, and across a plurality of Registry Financial Accounts **65** of the User, to more accurately reflect the payment preferences of the User. In this manner, more useful, pertinent rewards programs and benefits can be formulated for the User.

(598) In another example, the Master UAR **15** and the RMN **14** operate in conjunction with a software neural network and may electronically aggregate and project usage patterns within a Registry Financial Account **65**, and across a plurality of Registry Financial Accounts **65** of the User, to more accurately reflect the payment patterns of the authentic User. In this way, fraud detection is accelerated and enhanced, given that abnormal or deviating payment patterns can be more readily prevented and detected, since an unauthorized user may attempt to disperse fraudulent purchases across several Registry Financial Accounts **65** in the hope that no unusual payment activity in any one Registry Financial Account **65** might be detected. Here, the unusual payment activity may be detected via an abnormal pan-portfolio payment pattern, involving a plurality of Registry Financial Accounts **65**.

(599) Pan-Portfolio Analytics: Reports/Queries, Security, Data-Mining

(600) FIG. **34** illustrates an embodiment for accessing a plurality of proprietary Financial Accounts **65** via online user account registry, wherein accessing comprises any one of the following: querying, editing, and transacting. In one embodiment, a User's query may be accompanied by invoking an arbitrage Rule-Module **50**, presenting advertisements or rewards incentives to the User while the User is querying their Financial Account(s) **65**.

(601) FIG. **35** illustrates an embodiment for processing pan-portfolio analytics, wherein a User may access their UAR **15** via a Rule-Module **50** comprising any one of the following: a key-word search; a report on activity in their Financial Accounts **65**, such as spending report by category of expenditures, by chronology of expenditures, by tax-deductibility, and the like.

(602) FIG. **35**: Pan-Portfolio (“Pan-Portfolio” means Accessing a Plurality of Financial Accounts within a User Account Registry) Analytics enable the development of mathematical models and neural networks for fraud-detection, rewards programs, micro-merchandising, data-mining, and the

like. Said Pan-Portfolio Analytics comprise data compiled via the Rule-Module Nexus from a plurality of otherwise inaccessible Proprietary Financial Accounts. As such, pan-portfolio analytics are compiled via a plurality of otherwise inaccessible Proprietary Financial Accounts being rendered accessible by the Rule-Module Nexus.

(603) FIG. 36 illustrates an embodiment wherein as determined by the RMN 14, the PVC Association Platform 909 may also forward the PVC 202, comprising any one of the following: forwarding the PIN to a debit processing Network 18, and authorizing a PIN-enabled debit financial transaction; forwarding the PVC 202 to a debit financial transaction processing Network 18, and authorizing a PIN-enabled debit financial transaction; forwarding the PVC 202 to a financial transaction processing Network 18 and authorizing a PIN-enabled financial transaction; forwarding the PVC 202 to a financial transaction processing Network 18 and authorizing a non-PIN enabled financial transaction. As such, the User may “pre-PIN” any or all of their Financial Accounts 65, enabling cash-back and cash advance financial transactions. The RMN 14 may be enable a PIN-validated debit transaction without the User providing their PIN to the RMN 14, such that if the User's accessed to their UAR 15 is protected by requiring a PVC 202, a debit Financial Account 65 therein may be forwarded to process a financial transaction with the RMN 14 appending a confirmation that the User's transaction was validated by having submitting their PVC 202 to the VP 12 for verification of their authority to access the UAR 15. As such, a PIN-less debit transaction may be authorized for processing.

(604) FIG. 36A illustrates an embodiment wherein the RMN 14 translates a PVC 202 into a “live” PIN to forward with the Transaction Request 251. In this example, the PVC Association Platform 909 comprises a PIN Roster 171, matching the pre-registered “live” PIN to its associated Financial Account 65. As used herein, “live” PIN refers to a PIN which the User has already associated to a Financial Account 65 via an Account Issuer Third-Party Platform 28, such that the PIN enables approval, processing, and settlement of a financial transaction for its associated Financial Account 65 (e.g., a Wells Fargo® Visa® debit transaction; a Bank of America® debit transaction, etc.). Therefore, when the User enters a PVC 202 for executing a financial transaction via the RMN 14, upon the User's selection of an Account Signature 81 (e.g., Acc't 6 Debit), the PVC Association Platform 909 accesses the PIN Roster 171 and invokes the appropriate PIN for Acc't 6 Debit (e.g., 5712), whereby the “live” PIN is automatically forwarded with the Transaction Request 251 to the Acquirer Third Party Platform 28.

(605) In general, as used herein, when the RMN 14 “translates” data, this means the RMN 14 takes inbound information and converts it into, and/or it may be replaced by, alternative data which is associated with the original inbound information pursuant to Rule-Modules 50, the UAR 15, or other relational databases with the RMN 14. In addition, this may means that data proprietary to the RMN 14, such as a UUC 200 or a PVC 202, may be respectively converted into, and/or replaced by, a “live” Financial Account 65 number with it associated Financial Account-ID, and a “live” PIN, for including the Financial Request Packet 251 to forward to an Account Issuer or Acquirer Third-Party Platform 28.

(606) FIG. 38A illustrates an embodiment of an overall flowchart illustrating the major functions and operations of the pan-portfolio Neural Subsection 99 within the RMN 14. First Neural (or Network) Model 801 is trained (Step 701) using data describing past transactions from the UAR 15. Then data describing the Neural (or Network) Model 808 is stored (Step 702). Once the Neural (or Network) Model 801 description is stored, system RMN 14 is able to process current transactions. RMN 14 obtains data for a current transaction (Step 703), and applies the current transaction data to stored the Neural (or Network) Model 801 (Step 704). The Neural (or Network) Model 801 determines a fraud score and reason codes, which are output (Step 705) to the User, the Prior Pattern Platform 806, or to Third-Party Platform 28.

(607) FIG. 38B illustrates an embodiment of overall functional architecture of the pan-portfolio Neural Subsection (NS) 99 of the RMN 14. This Neural Subsection 99 of the RMN 14 is broken

down into two major components: Neural (or Network) Model **801** and Transaction Processing Platform **802**. Neural (or Network) Model **801** uses Past Data **804** to build Neural Network **808** containing information representing learned relationships among a number of variables. Together, the learned relationships form a model of the behavior of the variables. Although a neural network is used in the preferred embodiment, any type of predictive modeling technique may be used. For purposes of illustration, the invention is described here in terms of a neural network. Transaction Processing Platform **802** performs three functions: 1) it determines the likelihood of fraud for each transaction by feeding data from various sources Property Data **805**, Area Data **806** into Neural Network **808**, obtaining results, and outputting them via Output Platform **807**; 2) when applicable, it creates a record in the Prior Pattern (PP) Platform **806** summarizing past transactional patterns of the customer; and 3) when applicable, it updates the appropriate record in the Prior Pattern Platform **806**.

(608) The Neural (or Network) Model **801** further comprises use software for “learning” relationships through repeated exposure to data and adjustment of internal weights. They allow rapid model development and automated data analysis. Essentially, such networks represent a statistical modeling technique that is capable of building models from data containing both linear and non-linear relationships. While similar in concept to regression analysis, neural networks are able to capture nonlinearity and interactions among independent variables without pre-specification. In other words, while traditional regression analysis requires that nonlinearities and interactions be detected and specified manually, neural networks perform these tasks automatically. For a more detailed description of neural networks, see D. E. Rumelhart et al, “Learning Representations by Back-Propagating Errors”, *Nature* v. 323, pp. 533-36 (1986), and R. Hecht-Nielsen, “Theory of the Backpropagation Neural Network”, in *Neural Networks for Perception*, pp. 65-93 (1992), the teachings of which are incorporated herein by reference.

(609) Neural networks learn from the “training” process, the general techniques of which involves the following steps: 1) Repeatedly presenting examples of a particular input/output task to the neural network model; 2) Comparing the model output and desired output to measure error; and 3) Modifying model weights to reduce the error. This set of steps is repeated until further iteration fails to decrease the error. Then, the network is said to be “trained.” Once training is completed, the network can predict outcomes for new data inputs.

(610) Examples of software systems for running platforms in the Neural Subsection of the RMN **14** include the Internet Fraud Screen (IFS) system developed by CyberSource Corporation, of Mountain View, Calif., and the FraudShield™ system, developed by ClearCommerce Corporation, of Austin, Tex. An example of a case-based fraud detection system is the Minotaur™ system, developed by Neuralt Technologies, of Hampshire, UK.

(611) FIG. **38C** illustrates an embodiment of a multiple-output Predictive Model **600** of the pan-portfolio Neural Subsection (NS) **99** subsection of the RMN **14**, which provide Rule-Modules **50**, comprising any one of the following: predictive fraud, predictive purchasing patterns, predictive rewards redemption, predictive risk-management tools, predictive Financial Account **65** preferences, predictive advertising impact on the User, and the like. Predictive Model **600** has a set of “n” input variables (Var1-Var(n)) representing a set of “n” possible single actions. The Predictive Model **600** provides as output the marginal probability “P” of a given complex action (actions “A”, “Z”) being applied to the delinquent account. Since actions “A”, “B”, . . . , “Z” represent an exhaustive enumeration (i.e., partitioning) of all the possible complex actions to be taken, a constrained optimization is performed to ensure that the provided probabilities are normalized (sum up to “1”). This normalization may be avoided by constructing individual predictive models to estimate the marginal probability for each individual complex action. An implicit segmentation is imposed by setting a low threshold for each marginal probability (e.g., “R” could be defined as representing the segment of the population for which P(action A) is less than a specified threshold T.sub.A). By scoring the pan-portfolio (or, plurality proprietary Financial Accounts **65** in the UAR

15) data, the desired segmentation is obtained. The probability to pay given a complex action is computed either by using the predictive model estimate or by using the previously discussed prior probability. The prior probability for a population segment is computed as the probability to pay given all possible actions whose marginal probabilities exceed the corresponding thresholds $T_{sub.I}, T_{sub.J}, \dots, T_{sub.K}$. As sufficient data is gathered for sparse complex actions, the prior probability can be modified to reflect the success or failure of the complex action.

(612) In one embodiment, the RMN **14** identifies a pan-portfolio nomenclature, or numbering convention, via the Neural Subsection **99** for risk-management tool improvement using data from a plurality of proprietary Financial Accounts **65**. In this embodiment, the RMN **14** uses a 16-digit alphanumeric code for UPC **67** for each merchant wherein there would also be a uniform numbering or coding sequence assigned to associated categories, such that fast food restaurants or quick-service convenience stores. Further, when an acquirer processor Third-Party Platform **28** establishes connectivity with RMN **14**, the UPC **67** would be assigned with a cleansed list of merchant names being used by an acquirer's Third-Party Platform **28** (e.g., UPC **67** unifies all nomenclature being used for "McDonald's®" such as McD's, MickeyD's, McDonald's, McDonald's Hamburgers, etc.). Further, the UPC **67** may comprise a code identifying a payee's retail site location, its Financial Account **65** acceptance policies, the Display **7** capabilities of its Terminal **2-UTA 16** at different payee retail sites. For example, a McDonald's® in New York may accept American Express®, whereas a McDonald's in Iowa only accepts Discover®. In this way, RMN **14** more constructs more accurate Financial Transaction Response Packets **252** comprising presentation of Financial Accounts **65** which are applicable to the matrix of Rule-Module **50** applicable to a particular financial transaction, for a particular User, and for a particular payee, and the like. Further, the fraud detection capabilities can be more precise with respect to specific payee(s), geographic locale(s), and merchant categories.

(613) In one embodiment, the RMN **14** executes a thin-client poll (or pull) of a User's credit score, in order to make a determination comprising any of the following: a) if the User is sufficiently creditworthy for the RMN **14**, or a merchant's Third-Party Platform **28**, to offer said User a new Financial Account **65**, and/or; b) if the User's credit score would be adversely impacted by issuance of a new Financial Account **65**, whereby the RMN **14** may inform the User of the results of the inquiry prior to offering the User a new Financial Account **65**.

(614) In another embodiment, the RMN **14** enables the User to download or export data from the User's UAR **15** to an accounting software program such as Quickbooks.

(615) On-Line Transmissions

(616) There are also several embodiments of User-customized Execution Commands **52** that govern access to electronic online transmissions and associated data such as web sites, web site content and platforms. An electronic transmission is the online communication of content which is non-Financial Data and is not a financial transaction.

(617) In one embodiment, an Execution Command **52** governing electronic transmissions for data access is a Global Execution Command (GEC) **55** that is unique to the User, and said GEC **55** is not invoked as a default for all Users and for all online transmissions of the RMN **14**. Each such Execution Command **52** is optionally invoked by the User Verification Approval Code **206** serving as the Pattern Data **54**. This Execution Command **52** is a software command that provides an authorized User access to any secured electronic non-Financial Data, such as those on third-party **28** platforms. Invoking this Execution Command **52** enables the User to simultaneously access all Internet chat or messaging forums, web sites and online platform content to which the User has authorization.

(618) In another embodiment, the third-party **28** being contacted by the User for data access is also identified by the Verification Platform **12** using public/private key cryptography. Once the third-party is successfully identified by the Verification Platform **12**, this invokes a Rule-Module **50** in the Nexus which is unique to this third-party and which is used to confirm to the User that the

correct Third-Party Platform **28** was accessed.

(619) In another embodiment, the Global Execution Command **55** is an Execution Command **52** that activates any of the following: an online or Internet-connected device, such as a wireless pager, a wireless or tethered telephone, a network computer, an exercise machine that is connected to the Internet, an electronic book; an online public access Internet computer, an automobile or household appliance that is connected to the Internet, an Internet-connected personal digital assistant such as a Palm Pilot™; an online photocopy machine; an Internet-connected digital audio player such as the Rio™. In several such instances, the executed Rule-Module **50** renders the online or Internet connected device operational and permits the User that has gained access using their bid UUC **200** (or optionally their bid UUC **200-PVC 202**) to conduct online activity to control or otherwise access the above mentioned Internet connected devices. For example, in one embodiment, an exercise machine incorporates a UIA **16** and is connected to the Internet. A User of the exercise machine provides their bid UUC **200** (or optionally their bid UUC **200-PVC 202**) to the UIA **16**, which is transmitted to the Verification Platform **12** for a matching determination. Upon the User being verified by the VP **12**, and the exercise device is identified using its UIA-VC **204**, the Rule-Module **50** executes a command allowing the User to activate the exercise device. Optionally, additional Rule-Modules **50** allow a User to save the details of their exercise activity (number of times, weight amount, date of exercise, etc.) on that exercise device as Pattern Data **54**, in order to keep track of past performance and as a template for future exercise routines.

(620) In another embodiment, an Internet-connected electronic book that incorporates a UIA **16**, is activated when the Verification Platform **12** successfully identifies the User. This allows the User to download text and graphics of complete novels or films for which they have previously paid.

(621) In another embodiment, a personal digital assistant, such as the Palm Pilot™, incorporates a UIA **16**. When activated after the Verification Platform **12** has successfully identified the User, the personal digital assistant permits the User to download and take online academic examinations. In another embodiment, an Internet-connected digital audio player such as the Rio™, incorporates a UIA **16**. When activated as a result of successfully verification of the User by the Verification Platform **12**, the audio player permits the User to download music for which they have authorization. Optionally, additional Rule-Modules **50** can track how many pages of the electronic book have been displayed and can retain a bookmark for the most recently read page. Optionally, additional Rule-Modules **50** can track how many times a downloaded electronic audio track has been played.

(622) Upon the Verification Platform's **12** successful verification of the User from their bid UUC **200** (or optionally their bid UUC **200-PVC 202**), other illustrative embodiments of Execution Commands **52** governing electronic transmission access comprise permitting the User: to access their driver's license online and validate to an authority their eligibility to drive a car, to purchase restricted products like alcohol or tobacco, to pick up a prescription at a pharmacy, or to access a restricted entertainment event such as an R-rated film being shown in theatres; to access their credit-rating and validate to a cashier their eligibility for check-writing privileges; to access an Internet web site and enter a real-time chat room with other people on-line.

(623) In another illustrative embodiment of the invention, User verification further comprises a User facial photograph display, wherein upon the VP **12** outputting a positive matching determination that the User is authorized to access the rule-module database, the RMN **14** transmits the User's registered facial photograph **7** for the UTA **16** or the Transaction Terminal **2** to present via a Display **7** for a third-party present, such as a pharmacist, in real-time during the financial transaction, to visually compare the User's actual face with the User's displayed facial photograph **7**, for affirming or denying that the authentic User is providing the UUC **200** and/or PVC **202**.

(624) Further embodiments of Execution Commands **52** governing electronic transmission access include entitling a User to extend an online User-customized session by repeating their User-customized session log-in by providing their bid UUC **200** (or optionally their bid UUC **200-PVC**

202) when periodically queried to do so by the Verification Platform **12** or Rule-Module Nexus **14**, to access customized radio or television programming, wherein the User can be provided with customized programming, with or without time restrictions, that reflects predetermined preferences, such as a channel broadcasting only news on companies in which the User has an investment or a channel broadcasting only music from Broadway theater shows which the User has seen or indicated a desire to see, to access restricted portions of corporate intranet Network **18** platforms on a selective basis, based upon predetermined Pattern Data **54**, such as the User's job title or company division, to access their travel reservations and validate to the admittance attendant that the User is eligible to travel, such as boarding a particular flight or a specific train, on a predetermined day, at a predetermined time, and to sit in a predetermined seat, to access online position "papers" of User-customized political candidates and electoral ballot initiatives, and validate to an authorized third-party that the User is eligible to vote in particular elections, such as voting for a particular candidate running from a particular User-customized district.

(625) There are several embodiments of User-customized Execution Commands **52** governing the processing of electronic data and electronic transmissions. Such Execution Commands **52** can govern: User-customized notification preferences for such electronic transmissions as real-time medical updates, pending Internet auctions, electronic stock trades and the like; User-customized instructions for User-location designating, for example, that the User may be located by third parties via whichever UTA **16** the User is using during an indicated time period, whereby the User can automatically receive their e-mails, instant messages, phone calls, faxes, and the like in real-time at the particular UTA **16** in use by him; User-customized travel customizations such as the User's preferences for lodging accommodations, travel costs, food, travel locations, and the like.

(626) In other illustrative embodiments, upon verifying the User's UUC **200** (or optionally a UUC **200**-PVC **202**) packet via the Verification Platform **12**, the RMN **14** enables the User to access stored electronic non-Financial Data comprising accessing any of the following: word-processing files; spreadsheet files; software code; graphics files; audio files; medical records; internet web sites; online audio or graphical content; electronic game content; online chat content; online messaging content; online educational content; online academic examination-taking; online personalized medical and health content, and; server-based computer software programs and hardware drivers.

(627) Further embodiments of User-customized Execution Commands **52** governing the processing of electronic non-Financial Data and electronic transmissions include: User-customized verification presentation preferences depending upon various predetermined criteria such as the verification of a particular counter-parties, the User's sending location, and the like, whereby a User's pre-selected personal identifier, such as a distinct audio or visible sample, is electronically presented to a third-party counter-party of the User's electronic transmission; invocation of User-customized Internet environment preferences, whereby a User's preferences are used to create a customized Internet web portal with the User's preferred search engines, bookmarks, and the like; User-customized data presentation preferences, whereby the priority, formatting and organization of displaying data is predetermined by the User; User-customized Internet search engines, and; User-customized intelligent data tracking and extrapolating agents.

(628) In one embodiment of an Execution Command **52** governing the processing of an electronic transmission, the User-customized Internet search engine is customized to locate, retrieve and present electronic transmissions of the User using an intelligent tracking and extrapolating agent. In one embodiment, the User's customized Rule-Modules **50** provide instructions that even when the User is not logged onto a network, the Pattern Data **54** and Execution Commands **52** are periodically and automatically executed, added, changed or deleted based on the User's previous UTA **16** and online usage patterns. As a result, the User-customized search engine is automatically and progressively refined and customized to the User's evolving preferences and online activity patterns as tracked and interpreted by the User's own electronic, automated intelligent agent.

(629) As an example of the above, the User's intelligent agent can direct the User's search engine to automatically conduct periodic, customized online data retrievals reflecting User-customized priorities for: product or service promotional offers or discounts via email or instant messaging; User-customized investment updates; User-customized medical or health information; competitive product or service prices across a broad range of online Account Issuers; hobby or recreational interests; interactive User-customized online advertisements, wherein product or service providers are permitted to provide unsolicited information to a User based upon certain User-customized criteria; online event calendaring, wherein a User is automatically notified of upcoming events or activities reflecting their interests.

(630) Further, the intelligent agent can extrapolate from the User's existing preferences and online activity patterns to automatically and periodically recommend to the User new data that may expand or delete the User's Pattern Data **54** and Execution Commands **52** based upon the intelligent agent's algorithmic projection of what the User's online preferences and activities will be in the future.

(631) In another embodiment, an Execution Command **52** functioning as an intelligent tracking and extrapolating agent centrally integrates data on the User's Internet browsing to provide User-customized recommendations on new products and services available from any number of Internet web sites or Internet Account Issuers. Examples include the Execution Commands for retrieval of new types of music, books, and investment opportunities that reflect the User's preferences, but that such recommendations are pre-selected based on the Execution Command **52** having automatically conducted competitive price-comparisons from various Third-Party Platforms **28**. In another embodiment, an Execution Command **52** integrates User-customized data from a User's calendaring or scheduling software program to provide the User with customized recommendations on User-customized offering for products, services or upcoming events based on the User's pre-scheduled activities in their online calendar.

(632) In another embodiment, an Execution Command **52** appends a customized, User-customized audio or visible identifier which accompanies an electronic transmission for presentation to the counter-party. This identifier is appended to the User's electronic transmission as a form of "electronic personal signature" to readily notify the counter-party that the authenticated User sent the message. This identifier may be a unique image or sound sampled from the User, or it may be a non-unique, distinct graphical or audio selected by the User to reflect their personal preferences, such as a cartoon image or a favorite sound or audio tone.

(633) In another embodiment where greater security is required, an Execution Command **52** governs the appending of a User-unique network credential or digital certificate to an electronic transmission. If a User employing a UUC **200** (or optionally a UUC **200**-PVC **202**) seeks to append their digital certificate to an electronic transmission, the User stores at least one command to sign electronic documents using their private keys, which are themselves centrally stored on a Rule-Module Nexus **14** server. As such, the User's private keys are invoked as a header of the User's electronic transmission which, in combination with the electronic document itself and an MD5 calculation of the document, together form a digital signature. At a later time, an authorized counter-party can use the User's public key from the Master RMN **14** or a third-party certifier to verify the authenticity of the sender and the electronic document's contents to yield a secure, authenticated electronic transmission. In this way, Users do not have to manage their own private keys, nor do they have to retain physical possession of their digital certificates via smart cards or personal computers with resident User-customized data. In one embodiment, public keys of a particular certifying authority are initially stored in the UTA **16** at the time of construction.

(634) In another embodiment, an Execution Command **52** governs the processing of an on-line, User-customized calendaring program or Internet calendaring web site, wherein the User's online scheduling calendar is automatically updated by the User-customized search engine and the User-customized intelligent search and tracking agent based upon User-customized Pattern Data **54**. This

could include, but would not be limited to, automatically updating the User's online calendar based on upcoming: User-customized entertainment events, User-customized business seminars, User-customized airline discounts to the User's preferred destinations, User-customized candidate and elections bulletins, and the like.

(635) In another embodiment, the User pre-designates Execution Commands **52** governing the processing of electronic transmissions which filter the access and presentation of data when the User is subordinated User who is co-registrant or legal dependant of the primary User himself. Examples of such subordinated Users could be the children or the spouse of a User. Examples of such access and presentation, or viewing, filters may be restrictions predetermined by the primary User governing: subordinated User access to Internet web sites with adult or violent content; subordinated User access to online television or radio programming with adult or violent content; subordinated User access to the Internet **18** with restrictions covering online session length; subordinated User access to educational online resources which are automatically "pushed" to the subordinated User during a particular online session, as pre-determined by the primary User, in order to pro-actively circumscribe the content which a particular subordinated User is permitted to view or download.

(636) In another embodiment, an Execution Command **52** provided to the Nexus **14** by an authorized third-party, such as a User's employer, governs the processing and prioritization of electronic transmissions to the User on an intranet Network **18**. As such, the Execution Command **52** determines which electronic transmissions are automatically "pushed" to the User during a particular online session, as pre-determined by the authorized third-party, in order to pro-actively circumscribe the content which a particular User is permitted to view or download

(637) Embodiments of User-customized Execution Commands **52** governing the display or presentation of electronic transmissions include controlling the organization and prioritization of online content such that text, audio and graphics are displayed according to a User's pre-determined preferences. This includes displaying informational updates in a certain prioritization order, wherein User-customized regional news may be presented prior to national or international news, displaying expenditure records in User-customized categories which reflect anticipated tax deduction categories, such as home improvement expenses, charitable contributions, and the like, displaying customized User-customized Internet web sites or portals, including the User's predetermined bookmarks, preferred web links, calendaring programs, email mail addressing rosters, a plurality of email accounts with their accompanying inbox messages, User-customized instant messaging "buddy" lists.

(638) Other embodiments of User-customized Execution Commands **52** governing the display or presentation of electronic transmissions include: displaying accrued User-customized consumer rewards incentives or customized online advertising according to a User's prescribed priorities, such that skiing apparel is presented to the User at a time based on their calendaring program's designating their scheduled winter vacation or such that an advertisement for new coffee flavors from the User's preferred vendor is presented during the User's morning log-on session; displaying the User's customized fitness program on an Internet-connected exercise machine, whereby the User is reminded of the number of repetitions the User performed at what difficulty level during their last exercise session, and thereby also presents a recommended number of repetitions and a recommended difficulty level for the User's current session.

(639) Other embodiments include Execution Commands **52** governing: presentation or display filters which circumscribe what text, graphic or audio content the User is permitted to view; presentation or display filters which govern which products or services a User is permitted to purchase, such as a subordinated User whose parent is a primary User, and where the subordinated User is prohibited from purchasing cigarettes, is limited in their selection of online Account Issuers, is limited in the amount of online session time the User is permitted to have in a single day, and the like.

(640) In an illustrative embodiment, each verification request and each transmission request, whether successful or not, is logged in the Logging Platform (LF) **42**.

(641) In an illustrative embodiment, more than one Rule-Module Nexus **14** provides fault tolerance from either natural or man-made disasters. In this embodiment, each Verification Platform **12** uses a backup power generator, redundant hardware, mirrored platforms, and other standard fault tolerant equipment known in the industry

(642) Execution Platform

(643) In an illustrative embodiment, an Execution Command **52** of a Rule-Module **50** causes an electronic financial transaction to be executed by the Execution Platform **38**. The Execution Platform **38** may be on a platform which is located within the Master RMN **14** itself, or it may be co-located with an entity platform **28** that is external to the Master RMN **14**. In the event that a designated entity platform **28** cannot be contacted for the electronic financial transaction to be completed, the financial transaction is “declined”.

(644) In one embodiment, if the Account Issuer approves the transaction, the Execution Platform **38** returns a transaction number to the User Account Registry **15**, and the User's Financial Account **65** is thereby adjusted through either a credit or debit. The transaction number is returned to the UTA **16**, which lists the transaction on a daily transaction summary. The User need take no further action since financial transactions are automatically settled, at which point a calculation is made to automatically adjust the User's designated Financial Account **65**.

(645) In another embodiment, the Execution Platform **38** uses Rule-Modules **50** from the Master Rule-Module Nexus **14** which permit an Account Issuer to itself contribute financial directly to an alternative Account Issuer based upon a User's purchases. In such transactions, units of financial are electronically debited from the User Account controlled by the Account Issuer, and corresponding units of financial are electronically credited to the alternative Account Issuer's Financial Account **65**. An example would be that a purchase on a User's Citibank™ Visa™ Financial Account **65** will invoke a Rule-Module **50** wherein: Citibank™ contributes 1% of the User's purchase amount from the financial transaction to the American Lung Association™ registered Financial Account™, or; Citibank™ credits 1% to the User's purchase amount from the financial transaction towards frequent flier miles in the User's Star Alliance™ Rewards Financial Account **65**.

(646) Decryption Platform

(647) In an illustrative embodiment, all messages the Rule-Module Nexus **14** receives, with the exception of those not transmitted via a UTA **16**, comprise a UTA-VC **204**, a sequence number, and a Message Authentication Code (MAC). MACs, also known as cryptographic checksums, are well known in the computer industry, and are used to assure that any changes to the content of the message will be detectable by the entity receiving the financial transaction. The Decryption Platform (DP) **22** validates the message's MAC and checks the sequence number for that particular UIA **16**. If the Decryption Platform **22** determines that both the MAC and the sequence number are valid, the DP **22** uses the unique secret key for that particular UIA **16** to decrypt the message. For the decryption to function properly, the Decryption Platform **22** must comprise a copy of each UIA's **16** DUKPT key table.

(648) If the decryption operation fails, or if the MAC check fails, the message is considered an invalid message. The Decryption Platform **22** logs a warning to the logging facility (LF) **42**, terminates processing for the message, and returns an error message to the originating UIA **16**.

(649) Before the Decryption Platform **22** replies to a message that includes a response key, it encrypts the response message with that response key. The Decryption Platform **22** also generates a MAC for the response and appends it to the message.

(650) In an illustrative embodiment, error messages are not encrypted although the Decryption Platform **22** does include a MAC for message authentication. Such messages never include confidential information. However, most response messages include a status or response codes that

can indicate whether the request succeeded or not. For example, when the Execution Platform **38** declines a financial transaction for a specific reason, it does not return an error message, it returns a normal Financial Transaction Response Message **252** with a response code set to “failed”.

(651) Gateway Platform (GP)

(652) In an illustrative embodiment, the Gateway Platform **26** serves as an intermediary between redundant Verification Platform **12** and redundant in-house Registry **15** platforms, routing electronic financial transactions and/or electronic transmissions from platforms on overload to platforms that have available capacity. The Gateway Platform **26** also periodically queries platforms to ensure that are operative and to alert the RMN **14** or RMN-authorized Third-Party Platform **28** administrator is any server is inoperative.

(653) Prior Fraud Platform (PFP)

(654) In an illustrative embodiment, the PFP **27** is a central repository for UUCs **200**, PVCs **202**, and other Pattern Data **54** which have had prior fraud associated with them. Every new User's UUCs **200** (and optionally PVCs **202**) are checked against all PFP **27** records with the intent of reducing recidivism. In one embodiment, there is an automatic User prior fraud check step, wherein the User's bid UUCs **200** (and optionally PVCs **202**) are compared against previously registered UUCs **200** (and optionally PVCs **202**) in the PFP **27** wherein if a match occurs, the Rule-Module Nexus **14** and Verification Platform **12** are alerted to the fact that the User has attempted to re-register or re-access the Rule-Module Nexus **14** after having already been registered with the PFP **27**.

(655) Firewall (FW)

(656) In an illustrative embodiment, the Firewall (FW) **40** provides a first line of defense against network viruses and computer hackers. All communication links into or out of the Verification Platform **12** and Master Rule-Module Nexus **14** server sites first pass through a secure Firewall **40** Machine.

(657) In an illustrative embodiment, the firewall **40** Machine, an Internet-Local or Subset-Internet router, only handles messages destined for the Gateway Platform **26** machines.

(658) Intra-RMN Communications

(659) In an illustrative embodiment, UTA-equipped Transaction Terminals **2** send packets to Verification Platform **12** and Master Rule-Module Nexus **14** server sites via modem, X.25, or other communication medium. The Verification Platform **12** and Master Rule-Module Nexus **14** server sites rely on an entity to supply the modem banks required to handle the volume of calls and feed the data onto the Master RMN **14** backbone.

(660) In an illustrative embodiment, for communications between Verification Platform **12** and Master Rule-Module Nexus **14** server sites, the FW **40** Machines send out double-length DES encrypted packets. The server site LAN component handles the encryption and decryption: the firewall **40** does not have the ability to decrypt the packets.

(661) In an illustrative embodiment, a properly configured network sniffer acts as an intruder detector as backup for the FW **40**. If an anomalous message is detected, the intruding messages are recorded in their entirety, an operator is alerted, and the Firewall **40** is physically shut down by the sniffer.

(662) In an illustrative embodiment, the Firewall **40** disallows any financial transactions from the internal network to the rest of the Internet. An electronic financial transaction message requires about 400 bytes and registration packets require about 10 to 20 KB. To handle 1000 electronic financial transactions per second and 16 registration packet per second, the Firewall **40** machines are able to process about 400 KB per second.

(663) Logging Platform

(664) In an illustrative embodiment, the Logging Facility **42** logs all electronic financial transaction attempts, whether successful or not, to write-once media, so that a record is kept of each financial transaction and each error that has occurred during the operation of the Verification Platform **12**.

(665) Interconnections and Communications Among the Electronic Verification Platform, Rule-Module Nexus and User Account Registry

(666) A variety of conventional communications media and protocols may be used to connect or place the devices herein in communication. For example, the communications media may be an Internet Service Provider (ISP) configured to facilitate communications over a local loop as may be typically used in connection with standard modem communication, cable modem, dish networks, ISDN, Digital Subscriber Lines (DSL), or any wireless communication media. In one illustrative embodiment, the Retail Subset Platform (RSP) **130** is conjoined with a Subset RMN **17**. In another illustrative embodiment, the UTA **16** is conjoined with the Transaction Terminal **2** via a local area Network **18**, which in turn and interfaces to a remote network (not shown) for remote authorization via the RMN **14**. The Retail Subset Platform **130** may communicate with the remote network via a leased line, such as a T1, D3 line, or the like. Such communications lines are described in a variety of texts, such as, "Understanding Data Communications," by Gilbert Held, which may be incorporated herein by reference.

(667) In one embodiment depicted in FIG. **8**, the Verification Platform **12** platform is physically separate from, but associated and registered with, the Master Rule-Module Nexus **14**. In one embodiment depicted in FIG. **8A**, the User Account Registry **15** platforms are conjoined with the Verification Platform **12** within the RMN **14**, although they may be housed in independent platforms or joint platforms. In an embodiment shown in FIG. **18**, the Rule-Module Nexus **14** is separately located from a User Account Registry **15**, which is conjoined with a Third-Party Platform **28**, but associated and registered with the RMN **14**. In another embodiment depicted in FIGS. **1** and **3**, the Verification Platform **12** is physically integrated with the Rule-Module Nexus **14** and the User Account Registry **15**, whereby the Verification Platform **12**, Master Rule-Module Nexus **14** and the User Account Registry **15** are physically interconnected and integrated together within one server or platform. In these embodiments, communications among the Verification Platform **12**, the Master Rule-Module Nexus **14** and the User Account Registry **15**, in an illustrative embodiment, occur via many different methods and means that are well known in the art. Most depend on the particular communication networks already deployed by the organization or company that deploys the electronic financial transaction authorization system.

(668) In one embodiment, the Verification Platform **12**, the Master Rule-Module Nexus **14** and the User Account Registry **15** are connected via Ethernet **18** to a Local or Subset router, which is connected to a network operations center (NOC) via frame relay lines. Messages are sent among the Verification Platform **12**, the Master Rule-Module Nexus **14** and the User Account Registry **15** using TCP/IP over this Network **18**. In another embodiment, the Verification Platform **12**, the Master Rule-Module Nexus **14** and the User Account Registry **15** are connected via a cellular digital packet data (CDPD) modem Network **18** to a CDPD provider, who provides TCP/IP connectivity from the Verification Platform to an intranet **18** to which at least one Master Rule-Module Nexus **14** is attached.

(669) In yet another embodiment, a Verification Platform **12** is connected via the Internet **18**, as is at least one Master Rule-Module Nexus **14** and at least one User Account Registry **15**. TCP/IP is used to transmit messages from among the Verification Platform **12**, the Master Rule-Module Nexus **14** and the User Account Registry **15**. There are many different ways to connect the Verification Platform **12**, the Master Rule-Module Nexus **14** and the User Account Registry **15** that are well understood in the industry, such as cable networks, wireless networks, telephone networks, the Internet, an intranet, a local area network (LAN), a wide area network (WAN), and an X.25 network.

(670) The Verification Platform **12**, the Master Rule-Module Nexus **14** and the User Account Registry **15** hardware platforms are, in an illustrative embodiment, high-reliability platforms, well known in the art, such as those available from Sun™, Compaq™, Tandem™, IBM™ and the like. Further, the Verification Platform **12**, the Master Rule-Module Nexus **14** and the User Account

Registry **15** software may, in an illustrative embodiment, incorporate scalable platform architecture, well known in the art, such as those available from Oracle™, Sybase™, Informix™, Red Hat™ and the like.

(671) Electronic Verification Platform, Rule-Module Nexus and User Account Registry: Master Platforms and Local or Subset Platforms

(672) In certain embodiments, a Master Verification Platform **12** is responsible for storage of the entire set of UUC **200**, PVCs **202**, digital certificates and the like registered for use with this invention. In an illustrative embodiment, an electronic Master Rule-Module Nexus **14** is responsible for storage of the entire set of Pattern Data **54**, Execution Commands **52**, and Rule-Modules **50** registered for use with this invention. An electronic Master User Account Registry **15** is responsible for storage of the entire set of Financial Accounts **65** of a User and Financial Accounts **65** of an Account Issuer registered for use with this invention.

(673) Each Master Verification Platform **12**, Master Rule-Module Nexus **14** site is, in an illustrative embodiment, made up of a number of computers and platforms connected together over a LAN (known in the industry). Multiple and redundant Master computer sites ensure reliable service in the face of disaster or serious hardware failure at any single central computer site.

(674) In another embodiment, there is at least one Local or Subset Verification Platform **13** server which stores a subset of the entire set of UUC **200**, PVCs **202**, and digital certificates registered for use with this invention. In another embodiment, there is at least one Local or Subset Rule-Module Nexus **17** server which stores a subset of the entire set of Pattern Data **54**, Execution Commands **52**, and Rule-Modules **50** registered for use with this invention. Such Pattern Data **54** and Execution Commands **52** subsets may be circumscribed by any number of criteria including, usage location, usage frequency, usage recency, usage demographics and usage volume of electronic financial transactions. In another embodiment, there is at least one Local or Subset User Account Registry **19** server which stores a subset of the entire set of User Accounts and Account Issuer Accounts registered for use with this invention.

(675) It is preferred that the Master platforms have a Firewall **40** machine which is the entry point of data and messages into these computers, and a Gateway Platform **26** which is a RMN **14** coordinator and message processor.

(676) FIG. **33A** illustrates an embodiment of a distributed system architecture for processing financial transactions via Subset RMN's **17** (and/or Subset UAR's **19**), with use of a National/International (or Global) Gateway Platform **28**, optionally with co-located Master RMN **14** (and/or Master UAR **15**). This distributed system architecture enables payment processors Third-Party Platforms **28** to control returning authorization messages to the Acquirer Platform **28** for it to gateway to the Payment Networks **18** and/or the Account Issuer Platforms **28**. Further, Acquirer Platforms **28** can view the resolution/settlement of the translation of the RMN **14** message, comprising the UUC **200**, before the message gets passed to the Payment Networks **18** and/or the Account Issuer Platforms **28**.

(677) Use-Sensitive Configurations for Verification Platform, Rule-Module Nexus and User Account Registry

(678) As shown in FIG. **3** and FIG. **8A**, in some embodiments the invention has use-sensitive data processing capabilities, wherein at least two Verification Platforms **12**, at least two Rule-Module Nexuses **14**, or at least two electronic Rule-Module Nexuses **14** exist, some of which respectively store a subset of the total data registered with the RMN **14** or RMN-authorized Third-Party Platform **28**.

(679) FIG. **19** through FIG. **21** show use-sensitive embodiments, with Master, Intermediary, and Local (or Subset) configurations.

(680) One embodiment comprises at least one Master Verification Platform **12**, one Master Rule-Module Nexus **14**, which respectively comprise the entire set of all data registered with the RMN **14** or RMN-authorized Third-Party Platform **28**. This embodiment further comprises at least two

Local or Subset Verification Platforms **13**, at least two Local or Subset Rule-Module Nexuses **17**, that are physically apart from each other. Each Local or Subset Verification Platform **13**, Local or Subset Rule-Module Nexus **17** and Local or Subset User Account Registry **19** may comprise a subset of the data contained respectively within the Master Verification Platform **12**, Master Rule-Module Nexus **14**. Data communications lines may allow electronic financial transactions to flow between each Local or Subset Verification Platform **13**, Local or Subset Rule-Module Nexus **17** or Local or Subset User Account Registry **19**, and the Master Verification Platform **12**, Master Rule-Module Nexus **14** or Master User Account Registry **15**.

(681) In this embodiment, electronic financial transactions may first be sent to the Local or Subset Verification Platform **13**, Local or Subset Rule-Module Nexus **17** or Local or Subset User Account Registry **19** for processing. If a party cannot be identified by the Local or Subset Verification Platform **13** or if the requisite Rule-Module **50** or Financial Account **65** is not contained, respectively, in the Local or Subset Rule-Module Nexus **17** or the Local or Subset User Account Registry **19**, the electronic financial transaction is forwarded to the Master Verification Platform **12**, the Master Rule-Module Nexus **14** or the Master User Account Registry **15**. If the parties are identified properly by the Master Verification Platform **12** or if the requisite Rule-Module **50** or Financial Account **65** is located, respectively, in the Master Rule-Module Nexus **14** or the Master User Account Registry **15**, the electronic financial transaction is processed appropriately. In addition, the User's verification information can be transmitted from the Master Verification Platform **12** to the Local or Subset Verification Platform **13**, so that the next time the User will be successfully identified by the Local or Subset Verification Platform **13**. This can likewise occur for the Master Rule-Module Nexus **14** and Local or Subset Rule-Module Nexuses **17**, and Local or Subset Registries **17**.

(682) In another embodiment of a use-sensitive RMN **14** or RMN-authorized Third-Party Platform **28**, the RMN **14** or RMN-authorized Third-Party Platform **28** further comprise a purge engine for deleting a party's User-customized information from the Local or Subset Verification Platform **13**, the Local or Subset Rule-Module Nexus **17** or the Local or Subset User Account Registry **19** platforms. In order to store only records for those parties who use the RMN **14** or RMN-authorized Third-Party Platform **28** more than a prescribed frequency and prevent the overload of platforms with records from parties who use the RMN **14** or RMN-authorized Third-Party Platform **28** only occasionally, the record of a party is deleted from the Local or Subset Verification Platform **13**, Local or Subset Rule-Module Nexus **17** or Local or Subset User Account Registry **19** platforms if there has been no attempt to verify the party upon expiration of a predetermined time limit.

(683) In order to make communications between the Master RMN platforms **14** and the Local or Subset RMN platforms **17** secure, the RMN **14** or RMN-authorized Third-Party Platform **28** further comprises encryption and decryption means, wherein communications between the Master RMN platforms **14** and Local or Subset RMN platforms **17** are encrypted.

(684) In an illustrative embodiment, within the Rule-Module Nexus of the use-sensitive embodiment, each Master Platform (or Master Computer) **10**, each Intermediary Platform (or Intermediary Computer) **60**, and each Local or Subset Platform (or Local or Subset Computer) **34**, has electrical power backup and multiple redundancy in all of its critical hardware and platform systems.

(685) External Computers or External Entity Platforms

(686) In one embodiment, an Execution Command **52** optionally requires the RMN **14**, including the Master Rule-Module Nexus **14** and the Execution Platform **38**, to communicate with at least one external or Third-Party computer or platform **28** to conduct a User's financial transaction. For example, the Execution Platform **38** may need to communicate with: a banking or credit card entity; a merchant's purchasing incentives platform for generating financial; a financial counter-party's computers to determine the correct financial counter-party account for financial disbursement. In this embodiment, at least one Local or Subset Rule-Module Nexus **17** or at least one Local or

Subset User Account Registry **19** is located within an external, Third-Party platform **28**.

(687) On-Line Network and Communications for Financial Transactions

(688) Network transactions are characterized by verifying the User using a communications network such as the Internet, an intranet, or an extranet. The User's bid UUC **200** is submitted through the User's personal UIA **16**, or through a public UIA **16** attached to an ATM or other public Terminal **2**. Parties identified through a digital certificate are registered network entities, such as either the Account Issuer or the Account Issuer. The User is identified through UUCs **200** (or optionally UUCs **200**-PVCs **202**), while the Account Issuer or the Account Issuer, may be identified through the verification of a digital certificate issued by an authorized certifying authority.

(689) In an illustrative embodiment, the User locates the Account Issuer by locating the participating Account Issuer's place of business on the network: the web site, using the network address of the Account Issuer. The User downloads the Account Issuer's digital certificate to the UIA **16** that the User is using. The UIA **16** verifies that the digital certificate provided by the Account Issuer is a valid certificate.

(690) In one embodiment, the UIA **16** transmits the UUC **200** (or optionally the UUC **200**-PVC **202**) to the Master RMN **14** for verification, along with the Account Issuer's digital certificate.

(691) Both parties verify the Financial Accounts **65** to be involved in the transaction. In an illustrative embodiment, this occurs at the Master RMN **14** using the selected Financial Account **65** included in the transaction by the User. The User's Financial Account **65** is thereby selected by the RMN **14**.

(692) In one embodiment, the UIA **16** is actually built-in and/or integrated with a personal Transaction Terminal **2** such as a home computer, or a public Transaction Terminal **2**, for example an airport kiosk. In such an embodiment, the UIA-VCs **204** are not required to verify either party in a transaction.

(693) FIG. **18F** is illustrative of an embodiment wherein connectivity for a contactless communication between a NAT **62** and a UIA **16**-Terminal **2** is optimized using Rule-Modules **50** and a Contactless Connectivity Registry (CCR) **30** via the NAT **62**. A Subscriber Identification Module (SIM) card in the NAT **62** may be used to detect real-time availability of connectivity options, in order to select a default option based upon Rankings **96** of Rule-Modules **50** which take into account speed, cost, and security of connectivity options. In an illustrative embodiment, a NAT **62** may be dual-mode or multi-mode connectivity enabled (e.g., RFID and NFC; RFID, NFC and Cell network; etc.).

(694) FIG. **18G** is illustrative of an embodiment wherein routing for processing a financial transaction is optimized using Rule-Modules **50** and a Network Routing Registry (NRR) **31** via the RMN **14**. Optionally, the RMN **14** provides "best routing" (e.g., lowest cost) for payment processing by directing the bank-owned "on us" transactions directly to the institution's host system for zero-cost payment fulfillment. Optionally, the RMN **14** also supports multiple payment processor options to reduce the risk of getting locked into a single processor.

(695) In another embodiment, the User can be a representative of a business entity that has permission to access the business entity's Financial Accounts **65** to conduct direct transactions with a financial counter-party.

(696) From the foregoing, it will be appreciated how the objectives and features of the invention are met. First, the invention provides an improved financial transaction computer system and method that eliminates the need for a User to possess and manage multiple tokens, in order to authorize a transaction.

(697) Second, the invention provides a financial transaction computer system that is capable of online verification of a User's unique authority to access the Rule-Module Nexus by algorithmically combining their unique User code and personal verification code.

(698) Third, the invention does not require any portable token to store any data which may directly

identify or may directly access an online Financial Account of the User.

(699) Fourth, the invention provides a cost-effective financial transaction system that is practical, convenient, and cost-effective to use and integrate with existing financial transaction systems.

(700) Fifth, the invention provides a system of secured access to a financial computer system that is highly resistant to fraudulent transaction authorization attempts by unauthorized Users.

(701) Sixth, the invention provides a financial transaction authorization system that enables a User to notify authorities that a particular access request is being coerced by a third party without giving notice to the third party of the notification.

(702) Seventh, the invention provides a financial transaction system which verifies itself to the User for added security.

(703) Although the description above contains many specificities, these should not be construed as limiting the scope of the invention but as merely providing illustrations of some of the presently illustrated embodiments of this invention. Thus the scope of this invention should be considered in light of the appended claims and their legal equivalents. Therefore, it will be appreciated that the scope of the present invention fully encompasses other embodiments which may become obvious to those skilled in the art, and that the scope of the present invention is accordingly to be limited by nothing other than the appended claims, in which reference to an element in the singular is not intended to mean "one and only one" unless explicitly so stated, but rather "one or more." All structural and functional equivalents to the elements of the above-described preferred embodiment that are known to those of ordinary skill in the art are expressly incorporated herein by reference and are intended to be encompassed by the present claims. Moreover, it is not necessary for a device or method to address each and every problem sought to be solved by the present invention, for it to be encompassed by the present claims. Furthermore, no element, component, or method step in the present disclosure is intended to be dedicated to the public regardless of whether the element, component, or method step is explicitly recited in the claims. No claim element herein is to be construed under the provisions of 35 U.S.C. 112, sixth paragraph, unless the element is expressly recited using the phrase "means for."

(704) Although the invention has been described with respect to a particular system and method for its use, it will be appreciated that various modifications of the apparatus and method are possible without departing from the invention, which is defined by the claims set forth below.

Claims

1. A computer implemented method for enabling a payment transaction in a computer system executed at least partially by software programs tangibly embodied on a computer readable medium of a computer of a user connected via a communication network to a remote computer comprising a database of payment-related data, said method comprising the steps of: a) accessing the payment-related data of the user stored within the remote computer database, wherein a web browser cross-extension on the user's computer is invoked by the user clicking on a graphical user interface (GUI) widget of the web cross-browser extension as displayed on the user's computer, and wherein secure network access is established to access the user's payment-related data via the remote computer database; b) presenting the user's payment-related data via a pop-up window displayed on the user's computer, without requiring application programming interface integration with a website or a web application of a payee, and without requiring a form re-direct away from a webpage or the web application of the payee, and; c) enabling the payment transaction, wherein the web cross-browser extension invokes an auto-complete function to automatically populate fields of a checkout form presented from the website or the web application of the payee with the user's payment-related data.

2. The method of claim 1, wherein the payment-related data comprises any of the following: an account number; a visible signature; an audible signature; the user's first and/or last name; an

expiration date of the account number; a mailing address associated with the account number; a card verification value (CVV) number; a shipping address associated with the user; browsing data and/or purchasing data of the user; an advertisement customized for the user based on the browsing data and/or the purchasing data of the user, and/or; a rewards incentive customized for the user based on browsing data and/or purchasing data of the user.

3. The method of claim 1, further comprising verifying the user, wherein to access the user's payment-related data in the remote computer database, any of the following data is provided via the web cross-browser extension: a telephone number; an email address; a subscriber identification module (SIM) card number; an internet protocol address; a media access control (MAC) address, and/or; a globally unique identifier (GUID).

4. The method of claim 1, further comprising automatically populating the user's payment-related data on the remote computer database, wherein automatically populating comprises any of the following: a) electronically accessing a credit report of the user, automatically detecting at least one account number and other data of the user from within said credit report, and storing in the remote computer database said account number and said other data of the user, and/or; b) taking a picture of a payment or rewards card, invoking an optical character reading software program from within the remote computer database, and extracting payment-related data from the picture, said payment-related data comprising at least an account number.

5. The method of claim 1, further comprising transmitting the payment-related data via the communication network as secure socket layer data from the remote computer database to the user's computer.

6. The method of claim 1, wherein the remote database computer further comprises an intelligent software agent operable to track the user's Internet browsing via the cross-browser extension and operable to automatically conduct on-line data retrievals reflecting user-customized Web browsing priorities and/or preferences for any of the following: user-customized product and/or service promotional offers and/or discounts; user-customized investment updates; user-customized medical and/or health information, and/or; competitive product and/or service prices across a plurality of on-line merchants, reward incentives providers, banks, internet payment providers, barter providers, insurance companies, credit account organizations, and/or brokerage firms.

7. The method of claim 1, further comprising verifying the remote computer database to the user, wherein a private code, previously registered with the remote computer database and associated with the payment-related data of the user, is transmitted from the remote computer database and displayed to the user via the user's computer.

8. The method of claim 1, further comprising processing browsing and/or purchasing analytics by the remote computer database, comprising: a) collecting the user's browsing data and/or purchasing data via the web cross-browser extension and storing said browsing and/or said purchasing data in the remote computer database; b) the remote computer database surveying said browsing data and/or said purchasing data and detecting a pattern, and; c) presenting analytics, wherein an analysis by the remote computer database of the pattern detected in said browsing data and/or said purchasing data is presented to the user via the pop-up window on the display of the user's computer.

9. The method of claim 8, wherein the pattern further comprises any of the following: a prior pattern; a real-time pattern, and/or; a predictive pattern.

10. The method of claim 8, wherein the analytics further comprises any of the following: a pattern of fraudulent use of the user's payment-related data; a pattern of product purchasing by the user; a pattern of the user's preferences for enabling a payment transaction; a pattern of payment account selection by the user; a pattern of rewards redemption by the user; a pattern of user geographic locations of the user; a pattern retailer locations; a pattern of costs in processing a payment transaction; a pattern by the user of viewing products or services on the web, and/or; a pattern by the user of viewing website content on the web.

11. The method of claim 8, wherein the analysis comprises any of the following: an advertisement customized by the remote computer database for the user, wherein said advertisement comprises a clickable GUI widget; a reward incentive customized by the remote computer database for the user, and/or; an informational notification customized by the remote computer database for the user.
12. The method of claim 1, wherein the web cross-browser extension is a computer program using an application programming interface from the remote computer database to integrate with any of the following web browsers: Google Chrome; Opera; Microsoft Explorer; Apple Safari, and/or; Mozilla Firefox.
13. A system for processing a payment transaction, wherein at least one computer executable program is tangibly embodied on a computer readable medium, comprising: a) a database of payment-related data stored on a remote computer; b) a communication network connecting the remote computer with a user's computer, wherein the user's computer is configured to comprise a browser application that includes a web cross-browser extension that is invoked by the user clicking on a web cross-browser extension graphical user interface (GUI) widget displayed on the user's computer, and wherein secure network access is established to access the user's payment-related data via the remote computer database, and; c) a display on the user's computer, wherein the remote computer is configured to: (i) upon accessing the remote computer database, present the payment-related data to the user via a pop-up window on the display, without requiring application programming interface integration with a website or a web application of a payee, and without requiring a form re-direct away from a webpage or the web application, and; (ii) upon a checkout form from the website or web application of the payee being presented on the display, invoke via the web cross-browser extension an auto-complete function to automatically populate fields of the checkout form with the user's payment-related data.
14. The system of claim 13, wherein the payment-related data comprises any of the following: an account number; the user's first and last name; an expiration date of the account number; a mailing address associated with the account number; a card verification value (CVV) number; a shipping address associated with the user; browsing data and/or purchasing data of the user; an advertisement customized for the user based on the browsing data and/or the purchasing data of the user, and/or; a rewards incentive customized for the user based on browsing data and/or purchasing data of the user.
15. The system of claim 13, further comprising operability to verify the user, wherein to access the user's payment-related data in the remote computer database, any of the following data is provided to the remote computer database via the web cross-browser extension: a telephone number; a browser cookie; an email address; a subscriber identification module (SIM) card number; a media access control (MAC) address, and/or; a globally unique identifier (GUID).
16. The system of claim 13, further comprising operability to automatically populate the user's payment related data on the remote computer database, wherein automatically populating comprises any of the following: a) electronically accessing a credit report of the user, automatically detecting at least one account number and other data of the user from within said credit report, and storing in the remote computer database said account number and said other data of the user, and/or; b) taking a picture of a payment or rewards card, invoking an optical character reading software program from within the remote computer database, and extracting payment-related data from the picture, said payment-related data comprising at least an account number.
17. The system of claim 13, further comprising operability to transmit the payment-related data via the communication network as secure socket layer data from the remote computer database to the user's computer.
18. The system of claim 13, wherein the remote database computer further comprises an intelligent software agent operable to track the user's Internet browsing via the cross-browser extension and operable to direct the user's search engine to automatically conduct on-line data retrievals reflecting user-customized Web browsing priorities and/or preferences for any of the following: user-

customized product and/or service promotional offers and/or discounts; user-customized investment updates; user-customized medical and/or health information, and/or; competitive product and/or service prices across a plurality of on-line merchants, reward incentives providers, banks, internet payment providers, barter providers, insurance companies, credit account organizations, and/or brokerage firms.

19. The system of claim 18, wherein the intelligent software agent further comprises an extrapolating software agent operable to centrally integrate data on the user's Internet browsing preferences as tracked via the cross-browser extension to provide user-customized recommendations on new products and/or new services available from a plurality of on-line merchants, reward incentives providers, banks, internet payment providers, barter providers, insurance companies, credit account organizations, and/or brokerage firms.

20. The system of claim 19, wherein the user-customized recommendations are based on the extrapolating software agent being further operable to: automatically conduct competitive price-comparisons from a plurality of third party platforms, and/or; integrate user-customized data from the user's on-line calendaring and/or scheduling software program(s) to provide the user with user-customized offering for products, services, and/or upcoming events based on the user's pre-scheduled activities in their on-line calendar.

21. The system of claim 13, further comprising operability to verify the remote computer database to the user, wherein a private code, previously registered with the remote computer database and associated with the payment-related data of the user, is transmitted from the remote computer database and displayed to the user via the user's computer.

22. The system of claim 13, further comprising operability to process browsing and/or purchasing analytics by the remote computer database, comprising: a) collecting the user's browsing data and/or purchasing data via the web cross-browser extension and storing said browsing and/or said purchasing data in the remote computer database; b) the remote computer database surveying said browsing data and/or said purchasing data and detecting a pattern, and; c) presenting analytics, wherein an analysis by the remote computer database of the pattern detected in said browsing data and/or said purchasing data is presented to the user via the pop-up window on the display of the user's computer.

23. The system of claim 13, wherein the web cross-browser extension is a computer program using an application programming interface from the remote computer database to integrate with any of the following web browsers: Google Chrome; Opera; Microsoft Explorer; Apple Safari, and/or; Mozilla Firefox.

24. The system of claim 22, wherein the pattern further comprises any of the following: a prior pattern; a real-time pattern, and/or; a predictive pattern.

25. The system of claim 22, wherein the analytics further comprises any of the following: a pattern of fraudulent use of the user's payment-related data; a pattern of product purchasing by the user; a pattern of the user's preferences for enabling a payment transaction; a pattern of payment account selection by the user; a pattern of rewards redemption by the user; a pattern of user geographic locations of the user; a pattern retailer locations; a pattern of costs in processing a payment transaction; a pattern by the user of viewing products or services on the web, and/or; a pattern by the user of viewing website content on the web.

26. The system of claim 22, wherein the analysis comprises any of the following: an advertisement customized by the remote computer database for the user; a reward incentive customized by the remote computer database for the user, and/or; an informational notification customized by the remote computer database for the user.
